Fondamenti di Fisica Matematica
Introduzione alla Teoria delle Equazioni alle Derivate Parziali del Secondo Ordine

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Abstract

These notes are based on the course “Fondamenti di Fisica Matematica, mod. 2” held in Trento between February and June 2024. Parts with the symbol will not be the subject of the exam, however, students are assumed to be familiar with the results therein —notice that these results may be used in other parts of the notes.

I particularly thankful to V. Moretti, whose lecture notes have been the backbone of these. Although these notes are a revised version of those written in 2023 they may contain an outrageous (but hopefully decreasing in time) number of typos and more serious mistakes. Students’ feedback is the best way to address these issues and should be warmly welcomed. For this reason I am grateful to A. Barbieri, E. Barbieri, G. Belletti, R. Bertelli, M. Calabri, E. Floris, S. Morsi, A. Rigotti for their comments and questions, which have lead to a significant improvement of these notes.

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1 Examples of semi-linear PDEs

In this section we will provide a few examples of physically motivated models involving partial differential equations, see [11, 18]. These motivating examples will lead us to a better understanding of the phenomena described by the solutions to these equations.

For later convenience we recall that, by definition, $u \in \boxed{C^k([a,b])}$ —here $a, b \in \mathbb{R}$, $a < b$, while $k \in \mathbb{N} \cup \{0\}$— if $u \in C^k(a,b)$ and for all $\ell \in \{0,\ldots,k\}$, the derivative $u^{(\ell)} \in C^{k-\ell}(a,b)$ extends to a continuous function on $[a,b]$.

The wave equation

In this section we discuss a model for the evolution of the longitudinal vibrations of a string. The latter can be described as a model of $N+1$ interacting particles in the limit $N \to \infty$.

We consider $N+1$ particles with mass $m > 0$ constrained to move without friction on a straight line identified with the x-axis in a fixed inertial frame. We also assume that two nearby particles are connected by a spring of constant $k > 0$ and *rest length* ℓ_0 —*cf.* Figure 1. This

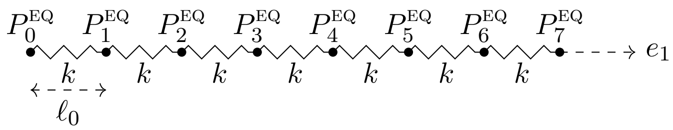


Figure 1: The system of $N = 7$ particles connected by springs at equilibrium.

implies that the force acting on P_j and due to the spring attached at P_{j-1} is

$$\boldsymbol{F}_{j,j-1} = -k(P_j - P_{j-1} - \ell_0 e_1)\,.$$

The unique (up to translations) stable equilibrium configuration of the system is

$$P_0^{\mathrm{EQ}} = O\,, \qquad P_1^{\mathrm{EQ}} = O + \ell_0 e_1\,, \qquad \ldots \qquad P_N^{\mathrm{EQ}} = O + N\ell_0 e_1\,,$$

where we have identified the position of P_0 with the origin O of the chosen inertial frame.

Out of equilibrium, we denote by $P_j(t) = P_j^{\mathrm{EQ}} + u_j(t)e_1$ the position of the j-th particle, $j \in \{0,\ldots,N\}$. Notice that u_j is the displacement of the j-th particle from equilibrium.

The pure equations of motion for the points $P_0,\ldots,P_N$ are given by

$$\begin{aligned} m\ddot{u}_0(t) &= k[u_1(t) - u_0(t)] \\ m\ddot{u}_j(t) &= k[u_{j+1}(t) + u_{j-1}(t) - 2u_j(t)] \qquad j = 1,\ldots,N-1 \\ m\ddot{u}_N(t) &= k[u_{N-1}(t) - u_N(t)]\,, \end{aligned}$$

where $\dot{}$ denotes time derivative while we computed

$$\boldsymbol{F}_{j,j-1} = -k(P_j - P_{j-1} - \ell_0 e_1) = -k(u_j - u_{j-1})\,.$$

The solution to system (1.1) is uniquely determined by assigning initial conditions

$$u_j(0) = u_{j,0}\,, \qquad \dot{u}_j(0) = u_{j,1}\,, \qquad j \in \{0, \ldots, N\}\,. \tag{1.2}$$

At this stage we would like to perform a "continuum limit" $N \to \infty$. As we shall see, this will turn the system of ordinary differential equations (1.1) into a single partial differential equation for a single function u.

The derivation requires to impose a suitable N-dependence on the parameters k and ℓ_0. In particular, for an arbitrary but fixed $L > 0$ we assume $\ell_0 := L/N$ and set $x_j := jL/N$, $j \in \{0, \ldots, N\}$. This implies that the rest length of the springs vanishes in the limit $N \to \infty$.

Moreover, we assume that there exist $v_0 \in C^2([0, L])$, $v_1 \in C^1([0, L])$ and $u \in C^2([0, L] \times \mathbb{R})$ fulfilling

$$v_0(x_j) = u_{j,0}\,, \qquad v_1(x_j) = u_{j,1}\,, \qquad u(x_j, t) = u_j(t)\,, \qquad j \in \{0, \ldots, N\}\,.$$

From a physical point of view $u(x, t)$ represents the longitudinal displacement of the point $x \in [0, L]$ at time t. Within this assumption the initial conditions for the system (1.2) as well as the time derivatives $\ddot{u}_j$ may be written as

$$u(x_j, 0) = v_0(x_j) \qquad \frac{\partial u}{\partial t}(x_j, 0) = v_1(x_j)\,, \qquad \ddot{u}_j(t) = \frac{\partial^2 u}{\partial t^2}(x_j, t)\,, \qquad j \in \{0, \ldots, N\}\,.$$

Moreover, a Taylor series expansion leads to

$$\begin{aligned}
u_1(t) - u_0(t) &= \frac{L}{N}\frac{\partial u}{\partial x}(x_0, t) + O(1/N^2)\\
u_{j+1}(t) + u_{j-1}(t) - 2u_j(t) &= \frac{L^2}{N^2}\frac{\partial^2 u}{\partial x^2}(x_j, t) + o(1/N^2)\\
u_{N-1}(t) - u_N(t) &= -\frac{L}{N}\frac{\partial u}{\partial x}(L, t) + O(1/N^2)\,.
\end{aligned}$$

Notice that these expressions have vanishing limit as $N \to \infty$: In order to avoid a trivial limit, we have to change k in a N-dependent fashion. In particular, if $k = \kappa N^2$ we find that the limit $N \to \infty$ of the system (1.1) can be seen as the discrete version of the following continuous system:

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} \qquad x \in [0, L]\,,\ t \in \mathbb{R}\,, \tag{1.3a}$$

$$\frac{\partial u}{\partial x}(0, t) = \frac{\partial u}{\partial x}(L, t) = 0 \qquad t \in \mathbb{R} \tag{1.3b}$$

$$u(x, 0) = v_0(x) \qquad \frac{\partial u}{\partial t}(x, 0) = v_1(x) \qquad x \in [0, L]\,, \tag{1.3c}$$

where we set $c^2 := \kappa L^2/m$. Notice that c has the dimension of a velocity: As we shall see in Section 17, c is interpreted as the velocity of propagation for the longitudinal vibrations. Equation (1.3a) is called **wave equation** (in 1 + 1-dimension). Equations (1.3c) are called **initial conditions** while Equations (1.3b) are called **(Neumann) boundary conditions**.

Remark 1.1: In [18, §3] Equation (1.3a) is derived from a model describing the transversal vibrations of a string. Within this setting, the function $u(x,t)$ represents the displacement along the z-axis of the point of the string which occupies $x \in [0,L]$ at time t. We shall see a few applications of this model to musical instruments in Sections 20-21. Both models (longitudinal and transversal vibrations) generalize to higher dimension: In the latter case the function u represents the displacement of the membrane (of *e.g.* a drum) in the normal direction with respect to its rest position [11, 18].

The heat/diffusion equation

In this section we will study diffusion phenomena. This will lead to an equation describing the variation of the density u of a given chemical or population. The same equation is used to model the diffusion of heat in a continuum: Within this setting $u(x,t)$ is interpreted as the temperature at x and at time t.

For the sake of simplicity we will consider a 1-dimensional situation: In the "heat interpretation" one may consider the temperature of a thin copper rod. At initial time a certain population has density (per unit length) $u_0(x)$. A physically reasonable condition on u_0 is

$$\int_{\mathbb{R}} |u_0(x)|\mathrm{d}x < \infty\,,$$

which ensures that we are considering a finite population. This allows for rather irregular initial population densities: For the sake of simplicity in what follows we will also assume $u_0 \in C^0(\mathbb{R})$.

To derive the evolution of the population density we will apply the diffusion principle. For that, let $(a,b) \subset \mathbb{R}$ be an arbitrary: The diffusion principle states that

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{matrix}\text{Population}\\ \text{in } (a,b)\end{matrix} = \begin{matrix}\text{Population produced}\\ \text{in } (a,b)\end{matrix} + \begin{matrix}\text{Population crossing}\\ \text{either } a \text{ or } b\end{matrix}\,. \tag{1.4}$$

Thus, the variation of the density u within (a,b) depends on the possible sources within (a,b) together with a contribution from $\partial(a,b) = \{a,b\}$ related to the incoming/outgoing population.

The left-hand side of Equation (1.4) is described by the integral

$$\text{LHS of (1.4)} = \frac{\mathrm{d}}{\mathrm{d}t}\int_a^b u(x,t)\mathrm{d}x = \int_a^b \frac{\partial u}{\partial t}(x,t)\,\mathrm{d}x\,,$$

where we assumed that $u \in C^2(\mathbb{R}\times\mathbb{R}_+)$, $\boxed{\mathbb{R}_+} := (0,+\infty)$ —exchanging derivative and integral can be made rigorous, but we will take it for granted for the time being.

Concerning the right-hand side of (1.4), the first term is modelled by an integral of the form

$$\begin{matrix}\text{Population produced}\\ \text{in } (a,b)\end{matrix} = \int_a^b f(x,t)\mathrm{d}x\,,$$

where $f \in C^0(\mathbb{R}\times\mathbb{R}_+)$ is the density (per unit time and per unit length) of the population source in (a, b). For the second term we write

$$\begin{matrix}\text{Population crossing}\\ \text{either } a \text{ or } b\end{matrix} = q(a,t) - q(b,t)\,,$$

where $q \in C^1(\mathbb{R}\times\mathbb{R}_+)$ is the x-component of the population flow (per unit length). In particular, $q(x, t)$ describes the population passing through the point x at time t in the direction of increasing x. Thus, $q(a, t)$ is the *incoming* population at a whereas $q(b, t)$ is the *outgoing* population at b: This explains the minus sign in the previous equation.

Summing up we find that (1.4) can be written as

$$0 = \int_a^b \left[\frac{\partial u}{\partial t} - f\right] \mathrm{d}x + q(b,t) - q(a,t) = \int_a^b \left[\frac{\partial u}{\partial t} - f + \frac{\partial q}{\partial x}\right] \mathrm{d}x\,,$$

where we applied the fundamental theorem of calculus. The assumed continuity of the integrand and the arbitrariness of (a, b) lead to

$$\frac{\partial u}{\partial t} + \frac{\partial q}{\partial x} = f \qquad \forall (x,t) \in \mathbb{R}\times\mathbb{R}_+\,. \tag{1.5}$$

Equation (1.5) is the general form of a diffusion equation. Notice that, although f is a given data of the problem, the actual form of q requires further information. Commonly one assumes the renown **Fourier law**

$$q = -\varsigma^2 \frac{\partial u}{\partial x}\,, \tag{1.6}$$

where $\varsigma > 0$ is a suitable constant. (When u represents a temperature, q has the interpretation of a heat flow: Within this setting the Fourier law expresses the fact that heat flows towards points where the temperature is lower.) The diffusion equation (1.5) simplifies to

$$\frac{\partial u}{\partial t} - \varsigma^2 \frac{\partial^2 u}{\partial x^2} = f \qquad \forall (x,t) \in \mathbb{R}\times\mathbb{R}_+\,. \tag{1.7}$$

where $\varsigma^2 > 0$ is known with the name of **diffusion constant**: The higher ς^2 is, the faster the diffusion process goes —notice that ς^2 has the dimension of length^2 $\times$ time^{-1}.

The initial-value problem (on $\mathbb{R}\times\mathbb{R}_+$) for Equation (1.7) consists of

$$\frac{\partial u}{\partial t} - \varsigma^2 \frac{\partial^2 u}{\partial x^2} = f \qquad \forall (x,t) \in \mathbb{R}\times\mathbb{R}_+ \tag{1.8a}$$
$$u(x,0) = u_0(x)\,, \qquad \forall x \in \mathbb{R}\,, \tag{1.8b}$$

where $f \in C^0(\mathbb{R}\times\mathbb{R}_+)$ and $u_0 \in C^0(\mathbb{R}) \cap L^1(\mathbb{R})$ are assigned while $u \in C^2(\mathbb{R}\times\mathbb{R}_+) \cap C^0(\mathbb{R}\times\overline{\mathbb{R}_+})$ is the unknown. These conditions ensure that all expressions in (1.8) are well-defined.

Remark 1.2:

(i) In our derivation of Equation (1.7) we assumed the Fourier law (1.6) with a constant factor $\varsigma > 0$. However, there are physically interesting situations in which ς is a function of x, t: In particular this occurs when modelling phase transitions. To wit, imagine that the continuum under investigation undergoes a phase transition (*e.g.* a block of ice is melting into water) —within this setting $u(x, t)$ is the temperature of the continuum. Therein at each fixed time there is a "melting point" $x(t) \in \mathbb{R}$ such that: (a) the continuum has undergone a phase transition (*e.g.* it has melted) for $x < x(t)$; (b) the continuum has not undergone a phase transition for $x > x(t)$.

In this situation ς is modelled by a step function

$$\varsigma(x,t) = \begin{cases} \varsigma_- & x < x(t) \\ \varsigma_+ & x > x(t) \end{cases},$$

for suitable positive constants $\varsigma_\pm$. It follows that Equation (1.7) is replaced by

$$\frac{\partial u}{\partial t} - \varsigma_-^2 \frac{\partial^2 u}{\partial x^2} = 0 \quad x < x(t)\,, \qquad \frac{\partial u}{\partial t} - \varsigma_+^2 \frac{\partial^2 u}{\partial x^2} = 0 \quad x > x(t)\,. \tag{1.9}$$

Notice that in this situation we have no reason to assume that u is regular across $x(t)$ —*cf.* Equations (1.10e)-(1.10f).

The task is to find u as well as the curve $t \mapsto x(t)$. These two data are not independent, because the "melting point" $x(t)$ depends on the temperature u and conversely u depends on $x(t)$ on account of Equation (1.9): This means that additional conditions are needed. The final problem reads as follows:

$$\frac{\partial u}{\partial t} - \varsigma_-^2 \frac{\partial^2 u}{\partial x^2} = 0 \qquad \forall t \in \mathbb{R}_+\,,\ \forall x < x(t)\,, \tag{1.10a}$$

$$\frac{\partial u}{\partial t} - \varsigma_+^2 \frac{\partial^2 u}{\partial x^2} = 0 \qquad \forall t \in \mathbb{R}_+\,,\ \forall x > x(t)\,, \tag{1.10b}$$

$$u(x,0) = u_0(x) \qquad \forall x \in \mathbb{R}\,, \tag{1.10c}$$

$$x(0) = x_0\,, \tag{1.10d}$$

$$u(x(t)^+, t) = u(x(t)^-, t) \qquad \forall t \in \mathbb{R}_+ \tag{1.10e}$$

$$\nu \dot{x}(t) = \varsigma_-^2 \frac{\partial u}{\partial x}(x(t)^-, t) - \varsigma_+^2 \frac{\partial u}{\partial x}(x(t)^+, t) \qquad \forall t \in \mathbb{R}_+\,. \tag{1.10f}$$

where we denoted $f(x^\pm) := \lim_{y \to x^\pm} f(y)$. Equations (1.10a)-(1.10b) are nothing but the heat equation for the continuum before and after the phase transition. Equations (1.10c) and (1.10d) provide initial data for the temperature $u(x, t)$ and the "melting" curve $x(t)$. Equation (1.10e) imposes a continuity condition on the temperature at the "melting point" $x(t)$. Equation (1.10f) is crucial: It states that the jump of $\varsigma^2 \frac{\partial u}{\partial x}$ across $x(t)$ is proportional to $\dot{x}$ —here $\nu > 0$ is a given constant with the same dimension of u. Since by Equation (1.6), $-\varsigma^2 \frac{\partial u}{\partial x}$ is interpreted as an heat flow, this condition expresses the fact that the heat dissipated at the phase transition is determining the evolution of the "melting point". In particular, $\dot{x}(t) > 0$ whenever the heat flow from the right region ($x > x(t)$) is larger than the heat flow from the left region ($x < x(t)$).

The system (1.10) is called **Stefan problem**. For more details, see [20].

(ii) The diffusion equation (1.7) has been derived from Equation (1.5) within the assumption of (1.6). Another possibility is to set, considering $n = 1$, $q = \nu^2 u^2/2$. In this case Equation (1.5) reduces to the renown **Burgers' equation**

$$\frac{\partial u}{\partial t} + \nu^2 u \frac{\partial u}{\partial x} = 0\,. \tag{1.11}$$

The latter is an example of a quasi-linear first order PDE —*cf.* Remark 2.3. The relevant phenomenon associated with this equation is that solutions with regular initial data may develop singularities in finite time —these solutions are called **shocks**. The problem of making sense of these solutions after the development of a singularity is of paramount interest, *cf.* [3, §3.4].

Sometimes Equation (1.11) is called inviscid Burgers' equation: The **viscid Burgers' equation** is instead given by

$$\frac{\partial u}{\partial t} - \varsigma^2 \frac{\partial^2 u}{\partial x^2} + u \frac{\partial u}{\partial x} = 0\,. \tag{1.12}$$

Notice that, with this choice of constants, u has necessarily the dimension of a velocity. Equation (1.12) combines the diffusive feature of the heat equation (1.7) with the shock solutions generated by the Burgers' equation (1.11).

2 Semi-linear PDEs: General results

In this section we will introduce the relevant class of partial differential equations of interest. The main references for this section are [3, 11, 21].

For later convenience we recall a few basic notion on differentiable functions. Let X be a topological space, $f\colon X \to \mathbb{C}$: The **support** of f is the closed subset of X defined by

$$\mathrm{supp}(f) := \overline{\{x \in X \mid f(x) \neq 0\}}\,, \tag{2.1}$$

where $\boxed{\overline{A}}$ denotes the closure of $A \subseteq X$ in X. Notice that $x \in \mathrm{supp}(f)$ does not imply that $f(x) \neq 0$.

Let $\Omega \subseteq \mathbb{R}^n$ be a non-empty open subset of $\mathbb{R}^n$. Given $f\colon \Omega \to \mathbb{R}^m$ and $x_0 \in \Omega$, we say that f is **differentiable at** x_0 if there exists a linear map $L_{x_0}\colon \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h\to 0} \frac{\|f(x_0+h) - f(x_0) - L_{x_0}(h)\|_m}{\|h\|_n} = 0\,, \tag{2.2}$$

where $\|\ \|_k$ denotes the Euclidean norm of $\mathbb{R}^k$.

Let $\{e_j\}_j$, $j \in \{1,\ldots,n\}$, be the standard Euclidean basis of $\mathbb{R}^n$. Setting $h = e_j$, Equation (2.2) implies the existence of the j-th partial derivative $\frac{\partial f}{\partial x^j}\big|_{x_0}$ of f at x_0, moreover,

$$L_{x_0}(h) = \sum_{j=1}^n h^j \frac{\partial f}{\partial x^j}(x_0) = h^j \frac{\partial f}{\partial x^j}(x_0)\,,$$

where we used Einstein notation —this means that a sum like $\sum_{k=1}^n a^k b_k$ is replaced by $a^k b_k$ where the sum over k is *implicitly* taken over all possible values of k.

The existence of $\frac{\partial f}{\partial x^j}\big|_{x_0}$ for all $j = 1,\ldots,n$ does not imply that f is differentiable at x_0, however, the following holds true [6].

Proposition 2.1: Let $f\colon \Omega \to \mathbb{R}^m$, $x_0 \in \Omega$ be such that there exists an open neighbourhood $N_{x_0} \subseteq \Omega$ of x_0 such that forall $x \in N_{x_0}$ $\frac{\partial f}{\partial x^j}\big|_x$ exists and it is continuous for all $j = 1,\ldots,n$. Then f is differentiable at x_0.

Out of Proposition 2.1 the notion of differentiability is extended at higher order. In particular a function $f\colon \Omega \to \mathbb{R}^m$ is called C^k**-differentiable in** Ω —and we write $f \in C^k(\Omega,\mathbb{R}^m)$— if for all $\alpha \in \mathbb{Z}^n_+$ with $\boxed{|\alpha|} := \alpha_1 + \ldots + \alpha_n \leqslant k$ the (multi-index) partial derivative

$$\partial^\alpha f := \frac{\partial^{|\alpha|} f}{(\partial x^1)^{\alpha_1}\cdots(\partial x^n)^{\alpha_n}}\,, \tag{2.3}$$

exists and it is continuous in Ω. For $k = 0$ the definition identifies continuous functions on Ω. For the sake of simplicity we shall denote by $\boxed{C^k(\Omega)} := C^k(\Omega,\mathbb{R})$. We will denote by $C^k_c(\Omega,\mathbb{R}^m)$

the space of functions $f \in C^k(\Omega, \mathbb{R}^m)$ with $\mathrm{supp}(f)$ compact. Finally we shall refer to elements of

$$C^\infty(\Omega, \mathbb{R}^m) := \bigcap_{k \geqslant 0} C^k(\Omega, \mathbb{R}^m)\,, \qquad (\textit{resp.}\ C^\infty_c(\Omega, \mathbb{R}^m) := \bigcap_{k \geqslant 0} C^k_c(\Omega, \mathbb{R}^m))\,,$$

as **smooth functions** (*resp.* **smooth functions with compact support**).

For future convenience we recall the notion of C^k**-differentiable function on** $\overline{\Omega}$, $\overline{\Omega}$ being the closure of Ω. In particular $f\colon \overline{\Omega} \to \mathbb{R}^m$ belongs to the space $f \in \boxed{C^k(\overline{\Omega}, \mathbb{R}^m)}$ if:

1. $f|_\Omega \in C^k(\Omega, \mathbb{R}^m)$;
2. for all $\alpha \in \mathbb{Z}^n_+$ with $|\alpha| \leqslant k$ the partial derivative $\partial^\alpha[f|_\Omega]$ extends by continuity to $\overline{\Omega}$.

We are now ready to introduce the relevant class of differential operators we will consider.

Definition 2.2: An operator $D\colon C^2(\Omega) \to C^0(\Omega)$ is called **semi-linear of second order** if

$$(Du)(x) = a^{jk}(x)\frac{\partial^2 u}{\partial x^j \partial x^k} + \Phi\left(x, u(x), \frac{\partial u}{\partial x^k}\right)\,, \qquad \forall u \in C^2(\Omega)\,, \tag{2.4}$$

where $a^{jk} \in C^0(\Omega)$ for all $j, k = 1, \ldots, n$ and $\Phi \in C^0(\Omega \times \mathbb{R} \times \mathbb{R}^n)$ —we may recollect a^{jk} in a unique matrix-valued function $a \in C^0(\Omega, \mathbb{R}^{n(n+1)/2})$ where we stressed that a^{jk} is symmetric. A **semi-linear partial differential equation** (PDE for short) is an equation of the form $Du = f$ for a semi-linear operator D —here D, f are given and u is the unknown.

Remark 2.3:

(i) Definition 2.2 generalizes to: (a) vector-valued functions, possibly complex valued —in this latter case $a = a^{jk}, \Phi$ are required to be real-valued; (b) order k different from $k = 2$. In particular $D\colon C^k(\Omega) \to C^0(\Omega)$ is called semi-linear of order k if

$$(Du)(x) = \sum_{\substack{\alpha \in \mathbb{Z}^n_+ \\ |\alpha| = k}} a_\alpha(x)\partial^\alpha u + \Phi(x, \{\partial^\beta u(x)\}_{\beta \in \mathbb{Z}^n\ |\beta| < k})\,.$$

For example, a semilinear-operator of order 1 reads $(Du)(x) = a^k(x)\frac{\partial u}{\partial x^k} + \Phi(x, u(x))$.

(ii) The form (2.4) allows for a mild non-linearity which requires a linear dependence in u in the term involving the highest order derivatives. This class of operators includes in particular the case where Φ depends linearly on u and ∇u, that is, when

$$\Phi(x, u(x), \nabla u(x)) = a^k(x)\frac{\partial u}{\partial x^k} + b(x)u(x) + c(x)\,,$$

for $a^k, b, c \in C^0(\Omega)$. In this case the operator and the associated PDE are called **linear**. On the other side, semi-linear operators are included in the class of **quasi-linear** [1] operators which reads

$$(Du)(x) = a^{jk}\left(x, u(x), \frac{\partial u}{\partial x^k}\right)\frac{\partial^2 u}{\partial x^j \partial x^k} + \Phi\left(x, u(x), \frac{\partial u}{\partial x^k}\right).$$

Notice that these operators are **local**, namely, $(Du)(x)$ only depends on the values of $u, \nabla u, \ldots$ at x. This excludes operators of the form

$$a^{jk}(x)\frac{\partial^2 u}{\partial x^j \partial x^k} + \int_\Omega K(x,y)u(y)\mathrm{d}^n y\,,$$

for some $K \in C(\Omega^2)$ —here $\mathrm{d}^n y$ denotes the Lebesgue measure.

EXAMPLE 2.4:

(i) Extending the derivation of the wave equation (1.3a) to higher dimension we find

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \Delta_x u = 0\,, \tag{2.5}$$

where $\Delta_x u$ denotes the **Laplace operator** of u (in the x-variable only)

$$\Delta_x u := \sum_{j=1}^n \frac{\partial^2 u}{(\partial x^j)^2} = \delta^{jk}\frac{\partial^2 u}{\partial x^j \partial x^k}\,. \tag{2.6}$$

In the model described in [11, 18] the function u represents the displacement of the membrane of the drum in the normal direction with respect to its rest position.

(ii) Considering solutions to Equation (2.5) with a special form leads to interesting PDEs per se. To wit, let consider u of the form $u(x,t) = u(x)e^{i\omega t}$, where $\omega \in \mathbb{R}$: Then Equation (2.5) reduces to

$$\lambda u + \Delta u = 0\,, \qquad \lambda = \omega^2/c^2\,. \tag{2.7}$$

This latter equation is known with the name of **Helmholtz equation**. In the particular case $\omega = 0$ we find the renown **Laplace equation**

$$\Delta u = 0\,. \tag{2.8}$$

[1] It is an unfortunate fact of life that [21] and [3] have different (opposite) conventions concerning quasi-linear and semi-linear PDEs: Here we shall stick with the convention of [3].

For further references, see [3]. Finally, in Quantum Field Theory it is quite common to consider a slight variation of Equation (2.5): Therein one consider the **Klein-Gordon equation**

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \Delta_x u + \frac{m^2c^2}{\hbar^2}u = 0\,, \tag{2.9}$$

where $m \geqslant 0$ is referred to as the mass (of the particle under investigation), c is the speed of light and $\hbar$ is the Planck constant. Time-independent solutions to the latter equations reduces to the solutions to the **massive Laplace equation**

$$-\Delta_x u + \mu^2 u = 0\,, \tag{2.10}$$

where $\mu^2 > 0$.

(iii) The derivation of the diffusion equation (1.7) described in Section 1 can be generalized to higher dimensions —*cf.* [11]. To make a long story short the resulting equation is (for $f = 0$)

$$\frac{\partial u}{\partial t} - \varsigma^2 \Delta_x u = 0\,. \tag{2.11}$$

We shall come back to this derivation in Section 8 —*cf.* Remark 8.2.

We are interested in the general properties of semi-linear PDEs. To begin with, we will investigate the invariance property of the semi-linear differential operator D introduced in Equation (2.4): This will lead to a classification of these operators —*cf.* Definition 2.9.

To this avail we recall without proof a few important theorems from Calculus —*cf.* [6, §16], [3, §C.6]. We recall that, given two open sets Ω, Ω', $f\colon \Omega \to \Omega'$ is called a C^k**-diffeomorphism** if: (a) $f \in C^k(\Omega, \Omega')$; (b) f is invertible with $f^{-1} \in C^k(\Omega', \Omega)$.

Theorem 2.5 (Inverse function theorem): Let $\Omega \subset \mathbb{R}^n$ be open, $x_0 \in \Omega$, $k > 1$ and $f \in C^k(\Omega, \mathbb{R}^n)$ with

$$\det \frac{\partial f^a}{\partial x^b}\bigg|_{x_0} \neq 0\,. \tag{2.12}$$

Then there exists a neighbourhood $N_{x_0} \subseteq \Omega$ of x_0 such that $f(N_{x_0})$ is open and $f|_{N_{x_0}}$ is a C^k-diffeomorphism.

Theorem 2.5 is commonly used in the following form:

COROLLARY 2.6: Let $\Omega \subseteq \mathbb{R}^n$ be open and let $f \in C^k(\Omega, \mathbb{R}^n)$, $k > 1$, be such that $\forall x \in \Omega$, $\det \frac{\partial f^a}{\partial x^b}\Big|_x \neq 0$. Then f is an open map, that is, $f(\Omega')$ is open for all open $\Omega' \subseteq \Omega$. Moreover, if f is injective then $f\colon \Omega \to f(\Omega)$ is a C^k-diffeomorphism.

LEMMA 2.7: Let $D\colon C^2(\Omega) \to C^0(\Omega)$ be a second order semi-linear differential operator as by Definition 2.2. Let $\Omega \ni x \mapsto \tilde{x}(x) \in \tilde{x}(\Omega) =: \widetilde{\Omega}$ be an injective C^2-function such that $\det \frac{\partial \tilde{x}^a}{\partial x^b} \neq 0$. For all $u \in C^2(\Omega)$, let $\widetilde{u} \in C^2(\widetilde{\Omega})$ be defined by $\widetilde{u}(\tilde{x}) = u(x(\tilde{x}))$. Then $Du = 0$ if and only if $\widetilde{D}\widetilde{u} = 0$ where

$$\widetilde{D}\widetilde{u}(\tilde{x}) = \widetilde{a}^{kj}(\tilde{x})\frac{\partial^2 \widetilde{u}}{\partial \tilde{x}^k \partial \tilde{x}^j} + \widetilde{\Phi}\left(\tilde{x}, \widetilde{u}(\tilde{x}), \frac{\partial \widetilde{u}}{\partial \tilde{x}^k}\right),$$

where $\widetilde{a}^{kj} \in C^0(\widetilde{\Omega})$ and $\widetilde{\Phi} \in C^0(\widetilde{\Omega} \times \mathbb{R} \times \mathbb{R}^n)$ are defined by

$$\widetilde{a}^{kj}(\tilde{x}) := a^{\ell m}(x(\tilde{x}))\frac{\partial \tilde{x}^k}{\partial x^\ell}\frac{\partial \tilde{x}^j}{\partial x^m}, \tag{2.13}$$

$$\widetilde{\Phi}\left(\tilde{x}, \widetilde{u}(\tilde{x}), \frac{\partial \widetilde{u}}{\partial \tilde{x}^k}\right) := \Phi\left(x(\tilde{x}), u(x(\tilde{x})), \frac{\partial \tilde{x}^k}{\partial x^\ell}\frac{\partial \widetilde{u}}{\partial \tilde{x}^k}\right) + a^{kj}(x(\tilde{x}))\frac{\partial^2 \tilde{x}^\ell}{\partial x^k \partial x^j}\frac{\partial \widetilde{u}}{\partial \tilde{x}^\ell}. \tag{2.14}$$

Proof. Notice that, on account of Corollary 2.6, $\widetilde{\Omega} := \tilde{x}(\Omega)$ is open and $\tilde{x} \in C^2(\Omega, \widetilde{\Omega})$ is a C^2-diffeomorphism. Since $\widetilde{u}(\tilde{x}) = u(x(\tilde{x}))$ and conversely $u(x) = \widetilde{u}(\tilde{x}(x))$, the chain rule implies

$$\begin{aligned}
\frac{\partial u}{\partial x^\ell} &= \frac{\partial \widetilde{u}}{\partial \tilde{x}^k}\frac{\partial \tilde{x}^k}{\partial x^\ell}, \\
\frac{\partial^2 u}{\partial x^\ell \partial x^m} &= \frac{\partial}{\partial x^m}\left[\frac{\partial \widetilde{u}}{\partial \tilde{x}^k}\frac{\partial \tilde{x}^k}{\partial x^\ell}\right] = \frac{\partial^2 \widetilde{u}}{\partial \tilde{x}^k \partial \tilde{x}^j}\frac{\partial \tilde{x}^j}{\partial x^m}\frac{\partial \tilde{x}^k}{\partial x^\ell} + \frac{\partial \widetilde{u}}{\partial \tilde{x}^k}\frac{\partial^2 \tilde{x}^k}{\partial x^\ell \partial x^m}.
\end{aligned}$$

By inserting these expressions in $Du = 0$ we find

$$\begin{aligned}
0 = (Du)(x) &= a^{\ell m}(x)\frac{\partial^2 u}{\partial x^m \partial x^\ell} + \Phi\left(x, u(x), \frac{\partial u}{\partial x^\ell}\right) \\
&= a^{\ell m}(x)\frac{\partial^2 \widetilde{u}}{\partial \tilde{x}^k \partial \tilde{x}^j}\frac{\partial \tilde{x}^j}{\partial x^m}\frac{\partial \tilde{x}^k}{\partial x^\ell} + a^{\ell m}(x)\frac{\partial \widetilde{u}}{\partial \tilde{x}^k}\frac{\partial^2 \tilde{x}^k}{\partial x^\ell \partial x^m} + \Phi\left(x, u(x), \frac{\partial \widetilde{u}}{\partial \tilde{x}^k}\frac{\partial \tilde{x}^k}{\partial x^\ell}\right) \\
&= \widetilde{a}^{jk}(\tilde{x})\frac{\partial \widetilde{u}}{\partial \tilde{x}^j \partial \tilde{x}^\ell} + \widetilde{\Phi}\left(\tilde{x}, \widetilde{u}(\tilde{x}), \frac{\partial \widetilde{u}}{\partial \tilde{x}^k}\right),
\end{aligned}$$

as claimed. Notice that the continuity properties of $\widetilde{a}^{kj}$ and $\widetilde{\Phi}$ follow from the regularity assumption on the map $x \mapsto \tilde{x}(x)$. □

REMARK 2.8:

(i) The transformation laws (2.13) and (2.14) are rather different. Indeed, Equation (2.13) expresses the tensorial nature of the matrix-valued function $a = (a^{jk}) \in C^0(\Omega, \mathbb{R}^{n(n+1)/2})$: Roughly speaking this means that a is coordinate independent. Conversely, Φ does not have any sensible tensorial property. For these reasons the function $a = (a^{jk})$ appearing in Equation (2.4) is usually referred to as the **principal symbol** of D.

(ii) Up to minor adjustments, the proof of Lemma 2.7 applies also for semi-linear differential operators of order different from 2. For example, let $D\colon C^1(\Omega) \to C^0(\Omega)$ be a first order semi-linear differential operator —*cf.* Remark 2.3—

$$(Du)(x) = a^k(x)\frac{\partial u}{\partial x^k} + \Phi(x, u(x))\,.$$

Then, under the assumptions of Lemma 2.7 —this time we may assume that $x \mapsto \tilde{x}(x)$ is only C^1— we find that

$$(\widetilde{D}\widetilde{u})(\tilde{x}) = \widetilde{a}^k(\tilde{x})\frac{\partial \widetilde{u}}{\partial \tilde{x}^k} + \widetilde{\Phi}(\tilde{x}, \widetilde{u}(\tilde{x}))\,,$$
$$\widetilde{a}^k(\tilde{x}) = a^\ell(x(\tilde{x}))\frac{\partial \tilde{x}^k}{\partial x^\ell}\,, \qquad \widetilde{\Phi}(\tilde{x}, \widetilde{u}(\tilde{x})) = \Phi(x(\tilde{x}), u(x(\tilde{x})))\,.$$

Notice that in this particular case Φ transforms as a scalar function.

Let now $x_0 \in \Omega$ be fixed. If we consider the linear map $x \mapsto \tilde{x}$ defined by $\tilde{x}^a = J^a_b x^b$, Equation (2.13) leads to

$$\widetilde{a}^{jk}(\tilde{x}_0) = a^{\ell m}(x_0) J^j_\ell J^k_m = [J^* a J]^{jk}(x_0)\,,$$

where $\tilde{x}_0 := \tilde{x}(x_0)$ while we denoted by $J^* \in M_n(\mathbb{R})$ the transpose of the $n \times n$-matrix $J \in M_n(\mathbb{R})$ —which we have to choose so that $\det J \neq 0$. Since a^{jk} is symmetric the Sylvester Theorem applies and there exists $J \in M_n(\mathbb{R})$ with $\det J \neq 0$ such that

$$[J^* a J](\tilde{x}_0) = \mathrm{diag}\left(\underbrace{1, \ldots, 1}_{n_+}, \underbrace{-1, \ldots, -1}_{n_-}, \underbrace{0, \ldots, 0}_{n_0}\right), \tag{2.15}$$

where $n_+, n_-, n_0 \in \mathbb{N} \cup \{0\}$ are such that $n_+ + n_- + n_0 = n$. Notice that J will depend on the chosen point x_0. The numbers n_+, n_-, n_0 are invariants of D at x_0: depending on their value we have the following classification —*cf.* [21, §1, §3]

Definition 2.9: Let $D\colon C^2(\Omega) \to C^0(\Omega)$ be a second-order semi-linear differential operator and let $x_0 \in \Omega$. Then D is called:

1. **elliptic** at x_0 if $(n_+, n_-, n_0) = (n, 0, 0)$ or $(n_+, n_-, n_0) = (0, n, 0)$;

2. **hyperbolic** at x_0 if $(n_+, n_-, n_0) = (n_+, n_-, 0)$ for $n_+, n_- \in \mathbb{N}$; In particular D is **normally hyperbolic** if $(n_+, n_-, 0) = (1, n-1, 0)$ or $(n_+, n_-, 0) = (n-1, 1, 0)$.

3. **parabolic** at x_0 if $n_0 \neq 0$ and either $n_+ = 0$ or $n_- = 0$; In particular D is **normally parabolic** if $(n_+, n_-, n_0) = (n-1, 0, 1)$ or $(n_+, n_-, n_0) = (0, n-1, 1)$.

The operator D is called elliptic, hyperbolic or parabolic if it is elliptic, hyperbolic or parabolic at all $x \in \Omega$.

Example 2.10:

(i) The Laplace operator Δ, the wave operator $\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta_x$ and the heat operator $\frac{\partial}{\partial t} - \varsigma^2\Delta_x$ are the prototypes of elliptic, (normally) hyperbolic and (normally) parabolic operators.

(ii) The Laplace equation $\Delta u = 0$, the Helmholtz equation $\Delta u + \omega^2 u = 0$ and the massive Laplace equation $(\mu^2 - \Delta)u = 0$ fall into the class of elliptic operators. However, the presence of a lower order contribution changes the behaviour of the solution for large value of x. This is particularly clear if $n = 1$, where explicit formulas are at disposal: In that case we have

$$
\begin{aligned}
&0 = \Delta u = u'' \quad \Rightarrow \quad u(x) = u_0 + u_1 x\,,\\
&0 = (\Delta u + \omega^2) = u'' + \omega^2 u \quad \Rightarrow \quad u(x) = u_0\cos(\omega x) + u'(0)\frac{\sin(\omega x)}{\omega}\,,\\
&0 = (\Delta u - \mu^2) = u'' - \mu^2 u \quad \Rightarrow \quad u(x) = \alpha_+ e^{\mu x} + \alpha_- e^{-\mu x} = u_0\cosh(\mu t) + u'(0)\frac{\sinh(\mu t)}{\mu}\,,
\end{aligned}
$$

where $\boxed{\cosh(z)} := (e^z + e^{-z})/2$ and $\boxed{\sinh(z)} := (e^z - e^{-z})/2$ are called hyperbolic sine and cosine.

(iii) The **Tricomi operator** $D\colon C^2(\mathbb{R}^2) \to C^0(\mathbb{R}^2)$

$$
Du := y\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\,, \tag{2.16}
$$

is a second order linear differential operator. Referring to Equation (2.4) we have

$$
a(x, y) = \begin{pmatrix} y & 0 \\ 0 & 1 \end{pmatrix}.
$$

It follows that D is: (a) elliptic at (x, y) for all $y > 0$ and $x \in \mathbb{R}$; (b) parabolic at $(x, 0)$ for all $x \in \mathbb{R}$; (c) hyperbolic at (x, y) for all $y < 0$ and $x \in \mathbb{R}$.

(iv) The **Schrödinger operator** $D\colon C^2(\Omega,\mathbb{C}) \to C^0(\Omega,\mathbb{C})$ is defined by

$$Du := i\hbar\frac{\partial u}{\partial t} + \frac{\hbar^2}{2m}\Delta_x u\,, \tag{2.17}$$

where $m, \hbar \in \mathbb{R}$. This operator is of paramount interest in Quantum Mechanics, where $\hbar, m$ are interpreted as the Planck' constant and as the mass of the (quantum) particle we are investigating.

This operator does not fall within the previous classification because, strictly speaking, it is not a second order semi-linear differential operator as by Definition 2.2 —the function Φ is now complex-valued. This sounds a little bit artificial and one may wonder about extending Definitions 2.2 and 2.9 by allowing $\mathbb{C}$-valued functions, with the idea that the Schrödinger operator should be classified as of parabolic-type. However, it is important to point out that the behaviour of the solutions to the Schrödinger equation $Du = 0$ is similar to the one of the solutions to the wave equation (2.5) rather than the one of the solutions to the heat equation (2.11).

To show this, let $n = 1$ and set $u(x,t) = e^{ikx+\omega t}$ for $k \in \mathbb{R}$ and $\omega \in \mathbb{C}$ —this entails that $u(x,0) = e^{ikx}$ is oscillating in space. Inserting such form of u into the wave equation (2.5) leads to

$$\omega^2 + c^2k^2 = 0 \quad \Rightarrow \quad u(x,t) = \exp\left[i(kx + c|k|t)\right]\,, \tag{2.18}$$

where we chose $\omega = c|k|$ (we could have chosen $\omega = -c|k|$ as well). This solution has an oscillating behaviour also in time, as it is expected from a wave phenomenon —actually the solution $e^{i(kx+c|k|t)}$ is called plain wave with direction k and wave number $|k|$. Similarly, imposing the heat equation (1.7) leads to

$$\omega + \varsigma^2k^2 = 0 \quad \Rightarrow \quad u(x,t) = \exp[ikx - \varsigma^2k^2t]\,.$$

This time the solution rapidly decays as $t \to +\infty$. Finally, let impose that u solves the Schrödinger equation: This implies

$$i\hbar\omega + \frac{\hbar^2}{2m}k^2 = 0 \quad \Rightarrow \quad u(x,t) = \exp\left[i\left(kx + \frac{\hbar}{2m}k^2t\right)\right]\,,$$

which shows an oscillating behaviour more similar to the one found when imposing the wave equation.

The wave-like behaviour of the solutions to the Schrödinger equation is best understood when comparing them with the solutions to the Klein-Gordon equation —*cf.* Equation (2.9). Indeed, let again consider $u(x,t) = e^{ikx+\omega t}$: Then u solves Equation (2.9) if and only if

$$\omega^2 + c^2k^2 + \frac{m^2c^4}{\hbar^2} = 0 \quad \Rightarrow \quad u(x,t) = \exp\left[i\left(kx + t\sqrt{c^2k^2 + \frac{m^2c^4}{\hbar^2}}\right)\right]\,.$$

At this stage we need a slight clue from Physics: The Klein-Gordon equation describes a particle (actually, a field) of mass m in a quantum and relativistic setting —this is the reason why both $\hbar$ and c pop up in Equation (2.9). Solutions whose energy is way lower than the relativistic rest energy mc^2 of the classical particle can be approximated in the so-called non-relativistic regime. This is ultimately an approximation scheme which, for the case of the found solution u, is obtained considering situations where $c^2k^2 \ll m^2c^4/\hbar^2$, *i.e.* $\hbar^2k^2/m^2c^2 \ll 1$. Within this approximation we have

$$\sqrt{c^2k^2 + \frac{m^2c^4}{\hbar^2}} = \frac{mc^2}{\hbar}\sqrt{\frac{\hbar^2k^2}{m^2c^2} + 1} \approx \frac{mc^2}{\hbar} + \frac{\hbar k^2}{2m} \,.$$

Thus, in the non-relativistic approximation we find

$$u(x,t) = \exp\left[i\left(kx + t\sqrt{c^2k^2 + \frac{m^2c^4}{\hbar}}\right)\right] \approx \exp\left[i\left(kx + \frac{\hbar k^2}{2m}t + \frac{mc^2}{\hbar}t\right)\right].$$

Within the non-relativistic approximation, the phase $\exp\left[i\frac{mc^2}{\hbar}t\right]$ is usually dropped, the reason being that it represents a “fast oscillation”. We discovered that, within this approximation scheme, the solutions to the Schrödinger equation can be seen as non-relativistic approximation of the solutions to the Klein-Gordon equation.

Actually the above derivation should cope with the fact that solutions to the Klein-Gordon equation and to the Schrödinger equation have a completely different physical interpretation: We refer to [13] for a more precise treatment of this topic.

3 Intermezzo: A glimpse of Differential Geometry

In the forthcoming sections we will discuss the Cauchy problem for a local PDE of second order. To properly introduce this notion we have to recall/introduce a few basic facts on regular hypersurfaces on $\mathbb{R}^n$: This section is devoted to recollect useful material on this topic.

Given $S \in C^k(\mathbb{R}^n)$ we recall that $\boxed{\nabla S} \in C^{k-1}(\mathbb{R}^n, \mathbb{R}^n)$ denotes the **gradient** of S, defined by

$$\nabla S := \sum_{j=1}^{n} \frac{\partial S}{\partial x^j} e_j = \delta^{jk} \frac{\partial S}{\partial x^j} e_k \,,$$

$\{e_j\}_j$ being the canonical Euclidean basis of $\mathbb{R}^n$.

Definition 3.1: A subset $\Sigma \subset \mathbb{R}^n$ is called a C^k**-regular hypersurface** if for all $x_0 \in \Sigma$ there exists an open neighbourhood $N_{x_0} \subset \mathbb{R}^n$ of x_0 and a function $S \in C^k(N_{x_0})$ such that

$$\Sigma \cap N_{x_0} = \{x \in N_{x_0} \,|\, S(x) = 0\} \qquad \nabla S(x) \neq 0 \;\forall x \in \Sigma \cap N_{x_0} \,. \tag{3.1}$$

Example 3.2: Through this section we will specialize our definitions and constructions to the simplest case ever, namely the unit sphere $\boxed{\partial B_1(0)}$ centred at $0 \in \mathbb{R}^n$. The latter is the C^∞-regular hypersurface defined by

$$\partial B_1(0) := \{x \in \mathbb{R}^n \,|\, \|x\| = 1\} \,.$$

Condition (3.1) is fulfilled by considering the smooth function $S(x) := \|x\| - 1$ for which $\nabla S(x) = x/\|x\| = x$ for $x \in \partial B_1(0)$. This proves that $\partial B_1(0)$ is a smooth regular hypersurface. Denoting by $\boxed{B_1(0)} := \{x \in \mathbb{R}^n \,|\, \|x\| \leqslant 1\}$ the unit ball of radius 1 centred at $0 \in \mathbb{R}^n$ one immediately finds that $\partial B_1(0) = \overline{B_1(0)} \setminus B_1(0)$, that is, $\partial B_1(0)$ consists of the topological boundary of $B_1(0)$.

Remark 3.3:

(i) Notice that the condition $\nabla S \neq 0$ is invariant under coordinates transformations. Indeed if $N_{x_0} \ni x \mapsto \tilde{x}(x) \in \tilde{x}(N_{x_0})$ is an injective C^k function such that $\det \frac{\partial \tilde{x}^k}{\partial x^\ell}(x) \neq 0$ for all $x \in N_{x_0}$ then: (a) $\widetilde{N}_{x_0} := \tilde{x}(N_{x_0})$ is open and $\tilde{x} \in C^k(N_{x_0}, \widetilde{N}_{x_0})$ is a C^k-diffeomorphism by Corollary 2.6; (b) setting $\widetilde{S}(\tilde{x}) := S(x(\tilde{x}))$ —thus $S(x) = \widetilde{S}(\tilde{x}(x))$— we find that

$$\frac{\partial S}{\partial x^a} = \frac{\partial \widetilde{S}}{\partial \tilde{x}^b} \frac{\partial \tilde{x}^b}{\partial x^a} \,,$$

which entails that, since $\frac{\partial \tilde{x}^b}{\partial x^a}$ is a non-singular matrix, $\frac{\partial S}{\partial x^a}$ vanishes if and only if so does $\frac{\partial \widetilde{S}}{\partial \tilde{x}^b}$.

(ii) From a geometrical point of view, for all $x \in \Sigma \cap N_{x_0}$ the vector $\nabla S(x)$ is normal to Σ at x: Indeed, for any curve $(-\varepsilon, \varepsilon) \ni t \mapsto x(t)$, $\varepsilon > 0$, such that $x(t) \in \Sigma$ for all $t \in (-\varepsilon, \varepsilon)$ and $x(0) = x$ we have

$$0 = \frac{\mathrm{d}}{\mathrm{d}t} S(x(t))\Big|_{t=0} = \dot{x}^k(0) \frac{\partial S}{\partial x^k}\Big|_x = \nabla S(x) \cdot \dot{x}(0)\,.$$

where $\dot{x}(0) := \frac{\mathrm{d}x}{\mathrm{d}t}\Big|_{t=0}$ is the tangent vector to γ at $x(0)$. Since $\dot{x}(0)$ ranges among all possible tangent vectors to Σ at x, it follows that $\nabla S(x)$ is normal to Σ.

The hypersurface Σ is called **orientable** if there exists $\nu \in C^0(\Sigma, \mathbb{R}^n)$ such that: (a) $\|\nu(x)\| = 1$ for all $x \in \Sigma$; (b) $\nu(x)$ is normal to Σ for all $x \in \Sigma$.

(iii) Let $\Omega \subset \mathbb{R}^n$ be a non-empty, open subset such that its boundary $\partial\Omega := \overline{\Omega}\backslash\Omega$ is an orientable C^1-regular hypersurface. We say that ν is **outward pointing** if for all $x_0 \in \partial\Omega$ there exists an open neighbourhood N_{x_0} of x_0 and $S \in C^1(N_{x_0})$ such that: (a) $\partial\Omega \cap N_{x_0} = \{x \in N_{x_0} \,|\, S(x) = 0\}$; (b) $\nabla S(x) \neq 0$ for all $x \in \Sigma \cap N_{x_0}$; (c) $\nu(x) = \nabla S(x)/\|\nabla S(x)\|$; (d) $\Omega \cap N_{x_0} = \{x \in N_{x_0} \,|\, S(x) < 0\}$. Roughly speaking, ν points outside Ω.

Example 3.4: By direct inspection $\partial B_1(0)$ is orientable with outward pointing normal $\nu(x) = x$ for all $x \in \partial B_1(0)$. Notice that the latter is the outward pointing vector if we consider $\Omega = B_1(0)$: Had we considered $\Omega = \mathbb{R}^n \backslash \overline{B_1(0)}$ the outward pointing normal would have been $\nu(x) := -x$.

The following theorem [6, §15] is crucial to endow a C^k-regular hypersurface, $k \geqslant 2$, with a special set of coordinates, called Riemannian normal coordinates.

Theorem 3.5 (Implicit function theorem): Let $x_0 \in \Omega$, $\Omega \subset \mathbb{R}^n$ being open, $k > 1$, $f \in C^k(\Omega)$ be such that $\frac{\partial f}{\partial x^1}\Big|_{x_0} \neq 0$. For $x \in \mathbb{R}^n$ we write $x = x^j e_j = x^1 e_1 + \hat{x}$. Then there exists:

(i) an open neighbourhood $N_{x_0^1} \subseteq \mathbb{R}$ of x_0^1 — here $x_0 = x_0^1 e_1 + \hat{x}_0$;

(ii) an open neighbourhood $N_{\hat{x}_0} \subseteq \mathbb{R}^{n-1}$ of $\hat{x}_0$ such that $N_{x_0^1} \times N_{\hat{x}_0} \subseteq \Omega$;

(iii) a unique $g \in C^k(N_{\hat{x}_0}, N_{x_0^1})$, such that

$$\{(g(\hat{x}), \hat{x}) \,|\, \hat{x} \in N_{\hat{x}_0}\} = \{x \in N_{x_0^1} \times N_{\hat{x}_0} \,|\, f(x) = f(x_0)\}\,.$$

Let $\Sigma \subseteq \mathbb{R}^n$ be a C^k-regular hypersurface: For later convenience we consider $k \geqslant 2$. On account of condition (3.1), for all $x_0 \in \Sigma$ there exists an open neighbourhood N_{x_0} of x_0 and $S \in C^k(N_{x_0})$ such that $\nabla S(x) \neq 0$ for all $x \in \Sigma \cap N_{x_0}$, moreover, $\Sigma \cap N_{x_0} = \{x \in N_{x_0} \,|\, S(x) = 0\}$.

In particular there exists $k = 1, \ldots, n$ such that $\frac{\partial S}{\partial x^k}\Big|_{x_0} \neq 0$ —without loss of generality we will assume $k = 1$. On account of Theorem 3.5 there exist two open neighbourhoods $N_{x_0^1} \subset \mathbb{R}$ and $N_{\hat{x}_0} \subset \mathbb{R}^{n-1}$ and $g \in C^k(N_{\hat{x}_0}, N_{x_0^1})$ such that, $N_{x_0^1} \times N_{\hat{x}_0} \subset N_{x_0}$ and

$$(N_{x_0^1} \times N_{\hat{x}_0}) \cap \Sigma = \{(g(\hat{x}), \hat{x}) \,|\, \hat{x} \in N_{\hat{x}_0}\}\,.$$

We can regard $\hat{x}$ as locally defined coordinates for Σ. Moreover, the function $S_g(x) := x^1 - g(\hat{x})$ fulfils all the properties of condition (3.1). In particular, on account of Remark 3.3 we may identify a normal to Σ as

$$\nu = \frac{1}{\sqrt{1 + \delta^{jk} \frac{\partial g}{\partial \hat{x}^j} \frac{\partial g}{\partial \hat{x}^k}}} \left[e_1 - \frac{\partial g}{\partial \hat{x}^k} \delta^{kj} e_j \right] . \tag{3.2}$$

EXAMPLE 3.6: Let $x_0 \in \partial B_1(0)$ be characterized by $x_0 \cdot e_1 = 1$. Letting $N_{x_0} := \{x \in \partial B_1(0) \,|\, x \cdot e_1 > 0\}$ we have an open neighbourhood of x_0 such that

$$\frac{\partial S}{\partial x^1} = x^1 > 0 \quad \forall x \in \partial B_1(0) \cap N_{x_0}\,.$$

The hypotheses of Theorem 3.5 are met and we may identify the data $N_{\hat{x}_0}$, $N_{x_0^1}$ and g by

$$N_{\hat{x}_0} = \{\hat{x} \in \mathbb{R}^{n-1} \,|\, \|\hat{x}\| < 1\}\,, \qquad N_{x_0^1} = (0, 1/2)\,, \qquad g(\hat{x}) = \sqrt{1 - \|\hat{x}\|}\,.$$

We shall now discuss a special class of coordinates on Σ called **Riemannian normal coordinates**. To this avail we recall that the condition $\nabla S(x_0) \neq 0$ is invariant with respect to coordinates transformations —*cf.* Remark 3.3. Thus, without loss of generality we may assume that $x_0 = 0 \in \mathbb{R}^n$ and that $\nabla S(x_0)$ is parallel to e_1 —this can be a achieved by rotations and translations. In particular this entails that $g(0) = 0$ and that $\frac{\partial g}{\partial \hat{x}^k}\Big|_{\hat{x}=0} = 0$ —*cf.* Equation (3.2).

Let $N_{x_0^1}$, $N_{\hat{x}_0}$ and g as above: Our task is to introduce different coordinates, say $t, \xi_2, \ldots, \xi_n$ possibly defined on a smaller open neighbourhood $\widetilde{N}_{x_0} \subset N_{x_0^1} \times N_{\hat{x}_0}$ of x_0 such that

$$\Sigma \cap \widetilde{N}_{x_0} = \{(t, \xi_2, \ldots, \xi_n) \in \widetilde{N}_{x_0} \,|\, t = 0\}\,. \tag{3.3}$$

To this avail, let $x = (g(\hat{x}), \hat{x}) \in (N_{x_0^1} \times N_{\hat{x}_0}) \cap \Sigma$ and let

$$\mathbb{R} \ni t \mapsto x + t\nu = g(\hat{x})e_1 + \hat{x}^k e_k + t\nu\,,$$

be the straight line passing through x and with direction given by the unit normal ν —*cf.* Equation (3.2). We now consider the function

$$\mathbb{R} \times N_{\hat{x}_0} \ni (t, \hat{x}) \mapsto f(t, \hat{x}) := g(\hat{x}) e_1 + \hat{x}^k e_k + t\nu\,;$$

On account of Equation (3.2), f is C^{k-1}-differentiable, moreover, $f(0, \hat{x}) \in (N_{x_0^1} \times N_{\hat{x}_0}) \cap \Sigma$ —notice the loss of regularity of f: This is ultimately the reason for assuming $k \geqslant 2$.

We shall prove that

$$\det \frac{\partial f^j}{\partial z^k}\bigg|_{(t,\hat{x})=(0,\hat{x}_0)} \neq 0\,. \tag{3.4}$$

On account of Theorem 2.5 this would imply that: (a) there exists an open neighbourhood $\widetilde{N}_{x_0} \subset \mathbb{R} \times N_{\hat{x}_0}$ of x_0 such that $f(\widetilde{N}_{x_0})$ is open (and $x_0 \in f(\widetilde{N}_{x_0})$ since $f(0, \hat{x}_0) = x_0$); (b) $f \in C^{k-1}(\widetilde{N}_{x_0}, f(\widetilde{N}_{x_0}))$ is a C^{k-1}-diffeomorphism. This way one obtains new coordinates $(t, \hat{x})$ which are called Riemannian normal coordinates and fulfil property (3.3).

Thus, we have to prove Equation (3.4): To this avail we recall that $\hat{x}_0 = 0$, $\nu = e_1$, $g(0) = 0$ and that $\frac{\partial g}{\partial \hat{x}^k}\Big|_{\hat{x}=0} = 0$. This implies

$$\begin{aligned}
\frac{\partial f^j}{\partial t}\bigg|_{(t,\hat{x})=(0,\hat{x}_0)} &= \nu^j = \delta^j_1\,,\\
\frac{\partial f^j}{\partial \hat{x}^k}\bigg|_{(t,\hat{x})=(0,\hat{x}_0)} &= \left[\frac{\partial g}{\partial \hat{x}^k} e_1 + \delta^j_k e_j + t\frac{\partial \nu}{\partial \hat{x}^k}\right]\bigg|_{(t,\hat{x})=(0,\hat{x}_0)} = \delta^j_k\,.
\end{aligned}$$

It follows that the matrix $\frac{\partial f^j}{\partial \hat{x}^k}\Big|_{(t,\hat{x})=(0,\hat{x}_0)}$ is the identity matrix.

Riemannian normal coordinates lead to the notion of the **induced Lebesgue measure** on Σ. To wit, let $(t, \xi) = (t, \xi_2, \ldots, \xi_n)$ be Riemannian normal coordinates defined in an open neighbourhood $U \subset \mathbb{R}^n$: Without loss of generality we may also assume that $U = I \times U_\Sigma$, where I is an open interval and $U_\Sigma \subseteq \mathbb{R}^{n-1}$ is open. Let $f \in C^0_c(U)$ be a continuous function with compact support contained in U. Denoting with $\widetilde{f}(t, \xi) := f(x(t, \xi))$ we find

$$\int_{\mathbb{R}^n} f(x) \mathrm{d}^n x = \int_I \int_{U_\Sigma} \widetilde{f}(t, \xi) J(t, \xi) \mathrm{d}\xi \mathrm{d}t\,, \tag{3.5}$$

where $J(t, \xi)$ denotes the Jacobian of the map $t, \xi \mapsto x(t, \xi)$. Since f is continuous we may consider the integral

$$\int_\Sigma f|_\Sigma(y) \mathrm{d}S(y) := \int_{U_\Sigma} \widetilde{f}(0, \xi) J(0, \xi) \mathrm{d}\xi\,. \tag{3.6}$$

By a gluing procedure, Equation (3.6) defines a measure on the Borel σ-algebra over Σ which is called the **induced Lebesgue measure on** Σ —We refer to [9] for further details. Whenever $\Sigma = \partial\Omega$ the corresponding integral will denoted by $\boxed{\oint_{+\partial\Omega}}$, the plus sign denoting the chosen orientation.

Example 3.7: When applying the previous machinery to $\partial B_1(0)$ one may identify Riemannian normal coordinates with polar coordinates (on $\mathbb{R}^n$). The latter coordinates system on $\mathbb{R}^n$ can be defined directly and generalizes what is known for the case $n = 2, 3$. In these cases we have

$$\mathbb{R}^2\colon x = R\cos\theta\,,\quad y = R\sin\theta\,,\quad R > 0\,,\, \theta \in (-\pi, \pi)\,, \tag{3.7}$$

$$\mathbb{R}^3\colon z = R\cos\theta\,,\quad x = R\sin\theta\cos\varphi\,,\quad y = R\sin\theta\sin\varphi\,,\quad R > 0,\, \theta \in (0, \pi)\,,\, \varphi \in (-\pi, \pi)\,. \tag{3.8}$$

For the general case, let $x = (x^1, \ldots, x^n) \in \mathbb{R}^n$. We define $R := \|x\|$ and $\theta_{n-1} \in (0, \pi)$ by

$$x^n = R\cos\theta_{n-1}\,.$$

The angle $\theta_{n-1} \in (0, \pi)$ is nothing but the angle between x (though as a vector) and e_n. Notice

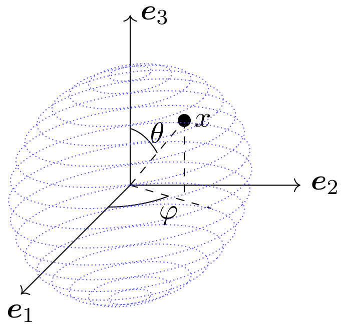


Figure 2: Polar coordinates for $x \in \mathbb{R}^3$.

that, since $\theta_{n-1} \in (0, \pi)$, we are excluding the case where x is parallel to e_n: This will not be a problem as the resulting points form a closed set —which, moreover, has null Lebesgue measure. Projecting x into the plane $\mathbb{R}^{n-1}$ orthogonal to e_n we find a new vector $x_{n-1} \in \mathbb{R}^{n-1}$ whose norm will be equal to $R\sin\theta_{n-1}$. We now repeat the procedure, introducing a new angle $\theta_{n-2} \in (0, \pi)$ such that

$$x^{n-1} = R\sin\theta_{n-1}\cos\theta_{n-2}\,.$$

Mutatis mutandis, the angle $\theta_{n-2} \in (0, \pi)$ has the same interpretation of θ_{n-1} —notice that we are again losing a Lebesgue null-measure closed set because we are excluding the case of x_{n-1} parallel to e_{n-1}. Proceeding by induction we find $R > 0$, angles $\theta_{n-1}, \ldots, \theta_2 \in (0, \pi)$ and

$\theta_1 \in (-\pi, \pi)$ defined by

$$
\begin{aligned}
x^n &= R\cos\theta_{n-1}\,, && (3.9a)\\
x^{n-1} &= R\sin\theta_{n-1}\cos\theta_{n-2}\,, && (3.9b)\\
&\vdots \\
x^3 &= R\prod_{\ell=1}^{n-3}\sin\theta_{n-\ell}\cos\theta_2\,, && (3.9c)\\
x^1 &= R\prod_{\ell=1}^{n-2}\sin\theta_{n-\ell}\cos\theta_1\,, && (3.9d)\\
x^2 &= R\prod_{\ell=1}^{n-2}\sin\theta_{n-\ell}\sin\theta_1\,. && (3.9e)
\end{aligned}
$$

The swap between the components x^1, x^2 is made to ensure that, for $n = 2$, Equation (3.9) reduces to Equation (3.7). Also, notice the different range of $\theta_1 \in (-\pi, \pi)$ with respect to the other angles —this matches with the $n = 3$-dimensional case, *c.f.* Equation (3.8).

Thus, we have defined a bijective map

$$
f\colon (0, +\infty) \times (0, \pi)^{n-1} \times (-\pi, \pi) \ni (R, \theta) := (R, \theta_{n-1}, \dots, \theta_1) \mapsto x \in \Omega\,,
$$

where $f(R, \theta) := x$ is defined via Equation (3.9) while $\Omega \subset \mathbb{R}^n$ is the open set defined by

$$
\Omega := \mathbb{R}^n \Big\backslash \Big[\{0\} \cup \{x \in \mathbb{R}^n \,|\, \exists k \in \{1, \dots, n\}\colon\ x \parallel e_k\} \cup \{x \in \mathbb{R}^n \,|\, x^1 \leqslant 0\,, x^2 = 0\}\Big]\,,
$$

where $x \parallel e_k$ means that x is parallel to e_k. Notice that Ω is the complement of a Lebesgue null-measure closed set.

We shall now prove that the Jacobian of f is non singular at all point (R, θ). Since f is C^∞, Corollary 2.6 would then imply that f defines a C^∞-diffeomorphism. To this avail we observe that, denoting $x = f(R, \theta)$,

$$
\frac{\partial x^k}{\partial R} = \frac{x^k}{R}\,, \qquad \frac{\partial x^k}{\partial \theta_{n-\ell}} = \begin{cases} 0 & k > n - \ell + 1 \\ -x^k \tan\theta_{n-\ell} & k = n - \ell + 1 \\ x^k \cot\theta_{n-\ell} & k < n - \ell + 1 \end{cases}
$$

It follows that, denoting by J_n the Jacobian of f at (R, θ), it holds

$$
J_n = \begin{vmatrix}
x^n/R & -x^n\tan\theta_{n-1} & 0 & \dots & 0 \\
x^{n-1}/R & x^{n-1}\cot\theta_{n-1} & -x^{n-1}\tan\theta_{n-2} & \dots & 0 \\
\vdots & & & & \\
x^1/R & x^1\cot\theta_{n-1} & \dots & x^1\cot\theta_2 & -x^1\tan\theta_1 \\
x^2/R & x^2\cot\theta_{n-1} & \dots & x^2\cot\theta_2 & x^2\cot\theta_1
\end{vmatrix}.
$$

Laplace's rule applied to the first line leads to

$$
\begin{aligned}
J_n &= \frac{x^n}{R}\begin{vmatrix} x^{n-1}\cot\theta_{n-1} & -x^{n-1}\tan\theta_{n-2} & \ldots & 0 \\ \vdots & & & \\ x^1\cot\theta_{n-1} & x^1\cot\theta_{n-2} & \ldots\; x^1\cot\theta_2 & -x^1\tan\theta_1 \\ x^2\cot\theta_{n-1} & x^2\cot\theta_{n-2} & \ldots\; x^2\cot\theta_2 & x^2\cot\theta_1 \end{vmatrix} \\
&+ x^n\tan\theta_{n-1}\begin{vmatrix} x^{n-1}/R & -x^{n-1}\tan\theta_{n-2} & \ldots & \ldots & 0 \\ \vdots & & & & \\ x^1/R & x^1\cot\theta_{n-2} & \ldots & x^1\cot\theta_2 & -x^1\tan\theta_1 \\ x^2/R & x^2\cot\theta_{n-2} & \ldots & x^2\cot\theta_2 & x^2\cot\theta_1 \end{vmatrix} \\
&= x^n\cot\theta_{n-1}(\sin\theta_{n-1})^{n-1}|J_{n-1}| \qquad \begin{array}{c}\text{collecting } R\cot\theta_{n-1} \text{ from the first column} \\ \text{collecting a factor } \sin\theta_{n-1} \text{ from each colomn}\end{array} \\
&+ x^n\tan\theta_{n-1}(\sin\theta_{n-1})^{n-1}|J_{n-1}| \qquad \text{collecting a factor } \sin\theta_{n-1} \text{ from each column} \\
&= R(\sin\theta_{n-1})^{n-2}|J_{n-1}|\,.
\end{aligned}
$$

Since $J_2 = R$ we find by induction

$$J_n = R^{n-1}(\sin\theta_{n-1})^{n-2}(\sin\theta_{n-2})^{n-3}\cdots\sin\theta_2 > 0\,. \tag{3.10}$$

Thus f is a C^∞-diffeomorphism: The coordinates (R,θ) are called **polar coordinates** in $\mathbb{R}^n$.

We observe that the outward pointing normal ν on $\partial B_1(0)$ is given by $\nu = x/R$, therefore, it holds

$$\frac{\partial u}{\partial\nu} = \frac{x^k}{R}\frac{\partial u}{\partial x^k} = \frac{\partial\widetilde{u}}{\partial R}\,, \tag{3.11}$$

for all $u \in C^1(\mathbb{R}^n)$, where $\widetilde{u}(R,\theta) := u(x(R,\theta))$.

Finally we specialize Equations (3.5)-(3.6) to the case of polar coordinates. For that, let $f \in C^1_c(\mathbb{R}^n)$ and let $\widetilde{f}$ be the function defined by $\widetilde{f}(R,\theta) = f(x(R,\theta))$. It follows that

$$\int_{\mathbb{R}^n} f(x)\mathrm{d}^n x = \int_0^{+\infty}\int_{(0,\pi)^{n-2}\times(-\pi,\pi)}\widetilde{f}(R,\theta)J_n\mathrm{d}\theta\mathrm{d}R \tag{3.12a}$$

$$= \int_0^{+\infty}\oint_{+\partial B_1(0)}\widetilde{f}(R,\theta)\mathrm{d}\Omega_n R^{n-1}\mathrm{d}R \tag{3.12b}$$

$$= \int_0^{+\infty}\oint_{+\partial B_R(0)}\widetilde{f}(R,\theta)\mathrm{d}S\mathrm{d}R\,, \tag{3.12c}$$

where $\mathrm{d}\theta = \mathrm{d}\theta_{n-1}\cdots\mathrm{d}\theta_1$ is a short notation while $\boxed{\mathrm{d}\Omega_n} := (\sin\theta_{n-1})^{n-2}(\sin\theta_{n-2})^{n-3}\cdots\sin\theta_2\mathrm{d}\theta$ —*cf.* Equation (3.10)— is the induced Lebesgue measure on the unit sphere $\partial B_1(0)$. Similarly $\mathrm{d}S = J_n\mathrm{d}\theta = R^{n-1}\mathrm{d}\Omega_n$ is the induced Lebesgue measure on $\partial B_R(0)$. Since the Lebesgue measure is invariant under translations, Equation (3.12) holds if we replace $\partial B_R(0)$ with $\partial B_R(x_0)$ for any $x_0 \in \mathbb{R}^n$ —the resulting induced measure $\mathrm{d}S$ is the same.

Remark 3.8: Applying Equation (3.12) for $f = 1_{B_R(0)}$ we obtain

$$|B_R(0)| = \int_0^R |\partial B_{R'}(0)| \mathrm{d}R' = \int_0^R |\partial B_1(0)|(R')^{n-1} \mathrm{d}R' = \frac{R^n}{n}|\partial B_1(0)| = \frac{R}{n}|\partial B_R(0)| \,. \tag{3.13}$$

In particular it is interesting to compute the exact value of $\boxed{\omega_n} := |\partial B_1(0)|$. To this avail we recall the Gaussian integral

$$\int_{\mathbb{R}} e^{-x^2} \mathrm{d}x = \sqrt{\pi} \,. \tag{3.14}$$

We will also need the **Euler-Gamma function** which is defined by

$$\Gamma(\alpha) := \int_0^{+\infty} e^{-z} z^{\alpha-1} \mathrm{d}z \,, \qquad \alpha > 0 \,. \tag{3.15}$$

By direct inspection one has $\Gamma(1) = \Gamma(2) = 1$ and

$$\alpha\Gamma(\alpha) = \int_0^{+\infty} e^{-z} \frac{\mathrm{d}z^\alpha}{\mathrm{d}z} \mathrm{d}z = \Gamma(\alpha+1) \,.$$

from which it follows that $\Gamma(k) = (k-1)!$ as well as $\lim_{\alpha \to 0^+} \alpha\Gamma(\alpha) = 1$. Finally

$$\Gamma(1/2) = \int_0^{+\infty} e^{-z} z^{-1/2} \mathrm{d}z \overset{z=x^2}{=} 2\int_0^{+\infty} e^{-x^2} \mathrm{d}x \overset{(3.14)}{=} \sqrt{\pi} \,.$$

With these preliminaries we observe that

$$\pi^{n/2} = \left(\int_{\mathbb{R}} e^{-x^2} \mathrm{d}x \right)^n = \int_{\mathbb{R}^n} e^{-\|x\|^2} \mathrm{d}^n x \overset{(3.12)}{=} \omega_n \int_0^{+\infty} e^{-R^2} R^{n-1} \mathrm{d}R = \frac{\omega_n}{2} \Gamma(n/2) \,.$$

Thus we conclude

$$\omega_n = \frac{2\pi^{n/2}}{\Gamma(n/2)} = \begin{cases} \dfrac{2\pi^k}{(k-1)!} & n = 2k \geqslant 2 \\ \dfrac{2^{k+1}\pi^k}{(2k-1)!!} & n = 2k+1 \geqslant 3 \end{cases} , \tag{3.16}$$

where $n!!$ denotes the **double factorial** $n!! := n(n-2)\cdots 1$. Notice that $\lim_{n\to+\infty} \omega_n = 0$.

4 The Cauchy problem for local PDEs

In this section we will introduce the notion of Cauchy problem for a semi-linear PDE as by Definition 2.2 —in fact, the definition will be stated for a slightly larger class of PDEs. The main references for this section are [3, §1],[11] and [21, §1.3].

The Cauchy problem for local PDEs

Through the whole section we will consider a general **local PDE of second order** which reads

$$F\left(x, u(x), \frac{\partial u}{\partial x^k}, \frac{\partial^2 u}{\partial x^k \partial x^j}\right) = 0\,, \tag{4.1}$$

for $u \in C^2(\Omega)$ where $F \in C^0(\Omega \times \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^{n(n+1)/2})$.

Definition 4.1: Let $\Omega \subset \mathbb{R}^n$ be a non-empty open set and let $\Sigma \subseteq \Omega$ be an orientable C^2-regular hypersurface with unit normal ν, *cf.* Definition (3.1). The **Cauchy problem** for Equation (4.1) with initial data on Σ consists of determining $u \in C^2(\Omega)$ such that

$$F\left(x, u(x), \frac{\partial u}{\partial x^k}, \frac{\partial^2 u}{\partial x^k \partial x^j}\right) = 0\,, \tag{4.2a}$$

$$u|_\Sigma = u_0\,, \qquad \frac{\partial u}{\partial \nu} = u_1\,, \tag{4.2b}$$

where $u_0 \in C^2(\Sigma)$, $u_1 \in C^1(\Sigma)$ are given while we set $\frac{\partial u}{\partial \nu} = \nu \cdot \nabla u = \nu^k \frac{\partial u}{\partial x^k}$ —notice that this expression is defined only on Σ because ν is defined only there, *cf.* Remark 3.3.

Observe that the regularity of u_0 and u_1 is compatible with the prescribed regularity of u. The Cauchy problem (4.2) is said **well-posed in the sense of Hadamard** if:

C1 There exists a solution u to (4.2);

C2 The found solution u is unique;

C3 A "slight variation" of the data u_0, u_1 leads to a "small change" in the resulting solution u. This formulation is purposely vague, but it can be formalized as follows. Let assume that $u_0^{(n)} \in C^2(\Sigma)$, $u_1^{(n)} \in C^1(\Sigma)$ are sequences such that $\lim_{n\to+\infty} u_0^{(n)} = u_0 \in C^2(\Sigma)$ and $\lim_{n\to+\infty} u_1^{(n)} = u_1 \in C^1(\Sigma)$ in a fixed topology, *e.g.* in the pointwise convergence. Let $u^{(n)} \in C^2(\Omega)$ (*resp.* $u \in C^2(\Omega)$) be the sequence of solutions to the Cauchy problem (4.2) associated with the data $u_0^{(n)}, u_1^{(n)}$ (*resp.* u_0, u_1). Then the Cauchy problem (4.2) fulfils Condition 3 if $\lim_{n\to+\infty} u^{(n)} = u$.

The Cauchy problem is said **ill-posed** if at least one among 1-2-3 fails.

Remark 4.2:

(i) Condition 3 is somehow ambiguous, because it depends on the choice of the topologies for the data u_0, u_1 and also for the solution u. Ultimately, this choice is linked to the concrete situation under investigation.

(ii) The concept of well-posedness is a formalization of the natural expectation that the data of the Cauchy problem (4.2) should determine the solution u uniquely —this is basically the content of Conditions 1-2. Moreover, Condition 3 entails that small perturbations of the initial data u_0, u_1 do not alter significantly the resulting solution u: This is important in applications because there is always a slight uncertainty due to our measurements.

However, we stress that this expectation may fail in concrete situations: An important class of ill-posed problems is the one of inverse problems, among others the X-rays tomography (TAC) imaging —*cf.* [15].

(iii) Let consider a concrete example of an ill-posed Cauchy problem. We consider the Cauchy problem on $\mathbb{R}^2$ given by

$$\Delta u = 0\,, \qquad u(x,0) = u_0(x)\,, \qquad \frac{\partial u}{\partial y}(x,0) = u_1(x)\,. \tag{4.3}$$

Let us consider the data

$$u_0^{(n)}(x) := 0\,, \qquad u_1^{(n)}(x) := \frac{1}{n}\sin(nx)\,.$$

In this situation the solution is unique, as we shall prove in Section 7, and it is given by

$$u^{(n)}(x,y) = \frac{1}{n^2}\sin(nx)\sinh(ny)\,,$$

where $\sinh(z) := (e^z - e^{-z})/2$ is the hyperbolic sine.

We notice that $u_0^{(n)}, u_1^{(n)} \to 0$ in the topology of pointwise convergence. Moreover, the Cauchy problem (4.3) with data $u_0 = u_1 = 0$ has the unique solution $u = 0$. Nevertheless, $u^{(n)}$ does not converge to u, therefore, the Cauchy problem (4.3) is ill-posed.

This example shows that the Cauchy problem of the Laplace operator (2.6) is ill-posed on general domains. For this reason one resorts to a slightly different class of problems, called **boundary-value problems**, which we will discuss in Sections 7-8. To wit, let $\Omega \subset \mathbb{R}^n$ be open with $\overline{\Omega}$ compact and with a C^1-regular and orientable boundary $\partial\Omega$. Within this setting, the boundary-value problem for the Laplace operator with (Dirichlet) boundary conditions is formulated as follows. The task is to find $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ such that

$$\Delta u = f \quad \text{in } \Omega\,, \qquad u|_{\partial\Omega} = u_0\,,$$

where $u_0 \in C^0(\partial\Omega)$ and $f \in C^0(\Omega)$ are assigned.

The Cauchy-Kovalevskaja Theorem for semi-linear PDEs

We will now discuss the Cauchy-Kovalevskaja Theorem, an important result stating the local existence and uniqueness of the Cauchy problem (4.2) for the case of semi-linear PDEs. The same result applies for the larger class of quasi-linear PDE, *cf.* Remark 2.3-ii. The main reference for this argument is [3, §4.6].

The hypotheses of Cauchy-Kovalevskaja Theorem are rather strong, making its application difficult in concrete situations. However, this theorem is a crucial building block in the proof of many of the more general results at disposal in the literature. Moreover, a simple application of this theorem leads to the notion of characteristic hypersurface, which is another important ingredient providing information on the PDE of interest, *cf.* Section 6.

For that sake of simplicity we begin by formulating the Cauchy-Kovalevskaja theorem for the case of the Cauchy problem (4.2) on $\Omega = \mathbb{R}^n$ and $\Sigma = \{x \in \mathbb{R}^n \,|\, x^1 = 0\}$ —notice that in this case $\nu = e_1$. Here we adopted the notation $x = x^k e_k = x^1 e_1 + \hat{x}$ for all $x \in \mathbb{R}^n$, where $\hat{x} \in \mathbb{R}^{n-1}$. For the rest of the section we will state the Cauchy-Kovalevskaja Theorem and recollect a few useful result: The proof is postponed to Section 5.

The following definition recollect the first main assumption of the Cauchy-Kovalevskaja theorem.

Definition 4.3: The Cauchy problem (4.2) for a semi-linear PDE is called in **normal form** if there exists $a^{1k}, a^{jk} \in C^0(\mathbb{R}^n)$ such that the problem (4.2) can be written as

$$\frac{\partial^2 u}{(\partial x^1)^2} = a^{1k}(x)\frac{\partial^2 u}{\partial x^1 \partial \hat{x}^k} + a^{jk}(x)\frac{\partial^2 u}{\partial \hat{x}^k \partial \hat{x}^j} + \Phi\left(x, u(x), \frac{\partial u}{\partial x^1}, \frac{\partial u}{\partial \hat{x}^k}\right), \tag{4.4a}$$

$$u(0,\hat{x}) = u_0(\hat{x}) \qquad \left.\frac{\partial u}{\partial x^1}\right|_{(0,\hat{x})} = u_1(\hat{x}). \tag{4.4b}$$

Notice that in Equation (4.4a) the highest derivative in x^1 appears factorized in the right-hand side of the PDE. The possibility of recasting a semi-linear PDE in normal form (4.4) is not granted a priori: We will discuss this assumption in more details in Section 6.

The second main ingredient of the Cauchy-Kovalevskaja theorem is analyticity.

Definition 4.4: A function $f \in C^\infty(\Omega)$, $\Omega \subset \mathbb{R}^n$ being open, is called **real-analytic** at $x_0 \in \Omega$ if the Taylor series of f centred at x_0 converges absolutely in a neighbourhood of x_0 and coincides with f:

$$\exists N_{x_0} \overset{\text{open}}{\subseteq} \Omega\,, x_0 \in N_{x_0} : \forall x \in N_{x_0}\,,\ f(x) = \sum_{N=0}^{+\infty} \sum_{|\alpha|=N} \frac{1}{\alpha_1! \cdots \alpha_n!} \partial^\alpha f(x_0) \prod_{\ell=1}^{n} (x^\ell - x_0^\ell)^{\alpha_\ell}\,, \tag{4.5}$$

where the above series converges absolutely. In particular, f is real-analytic if it is a real-analytic at every point $x_0 \in \Omega$.

REMARK 4.5:

(i) The formula in condition (4.5) is obtained by considering the function $g_{x_0}(t) := f(x_0 + t(x - x_0))$ for fixed $x, x_0 \in \mathbb{R}^n$. Then the Taylor series of f at x_0 evaluated at x coincides with the Taylor series of g_{x_0} at $t = 0$ evaluated at $t = 1$. The coefficients appearing in (4.5) arise from the identity

$$\frac{\mathrm{d}^N g_{x_0}}{\mathrm{d}t^N}\bigg|_{t=0} = \sum_{\substack{\alpha \in \mathbb{Z}^n_+ \\ |\alpha| = N}} \frac{N!}{\alpha_1! \cdots \alpha_n!} (\partial^\alpha f)(x_0) \prod_{\ell=1}^n (x^\ell - x_0^\ell)^{\alpha_\ell} ,$$

which can be proved by induction or by combinatorial arguments.

(ii) For $x, x_0 \in \mathbb{R}^n$ and $\alpha \in \mathbb{Z}^n_+$ we set

$$\alpha! := \alpha_1! \cdots \alpha_n! , \qquad (x - x_0)^\alpha := (x^1 - x_0^1)^{\alpha_1} \cdots (x^n - x_0^n)^{\alpha_n} ;$$

Thus the Taylor series of f at x_0 reads

$$\sum_{N=0}^{+\infty} \sum_{\substack{\alpha \in \mathbb{Z}^n_+ \\ |\alpha| = N}} \frac{1}{\alpha!} \partial^\alpha f(x_0)(x - x_0)^\alpha = \sum_{\alpha \in \mathbb{Z}^n_+} \frac{1}{\alpha!} \partial^\alpha f(x_0)(x - x_0)^\alpha .$$

With the same notation one may prove by induction the generalized binomial formulas

$$(x^1 + \ldots + x^n)^N = \sum_{\substack{\alpha \in \mathbb{Z}^n_+ \\ |\alpha| = N}} \frac{N!}{\alpha!} x^\alpha , \qquad (x + y)^\alpha = \sum_{\substack{\beta_1, \beta_2 \in \mathbb{Z}^n_+ \\ \beta_1 + \beta_2 = \alpha}} \frac{\alpha!}{\beta_1! \beta_2!} x^{\beta_1} y^{\beta_2} , \tag{4.6}$$

as well as the “generalized” Leibniz rule

$$\partial^\alpha (u_1 u_2) = \sum_{\substack{\beta_1, \beta_2 \in \mathbb{Z}^n_+ \\ \beta_1 + \beta_2 = \alpha}} \frac{\alpha!}{\beta_1! \beta_2!} (\partial^{\beta_1} u_1)(\partial^{\beta_2} u_2) . \tag{4.7}$$

We have now all ingredients to state the Cauchy-Kovalevskaja theorem.

THEOREM 4.6 (Cauchy-Kovalevskaja): Let consider the Cauchy problem (4.4). Let assume that u_0, u_1 are real-analytic at $\hat{x}_0 \in \Sigma$. Moreover, let assume that $\{a^{jk}\}_{j,k=1,\ldots,n}$ are real-analytic at $x_0 = (0, \hat{x})$ while Φ is real-analytic at X_0, where

$$X_0 = \left(x_0, u_0(\hat{x}_0), u_1(\hat{x}_0), \frac{\partial u_0}{\partial \hat{x}^k}\bigg|_{\hat{x}_0} \right) .$$

Then there is a neighbourhood $N_{x_0} \subset \Omega$ such that there exists a unique solution $u \in C^2(N_{x_0})$ to the Cauchy problem (4.4) which is real-analytic at x_0.

We stress that the existence and uniqueness result provided by Theorem 4.6 holds only locally and may fail globally.

As we shall see, the proof of Theorem 4.6 essentially boils down to the computation of the Taylor coefficients of the solution u: This implies uniqueness of the solution. Concerning existence, we will need to prove the convergence of the found Taylor series of u: To this avail we will apply the “method of majorants”.

DEFINITION 4.7: Let $\underline{f} = (f_\alpha)_{\alpha\in\mathbb{Z}^n_+}$ and $\underline{g} = (g_\alpha)_{\alpha\in\mathbb{Z}^n_+}$ where $f_\alpha \in \mathbb{R}$, $g_\alpha \geqslant 0$. We say that $\underline{f}$ is **majorized** by $\underline{g}$, and we write $\underline{f} \ll \underline{g}$, if $|f_\alpha| \leqslant g_\alpha$ for all $\alpha \in \mathbb{Z}^n_+$.

LEMMA 4.8 ():

(a) If $\underline{f} \ll \underline{g}$ and there exists $R > 0$ such that $g(x) := \sum_{\alpha\in\mathbb{Z}^n_+} g_\alpha x^\alpha$ converges absolutely for $\|x\| < R$ then also $f(x) := \sum_{\alpha\in\mathbb{Z}^n_+} f_\alpha x^\alpha$ converges absolutely for $\|x\| < R$;

(b) If there exists $R > 0$ such that $f(x) := \sum_{\alpha\in\mathbb{Z}^n_+} f_\alpha x^\alpha$ converges absolutely for $\|x\| < R$, then $\underline{f} = (f_\alpha)_{\alpha\in\mathbb{Z}^n_+}$ has a majorant $\underline{g}$. Moreover, for all $s \in (0, R/\sqrt{n})$ there exists a constant $C > 0$ such that

$$g_\alpha := C\frac{|\alpha|!}{\alpha!}\frac{1}{s^{|\alpha|}}\,,$$

defines a majorant $\underline{g}$ of $\underline{f}$ such that $g(x) := \sum_{\alpha\in\mathbb{Z}^n_+} g_\alpha x^\alpha$ converges absolutely for $\|x\| < s/n$.

Proof.

a By direct inspection we have

$$\sum_{\alpha\in\mathbb{Z}^n_+} |f_\alpha|\,|x^\alpha| \leqslant \sum_{\alpha\in\mathbb{Z}^n_+} |g_\alpha|\,|x^\alpha| < +\infty \qquad \forall x \in B_R(0)\,.$$

b Let $s > 0$ and set $y = (s, \ldots, s) \in \mathbb{R}^n$: Then $\|y\| = s\sqrt{n} < R$ provided $s < R/\sqrt{n}$. Thus, for $s \in (0, R/\sqrt{n})$ the series $f(y) = \sum_{\alpha\in\mathbb{Z}^n_+} f_\alpha y^\alpha$ converges absolutely. In particular there exists $C > 0$ such that $|f_\alpha||y^\alpha| \leqslant C$ for all $\alpha \in \mathbb{Z}^n_+$. Since $|y^\alpha| = |y^1|^{\alpha_1}\cdots|y^n|^{\alpha_n} = s^{|\alpha|}$ we find

$$|f_\alpha| \leqslant \frac{C}{s^{|\alpha|}} \leqslant C\frac{|\alpha|!}{\alpha!}\frac{1}{s^{|\alpha|}} =: g_\alpha\,,$$

where we used $\alpha! \leqslant |\alpha|!$ —*cf.* Remark 4.5. This proves that $\underline{f}$ has a majorant. We shall now prove that $\underline{g}$ leads to an absolutely convergent power series for $x \in B_{s/n}(0)$. To this avail we recall that $(1-z)^{-1} = \sum_{k\geqslant 0} z^k$ for $|z| < 1$ so that

$$\begin{aligned}
\sum_{\alpha\in\mathbb{Z}^n} g_\alpha |x^\alpha| &= C \sum_{\alpha\in\mathbb{Z}^n_+} \frac{|\alpha|!}{\alpha!}\frac{|x|^\alpha}{s^{|\alpha|}} && \underset{|x^\alpha|=|x^1|^{\alpha_1}\cdots|x^n|^{\alpha_n}=:|x|^\alpha}{\text{Def. } g_\alpha} \\
&= C \sum_{k\geqslant 0} \sum_{\substack{\alpha\in\mathbb{Z}^n_+ \\ |\alpha|=k}} \frac{|\alpha|!}{\alpha!}\frac{|x|^\alpha}{s^k} \\
&= C \sum_{k\geqslant 0} \left(\frac{|x^1| + \ldots + |x^n|}{s}\right)^k && \text{Eq. (4.6)} \\
&= \frac{C}{1 - (|x^1| + \ldots + |x^n|)/s} && |x^1| + \ldots + |x^n| \leqslant n\|x\| < s\,.
\end{aligned}$$

□

5 The proof of the Cauchy-Kovalevskaja Theorem

This section is entirely devoted to the proof of Theorem 4.6. To this avail, Lemma 4.8 will come in handy.

We begin by recasting the Cauchy problem (4.4) in a convenient form. Without loss of generality, we may assume $\hat{x}_0 = 0$ (*i.e.* $x_0 = 0$): This can be achieved with a suitable change of coordinates which preserves problem (4.4) on account of Lemma 2.7. Moreover, such coordinate change is a mere translation, therefore, the analyticity assumptions are preserved as well.

Additionally, we observe that we may assume $u_1 = u_0 = 0$ without loss of generality. Indeed let u be a solution to the Cauchy problem (4.4): Then setting

$$u(x^1, \hat{x}) := u'(x^1, \hat{x}) + u_0(\hat{x}) + x^1 u_1(\hat{x}),$$

we observe that the boundary conditions (4.4b) are fulfilled provided that

$$u'(0, \hat{x}) = 0, \qquad \frac{\partial u'}{\partial x^1}\Big|_{x^1=0} = 0.$$

Moreover, by direct inspection

$$\begin{aligned}\frac{\partial^2 u'}{\partial (x^1)^2} = \frac{\partial^2 u}{\partial (x^1)^2} &= a^{1k}(x)\frac{\partial^2 u}{\partial x^1 \partial \hat{x}^k} + a^{jk}(x)\frac{\partial^2 u}{\partial \hat{x}^k \partial \hat{x}^j} + \Phi\left(x, u(x), \frac{\partial u}{\partial x^1}, \frac{\partial u}{\partial \hat{x}^k}\right) \\ &= a^{1k}(x)\frac{\partial^2 u'}{\partial x^1 \partial \hat{x}^k} + a^{jk}(x)\frac{\partial^2 u'}{\partial \hat{x}^k \partial \hat{x}^j} + \Phi'\left(x, u'(x), \frac{\partial u'}{\partial x^1}, \frac{\partial u'}{\partial \hat{x}^k}\right),\end{aligned}$$

where Φ is again real-analytic. This proves that u is a solution to the Cauchy problem (4.4) if and only if u' is a solution to a quasilinear Cauchy problem (4.4) with vanishing boundary data. Moreover, in the process the analyticity of the coefficients is preserved.

The next step is to recast the Cauchy problem (4.4) (with $u_0 = u_1 = 0$) into a first order system. To this avail we set

$$\boldsymbol{u} = \left(x^1, u, \frac{\partial u}{\partial x^k}\right) \in \mathbb{R}^m, \qquad m = n + 2.$$

The Cauchy problem (4.4) for u is equivalent to the following Cauchy problem for $\boldsymbol{u}$:

$$\frac{\partial \boldsymbol{u}}{\partial x^1} = \sum_{j=2}^{n} \boldsymbol{B}_j(\hat{x}, \boldsymbol{u})\frac{\partial \boldsymbol{u}}{\partial \hat{x}^j} + \boldsymbol{c}(\hat{x}, \boldsymbol{u}), \tag{5.1a}$$

$$\boldsymbol{u}(0, \hat{x}) = 0. \tag{5.1b}$$

Here $\boldsymbol{B}_j$ and $\boldsymbol{c}$ are built out of a^{1k}, a^{jk} and Φ and have analytic components. Notice that the explicit dependence on x^1 on $\boldsymbol{B}_j$ and $\boldsymbol{c}$ is included in a further component of $\boldsymbol{u}$. Equation

(5.1a) codifies Equation (4.4a) together with a few more self-consistent relations due to Schwarz Theorem. To wit, componentwise Equation (5.1a) reads

$$\frac{\partial u^k}{\partial x^1} = \sum_{j=2}^{n} B_{j\ell}{}^{k}(\hat{x}, \boldsymbol{u}) \frac{\partial u^\ell}{\partial \hat{x}^j} + c^k(\hat{x}, \boldsymbol{u}) \,, \qquad k = 1, \ldots, m \,.$$

Equation (4.4a) is obtained for $k = 3$ whereas for $k \neq 3$ we have

$$\frac{\partial u^k}{\partial x^1} = \begin{cases} 1 & k = 1 \\ u^3 & k = 2 \\ \dfrac{\partial u^3}{\partial \hat{x}^k} & k \geqslant 4 \end{cases} .$$

(In fact, $\boldsymbol{B}_j$ depends on $\boldsymbol{u}$ only through the component $u^1 = x^1$.) Thus, it is enough to prove that there exists an analytic solution $\boldsymbol{u}$ to the Cauchy problem (5.1).

Uniqueness of u

Let us assume that there exists a solution $\boldsymbol{u}$ to the Cauchy problem (5.1) that is analytic in a neighbourhood of $x_0 = 0$. We now prove that $\boldsymbol{u}$ is uniquely determined. To this avail we observe that the analyticity condition implies that $\boldsymbol{u}$ is uniquely determined by its derivatives $\boldsymbol{u}_\alpha := \partial^\alpha \boldsymbol{u}(0)$ for all $\alpha \in \mathbb{Z}_+^n$. For later convenience we set $\boldsymbol{B}_{j\gamma} := \partial^\gamma \boldsymbol{B}_j(0)$, and $\boldsymbol{c}_\gamma := \partial^\gamma \boldsymbol{c}(0)$ where $\gamma \in \mathbb{Z}_+^{n+m}$.

In what follows we shall write $\alpha = (\alpha^1, \hat{\alpha}) \in \mathbb{Z}_+^n$ where $\hat{\alpha} \in \mathbb{Z}_+^{n-1}$. We observe that, for all $\hat{\alpha} \in \mathbb{Z}_+^{n-1}$ the boundary conditions (5.1b) implies

$$\boldsymbol{u}_{(0,\hat{\alpha})} = (\partial^{\hat{\alpha}} \boldsymbol{u})|_{x=0} = \partial^{\hat{\alpha}} (\boldsymbol{u}|_{x^1=0})|_{\hat{x}=0} = 0 \,.$$

Similarly, Equation (5.1a) implies

$$\begin{aligned}
(\partial^{(1,\hat{\alpha})} \boldsymbol{u})|_{x=0} &= \partial^{\hat{\alpha}} \left[\sum_{j=2}^{n} \boldsymbol{B}_j \frac{\partial \boldsymbol{u}}{\partial \hat{x}^j} + \boldsymbol{c} \right] \Bigg|_{x=0} = \boldsymbol{c}_{(0,\hat{\alpha})} \,, \\
(\partial^{(2,\hat{\alpha})} \boldsymbol{u})|_{x=0} &= \partial^{\hat{\alpha}} \left[\sum_{j=2}^{n} \left(\frac{\partial \boldsymbol{B}_j}{\partial u^\ell} \frac{\partial u^\ell}{\partial x^1} \frac{\partial \boldsymbol{u}}{\partial \hat{x}^j} + \boldsymbol{B}_j \frac{\partial^2 \boldsymbol{u}}{\partial x^1 \partial \hat{x}^k} \right) + \frac{\partial \boldsymbol{c}}{\partial u^\ell} \frac{\partial u^\ell}{\partial x^1} \right] \Bigg|_{x=0} \\
&= \partial^{\hat{\alpha}} \left[\sum_{j=2}^{n} \left(\frac{\partial \boldsymbol{B}_j}{\partial u^\ell} \frac{\partial u^\ell}{\partial x^1} \frac{\partial \boldsymbol{u}}{\partial \hat{x}^j} + \boldsymbol{B}_j \left(\sum_{j_2=2}^{n} \boldsymbol{B}_{j_2} \frac{\partial^2 \boldsymbol{u}}{\partial \hat{x}^{j_2} \partial \hat{x}^k} + \boldsymbol{c} \right) \right) + \frac{\partial \boldsymbol{c}}{\partial u^\ell} \frac{\partial u^\ell}{\partial x^1} \right] \Bigg|_{x=0} \\
&= \boldsymbol{p}_{(2,\hat{\alpha})}(\boldsymbol{B}_{j\gamma_j}, \boldsymbol{c}_\gamma, \boldsymbol{u}_\beta) \,,
\end{aligned}$$

where we used Equations (5.1a) and (4.7). Observe that the previous expressions allows to compute $\boldsymbol{u}_\alpha$ in terms of a vector-valued polynomial $\boldsymbol{p}_\alpha$ in $\boldsymbol{B}_{j\gamma_j}$, $\boldsymbol{c}_\gamma$ and $\boldsymbol{u}_\beta$ where $|\beta| \leqslant |\alpha|$ and

$\beta_1 < \alpha_1$ —In fact, iteration of this process allows to assume $\beta_1 \leqslant 1$. Moreover, the polynomial $\boldsymbol{p}_\alpha$ has positive coefficients. Proceeding by induction over α_1 one finds

$$\boldsymbol{u}_\alpha = \boldsymbol{p}_\alpha(\boldsymbol{B}_{j\gamma_j}, \boldsymbol{c}_\gamma, \boldsymbol{u}_\beta)\,, \tag{5.2}$$

where $\boldsymbol{p}_\alpha$ is a vector-valued polynomial with positive coefficients and $|\beta| \leqslant |\alpha|$ and $\beta_1 < \alpha_1$. By induction on α_1, this proves that $\boldsymbol{u}_\alpha$ are uniquely determined by $\boldsymbol{B}_{j\gamma_j}$ and $\boldsymbol{c}_\gamma$.

Existence of u

We shall now prove existence of an analytic solution $\boldsymbol{u}$ to (5.1). To this avail we observe that it is sufficient to prove that

$$\boldsymbol{u}(x) = \sum_{\alpha \in \mathbb{Z}^n_+} \frac{1}{\alpha!} \boldsymbol{u}_\alpha x^\alpha\,, \tag{5.3}$$

converges absolutely in a neighbourhood of $x_0 = 0$ and yields a solution to (5.1). Actually, since $\{\boldsymbol{u}_\alpha\}_{\alpha \in \mathbb{Z}^n_+}$ are defined as per Equation (5.2), the fulfilment of Equation (5.1a) is straightforward provided $\boldsymbol{u}(x)$ makes sense. Thus, we are left with showing the convergence of (5.3).

To this avail we recall that $\{\boldsymbol{B}_j\}_{j=2}^n$ and $\boldsymbol{c}$ are analytic (matrix-valued) functions. Since we are dealing with finitely many analytic functions, we may assume that their Taylor series centred at 0 converges absolutely in $B_\varepsilon(0)$, for a fixed $\varepsilon > 0$ —Notice that the Taylor series of $\boldsymbol{B}_j, \boldsymbol{c}$ are meant in x and $\boldsymbol{u}$. Applying Lemma 4.8-b to each component of $\{\boldsymbol{B}_j\}_j$ and $\boldsymbol{c}$ we obtain majorant sequences $\{\underline{\widetilde{\boldsymbol{B}}}_j\}_j$, $\underline{\widetilde{\boldsymbol{B}}}_j = \{\widetilde{\boldsymbol{B}}_{j\gamma}\}_{\gamma \in \mathbb{Z}^{n+m}_+}$ and $\underline{\widetilde{\boldsymbol{c}}} = \{\widetilde{\boldsymbol{c}}_\gamma\}_{\gamma \in \mathbb{Z}^{n+m}_+}$. Moreover, we may also assume that

$$\begin{aligned}
\widetilde{\boldsymbol{B}}_j(\hat{x}, \boldsymbol{u}) &= \sum_{\substack{\gamma = (\hat{\alpha}, \beta) \\ \hat{\alpha} \in \mathbb{Z}^{n-1}_+,\, \beta \in \mathbb{Z}^m_+}} \widetilde{\boldsymbol{B}}_{j\gamma} \hat{x}^{\hat{\alpha}} \boldsymbol{u}^\beta = \frac{Cs}{s - (x^2 + \ldots + x^n) - (u^1 + \ldots + u^m)} \begin{bmatrix} 1 & \ldots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \ldots & 1 \end{bmatrix} \\
\widetilde{\boldsymbol{c}}(\hat{x}, \boldsymbol{u}) &= \sum_{\substack{\gamma = (\hat{\alpha}, \beta) \\ \hat{\alpha} \in \mathbb{Z}^{n-1}_+,\, \beta \in \mathbb{Z}^m_+}} \boldsymbol{c}_\gamma \hat{x}^{\hat{\alpha}} \boldsymbol{u}^\beta = \frac{Cs}{s - (x^2 + \ldots + x^n) - (u^1 + \ldots + u^m)} \begin{bmatrix} 1 & \ldots & 1 \end{bmatrix}\,,
\end{aligned}$$

where $C > 0$ while $0 < s < \varepsilon/\sqrt{n}$, the series being absolutely convergent in a $B_{\varepsilon'}(0)$ for $0 < \varepsilon' < \varepsilon$.

We now consider the auxiliary Cauchy problem

$$\frac{\partial \widetilde{\boldsymbol{u}}}{\partial x^1} = \sum_{j=2}^n \widetilde{\boldsymbol{B}}_j(\hat{x}, \widetilde{\boldsymbol{u}}) \frac{\partial \widetilde{\boldsymbol{u}}}{\partial \hat{x}^j} + \widetilde{\boldsymbol{c}}(\hat{x}, \widetilde{\boldsymbol{u}})\,, \tag{5.4a}$$

$$\widetilde{\boldsymbol{u}}(0) = 0\,. \tag{5.4b}$$

Our goal is to prove that problem (5.4) admits an analytic solution $\widetilde{\boldsymbol{u}}$. Notice that the first part of the proof shows that such solution is unique, moreover, it fulfils Equation (5.2) with $\widetilde{\boldsymbol{B}}_{j\gamma}$ and

$\widetilde{\boldsymbol{c}}_\gamma$. Moreover, we have $\widetilde{u}^k_\alpha \geqslant 0$ for all $k \in \{1,\ldots,m\}$ and $\alpha \in \mathbb{Z}^n_+$ since, using Equation (5.2) and induction over α_1,

$$\widetilde{u}^k_\alpha = p^k_\alpha(\widetilde{\boldsymbol{B}}_{j\gamma_j}, \widetilde{\boldsymbol{c}}_\gamma, \widetilde{\boldsymbol{u}}_\beta) \geqslant 0\,,$$

where we used that $\boldsymbol{p}_\alpha$ has positive coefficients.

Assuming that we proved the existence of $\widetilde{\boldsymbol{u}}$, we have that $\underline{\widetilde{\boldsymbol{u}}} = \{\widetilde{\boldsymbol{u}}_\alpha\}_{\alpha\in\mathbb{Z}^n_+}$ is a majorant (componentwise) of $\underline{\boldsymbol{u}} = \{\boldsymbol{u}_\alpha\}_{\alpha\in\mathbb{Z}^n_+}$. The proof proceed by induction over α_1. Indeed, for all $k \in \{1,\ldots,m\}$ we have $u^k_{(0,\hat{\alpha})} = 0 = \widetilde{u}^k_{(0,\widetilde{\alpha})}$, proving $|u^k_\alpha| \leqslant \widetilde{u}^k_\alpha$ for $\alpha_1 = 0$. Assuming by induction that $|u^k_\alpha| \leqslant \widetilde{u}^k_\alpha$ for all $\alpha \in \mathbb{Z}^n_+$ with $\alpha_1 < h$, we now prove that $|u^k_\alpha| \leqslant \widetilde{u}^k_\alpha$ also for all $\alpha \in \mathbb{Z}^n_+$ with $\alpha_1 = h$. To this avail we observe that both $\boldsymbol{u}_\alpha$ and $\widetilde{\boldsymbol{u}}_\alpha$ fulfils Equation (5.2) with $\boldsymbol{B}_{j\gamma}, \boldsymbol{c}_\gamma$ and $\widetilde{\boldsymbol{B}}_{j\gamma}, \widetilde{\boldsymbol{c}}_\gamma$ respectively. Then we have, for $k \in \{1,\ldots,m\}$,

$$\begin{aligned}
|u^k_\alpha| &= |p^k_\alpha(\boldsymbol{B}_{j\gamma_j}, \boldsymbol{c}_\gamma, \boldsymbol{u}_\beta)| && \text{Eq. (5.2) component-wise} \\
&\leqslant p^k_\alpha(|\boldsymbol{B}_{j\gamma_j}|, |\boldsymbol{c}_\gamma|, |\boldsymbol{u}_\beta|) && p^k_\alpha \text{ has positive coefficients} \\
&\leqslant p^k_\alpha(\widetilde{\boldsymbol{B}}_{j\gamma_j}, \widetilde{\boldsymbol{c}}_\gamma, |\boldsymbol{u}_\beta|) && \underline{\boldsymbol{B}}_j \ll \underline{\widetilde{\boldsymbol{B}}}_j\,,\ \underline{\boldsymbol{c}} \ll \underline{\widetilde{\boldsymbol{c}}} \\
&\leqslant p^k_\alpha(\widetilde{\boldsymbol{B}}_{j\gamma_j}, \widetilde{\boldsymbol{c}}_\gamma, \widetilde{\boldsymbol{u}}_\beta) && \text{induction because } \beta_1 < \alpha_1 \\
&= |\widetilde{u}^k_\alpha| && \text{Eq. (5.2)}\,.
\end{aligned}$$

At this stage Lemma 4.8-a applies, showing that $\boldsymbol{u}(x)$ as per Equation (5.3) is well-defined and analytic in a neighbourhood of $x_0 = 0$.

It remains to prove that there exists an analytic solution $\widetilde{\boldsymbol{u}}$ to the Cauchy problem (5.4). To this avail, we observe that Equation (5.4a) explicitly reads

$$\frac{\partial \widetilde{u}^k}{\partial x^1} = \frac{Cs}{s - (x^2 + \ldots + x^n) - (\widetilde{u}^1 + \ldots + \widetilde{u}^m)} \left[\sum_{j=2}^n \sum_{\ell=1}^m \frac{\partial \widetilde{u}^\ell}{\partial \hat{x}^j} + 1\right], \qquad k = 1,\ldots,m\,.$$

Let $(\zeta, z) := (x^1, x^2 + \ldots + x^n)$: Considering the ansaltz

$$\widetilde{\boldsymbol{u}}(x) = v(\zeta, z)(1,\ldots,1)\,,$$

problem (5.4) boils down to a single PDE

$$\frac{\partial v}{\partial \zeta} = \frac{Cs}{s - z - mv}\left[(n-1)m\frac{\partial v}{\partial z} + 1\right], \qquad v(0,z) = 0\,. \tag{5.5}$$

To solve the latter problem we consider the curve $\eta(\zeta) := v(\zeta, z(\zeta))$ where the curve $\zeta \mapsto z(\zeta)$ is regular but arbitrary: Then

$$\dot{\eta} = \frac{\partial v}{\partial \zeta} + \dot{z}\frac{\partial v}{\partial z}\,,$$

where $\dot{}$ denotes differentiation with respect to ζ. The latter equation boils down to (5.5) provided $z(\zeta)$ and $\eta(\zeta)$ fulfils

$$\dot{\eta} = \frac{Cs}{s - z - m\eta} \tag{5.6a}$$

$$\dot{z} = -\frac{Cs(n-1)m}{s - z - m\eta} \,. \tag{5.6b}$$

Thus, a solution $v(\zeta, z)$ to (5.5) can be found by solving the system of ODE (5.6) and then defining v implicitly by

$$\eta(\zeta) = v(\zeta, z(\zeta)) \,. \tag{5.7}$$

Equation (5.7) allows to determine (implicitly) $v(\zeta, z_0)$ at any point (ζ, z_0) provided we may solve the system (5.6) with initial conditions $\eta(0) = 0$ and $z(0) = z_0$ —the former condition being a consequence of the boundary condition (5.4a).

To begin with we observe that any solution to (5.6) fulfils

$$\dot{z} + (n-1)m\dot{\eta} = 0 \quad \Rightarrow \quad z(\zeta) = z_0 - (n-1)m\eta(\zeta) \,,$$

which allows to simplify Equation (5.6b) to

$$\dot{\eta} = \frac{Cs}{s - z_0 + (n-2)m\eta} \,.$$

Integration by separation of variable yields

$$\begin{aligned}
Cs\zeta &= \int_0^\zeta [s - z_0 + (n-2)m\eta(\zeta')]\dot{\eta}(\zeta')\mathrm{d}\zeta = \int_0^{\eta(\zeta)} [s - z_0 + (n-2)m\eta]\mathrm{d}\eta \\
&= (s - z_0)\eta(\zeta) + \frac{1}{2}(n-2)m\eta(\zeta)^2 \\
&= (s - z(\zeta))\eta(\zeta) - \frac{mn}{2}\eta(\zeta)^2 \\
&\Rightarrow \eta(\zeta) = \frac{1}{mn}\left[s - z(\zeta) - \sqrt{(s - z(\zeta))^2 - 2Csmn\zeta}\right] .
\end{aligned}$$

Equation (5.7) leads to

$$v(\zeta, z) = \frac{1}{mn}\left[s - z - \sqrt{(s-z)^2 - 2Csmn\zeta}\right] .$$

The found solution is analytic in ζ, z if s is small enough: This essentially follows from the analyticity of the function

$$(1-z)^{1/2} = \sum_{k\geqslant 0} \frac{(-1)^k}{k!}\frac{1}{2}\left(\frac{1}{2} - 1\right)\cdots\left(\frac{1}{2} - k + 1\right)z^k = -\sum_{k\geqslant 0} \frac{(2k-3)!!}{2^k k!} z^k \,,$$

where $(2k-3)!! = (2k-3)(2k-5)\cdots 3\cdot 1$ is the **double factorial**. Notice that the above power series has radius of convergence

$$\lim_{k\to\infty} \frac{(2k-3)!!}{2^k k!}\left(\frac{(2(k+1)-3)!!}{2^{k+1}(k+1)!}\right)^{-1} = 1\,.$$

6 Characteristic surfaces for semi-linear PDEs

This section discuss the extension of the Cauchy-Kovalevskaja theorem on a general domain Ω. For the rest of this section $\Omega \subseteq \mathbb{R}^n$ will denote a non-empty open and connected subset while $\Sigma \subset \Omega$ will denote an orientable C^k-regular hypersurface ($k \geqslant 2$) whose unit normal is denoted by ν. In this setting the Cauchy-Kovalevskaja Theorem 4.6 can be applied by considering Riemannian coordinates in a neighbourhood of any $x_0 \in \Sigma$ —*cf.* Section 3. This allows to reduce the problem to the case $\Omega = \mathbb{R}^n$ and $\Sigma = \mathbb{R}^{n-1}$ for which we may apply the results of Section 4. Notice that an important technical assumption of Theorem 4.6 is that the semi-linear PDE can be written in normal form —*cf.* Equation (4.4). This will be possible provided Σ is "non-degenerate" in a suitable sense.

We shall combine Theorem 4.6 and the notion of Riemannian coordinates. For that we consider the Cauchy problem (4.2) for a second order semi-linear PDE

$$a^{jk}(x)\frac{\partial^2 u}{\partial x^k \partial x^j} + \Phi\left(x, u(x), \frac{\partial u}{\partial x^k}\right) = 0\,, \tag{6.1a}$$

$$u|_\Sigma = u_0\,, \qquad \frac{\partial u}{\partial \nu}\bigg|_\Sigma = u_1\,. \tag{6.1b}$$

In what follows we shall not discuss the analyticity assumptions on u_0, u_1 and $a = a^{ij}$, Φ: Ultimately, they have to be imposed to apply Theorem 4.6. However, we shall investigate whether the problem (6.1) can be recast into a normal form (4.4).

To this avail, we focus on $x_0 \in \Sigma$ and consider Riemannian normal coordinates $(t, \xi) = (t, \xi_2, \ldots, \xi_n)$ in a neighbourhood N_{x_0} of that point. This entails in particular that $x \in N_{x_0}$ can be parametrized by (t, ξ) so that $x = x(t, \xi) = x^k(t, \xi)e_k$, moreover, $x \in N_{x_0} \cap \Sigma$ if and only if $t = 0$ —in particular $x_0 = x(0, \xi_0)$ (as explained in Section 3 we may choose $\xi_0 = 0$ but this is not strictly necessary for what follows). By construction, keeping ξ fixed and considering the curve $t \mapsto x(t, \xi)$, we identify the straight line normal to Σ and starting at $x(0, \xi) \in \Sigma \cap N_{x_0}$, moreover, its tangent vector coincides with the unit normal ν at $x(0, \xi)$:

$$\nu(x(0, \xi)) = \frac{\mathrm{d}x^k}{\mathrm{d}t}\bigg|_{t=0} e_k\,.$$

This implies in particular

$$\frac{\partial u}{\partial \nu}\bigg|_{x_0} = \nu^k \frac{\partial u}{\partial x^k}\bigg|_{x_0} = \frac{\mathrm{d}x^k}{\mathrm{d}t}\bigg|_{(t,\xi)=(0,\xi_0)} \frac{\partial u}{\partial x^k}\bigg|_{x_0} = \frac{\partial \widetilde{u}}{\partial t}\bigg|_{t=0}\,,$$

where $\widetilde{u}$ is defined by $\widetilde{u}(t, \xi) = u(x(t, \xi))$.

Combining this observations with Lemma 2.7 we find that the Cauchy problem (6.1) can be written as

$$\widetilde{a}^{tt}(t,\xi)\frac{\partial^2\widetilde{u}}{\partial t^2}+\widetilde{a}^{tk}(t,\xi)\frac{\partial^2\widetilde{u}}{\partial t\partial\xi^k}+\widetilde{a}^{jk}(t,\xi)\frac{\partial^2\widetilde{u}}{\partial\xi^k\partial\xi^j}+\widetilde{\Phi}\left(t,\xi,\widetilde{u}(t,\xi),\frac{\partial\widetilde{u}}{\partial t},\frac{\partial\widetilde{u}}{\partial\xi^k}\right)=0\,, \tag{6.2a}$$

$$\widetilde{u}|_{t=0}=\widetilde{u}_0\,, \tag{6.2b}$$

$$\left.\frac{\partial\widetilde{u}}{\partial t}\right|_{t=0}=\widetilde{u}_1\,, \tag{6.2c}$$

where $\widetilde{u}_0,\widetilde{u}_1$ are defined by $\widetilde{u}_0(\xi)=u_0(x(0,\xi))$ and similarly for $\widetilde{u}_1$.

At this stage we see that a necessary and sufficient condition for the Cauchy problem (6.2) to be written in normal form (4.4) is $\widetilde{a}^{tt}(t,\xi)\neq 0$. By continuity of $\widetilde{a}$, if this holds true at x_0, then it holds true in an open neighbourhood of x_0, leading to the following result.

Corollary 6.1 (Cauchy-Kovalevskaja for semi-linear PDEs): Let $x_0\in\Sigma$ be such that

$$\widetilde{a}^{tt}(0,\xi_0)\neq 0\,,$$

and let assume that there exists:

(I) an open neighbourhood $N_{\xi_0}\subset\Sigma$ of ξ_0 such that $\widetilde{u}_0,\widetilde{u}_1$ are analytic in N_{ξ_0};

(II) an open neighbourhood $N_{(0,\xi_0)}\subseteq\Omega$ of $(0,\xi_0)$ such that $\widetilde{a}=\widetilde{a}^{jk}$ are analytic in $N_{(0,\xi_0)}$;

(III) an open neighbourhood N_{Ξ_0} of

$$\Xi_0=\left(0,\xi_0,\widetilde{u}_0(\xi_0),\widetilde{u}_1(\xi_0),\left.\frac{\partial\widetilde{u}_0}{\partial\xi^k}\right|_{\xi_0}\right)\,,$$

such that $\widetilde{\Phi}$ is analytic in N_{Ξ_0}.

Then there exists a neighbourhood $\widetilde{N}_{(0,\xi_0)}\subseteq\Omega$ of $(0,\xi_0)$ such that there exists a unique solution $\widetilde{u}$ to the Cauchy problem (6.2) —thus, of (6.1)— which is analytic in $\widetilde{N}_{x_0}$.

Corollary 6.1 cannot be applied at all points $x_0\in\Sigma$ such that $\widetilde{a}^{tt}(0,\xi_0)=0$. Those hypersurfaces Σ for which this happens for all $x_0\in\Sigma$ deserve a special name.

Definition 6.2: An orientable C^k-regular, $k\geqslant 2$, hypersurface Σ is called **characteristic hypersurface** for a semi-linear operator D of second order given by Equation (2.4) if:

$$\forall x_0\in\Sigma\,,\ \exists N_{x_0}\overset{\text{open}}{\subset}\mathbb{R}^n\,,\ x_0\in N_{x_0}\,,\ \exists S\in C^k(N_{x_0}):$$
$$\Sigma\cap N_{x_0}=\{x\in N_{x_0}\,|\,S(x)=0\}\,,\ \forall x\in\Sigma\cap N_{x_0}\ \nabla S(x)\neq 0\,,\ a^{kj}(x)\frac{\partial S}{\partial x^k}\frac{\partial S}{\partial x^j}=0\,. \tag{6.3}$$

The Cauchy problem (6.1) is called **characteristic** if Σ is characteristic for the associated semi-linear second order differential operator D.

REMARK 6.3:

(i) Notice that Definition 6.2 is consistent with the failure of the condition $\widetilde{a}^{tt}(0,\xi) \neq 0$.

Indeed, let $x_0 \in \Sigma$ and let consider Riemannian normal coordinates around x_0: Setting $\widetilde{S}(t,\xi) := S(x(t,\xi))$ we find $\widetilde{S}(0,\xi) = 0$ for all ξ —because $x(0,\xi) \in \Sigma \cap N_{x_0}$— and therefore $\frac{\partial \widetilde{S}}{\partial \xi^k}\Big|_{t=0} = 0$. However, since we also have $\nabla S(x) \neq 0$ for all $x \in \Sigma \cap N_{x_0}$, we must have $\frac{\partial \widetilde{S}}{\partial t}\Big|_{t=0} \neq 0$. Combining these identities with Equation (2.13), the last condition in (6.3) reduces to

$$
\begin{aligned}
0 &= a^{kj}(x)\frac{\partial S}{\partial x^k}\frac{\partial S}{\partial x^j} \\
&= a^{kj}(x)\left[\frac{\partial t}{\partial x^k}\frac{\partial \widetilde{S}}{\partial t} + \frac{\partial \xi^\ell}{\partial x^k}\cancel{\frac{\partial \widetilde{S}}{\partial \xi^\ell}}\right]\left[\frac{\partial t}{\partial x^j}\frac{\partial \widetilde{S}}{\partial t} + \frac{\partial \xi^\ell}{\partial x^j}\cancel{\frac{\partial \widetilde{S}}{\partial \xi^\ell}}\right]\Bigg|_{t=0} = \widetilde{a}^{tt}(0,\xi)\left(\frac{\partial \widetilde{S}}{\partial t}\right)^2\Bigg|_{t=0}.
\end{aligned}
$$

which is equivalent to $\widetilde{a}^{tt}(0,\xi) = 0$.

(ii) Since the Cauchy problem (6.1) cannot be recast in normal form (4.4), Corollary 6.1 cannot be applied —this has been the reason for introducing characteristic hypersurfaces. However, notice that Corollary 6.1 only provides sufficient conditions for the uniqueness and existence of a solution to (6.1) and there are cases where such problem can indeed be solved even if Σ is characteristic for D.

Nevertheless, the existence of solutions to (4.2) with initial data on a characteristic hypersurface is subjected to additional requirements on the data u_0, u_1. Indeed, let us consider Equation (6.2a): evaluation at $t = 0$ leads to

$$
\begin{aligned}
0 &= \cancel{\widetilde{a}^{tt}(0,\xi)}\frac{\partial^2 \widetilde{u}}{\partial t^2}\Big|_{t=0} + \widetilde{a}^{tk}(0,\xi)\frac{\partial^2 \widetilde{u}}{\partial t \partial \xi^k}\Big|_{t=0} + \widetilde{a}^{jk}(0,\xi)\frac{\partial^2 \widetilde{u}}{\partial \xi^k \partial \xi^j}\Big|_{t=0} \\
&\quad + \widetilde{\Phi}\left(0,\xi,\widetilde{u}(0,\xi),\frac{\partial \widetilde{u}}{\partial t}\Big|_{t=0},\frac{\partial \widetilde{u}}{\partial \xi^k}\Big|_{t=0}\right) \\
&= \widetilde{a}^{tk}(0,\xi)\frac{\partial \widetilde{u}_1}{\partial \xi^k} + \widetilde{a}^{jk}(0,\xi)\frac{\partial^2 \widetilde{u}_0}{\partial \xi^k \partial \xi^j} + \widetilde{\Phi}\left(0,\xi,\widetilde{u}_0(\xi),\widetilde{u}_1(\xi),\frac{\partial \widetilde{u}_0}{\partial \xi^k}\right). \qquad (6.4)
\end{aligned}
$$

The last equation is a consistency condition involving the data $\widetilde{u}_0, \widetilde{u}_1$: Thus, it is a necessary condition to obtain solutions to the Cauchy problem (6.2).

We close this section with some examples.

EXAMPLE 6.4:

(i) Let consider the Cauchy problem on $\mathbb{R}^2$ given by

$$\frac{\partial^2 u}{\partial x \partial y} = 0\,, \tag{6.5a}$$

$$u(x,0) = u_0(x)\,, \tag{6.5b}$$

$$\left.\frac{\partial u}{\partial y}\right|_{y=0} = u_1(x)\,, \tag{6.5c}$$

where u_0, u_1 are assigned. The operator $D = \frac{\partial^2}{\partial x \partial y}$ is linear of second order with constant principal symbol given by

$$a^{jk} = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\,.$$

By direct inspection the eigenvalues of a^{jk} are ± 1, therefore, D is of hyperbolic type according to Definition 2.9. The characteristic hypersurfaces(=curves) are identified by the condition

$$\frac{\partial S}{\partial x}\frac{\partial S}{\partial y} = 0\,.$$

It follows that any line $x = c$ or $y = c$ is a characteristic curve as by Definition 6.2: In particular the Cauchy problem (6.5) is formulated on a characteristic curve.

We shall now discuss the solutions to problem (6.5). To this avail, notice that evaluation of (6.5a) at $y = 0$ leads to

$$0 = \left.\frac{\partial^2 u}{\partial x \partial y}\right|_{y=0} = \frac{\partial u_1}{\partial x}\,,$$

which implies that u_1 has to be constant —this is a particular example of Equation (6.4). Moreover, the general solution to (6.5a) is

$$u(x,y) = f(x) + g(y)\,,$$

for functions f, g to be determined by imposing

$$u_0(x) = u(x,0) = f(x) + g(0)\,, \qquad u_1 = \left.\frac{\partial u}{\partial y}\right|_{y=0} = g'(0)\,.$$

The general form of the solution is thus

$$u(x,y) = u_0(x) + g(y) - g(0)\,,$$

where g is any function such that $g'(0) = u_1$: In particular we have

$$u(x,y) = u_0(x) + u_1 y + g(y) - g(0) - u_1 y =: u_0(x) + u_1 y + h(y)\,,$$

where h is any function such that $h(0) = 0$ and $h'(0) = 0$. It follows that the Cauchy problem (6.5) admits infinitely many solutions.

(ii) We shall now list some characteristic hypersurfaces of the wave equation (2.5). The latter are identified by —*cf.* condition (6.3)—

$$\frac{1}{c^2}\left(\frac{\partial S}{\partial t}\right)^2 - \delta^{jk}\frac{\partial S}{\partial x^k}\frac{\partial S}{\partial x^j} = 0\,.$$

An example of a function S fulfilling these requirements is given by

$$S(t,x) = c^2(t-t_0)^2 - \|x-x_0\|^2\,,$$

for fixed $t_0 \in \mathbb{R}$, $x_0 \in \mathbb{R}^n$. We then consider

$$V_{t_0,x_0} := \{(t,x) \in \mathbb{R}^{1+n}\backslash\{(t_0,x_0)\} \mid S(t,x) = 0\}\,. \tag{6.6}$$

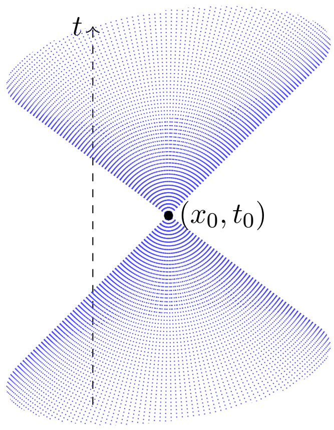


Figure 3: The lightcone V_{x_0,t_0} for $n = 2$.

Notice that

$$\nabla S(t,x) = 2c^2(t-t_0)e_1 - 2(x-x_0) \neq 0 \qquad \forall (t,x) \in V_{(t_0,x_0)}\,,$$

showing that $V_{(t_0,x_0)}$ is a C^∞-regular hypersurface, which is characteristic. This hypersurface is called the light-cone passing through (x_0,t_0) —*cf.* Figure 3. The characteristic problem on this hypersurface is called **Goursat problem**. Notably in 1+1-dimensions this problem coincides with (6.5): Indeed in that case we find

$$0 = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 \widetilde{u}}{\partial x_+ \partial x_-}\,,$$

where we introduced $x_\pm = x \pm ct$, and $\widetilde{u}(x_+,x_-) = u((x_+ + x_-)/2, (x_+ - x_-)/2c)$. Moreover in 1+1-dimension the light cone splits in two straight lines $c(t-t_0) \pm (x-x_0) = 0$ which corresponds to either $x_+ = 0$ or $x_- = 0$. We will come back to this trick in Section 17.

Another example of a class of characteristic surfaces can be obtained by considering

$$S(t,x) := ct + x \cdot b - d\,, \tag{6.7}$$

for fixed $b \in \mathbb{R}^n$, $d \in \mathbb{R}$. The latter function determines a characteristic hypersurface if $\|b\|^2 = 1$ and the corresponding hypersurface is a plane with orthogonal direction given by $ce_1 + b$.

(iii) Notably, there are no characteristic hypersurfaces for elliptic PDEs. Indeed, the non degeneracy of $a = a^{jk}$ —*cf.* Definition 2.9— leads to

$$a^{jk}\frac{\partial S}{\partial x^k}\frac{\partial S}{\partial x^j} = 0 \quad \Rightarrow \quad \frac{\partial S}{\partial x^k} = 0\,,$$

which contradicts the first condition in (6.3).

(iv) Considering the heat equation (1.7), condition (6.3) becomes

$$\delta^{jk}\frac{\partial S}{\partial x^j}\frac{\partial S}{\partial x^k} = 0 \quad \Rightarrow \quad \frac{\partial S}{\partial x^k} = 0 \quad \Rightarrow \quad S(x,t) = S(t)\,.$$

In particular the hyperplane $t = 0$ is a characteristic hypersurface. From a physical point of view this is rather unpleasant as it seems that we cannot assign the initial temperature $u(x,0)$. As a matter of fact this problem is easily circumvented by considering an initial-value problem with boundary condition rather than a Cauchy problem: An example in this direction is given by problem (1.8) introduced in Section 1. We will study these problems in more details in Sections 22.

7 Elliptic PDEs: Dirichlet problem

We shall now focus on elliptic PDEs: In particular, we will study the **Poisson equation**

$$-\Delta u = f\,. \tag{7.1}$$

For $f = 0$ the Poisson equation reduces to the Laplace equation (2.8) introduced in Section 1. We will consider Equation (7.1) either on Ω or on $\mathbb{R}^n\backslash\overline{\Omega}$: For the rest of this section, unless otherwise stated, $\Omega \subseteq \mathbb{R}^n$ is an open subset with compact closure $\overline{\Omega}$. The results discussed in this section are based on [3, 11, 18, 21].

From a physical point of view, Equation (7.1) arises in the study of stationary(= time independent) solutions to either the heat equation (2.11) or to the wave equation (2.5). In what follows we will mainly consider the first interpretation, where the function u is regarded as the temperature of a continuum —*e.g.* the temperature of an oven.

Remark 7.1 (): For the sake of completeness, we stress that Equation (7.1) also arises in electrostatics, where stationary electric fields $E\colon \mathbb{R}^3 \to \mathbb{R}^3$ are investigated. To make a long story short, in electrostatics the equations governing the electric field E are reduced to:

(a) $\nabla \wedge E = 0$ where $\nabla \wedge E$ is the **curl** of E, defined by

$$\begin{aligned}&\nabla\wedge\colon C^1(\Omega,\mathbb{R}^3) \to C^0(\Omega,\mathbb{R}^3)\,,\\ &\qquad\nabla \wedge E = \left(\frac{\partial E^2}{\partial x^3} - \frac{\partial E^3}{\partial x^2}\right)e_1 - \left(\frac{\partial E^1}{\partial x^3} - \frac{\partial E^3}{\partial x^1}\right)e_2 + \left(\frac{\partial E^1}{\partial x^2} - \frac{\partial E^2}{\partial x^1}\right)e_3\,.\end{aligned}$$

(b) $\nabla \cdot E = 4\pi\rho$, where ρ denotes the charge density generating the electric field E while $\nabla \cdot E$ denotes the **divergence** of E:

$$\nabla\cdot\colon C^1(\Omega,\mathbb{R}^3) \to C^0(\Omega)\,, \qquad \nabla \cdot E = \sum_{j=1}^3 \frac{\partial E^j}{\partial x^j} = \delta^k_j \frac{\partial E^j}{\partial x^k}\,. \tag{7.2}$$

On simply connected domains the first condition entails $E = -\nabla u$ for a scalar function $u\colon \mathbb{R}^3 \to \mathbb{R}$ called potential of E, whereas the second condition simplifies to

$$4\pi\rho = -\nabla\cdot\nabla u = -\sum_{j=1}^3 \frac{\partial^2 u}{(\partial x^j)^2} = -\Delta u\,,$$

which is exactly Equation (7.1).

Referring to Definitions 2.2,2.9,6.2, Equation (7.1) is a second order linear PDE of elliptic type, therefore, it has no characteristic surfaces. As seen in Remark 4.2, the associated Cauchy problem (6.1) is ill-posed in general: For this reason we will be interested in a slightly different version of problem (6.1).

DEFINITION 7.2 (Dirichlet problems for (7.1)): The **inner Dirichlet problem** consists of finding $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ such that

$$-\Delta u = f\,, \tag{7.3a}$$
$$u|_{\partial\Omega} = u_0\,, \tag{7.3b}$$

where $f \in C^0(\Omega)$ and $u_0 \in C^0(\partial\Omega)$ are assigned —Notice that the regularity imposed on f and u_0 is consistent with the one required on u. The **outer Dirichlet problem** consists of finding $u \in C^2(\mathbb{R}^n \backslash \overline{\Omega}) \cap C^0(\mathbb{R}^n \backslash \Omega)$ such that

$$-\Delta u = f\,, \tag{7.4a}$$
$$u|_{\partial\Omega} = u_0\,, \tag{7.4b}$$
$$\lim_{x\to\infty} |u(x) - u_\infty| = 0 \tag{7.4c}$$

where $u_\infty \in \mathbb{R}$, $f \in C^0(\mathbb{R}^n \backslash \overline{\Omega})$ and $u_0 \in C^0(\partial\Omega)$ are assigned —notice that $\partial(\mathbb{R}^n \backslash \Omega) = \partial\Omega$.

REMARK 7.3:

(i) The condition $u|_{\partial\Omega} = u_0$ is usually called **Dirichlet (inhomogeneous) boundary condition**. Here the term "boundary" condition refers to the fact that Equation (7.1) is non-dynamical —recall that u may be seen as a stationary solution to the heat equation (2.11). We point out that boundary conditions for the Poisson equation (7.1) include but are not limited to the Dirichlet boundary condition: In the forthcoming Section 8 we shall investigate two different examples of boundary conditions —*cf.* Definition 8.5.

(ii) Condition (7.4c) can be regarded as a Dirichlet boundary condition "at infinity". Notice that it implies $u(Rx) \underset{R\to\infty}{\longrightarrow} u_\infty$ uniformly in all directions $x \in \partial B_1(0)$, thus, in particular

$$\lim_{R\to\infty} \max_{\partial B_R(0)} |u - u_\infty| = 0\,.$$

(iii) From a physical point of view the inner Dirichlet problem 7.3 models the temperature density of an oven whose boundary is set to a fixed chosen temperature. Regarding the solutions to Equation (7.1) as stationary solutions to the wave equation (2.5), problem (7.3) can also be seen as describing the transversal vibrations of a membrane whose boundary is kept fixed. However, this latter interpretation is slightly troublesome as one may expect that *stationary* transversal vibrations should vanish.

Unfortunately, showing that problems (7.3)-(7.4) admit a solution is way out our league —*cf.* [19] for a comprehensive textbook which covers the topic. In what follows we shall content ourself in showing that problems (7.3)-(7.4) admit at most a unique solution.

Harmonic functions

The proof of the uniqueness result for the inner/outer Dirichlet problems (7.3)-(7.4) relies on the study of the boundary-value problem with zero data, $f = 0$, $u_0 = 0$ ($u_\infty = 0$). To this avail, we shall consider the remarkable properties of the solutions to the Laplace equation (2.8).

DEFINITION 7.4: A function $u \in C^2(\Omega)$ is called:

i. **subharmonic** (*resp.* **superharmonic**) in Ω if $-\Delta u \leqslant 0$ (*resp.* $-\Delta u \geqslant 0$);

ii. **harmonic** in Ω if it is both subharmonic and superharmonic in Ω, thus $-\Delta u = 0$.

REMARK 7.5 ():

(i) For $n = 1$ a function $u \in C^2(\mathbb{R})$ is harmonic if $u'' = 0$, that is, $u(x) = u(0) + u'(0)x$ is a linear function. Similarly a subharmonic function on $\mathbb{R}$ is a convex function.

(ii) For $n = 2$ there is a nice connection between harmonic functions and holomorphic functions [16, §10], [7, §3]. For that, in what follows we shall identify $\Omega \subseteq \mathbb{R}^2 \simeq \mathbb{C}$ with an open subset of $\mathbb{C}$, the identification being given by $\mathbb{R}^2 \ni (x, y) \mapsto z := x + iy \in \mathbb{C}$. By Definition a function $f : \Omega \to \mathbb{C}$ is called **holomorphic** if

$$\forall z_0 \in \Omega \quad \exists f'(z_0) := \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \in \mathbb{C} \,. \tag{7.5}$$

Remarkably, holomorphic functions are smooth, $f \in C^\infty(\Omega, \mathbb{C})$. Moreover, let

$$f(z) = u(x, y) + iv(x, y) \,,$$
$$u(x, y) := \frac{1}{2}\left[f(z) + \overline{f(z)}\right] , \qquad v(x, y) := \frac{1}{2i}\left[f(z) - \overline{f(z)}\right]$$

where $u, v : \Omega \to \mathbb{R}$ are called real and imaginary part of f. Since the limit $\lim_{z \to z_0}$ is meant as a limit in $\mathbb{R}^2$ we may restrict to $z = z_0 + h$ or to $z = z_0 + ih$ without affecting the final result. This leads to

$$\begin{aligned}
f'(z_0) &= \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} \\
&= \lim_{h \to 0} \left[\frac{u(x_0 + h, y_0) - u(x_0, y_0)}{h} + i \frac{v(x_0 + h, y_0) - v(x_0, y_0)}{h}\right] = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \,, \\
f'(z_0) &= \lim_{h \to 0} \frac{f(z_0 + ih) - f(z_0)}{ih} \\
&= \lim_{h \to 0} \left[\frac{u(x_0, y_0 + h) - u(x_0, y_0)}{ih} + \frac{v(x_0, y_0 + h) - v(x_0, y_0)}{h}\right] = \frac{1}{i} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \,.
\end{aligned}$$

Comparing the found results leads to the **Cauchy-Riemann conditions**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}\,, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}\,. \tag{7.6}$$

Notably, these conditions are necessary and sufficient for f to be holomorphic: Actually f is holomorphic in Ω if and only if $u, v \in C^1(\Omega)$ and Equations (7.6) are fulfilled.

Notice that, since $u, v \in C^\infty(\Omega)$, because so is f, differentiation of Equations (7.6) leads to $\Delta u = \Delta v = 0$. We obtain that the real and imaginary part of an holomorphic function are harmonic functions.

The connection between holomorphic and harmonic functions is particularly useful to produce examples of harmonic functions in $\mathbb{R}^2$. Indeed all polynomials in z are holomophic and so is the exponential function

$$e^z = e^{x+iy} = e^x e^{iy} = e^x \left[\cos y + i \sin y\right]\,,$$

where we used Euler formula. This entails that, for example, the following are harmonic functions:

$$e^x \cos y\,,\; e^x \sin y\,,\; \frac{e^z - e^{-z}}{2i} = \sinh(x) \sin y\,,$$

where the last function is similar to the one encountered in Remark 4.2.

We now discuss an important property of (sub-)harmonic functions. The latter generalizes the following situation: A linear or convex function on a closed interval $[a, b]$ attains its maximum at either a or b.

THEOREM 7.6 (Maximum principle —weak form): Let $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$.

1. if u is subharmonic in Ω then
$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u\,; \tag{7.7}$$
2. if u is harmonic in Ω then
$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u\,, \qquad \min_{\overline{\Omega}} u = \min_{\partial\Omega} u\,, \qquad \max_{\overline{\Omega}} |u| = \max_{\partial\Omega} |u|\,. \tag{7.8}$$

Proof. We observe that, since $u \in C^0(\overline{\Omega})$ and on account of the compactness of $\overline{\Omega}$, all maxima appearing in equations (7.7)-(7.8) are well-defined because of Weierstrass theorem [6] —notice that $\partial\Omega = \overline{\Omega} \cap (\mathbb{R}^n \backslash \Omega)$ is a closed set contained in a compact one, thus, it is compact as well.

$\boxed{1}$ Let u be subharmonic on Ω. Since $\partial\Omega \subseteq \overline{\Omega}$ we have

$$\max_{\partial\Omega} u \leqslant \max_{\overline{\Omega}} u\,,$$

so that only the other inequality has to be proven.

At first, let us assume that a stronger condition holds, namely $-\Delta u < 0$. Let x_0 be such that $u(x_0) := \max\limits_{\overline{\Omega}} u$. If $x_0 \in \partial\Omega$ we are done, otherwise we know that $\nabla u(x_0) = 0$ while the Hessian matrix

$$H[u, x_0] = \left(\frac{\partial^2 u}{\partial x^k \partial x^j}\bigg|_{x_0} \right)_{ij}\,,$$

is non-positive definite —that is, $(x, H[u, x_0]x) \leqslant 0$ for all $x \in \mathbb{R}^n$. However, this implies that

$$0 \geqslant \sum_{j=1}^{n} (e_j, H[u, x_0]e_j) = \operatorname{tr} \frac{\partial^2 u}{\partial x^k \partial x^j}\bigg|_{x_0} = \Delta u(x_0) > 0\,,$$

a contradiction. ☠

Let now u be a generic subharmonic function. For all $\varepsilon > 0$ let $u_\varepsilon \in C^2(\Omega) \cap C^0(\overline{\Omega})$ be defined by

$$u_\varepsilon(x) := u(x) + \varepsilon \|x\|^2\,.$$

Then $u \leqslant u_\varepsilon$, moreover,

$$\Delta u_\varepsilon = \Delta u + 2\varepsilon n > 0\,,$$

because $\Delta u \geqslant 0$. Thus, we may apply the previous result to get

$$\max_{\overline{\Omega}} u \leqslant \max_{\overline{\Omega}} u_\varepsilon = \max_{\partial\Omega} u_\varepsilon \leqslant \max_{\partial\Omega} u + \varepsilon R^2\,,$$

where $R^2 := \max\limits_{\partial\Omega} \|x\|^2$. The arbitrariness of $\varepsilon > 0$ leads to Equation (7.7).

$\boxed{2}$ Let u be harmonic, then so is $-u$. Since an harmonic function is also subharmonic, the first two equalities in (7.8) are consequences of (7.7) together with $\max u = -\min -u$. Concerning the last identity in (7.8) this follows from $|u| = \max\{u, -u\}$ together with

$$\begin{aligned}\max_{\overline{\Omega}} |u| = \max_{\overline{\Omega}} \max\{u, -u\} &= \max\{\max_{\overline{\Omega}} u, \max_{\overline{\Omega}} -u\} \\ &= \max\{\max_{\partial\Omega} u, \max_{\partial\Omega} -u\} = \max_{\partial\Omega} \max\{u, -u\} = \max_{\partial\Omega} |u|\,.\end{aligned}$$

□

Remark 7.7:

(i) Theorem 7.6 holds for a slightly more general class of differential operators. In particular let D be such that

$$(Du)(x) = a^{jk}(x)\frac{\partial^2 u}{\partial x^k \partial x^j} + b^k(x)\frac{\partial u}{\partial x^k}\,,$$

for a positive definite matrix-valued function $a \in C^0(\overline{\Omega}, \mathbb{R}^{n(n+1)/2})$ and $b \in C^0(\overline{\Omega}, \mathbb{R}^n)$. Then if $u \in C^0(\overline{\Omega}) \cap C^2(\Omega)$ is such that $Du \geqslant 0$ then (7.7) holds true: We refer to [11, Thm. 2.2] for further details.

(ii) Within the proof of Theorem 7.6 we proved the following: if $u \in C^0(\overline{\Omega}) \cap C^2(\Omega)$ is such that $\Delta u > 0$ then $u(x) < \max_{\overline{\Omega}} u$ for all $x \in \Omega$. For subharmonic and harmonic functions the identity (7.7) does not imply that $u(x) < \max_{\overline{\Omega}} u$ for all $x \in \Omega$. However, this is almost true, more precisely, it fails only in particular situations. Indeed, as we shall see in a Section 11, if Ω is connected and $u(x)$ reaches $\max_{\overline{\Omega}} u$ for a point $x \in \Omega$, then u is constant: This is the content of the strong form of the maximum principle —*cf.* Theorem 11.6.

Uniqueness results

We can now prove the claimed uniqueness result for the Dirichlet problems introduced in Definition 7.2.

Theorem 7.8 (Uniqueness for inner Dirichlet problem): There exists at most one solution $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ to the inner Dirichlet problem (7.3).

Proof. Let u, u' be two solutions to the inner Dirichlet problem (7.3) with the same data $f \in C^0(\Omega)$ and $u_0 \in C^0(\partial\Omega)$. Then $v := u - u' \in C^2(\Omega) \cap C^0(\overline{\Omega})$ solves the inner Dirichlet problem with data $f = 0$ and $u_0 = 0$. In particular v is harmonic, therefore, by Theorem 7.6 we have

$$\max_{\overline{\Omega}} |v| = \max_{\partial\Omega} |v| = 0\,,$$

which entails $u = u'$. □

Remark 7.9:

(i) In Remark 4.2 we provided an example of an ill-posed Cauchy problem on $\mathbb{R}^2$ which was based on the Laplace equation (2.8). Using Theorem 7.6 we can prove that generically there is no solution to the Cauchy problem

$$-\Delta u = f\,, \tag{7.9a}$$

$$u|_{\partial\Omega} = u_0\,, \tag{7.9b}$$

$$\frac{\partial u}{\partial \nu} = u_1\,, \tag{7.9c}$$

where $\Omega \subseteq \mathbb{R}^n$ is such that $\overline{\Omega}$ is compact and with C^1-regular and orientable boundary $\partial\Omega$. To this avail let consider the solutions u, u' to the problem (7.9) with data f, u_0, u_1 and f, u_0, u_1' respectively, where $u_1 \neq u_1'$. Then u, u' solve the Dirichlet inner problem (7.3) with the same data f, u_0, thus $u = u'$ because of Theorem 7.8. However, we also have

$$\frac{\partial u}{\partial \nu} = u_1 \neq u_1' = \frac{\partial u'}{\partial \nu} = \frac{\partial u}{\partial \nu}\,,$$

a contradiction ☠.

(ii) Out of Theorem 7.6 we may conclude that the inner Dirichlet problem (7.3) with fixed $f \in C^0(\Omega)$ depends continuously on the boundary data $u_0 \in C^0(\partial\Omega)$ — *cf.* Condition 3 in Section 4. In particular if we consider the sup norms on $C^0(\partial\Omega)$ and $C^2(\Omega) \cap C^0(\overline{\Omega})$ we find that two solutions u, u' for the inner Dirichlet problem (7.3) with the same f but possibly different u_0, u_0' fulfils

$$\|u - u'\| = \max_{\overline{\Omega}} |u - u'| = \max_{\partial\Omega} |u_0 - u_0'| = \|u_0 - u_0'\|\,,$$

where we applied Theorem 7.6 because $\Delta(u - u') = 0$. This implies Condition 3: if $u_0^{(n)} \in C^0(\partial\Omega)$ is a convergent sequence in $C^0(\partial\Omega)$ with $u_0^{(n)} \to u_0$, then the corresponding solutions $u^{(n)}, u$ to the inner Dirichlet problem (7.3) with data $u_0^{(n)}, u_0$ are such that

$$\lim_{n\to+\infty} \|u^{(n)} - u\| = \lim_{n\to+\infty} \max_{\overline{\Omega}} |u^{(n)} - u| = 0\,.$$

Together with Theorem 7.8, which proves Condition 2 and assuming Condition 1 —which is true but whose proof is out of reach for the moment— we may conclude that the inner Dirichlet problem is well-posed in the sense of Hadamard.

There is a big caveat in this result. While the space $C^0(\partial\Omega)$ is a complete normed space when equipped with the norm $\|u_0\| := \sup_{\partial\Omega} |u_0|$, the same does not apply when equipping $C^2(\Omega) \cap C^0(\overline{\Omega})$ with the sup norm $\|u\| := \sup_{\Omega} |u|$. As a matter of fact one may show that the completion of $C^2(\Omega) \cap C^0(\overline{\Omega})$ with respect to this norm coincides with $C^0(\overline{\Omega})$. Strictly speaking, this does not spoil the well-posedness described above, however, it raises the question of whether one may find a better suited norm for the space $C^2(\Omega) \cap C^0(\overline{\Omega})$. This is in fact possible, *cf.* [19, App. F] for further details.

Theorem 7.10 (Uniqueness for outer Dirichlet problem): There exists at most one solution $u \in C^2(\mathbb{R}^n\backslash\overline{\Omega}) \cap C^0(\mathbb{R}^n\backslash\Omega)$ to the outer Dirichlet problem (7.4).

Proof. We proceed as before: Let u, u' be two solutions to the outer Dirichlet problem (7.3) with the same data $f \in C^0(\mathbb{R}^n\backslash\overline{\Omega})$, $u_0 \in C^0(\partial\Omega)$ and $u_\infty \in \mathbb{R}$. Then $v := u-u' \in C^2(\mathbb{R}^n\backslash\overline{\Omega})\cap C^0(\mathbb{R}^n\backslash\Omega)$ solves the outer Dirichlet problem with vanishing data f, u_0 and u_∞, in particular v is harmonic on $\mathbb{R}^n\backslash\overline{\Omega}$.

Let $x \in \mathbb{R}^n\backslash\Omega$, we shall prove that $v(x) = 0$. Since $\overline{\Omega}$ is compact there exists $R > 0$ large enough such that $\overline{\Omega} \cup \{x\} \subset B_R(0)$. Let $\Omega_R := B_R(0)\backslash\overline{\Omega}$: Then Ω_R is open with compact closure $\overline{\Omega_R} = \overline{B_R(0)}\backslash\Omega$, moreover, $\partial\Omega_R = \partial B_R(0) \cup \partial\Omega$. Since v is harmonic in $\mathbb{R}^n\backslash\overline{\Omega}$, and thus in particular on Ω_R, Theorem 7.6 implies

$$|v(x)| \leqslant \max_{\overline{\Omega}_R} |v| = \max_{\partial B_R(0)\cup\partial\Omega} |v| = \max_{\partial B_R(0)} |v| \xrightarrow[R\to+\infty]{} 0\,,$$

where we used $v|_{\partial\Omega} = 0$ as well as Remark 7.3. □

8 Elliptic PDEs: Neumann and Robin problem

In this section we will prove a uniqueness result for the problem concerning the Poisson equation (7.1) with a boundary condition different from the one discussed in Section 7 —*cf.* [11, 18]. Unless otherwise stated, in what follows $\Omega \subset \mathbb{R}^n$ will denote a connected open subset with compact closure $\overline{\Omega}$ and such that $\partial\Omega$ is a C^1-regular and orientable hypersurface whose outward pointing normal is denoted by ν.

Before stating the boundary-value problem we are interested in, we shall briefly review a few results from Calculus. To this avail we recall the notion of gradient and divergence operators defined by

$$\nabla \colon C^1(\Omega) \to C^0(\Omega, \mathbb{R}^n) \qquad \nabla f := \frac{\partial f}{\partial x^j}\delta^{jk} e_k \,, \tag{8.1}$$

$$\nabla\cdot \colon C^1(\Omega, \mathbb{R}^n) \to C^0(\Omega) \qquad \nabla \cdot V := \sum_{j=1}^n \frac{\partial V^j}{\partial x^j} = \frac{\partial V^j}{\partial x^k}\delta^k_j \,. \tag{8.2}$$

Notably for all $f \in C^2(\Omega)$ we have

$$\nabla \cdot \nabla f = \frac{\partial^2 f}{\partial x^j \partial x^k}\delta^{jk} = \Delta f \,.$$

THEOREM 8.1 (Gauss' divergence theorem): Let $F \in C^1(\overline{\Omega}, \mathbb{R}^n)$. Then

$$\int_\Omega \nabla \cdot F \, \mathrm{d}^n x = \oint_{+\partial\Omega} F \cdot \nu \, \mathrm{d}S \,, \tag{8.3}$$

where $\mathrm{d}S$ is the induced Lebesgue measure on $\partial\Omega$.

REMARK 8.2:

(i) For $n = 1$ Theorem 8.1 reduces to the fundamental theorem of calculus. Indeed, letting $\Omega = (a, b)$ we have $\partial\Omega = \{a, b\}$ while $\nu|_a = -1$, $\nu|_b = 1$, moreover, $\nabla \cdot F = \frac{\mathrm{d}F}{\mathrm{d}x}$. Thus, Equation (8.3) boils down to

$$\int_a^b \frac{\mathrm{d}F}{\mathrm{d}x}\mathrm{d}x = F(b) - F(a) \,.$$

(ii) The hypothesis of Theorem 8.1 can be relaxed, in particular, one may consider $F \in C^1(\Omega, \mathbb{R}^n) \cap C^0(\overline{\Omega}, \mathbb{R}^n)$ [11, 21]. In this case the integral $\oint_{+\partial\Omega} F \cdot \nu \, \mathrm{d}S$ has the same meaning as above —due to the assumption $F \in C^0(\overline{\Omega}, \mathbb{R}^n)$— whereas $\int\limits_{\Omega} \nabla \cdot F \,\mathrm{d}^n x$ should be interpreted as an improper Riemannian integral —which coincides with the Lebesgue integral $\int\limits_{\Omega} \nabla \cdot F \,\mathrm{d}^n x$ if $\nabla \cdot F \in \mathcal{L}^1(\Omega)$.

In more details the result is obtained by considering a sequence $\{\Omega_m\}_{m\in\mathbb{N}}$ of non-empty, open subsets with the following properties: (a) $\overline{\Omega_m} \subset \Omega_{m+1} \subset \Omega$, which implies in particular that Ω_m has compact closure; (b) $\partial\Omega_m$ is an oriented regular hypersurface with $\partial\Omega_m \subseteq \{x \in \Omega \,|\, d(x, \partial\Omega) < 1/m\}$, where $d(x, \partial\Omega) = \inf\limits_{y\in\partial\Omega} d(x, y) = \min\limits_{y\in\partial\Omega} d(x, y)$. The result will not depend on the chosen sequence as long as the above properties are fulfilled.

Since $F \in C^1(\Omega, \mathbb{R}^n)$ we have $F \in C^1(\overline{\Omega_m}, \mathbb{R}^n)$ so that Equation (8.3) applies for Ω_m for all m. At this stage we observe that the integral $\int\limits_{\Omega_m} \nabla \cdot F \,\mathrm{d}^n x$ can be either interpreted as a Lebesgue integral over Ω_m or as a Riemann integral over $\overline{\Omega_m}$. Letting $m \to +\infty$ we have that, since $F \cdot \nu \in C^0(\partial\Omega)$,

$$\lim_{m\to+\infty} \oint\limits_{+\partial\Omega_m} F \cdot \nu_m \,\mathrm{d}S = \oint\limits_{+\partial\Omega} F \cdot \nu \,\mathrm{d}S\,.$$

This leads to

$$\oint\limits_{+\partial\Omega} F \cdot \nu \,\mathrm{d}S = \lim_{m\to+\infty} \int_{\Omega_m} \nabla \cdot F \,\mathrm{d}^n x\,,$$

where the right-hand side is seen as the limit of Riemann integrals $\int_{\Omega_m} \nabla \cdot F \,\mathrm{d}^n x$, that is, an improper Riemann integral.

(iii) A proof of Theorem 8.1 is based on the Gauss-Green formula —*cf.* [6, §16]. However, this theorem is a particular case of a more general Stokes' theorem formulated in the framework of differential geometry —*cf.* [14, Thm 3.5].

The following useful result is an immediate consequence of Theorem 8.1.

Theorem 8.3 (Green's identities): For all $u_1 \in C^1(\overline{\Omega})$ and $u_2 \in C^2(\overline{\Omega})$ it holds

$$\int_{\Omega} u_1 \Delta u_2 \,\mathrm{d}^n x = \oint\limits_{+\partial\Omega} u_1 \frac{\partial u_2}{\partial \nu} \,\mathrm{d}S - \int_{\Omega} \nabla u_1 \cdot \nabla u_2 \,\mathrm{d}^n x\,. \tag{8.4}$$

For all $u_1, u_2 \in C^2(\overline{\Omega})$ it holds

$$\int_{\Omega} [u_1 \Delta u_2 - u_2 \Delta u_1] \,\mathrm{d}^n x = \oint\limits_{+\partial\Omega} \left[u_1 \frac{\partial u_2}{\partial \nu} - u_2 \frac{\partial u_1}{\partial \nu} \right] \mathrm{d}S\,. \tag{8.5}$$

Proof. Both proofs are based on Theorem 8.1.

$\boxed{(8.4)}$ We observe that

$$\begin{aligned} u_1 \Delta u_2 &= u_1 \delta^{jk} \frac{\partial^2 u_2}{\partial x^j \partial x^k} = \frac{\partial}{\partial x^k}\left[\delta^{jk} u_1 \frac{\partial u_2}{\partial x^j}\right] - \delta^{jk} \frac{\partial u_1}{\partial x^k}\frac{\partial u_2}{\partial x^j} \\ &= \nabla \cdot [u_1 \nabla u_2] - \nabla u_1 \cdot \nabla u_2 \,. \end{aligned}$$

Integrating this equality on Ω we have

$$\begin{aligned} \int_\Omega u_1 \Delta u_2 \, \mathrm{d}^n x &= \int_\Omega \nabla \cdot [u_1 \nabla u_2] \, \mathrm{d}^n x - \int_\Omega \nabla u_1 \cdot \nabla u_2 \, \mathrm{d}^n x \\ &= \oint_{+\partial\Omega} u_1 \frac{\partial u_2}{\partial \nu} \, \mathrm{d}^n x - \int_\Omega \nabla u_1 \cdot \nabla u_2 \, \mathrm{d}^n x \,, \end{aligned}$$

where we applied Equation (8.3) together with $\frac{\partial u}{\partial \nu} = \nu \cdot \nabla u$.

$\boxed{(8.5)}$ Equation (8.5) is nothing but the skew-symmetric part in u_1, u_2 of Equation (8.4).

□

Remark 8.4 (☆):

(i) In the case $n = 1$ and setting $\Omega = (a, b)$, Equations (8.4)-(8.5) reduce to the integration by parts formula.

(ii) The regularity assumption on u_1, u_2 can be relaxed in a way similar to Remark 8.2.

We now consider the boundary-value problem of interest for this section.

Definition 8.5 (Neumann and Robin inner problems for (7.1)): The **Neumann inner problem** consists of finding $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ such that

$$-\Delta u = f \,, \tag{8.6a}$$

$$\frac{\partial u}{\partial \nu} = u_1 \,, \tag{8.6b}$$

where $f \in C^0(\Omega)$ and $u_1 \in C^0(\partial\Omega)$ are assigned —Notice that the regularity imposed on f and u_1 is consistent with the one required on u. Condition (8.6b) is called **(inhomogeneous) Neumann boundary condition.**

The **Robin inner problem** consists of finding $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ such that

$$-\Delta u = f \,, \tag{8.7a}$$

$$\left(\frac{\partial u}{\partial \nu} + \alpha u\right)\Big|_{\partial\Omega} = u_1 \,, \tag{8.7b}$$

where $\alpha > 0$, $f \in C^0(\Omega)$ and $u_1 \in C^0(\partial\Omega)$ are assigned. Condition (8.7b) is called **(inhomogeneous) Robin boundary condition.**

Remark 8.6:

(i) Notice that both the Neumann and the Robin boundary conditions (8.6b)-(8.7b) involve the normal derivative $\frac{\partial u}{\partial \nu} := \nu \cdot \nabla u$ of u along ν. This is ultimately the reason for requiring: (a) the orientability of $\partial\Omega$; (b) the higher regularity $u \in C^1(\overline{\Omega})$ —notice that for the Dirichlet problem (7.3) we only assumed $u \in C^0(\overline{\Omega})$.

Strictly speaking, assuming that $\partial\Omega$ is a C^1-regular hypersurface guarantees that $\frac{\partial u}{\partial \nu}$ is well-defined at any point of $\partial\Omega$. However, even if $u \in C^1(\overline{\Omega})$, none guarantees that $\frac{\partial u}{\partial \nu}$ is continuous on $\partial\Omega$: This is ultimately the reason for requiring $\partial\Omega$ to be orientable. Another technical reason is related to the applicability of Theorem 8.1.

(ii) From a physical point of view the boundary conditions (8.6b) and (8.7b) have the following interpretation. We recall that a solution u to the Poisson equation (7.1) can be interpreted as a stationary solution to the heat equation (1.7). Thus, we may think of u as the temperature of a continuum. From this point of view the boundary condition (7.3b) is interpreted as the assignment of the temperature on the boundary of the chosen continuum —*e.g.* u describes the temperature of an oven whose boundaries are set to a specific temperature.

N For what concerns the Neumann boundary condition (8.6b) we recall the Fourier law: The latter postulates that the heat flow $q \colon \Omega \to \mathbb{R}^n$, which describes the direction and intensity of the heat at each point of Ω, is related to the temperature u by the relation

$$q = -k\nabla u \,, \tag{8.8}$$

where, for the sake of simplicity, we shall assume k to be a positive constant —more generally k can be any positive function of x. Within these assumptions the Neumann boundary condition (8.6b) can be rewritten as

$$u_1 = \frac{\partial u}{\partial \nu} = -\frac{1}{k} q \cdot \nu \,.$$

Thus, the Neumann boundary condition is interpreted as a condition on the incoming/outgoing heat flux across $\partial\Omega$.

We stress that the physical content of Dirichlet and Neumann boundary conditions is rather different. On the one hand, the Dirichlet boundary condition (7.3b) restricts the temperature on the boundary of $\partial\Omega$: However, this does not forbid Ω from exchanging heat(=energy) with the outside $\mathbb{R}^n\backslash\Omega$. This is the reason why a kitchen warms up when we are baking: The oven, whose temperature is set, exchanges energy(=heat) with the outside.

Conversely, the Neumann boundary condition (8.6b) does not assign the value of the temperature on $\partial\Omega$, however, it prescribes the normal component of the heat flow on $\partial\Omega$. Assuming for simplicity that u_1 is a constant this means that: (a) the heat flow is ingoing if $u_1 > 0$; (b) the heat flow is outgoing if $u_1 < 0$; (c) there is no heat exchange if $u_1 = 0$. (We recall that ν is the outward pointing normal to $\partial\Omega$.) This latter situation is realized in a thermos: No matter what the temperature of the content of the thermos is, the heat flow across the boundary of the thermos should vanish, so that the inside of the thermos is thermically isolated from the outside.

R The thermodynamical interpretation of the Robin boundary condition (8.7b) is quite important to understand the assumption $\alpha > 0$. For that, let assume we consider a continuum Ω where we cannot prescribe neither the temperature at the boundary nor the ingoing/outgoing heat flow with the outside. In this situation a reasonable assumption is that

$$-q \cdot \nu = h(u_{\text{EXT}} - u)\,, \tag{8.9}$$

where $h > 0$ while u_{EXT} is the temperature of $\mathbb{R}^n\backslash\Omega$. Equation (8.9) describes the interaction between Ω and $\mathbb{R}^n\backslash\Omega$ at $\partial\Omega$. Notice that if $u_{\text{EXT}} > u$ —that is, if the outside has higher temperature than the inside— then $q \cdot \nu < 0$ which entails that the heat flow is incoming (with respect to Ω). Conversely, if $u_{\text{EXT}} < u$ then the heat flow is outgoing. Therefore, Equation (8.9) prescribes incoming heat flow if and only if the outside is warmer than the inside, as expected. Finally, recalling the Fourier law (8.8), Equation (8.9) reduces to

$$k\frac{\partial u}{\partial \nu} = h(u_{\text{EXT}} - u)\,,$$

which coincides with the Robin boundary condition (8.7b) for $\alpha = h/k$, $u_1 := hu_{\text{EXT}}/k$. Notice that $\alpha > 0$ is a rather important assumption: A Robin boundary condition with $\alpha < 0$ would prescribe incoming flow if the outside is colder than the inside.

Finally, notice that in this interpretation the temperature u_{EXT} of the outside is kept fixed. This is ultimately an assumption of our model, postulating that the effects of the temperature in Ω on the outside region are negligible. A more refined description may include the variation of the temperature on the whole $\mathbb{R}^n$.

(iii) As already mentioned, Dirichlet, Neumann and Robin boundary conditions (7.3b)-(8.6b)-(8.7b) do not exhaust all possible choices. For example, if $\partial\Omega$ is not connected, say

$\partial\Omega = U_1 \cup U_2$, then one may impose different boundary conditions on different connected components, say Dirichlet boundary conditions on U_1 and Neumann boundary conditions on U_2. Another possibility is to consider point dependent boundary conditions of the form

$$\alpha(x)u(x) + \beta(x)\frac{\partial u}{\partial \nu} = u_1(x) \quad x \in \partial\Omega\,,$$

for $\alpha, \beta, u_1 \in C^0(\partial\Omega)$. This boundary condition reduces to the Dirichlet, Neumann or Robin ones depending on the values of $\alpha(x), \beta(x)$.

(iv) The data $f \in C^0(\Omega)$ and $u_1 \in C^0(\partial\Omega)$ in the Neumann inner problem (8.6) are related by the following identity:

$$-\int_\Omega f\,\mathrm{d}^n x = \int_\Omega \Delta u\,\mathrm{d}^n x = \oint_{+\partial\Omega} \frac{\partial u}{\partial \nu}\,\mathrm{d}S = \oint_{+\partial\Omega} u_1\,\mathrm{d}S\,, \tag{8.10}$$

where we applied Theorem 8.1 —*cf.* Remark 8.2. Thus, the integral of f over Ω must be equal to the integral of u_1 over $\partial\Omega$. This is a necessary condition to find a solution u to problem (8.6).

As in Section 7, we shall not discuss the existence of the solution to the Neumann or Robin inner problems. We will focus on a uniqueness result: The latter is not based on the maximum principle but rather on an "energy method" based on the application of Theorem 8.1.

A function $\gamma\colon [a,b] \to \mathbb{R}^m$ is said C^k**-piecewise** if there exists finitely many points $\{t_j\}_{j=1}^{\ell}$ with $a = t_1 < \ldots < t_\ell = b$ such that $\gamma|_{[t_j,t_{j+1}]} \in C^k([t_j,t_{j+1}],\mathbb{R}^m)$ for all $j = 1,\ldots,\ell-1$. We recall that $\Omega \subseteq \mathbb{R}^n$ is said C^k**-piecewise path connected** if for all $x, y \in \Omega$ there exists a C^k-piecewise function $\gamma_{x,y}\colon [0,1] \to \Omega$ such that $\gamma_{x,y}(0) = x$ and $\gamma_{x,y}(1) = y$ —by Definition, for $k = 0$ we remove "piecewise" so that γ_{xy} has to be continuous.

Proposition 8.7 (): Let $\Omega \subseteq \mathbb{R}^n$ be non-empty, open and connected. Then Ω is C^∞-piecewise path connected.

Proof. We recall that Ω is connected if and only if the only simultaneously closed and open subsets of Ω are $\varnothing$ and Ω. Let $x \in \Omega$ and set

$$\begin{aligned} C_x &:= \{y \in \Omega \,|\, \exists \gamma_{x,y}\colon [0,1] \to \Omega\ C^\infty\text{-piecewise}\,,\ \gamma_{x,y}(0) = x\,,\ \gamma_{x,y}(1) = y\}\,,\\ N_x := \Omega\setminus C_x &:= \{y \in \Omega \,|\, \nexists \gamma_{x,y}\colon [0,1] \to \Omega\ C^\infty\text{-piecewise}\,,\ \gamma_{x,y}(0) = x\,,\ \gamma_{x,y}(1) = y\}\,, \end{aligned}$$

Since $x \in C_x$ we have $C_x \neq \varnothing$: We shall now prove that $C_x \subseteq \Omega$ is open and closed, thus $C_x = \Omega$ because of connectedness.

$\boxed{C_x \textbf{ is open}}$ Let $y \in C_x$, we wish to prove that there exists $\varepsilon > 0$ such that $B_\varepsilon(y) \subseteq C_x$, where $B_\varepsilon(y)$ denotes the open Euclidean ball centred at y and with radius ε. To this avail we observe that, since Ω is open and $y \in \Omega$, there exists $\varepsilon > 0$ such that $B_\varepsilon(y) \subseteq \Omega$. For all $z \in B_\varepsilon(y)$ there exists a C^∞ curve $\gamma_{y,z} \colon [0,1] \to \Omega$ such that $\gamma_{y,z}(0) = y$ and $\gamma_{y,z}(1) = z$: As a matter of fact $\gamma_{y,z}(t) := tz + (1-t)y$ fulfils all these properties. Let $\gamma_{x,y} \colon [0,1] \to \Omega$ be the C^∞-piecewise curve such that $\gamma_{x,y}(0) = x$ and $\gamma_{x,y}(1) = y$. We set

$$\gamma_{x,z}(t) := \begin{cases} \gamma_{x,y}(2t) & \text{if } t \in [0,1/2] \\ \gamma_{y,z}(2t-1) & \text{if } t \in [1/2,1] \end{cases} .$$

It follows that $\gamma_{x,z}$ is C^∞-piecewise, because so are its restrictions on $[0,1/2]$ and on $[1/2,1]$. Moreover, $\gamma_{x,z}(0) = x$ and $\gamma_{x,z}(1) = z$: Thus $z \in C_x$ for all $z \in B_\varepsilon(y)$.

$\boxed{C_x \textbf{ is closed}}$ We shall prove that N_x is open, thus implying that $C_x = \Omega \setminus N_x$ is closed. For that let $y \in N_x$: We claim that there exists $\varepsilon > 0$ such that $B_\varepsilon(y) \subset N_x$. By contradiction let assume that for all $\varepsilon > 0$ it holds $B_\varepsilon(y) \cap C_x \neq \varnothing$. Let $z \in B_\varepsilon(y) \cap C_x$ and let $\gamma_{x,z}$ be a C^∞-piecewise curve such that $\gamma_{x,z}(0) = x$ and $\gamma_{x,z}(1) = z$. Then the curve

$$\gamma_{x,y}(t) := \begin{cases} \gamma_{x,z}(2t) & \text{if } t \in [0,1/2] \\ (2t-1)y + 2(1-t)z & \text{if } t \in [1/2,1] \end{cases} ,$$

is C^∞-piecewise and fulfils $\gamma_{x,y}(0) = x$, $\gamma_{x,y}(1) = y$. It follows that $y \in C_x$, a contradiction. ☠

$\square$

Remark 8.8 (⛷):

(i) With minor adjustments one can show that Ω is C^∞-path connected, that is, $\gamma_{x,y}$ is a C^∞ curve.

(ii) For a generic topological space X the implication "C^0-connected $\Rightarrow$ connected" is always true, whereas the converse implication is false in general.

Lemma 8.9 (⛷): Let $\Omega \subseteq \mathbb{R}^n$ be open and connected and let $u \in C^1(\Omega)$ be such that $\nabla u = 0$. Then u is a constant.

Proof. Since Ω is connected, it is also C^∞-path connected by Proposition 8.7 and Remark 8.8. Let $x, y \in \Omega$ and let $\gamma_{x,y} \in C^\infty([0,1], \Omega)$ be such that $\gamma_{x,y}(0) = x$ and $\gamma_{x,y}(1) = y$. Let $g(t) := u(\gamma_{x,y}(t))$: A direct inspection leads to

$$\frac{\mathrm{d}g}{\mathrm{d}t} = \dot{\gamma} \cdot \nabla u = 0 \,,$$

which entails $u(x) = g(0) = g(1) = u(y)$. $\square$

Theorem 8.10 (Uniqueness for inner Neumann and Robin problem): There exists at most one solution, up to a constant, $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ to the inner Neumann problem (8.6). There exists at most one solution $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ to the inner Robin problem (8.7).

Proof. The proof is similar for both problems.

$\boxed{N}$ Let u, u' be two solutions to the Neumann inner problem (8.6) with the same data $f \in C^0(\Omega)$ and $u_1 \in C^0(\partial\Omega)$. Then $v := u - u'$ solves the Neumann inner problem with vanishing data. Applying Equation (8.4) we have

$$\int_\Omega v \cancel{\Delta v} \, \mathrm{d}^n x = \oint_{+\partial\Omega} v \frac{\partial \cancel{v}}{\cancel{\partial \nu}} \, \mathrm{d}S - \int_\Omega \|\nabla v\|^2 \, \mathrm{d}^n x \, .$$

Since the integrand is positive and continuous, it follows that $\|\nabla v\|^2 = 0$ in Ω. Indeed if there exists $x_0 \in \Omega$ such that $\|\nabla v(x_0)\|^2 > 0$ then there exists an open neighbourhood $N_{x_0} \subseteq \Omega$ of x_0 such that $\|\nabla v(y)\|^2 > 0$ for all $y \in N_{x_0}$. It follows that

$$0 = \int_\Omega \|\nabla v(y)\|^2 \, \mathrm{d}^n x > \int_{N_{x_0}} \|\nabla v(x)\|^2 \, \mathrm{d}^n x > 0 \, ,$$

a contradiction ☠. Since Ω is connected, Lemma 8.9 ensures that v is a constant, therefore, $u = u'$ up to a constant.

$\boxed{R}$ The proof proceed in a similar fashion. Let u, u' be two solutions to the Robin inner problem with the same data $f \in C^0(\Omega)$, $u_1 \in C^0(\partial\Omega)$ and $\alpha > 0$. Then $v := u - u'$ solves the Robin inner problem with vanishing data $f = 0$ and $u_1 = 0$. Applying Equation (8.4) we have

$$\int_\Omega v \cancel{\Delta v} \, \mathrm{d}^n x = \oint_{+\partial\Omega} v \frac{\partial v}{\partial \nu} \, \mathrm{d}S - \int_\Omega \|\nabla v\|^2 \, \mathrm{d}^n x = -\alpha \oint_{+\partial\Omega} v^2 \, \mathrm{d}S - \int_\Omega \|\nabla v\|^2 \, \mathrm{d}^n x \, .$$

Since both integrals are positive, they have to vanish separately, moreover, the positivity and continuity of the integrands leads to

$$\nabla v = 0 \, , \qquad v|_{\partial\Omega} = 0 \, .$$

By Lemma 8.9 v is a constant, moreover, $v|_{\partial\Omega} = 0$ implies that $v = 0$.

□

Remark 8.11:

(i) It is worth commenting the physical interpretation of the uniqueness result for the inner Neumann problem (8.6). Indeed, differently from the inner Dirichlet or Robin problems (7.3)-(8.7), in this case the solution u is unique only up to a constant. This is sensible once we consider a concrete model, like a thermos: As already discussed in Remark 8.6, this boils down to the inner Neumann problem with $f = 0$ and $u_1 = 0$. The fact that these data determine the solution u —*i.e.* the temperature of the content of the thermos— only up to a constant can be interpreted by saying that the thermos can be filled by either warm coffee or fresh water.

Adopting another point of view, one may interpret the solution u as the potential of an electric field $E = -\nabla u$, *cf.* Remark 7.1. In this situation, Theorem 8.10 ensures the uniqueness of the electric field E, which is the true physical object of interest.

(ii) As discussed in Remark 8.6 the condition $\alpha > 0$ in the Robin boundary condition (8.7b) is motivated on physical grounds. From a mathematical point of view, the proof of Theorem 8.10 will not work for $\alpha < 0$, however, one may still wonder whether the result is true or not. We shall now see a counterexample, namely an inner Robin (8.7) for $\alpha < 0$ with two different solutions for the same data.

Let consider $\Omega = B_1(0) \subset \mathbb{R}^2$ and $\alpha = -1$. We shall set $f = 0$ and $u_1 = 0$ so that the problem we are interested in reduces to

$$\Delta u = 0\,, \qquad \frac{\partial u}{\partial \nu} = u|_{\partial B_1(0)}\,. \tag{8.11}$$

The function $u = 0$ is clearly a solution to (8.11): We shall present infinitely many another solutions. For that it suffices to consider

$$u(x, y) := ax + by \qquad (x, y) \in B_1(0)\,. \tag{8.12}$$

By direct inspection $\Delta u = 0$, moreover, we have

$$\frac{\partial u}{\partial \nu} = \nu \cdot \nabla u = (x, y) \cdot (a, b) = u(x, y)\,,$$

where we observed that, for all $(x, y) \in \partial B_1(0)$ we have $\nu(x, y) = (x, y)$.

In fact, it can be show that all solutions to (8.11) are of the form (8.12) for some $a, b \in \mathbb{R}$.

9 Intermezzo: Dominated Convergence and its use

This section is devoted to discuss the dominated convergence Theorem and its corollaries: The latter will be used frequently.

In what follows $\boxed{(X,\Sigma,\mu)}$ will denote a **measurable space**, where X is a set, Σ a σ-algebra over X and $\mu\colon \Sigma \to [0,+\infty]$ a positive measure. A function $f\colon X \to \mathbb{C}$ is a **Σ-measurable** if $f^{-1}(U) \in \Sigma$ for all open $U \subseteq \mathbb{C}$. By definition $s\colon X \to [0,+\infty)$ is **simple** if there exists finitely many $E_1,\ldots,E_N \in \Sigma$ and $a_1,\ldots,a_N > 0$ such that $s = \sum_{j=1}^N a_j 1_{E_j}$, $\boxed{1_{E_j}}$ being the characteristic function of E_j. Finally, given a Σ-measurable function $f\colon X \to [0,+\infty)$ the **integral** of f with respect to μ is defined by

$$\int_X f\mathrm{d}\mu := \sup\left\{\int_X s\,\mathrm{d}\mu \,|\, s \text{ simple}\,,\ s \leqslant f\right\} \in [0,+\infty]\,, \tag{9.1}$$

where if $s = \sum_{j=1}^N a_j 1_{E_j}$ we set $\int_X s\,\mathrm{d}\mu := \sum_{j=1}^N a_j\mu(E_j)$.

For all $p \in [1,+\infty)$ we denote by $\boxed{\mathcal{L}^p(X,\mu)}$ the space of measurable functions $f\colon X \to \mathbb{C}$ such that $\int\limits_X |f|^p\mathrm{d}\mu < +\infty$. The quotient space $\boxed{L^p(X,\mu)} := \mathcal{L}^p(X,\mu)/\sim$, where $f \sim g$ if and only if $f(x) = g(x)$ for μ-almost all (μ-a.a.) $x \in X$, is a complete normed space with norm

$$\|[f]\|_{L^p(X,\mu)} := \Big(\int\limits_X |f|^p\mathrm{d}\mu\Big)^{1/p}\,.$$

In what follows we will sloppily write $\|f\|_{L^p(X,\mu)}$ in place of $\|[f]\|_{L^p(X,\mu)}$. For $p = \infty$ the space $\boxed{L^\infty(X,\mu)}$ consists of ($\sim$-equivalent classes of) functions $f\colon X \to \mathbb{C}$ which are μ-almost everywhere bounded: The norm in this space is

$$\|f\|_{L^\infty(X,\mu)} := \inf\{C > 0 \,|\, |f(x)| \leqslant C \text{ for } \mu\text{-a.a. } x \in X\}\,.$$

We refer to [16] for further details.

Example 9.1: The following examples will occur frequently in the forthcoming sections:

(i) Let consider $\mathbb{R}^n$ equipped with the Borel σ-algebra $\boxed{\mathcal{B}(\mathbb{R}^n)}$, that is, the smallest σ-algebra containing all open subsets of $\mathbb{R}^n$. Together with the Lebesgue measure, we have a measurable space whose corresponding L^p-spaces will be denoted by $\boxed{L^p(\mathbb{R}^n)}$. Choosing an arbitrary open $\Omega \subset \mathbb{R}^n$, leads to the notion of the L^p-spaces $\boxed{L^p(\Omega)}$.

(ii) Another case of interest is $(X,\Sigma,\mu) = (\mathbb{N}, 2^{\mathbb{N}}, \sharp)$, where $2^{\mathbb{N}}$ is the power set of $\mathbb{N}$ while $\sharp$ is the counting measure —one may substitute $\mathbb{N}$ with any numerable X, *e.g.* $\mathbb{Z}^d$, $d \geqslant 1$.

The corresponding L^p spaces are usually denoted by $\boxed{\ell^p(\mathbb{N})} := L^p(\mathbb{N}, 2^{\mathbb{N}}, \sharp)$. In this case Equation (9.1) reduces to

$$\int_{\mathbb{N}} a(n)\mathrm{d}\sharp(n) = \sum_{n=0}^{\infty} a_n\,,$$

for all $a = (a_n)_n \in \ell^1(\mathbb{N})$ with $a_n \geqslant 0$.

(iii) Finally, a slightly more exotic example: consider $\boxed{\ell^p(\mathbb{R})} := L^p(\mathbb{R}, 2^{\mathbb{R}}, \sharp)$, *i.e.* the L^p space built over $\mathbb{R}$ equipped with the power set σ-algebra and counting measure. By direct inspection for any $f\colon \mathbb{R} \to [0, +\infty)$ Equation (9.1) reduces to

$$\int_{\mathbb{R}} f(x)\mathrm{d}\sharp(x) := \sup\Big\{ \sum_{x\in\Lambda} f(x) \,|\, \Lambda \subset \mathbb{R}\,,\, \sharp\Lambda < \infty \Big\}\,. \tag{9.2}$$

Indeed for all $\Lambda \subset \mathbb{R}$ with $\sharp\Lambda < +\infty$ one finds that $s := \sum_{x\in\Lambda} f(x)1_x$ is simple with $s \leqslant f$, moreover, $\int_{\mathbb{R}} s\mathrm{d}\sharp = \sum_{x\in\Lambda} f(x)$. Thus, the right-hand side of Equation (9.2) is lower than the left-hand side. But for all simple function $s = \sum_{j=1}^{N} a_j 1_{E_j}$ with $s \leqslant f$ let $\Lambda := E_1 \cup \ldots E_N$: Then $s \leqslant s_\Lambda := \sum_{x\in\Lambda} f(x)1_x$, therefore, $\int_{\mathbb{R}} s\mathrm{d}\sharp \leqslant \int_{\mathbb{R}} s_\Lambda \mathrm{d}\sharp$. Considering the supremum over all s we find Equation (9.2).

It is an interesting exercise to prove that $\int_{\mathbb{R}} f(x)\mathrm{d}\sharp(x) < +\infty$ if and only $\sharp\{x \in \mathbb{R} \,|\, f(x) \neq 0\} \leqslant \aleph_0$, that is, if and only if there exists a sequence $(x_n)_n$ such that $f(x) \neq 0$ only for $x \in \{x_n\}_n$: In particular this implies that

$$\int_{\mathbb{R}} f(x)\mathrm{d}\sharp(x) = \sum_{n=0}^{\infty} f(x_n)$$

To prove the latter claim it suffices to observe that

$$\{x \in \mathbb{R} \,|\, f(x) \neq 0\} = \bigcup_{n\in\mathbb{N}} \{x \in \mathbb{R} \,|\, f(x) > 1/n\}\,,$$

$$\frac{1}{n}\sharp\{x \in \mathbb{R} \,|\, f(x) > 1/n\} \leqslant \int_{\mathbb{R}} f\mathrm{d}\sharp < +\infty\,.$$

Therefore $\{x \in \mathbb{R} \,|\, f(x) \neq 0\}$ is the countable union of countable sets, thus, it is at most countable.

We now state without proof the main theorem of this section.

Theorem 9.2 (dominated convergence): Let $(f_n)_n \subset \mathcal{L}^1(X, \mu)$ such that

(i) there exists $f\colon X \to \mathbb{C}$ such that $\lim_{n\to+\infty} f_n(x) = f(x)$ for μ-almost all (μ-a.a.) $x \in X$;

(ii) there exists $g \in \mathcal{L}^1(X,\mu)$ such that $|f(x)| \leqslant |g(x)|$ for μ-a.a. $x \in X$.

Then $f \in \mathcal{L}^1(X,\mu)$ and

$$\|f\|_{L^1(X,\mu)} \leqslant \|g\|_{L^1(X,\mu)}\,, \qquad \lim_{n\to+\infty} \|f_n - f\|_{L^1(X,\mu)} = 0\,,$$

$$\lim_{n\to+\infty} \int_X f_n \mathrm{d}\mu = \int_X f \mathrm{d}\mu\,. \quad (9.3)$$

Example 9.3: Let $f(x) := e^{-x^2/2} \in L^1(\mathbb{R})$ and set $f_n(x) := f(x-n)$. For the sequence $(f_n)_n$ assumption (i) is fulfilled because $f_n(x) \underset{n\to\infty}{\longrightarrow} 0$ for all $x \in \mathbb{R}$. However, conditions (9.3) fails: In particular

$$\int_{\mathbb{R}} f_n(x)\mathrm{d}x = \int_{\mathbb{R}} f(x)\mathrm{d}x = \sqrt{\pi}\,.$$

Thus, assumption (ii) of Theorem 9.2 must fail as well. Indeed, let $g \in L^1(\mathbb{R})$ be such that $f_n(x) \leqslant |g(x)|$ for all $x \in \mathbb{R}$ and $n \in \mathbb{N}$. Let $\delta > 0$ be such that $f(x) > 1/2$ for all $x \in (-\delta,\delta)$: Such δ exists because f is continuous and $f(0) = 1$. We may also choose δ small enough so that $(n-\delta,n+\delta)\cap(m-\delta,m+\delta) = \varnothing$ for all $n,m \in \mathbb{N}$ with $n \neq m$. It follows that $1/2 < f_n(x) \leqslant |g(x)|$ for all $x \in (n-\delta,n+\delta)$: But this implies

$$\int_{\mathbb{R}} |g(x)|\mathrm{d}x \geqslant \sum_{n=0}^{\infty} \int_{n-\delta}^{n+\delta} |g(x)|\mathrm{d}x \geqslant \sum_{n=0}^{\infty} \delta = +\infty\,,$$

a contradiction ☠.

Theorem 9.2 still holds true if we replace the sequence $(f_n)_{n\in\mathbb{N}}$ with a continuous family $(f_t)_{t\in\overline{\Omega}}$, where $\Omega \subseteq \mathbb{R}^n$ is open and non-empty: In this case one has the following result.

Corollary 9.4: Let $(f_t)_{t\in\overline{\Omega}} \subset \mathcal{L}^1(X,\mu)$ be such that $\overline{\Omega} \ni t \mapsto f_t(x) \in \mathbb{C}$ is continuous for μ-a.a. $x \in X$. Let assume that there exists $g \in \mathcal{L}^1(X,\mu)$ such that $|f_t(x)| \leqslant |g(x)|$ for all $t \in \overline{\Omega}$ and μ-a.a. $x \in X$. Then the function

$$F\colon \overline{\Omega} \ni t \mapsto \int_X f_t(x)\mathrm{d}\mu(x)\,,$$

is in $C^0(\overline{\Omega},\mathbb{C})$. (An analogous result holds by replacing $\overline{\Omega}$ with Ω everywhere.)

Proof. Since $\overline{\Omega}$ and $\mathbb{C}$ are closed subsets of metric spaces it follows that, for all $x \in X$, the function $\overline{\Omega} \ni t \mapsto f_t(x) \in \mathbb{C}$ has a limit $f_{t_0}(x)$ for $t \to t_0$ if and only if for all sequences $(t_n)_n \subset \overline{\Omega}$ such that $t_n \to t_0$ as $n \to +\infty$ it holds $\lim_{n\to+\infty} f_{t_n}(x) = f_{t_0}(x)$. It then suffices to apply Theorem 9.2 to $\int_X f_{t_n}(x)\mathrm{d}\mu(x)$. □

Theorem 9.2 leads to the following Corollary —*cf.* [11, Thm. B.5] for a more refined version.

Corollary 9.5: Let $(f_t)_{t\in\overline{\Omega}} \in \mathcal{L}^1(X,\mu)$ be such that

(i) The function $\overline{\Omega} \ni t \mapsto f_t(x) \in \mathbb{C}$ is $C^1(\overline{\Omega},\mathbb{C})$ for μ-a.a. $x \in X$;

(ii) there exists $g \in \mathcal{L}^1(X,\mu)$ such that $\left|\frac{\partial f_t}{\partial t^k}(x)\right| \leqslant |g(x)|$ for all $t \in \overline{\Omega}$, for all $k \in \{1,\dots,n\}$ and for μ-a.a. $x \in X$.

Then $\frac{\partial f_t}{\partial t^k} \in \mathcal{L}^1(X,\mu)$ for all $t \in \overline{\Omega}$ and for all $k \in \{1,\dots,n\}$. Furthermore, the function

$$F\colon \overline{\Omega} \ni t \mapsto \int_X f_t(x)\mathrm{d}\mu(x) \in \mathbb{C}\,,$$

is in $C^1(\overline{\Omega},\mathbb{C})$ and for all $k \in \{1,\dots,n\}$ we have

$$\frac{\partial F}{\partial t^k} = \frac{\partial}{\partial t^k}\int_X f_t(x)\mathrm{d}\mu(x) = \int_X \frac{\partial f_t}{\partial t^k}(x)\mathrm{d}\mu(x)\,. \tag{9.4}$$

(An analogous result holds by replacing $\overline{\Omega}$ with Ω everywhere.)

Proof. We apply Corollary 9.4 to the family $\left(\frac{\partial f_t}{\partial t^k}\right)_{t\in\overline{\Omega}}$. Therefore, we have that $\frac{\partial f_t}{\partial t^k} \in \mathcal{L}^1(X,\mu)$ for all $t \in \overline{\Omega}$, moreover, $\left\|\frac{\partial f_t}{\partial t^k}\right\|_{L^1(X,\mu)} \leqslant \|g\|_{L^1(X,\mu)}$ —*cf.* Equation (9.3). In addition, the function

$$\overline{\Omega} \ni t \mapsto \int_X \frac{\partial f_t}{\partial t^k}(x)\mathrm{d}\mu(x)\,,$$

is in $C^0(\overline{\Omega},\mathbb{C})$. For what concerns Equation (9.4) we observe that, for all $t \in \Omega$ and $k \in \{1,\dots,n\}$,

$$\begin{aligned}
\frac{\partial F}{\partial t^k} &= \lim_{\tau\to 0}\int_X \left[\frac{f_{t+\tau e_k}(x) - f_t(x)}{\tau}\right]\mathrm{d}\mu(x) \\
&= \lim_{\tau\to 0}\int_X \frac{\partial f_{t+\tau' e_k}}{\partial t^k}(x)\mathrm{d}\mu(x)\,, && \text{Lagrange Thm. } \tau' \in (0,\tau) \\
&= \int_X \frac{\partial f_t}{\partial t^k}(x)\mathrm{d}\mu(x)\,, && \text{Thm. 9.2 - Eq.(9.3)}\,.
\end{aligned}$$

This implies that $\frac{\partial F}{\partial t^k}$ exists for all $t \in \Omega$. With a similar argument Theorem 9.2 implies that $F \in C^0(\overline{\Omega},\mathbb{C})$. Thus, $F \in C^0(\Omega,\mathbb{C})$ admits continuous partial derivatives in all directions: Proposition 2.1 entails that $F \in C^1(\Omega,\mathbb{C})$. Finally we notice that $\frac{\partial F}{\partial t^k}$ coincides with the restriction to Ω of a function which is $C^0(\overline{\Omega},\mathbb{C})$. Thus $\frac{\partial F}{\partial t^k}$ extends to a continuous function on $\overline{\Omega}$ for all $k \in \{1,\dots,n\}$, therefore, $F \in C^1(\overline{\Omega},\mathbb{C})$. □

The rest of the section is devoted to illustrate the use of Theorem 9.2 and Corollaries 9.4-9.5 in several situations of practical interest.

Lemma 9.6: Let $f \in C^0(\mathbb{R}^n)$ and $x_0 \in \mathbb{R}^n$. Then

$$\lim_{\varepsilon \to 0^+} \frac{1}{|\partial B_\varepsilon(x_0)|} \oint\limits_{+\partial B_\varepsilon(x_0)} f(x) \mathrm{d}S(x) = f(x_0) \,. \tag{9.5}$$

Proof. Equation (3.12) together with $|\partial B_\varepsilon(x_0)| = \varepsilon^{n-1} |\partial B_1(0)|$ lead to

$$\frac{1}{|\partial B_\varepsilon(x_0)|} \oint\limits_{+\partial B_\varepsilon(x_0)} f(x) \mathrm{d}S(x) = \frac{1}{|\partial B_1(0)|} \oint\limits_{+\partial B_1(0)} f(x_0 + \varepsilon z) \mathrm{d}\Omega_n(z) \,.$$

Since f is continuous we have $\lim\limits_{\varepsilon \to 0^+} f(x_0 + \varepsilon z) = f(x_0)$, Moreover, continuity implies that there exists $C > 0$ such that $|f(x)| \leqslant C$ for all $x \in B_1(x_0)$. Since constant functions are L^1 on $\partial B_1(0)$ we may apply Corollary 9.4 so that

$$\lim_{\varepsilon \to 0^+} \frac{1}{|\partial B_\varepsilon(x_0)|} \oint\limits_{+\partial B_\varepsilon(x_0)} f(x) \mathrm{d}S(x) = \lim_{\varepsilon \to 0^+} \frac{1}{|\partial B_1(0)|} \oint\limits_{+\partial B_1(0)} f(x_0 + \varepsilon z) \mathrm{d}\Omega_n(z) = f(x_0) \,.$$

□

Lemma 9.7: Let $f \in C^2_c(\mathbb{R}^n)$ and set

$$u \colon \mathbb{R}^n \to \mathbb{R} \qquad u(x) := \int_{\mathbb{R}^n} \frac{f(y)}{\|x - y\|^{n-2}} \mathrm{d}y \,.$$

Then u is well-defined, in fact, $u \in C^2(\mathbb{R}^n)$, moreover

$$\frac{\partial^2 u}{\partial x^k \partial x^j} = \int_{\mathbb{R}^n} \frac{1}{\|x - y\|^{n-2}} \frac{\partial^2 f}{\partial y^k \partial y^j} \mathrm{d}y \,, \tag{9.6}$$

for all $k, j \in \{1, \ldots, n\}$.

Proof. Let $x \in \mathbb{R}^n$ and let $R_0 > 0$ be such that $\mathrm{supp}(f) \subset B_{R_0}(x)$: Then

$$\begin{aligned}\left| \int_{\mathbb{R}^n} \frac{f(y)}{\|x - y\|^{n-2}} \mathrm{d}y \right| &\leqslant \|f\|_\infty \int_{B_{R_0}(x)} \frac{1}{\|x - y\|^{n-2}} \mathrm{d}y \\ &= \|f\|_\infty \int_0^{R_0} \frac{\omega_n}{R^{n-2}} R^{n-1} \mathrm{d}R = \|f\|_\infty \omega_n R_0^2 / 2 \,,\end{aligned}$$

so that u is well-defined. Moreover, notice that

$$u(x) = \int_{\mathbb{R}^n} \frac{f(y)}{\|x-y\|^{n-2}} \mathrm{d}^n y = \int_{\mathbb{R}^n} \frac{f(x-z)}{\|z\|^{n-2}} \mathrm{d}^n z \,.$$

We prove that $u \in C^2(\mathbb{R}^n)$ by showing that u has continuous partial derivatives of second order at any point $x_0 \in \mathbb{R}^n$: This will suffice to conclude by Proposition 2.1.

Let $x_0 \in \mathbb{R}^n$. Our goal is to apply Corollary (9.5) twice —In passing, this will also prove Equation (9.6). To this avail we need to bound the x-second derivatives of $f(x-z)\|z\|^{2-n}$ with an L^1-function in z —*cf.* condition (ii)— uniformly for all $x \in B_\varepsilon(x_0)$ where $\varepsilon > 0$ is fixed.

For that let $R_0 > 0$ such that $\mathrm{supp}(f) \subset B_{R_0}(0)$. This implies that, for all $x \in B_\varepsilon(x_0)$, the function $f_x(z) := f(x-z)$ is compactly supported with $\mathrm{supp}(f_x) \subset B_{R_0}(x)$. Since the open set

$$\bigcup_{x \in B_\varepsilon(x_0)} B_{R_0}(x) \,,$$

has compact closure, there exists $R_0' > 0$ such that $\bigcup_{x \in B_\varepsilon(x_0)} B_{R_0}(x) \subseteq B_{R_0'}(0)$. Thus we have the estimate

$$\begin{aligned}
\|z\|^{2-n}|f(x-z)| &\leqslant \|f\|_\infty \|z\|^{2-n} 1_{B_{R_0'}(0)}(z) \,, \\
\|z\|^{2-n}\Big|\frac{\partial f}{\partial x^k}(x-z)\Big| &\leqslant \Big(\sup_{B_{R_0'}(0)} \max_{k\in\{0,\ldots,n\}} \Big|\frac{\partial f}{\partial x^k}\Big|\Big) |\|z\|^{2-n} 1_{B_{R_0'}(0)}(z) \,, \\
\|z\|^{2-n}\Big|\frac{\partial^2 f}{\partial x^k \partial x^j}(x-z)\Big| &\leqslant \Big(\sup_{B_{R_0'}(0)} \max_{j,k\in\{1,\ldots,n\}} \Big|\frac{\partial^2 f}{\partial x^k \partial x^j}\Big|\Big) \|z\|^{2-n} 1_{B_{R_0'}(0)}(z) \,,
\end{aligned}$$

where $\|z\|^{2-n} 1_{B_{R_0'}(0)}(z)$ is in $L^1(\mathbb{R}^n)$. □

LEMMA 9.8: Let $(a_n)_n \in \ell^1(\mathbb{Z})$ and set

$$u \colon \overline{\mathbb{R}_+} \to \mathbb{R} \qquad u(t) := \sum_{n\in\mathbb{Z}} a_n e^{-|n|t} \,.$$

Then $u \in C^0(\overline{\mathbb{R}_+}) \cap C^\infty(\mathbb{R}_+)$.

Proof. To begin with we observe that

$$\Big|\sum_{n\in\mathbb{Z}} a_n e^{-|n|t}\Big| \leqslant \sum_{n\in\mathbb{Z}} |a_n| < \infty \,,$$

which proves that u is well-defined.

Moreover, for all $t \geqslant 0$ and $n \in \mathbb{Z}$, let $u_n(t) := a_n e^{-|n|t}$. Then $u_n \in C^\infty(\overline{\mathbb{R}_+})$ and $(u_n(t))_n \in \ell^1(\mathbb{Z})$ for all $t \geqslant 0$, furthermore, $|u_n(t)| \leqslant |a_n|$. Thus, Corollary 9.4 applies and we can conclude that $u \in C^0(\overline{\mathbb{R}_+})$.

For what concerns smoothness of u on $\mathbb{R}_+$, we observe that it suffices to prove that u is smooth on $(a, +\infty)$ for all $a > 0$. But for $t \geqslant a$ we find

$$\left|\frac{\mathrm{d}^k}{\mathrm{d}t^k} u_n(t)\right| = (-|n|)^k a_n e^{-|n|t} \leqslant |n|^k |a_n| e^{-|n|a} =: b_n \,.$$

Since $a > 0$ we have that $(b_n)_n \in \ell^1(\mathbb{Z})$, therefore, we may apply (infinitely many times) Corollary 9.5 to conclude that u is smooth on $(a, +\infty)$, moreover,

$$\frac{\mathrm{d}^k u}{\mathrm{d}t^k} = \sum_{n \in \mathbb{Z}} (-|n|)^k a_n e^{-|n|t} \,.$$

$\square$

10 Elliptic PDEs: The fundamental solution

We shall now introduce and discuss the properties of the fundamental solution to the Laplace equation (2.8). The exposition profits of [3, 11, 18].

To begin with, we consider the problem of finding a solution $u \in C^2(\mathbb{R}^n)$ to the Laplace equation which only depends on $\|x\|$. More precisely we look for u such that $u(x) = \widetilde{u}(\|x\|^2)$ —here and in what follows we shall assume $n > 1$. We have

$$0 = \Delta u = \delta^{jk}\frac{\partial^2 u}{\partial x^k \partial x^j} = \frac{\partial}{\partial x^k} 2x^k \widetilde{u}'(\|x\|^2) = 2\left[n\widetilde{u}'(\|x\|^2) + 2\|x\|^2\widetilde{u}''(\|x\|^2)\right] .$$

Thus, we are left with the ordinary differential equation

$$n\widetilde{u}'(z) + 2z\widetilde{u}''(z) = 0 . \tag{10.1}$$

If $n > 2$ we consider solutions of the form $\widetilde{u}(z) = z^\alpha$: Inserting this ansatz in Equation (10.1) we have

$$0 = [n\alpha + 2\alpha(\alpha-1)]\, z^{\alpha-1} = \alpha(2\alpha - 2 + n)z^{\alpha-1} \quad \Rightarrow \quad \alpha = 0\,, \frac{2-n}{2}\,,$$

out of which we identify two independent solutions —as expected because (10.1) is a second order differential equation. If $n = 2$, Equation (10.1) simplifies to

$$0 = \widetilde{u}'(z) + z\widetilde{u}''(z) = [z\widetilde{u}'(z)]' \quad \Rightarrow \quad z\widetilde{u}'(z) = a \quad \Rightarrow \quad \widetilde{u}(z) = a\log z + b\,.$$

Overall we have found

$$u(x) = \begin{cases} a + b\|x\|^{2-n} & n > 2 \\ a + b\log\|x\| & n = 2 \end{cases} . \tag{10.2}$$

Notice that $u \in C^2(\mathbb{R}^n)$ if and only if u is constant: In all other cases we find an integrable singularity at $x = 0$. The fundamental solution to the Laplace equation (2.8) is defined by selecting a particular choice of a, b in Equation (10.2).

Definition 10.1: Let $x_0 \in \mathbb{R}^n$, $n \geqslant 2$. The **fundamental solution** to the Laplace equation (2.8) is the function

$$\mathbb{R}^n\backslash\{x_0\} \ni x \mapsto G_{n,x_0}(x) := G_n(x, x_0) := \widetilde{G}_n(\|x - x_0\|) \in \mathbb{R}\,,$$

$$\widetilde{G}_n(z) = \begin{cases} \dfrac{1}{(2-n)\omega_n} z^{2-n} & n > 2 \\ \dfrac{1}{2\pi}\log z & n = 2 \end{cases} , \tag{10.3}$$

where $\omega_n := |\partial B_1(0)|$ has been computed in (3.16).

REMARK 10.2:

(i) Notice that, differently from the case $n > 2$, G_2 is not everywhere negative and diverges at $x \to \infty$. Moreover, one may observe that, formally, $\widetilde{G}_2(z) = \lim_{\alpha\to 0^+} \frac{\widetilde{G}_\alpha(z)-1}{\alpha}$.

(ii) The constants a, b in (10.2) identifying the fundamental solution (10.3) are chosen to ensure the following identity: If ν is the outward-pointing normal to $\partial B_R(x_0)$ —explicitly $\nu = (x - x_0)/R$— we have

$$\frac{\partial G_n}{\partial \nu_x} \overset{(3.11)}{=} \frac{\partial \widetilde{G}_n}{\partial R} = \frac{1}{|\partial B_R(x_0)|}\,, \tag{10.4}$$

where ν_x stresses that ν and the partial derivatives are computed with respect to the x-variable. Notice that $\frac{\partial G_n}{\partial \nu_x}(x, x_0) = \frac{\partial G_n}{\partial \nu_{x_0}}(x, x_0)$ because $\nabla_x G_n(x, x_0) = -\nabla_{x_0} G_n(x, x_0)$ as well as $\nu_x = -\nu_{x_0}$.

The importance of Definition 10.1 is revealed by Theorem 10.3 —*cf.* [11, Thm. 3.1]-[3, Thm.1]. We recall that, by Definition, $f \in \mathcal{L}^1_{\mathrm{LOC}}(\mathbb{R}^n)$ if and only if $f \in \mathcal{L}^1(\Omega)$ for all open $\Omega \subseteq \mathbb{R}^n$ with compact closure.

THEOREM 10.3: Let $n \geqslant 2$, $x_0 \in \mathbb{R}^n$. Then:

(a) For all $x \in \mathbb{R}^n \backslash \{x_0\}$ it holds $G_n(x, x_0) = G_n(x_0, x)$, moreover $G_{n,x_0} \in C^\infty(\mathbb{R}^n \backslash \{x_0\}) \cap \mathcal{L}^1_{\mathrm{LOC}}(\mathbb{R}^n)$;

(b) For all $x \in \mathbb{R}^n \backslash \{x_0\}$ we have $\Delta_x G_n(x, x_0) = 0$;

(c) For all $f \in C^2_c(\mathbb{R}^n)$ and $x \in \mathbb{R}^n$ we have

$$\int_{\mathbb{R}^n} G_n(x, y) \Delta_y f(y) \mathrm{d}^n y = f(x)\,. \tag{10.5}$$

(d) For all $f \in C^2_c(\mathbb{R}^n)$ the function

$$u(x) := \int_{\mathbb{R}^n} G_n(x, y) f(y) \mathrm{d}^n y\,, \tag{10.6}$$

defines a solution $u \in C^2(\mathbb{R}^n)$ to the Poisson equation (7.1) on $\mathbb{R}^n$ with source $-f$.

Proof. We only consider the case $n > 2$: The exact same arguments work for $n = 2$, although the formulas are slightly different.

(a) All properties follow from Equation (10.3). In particular, the local integrability can be shown by integrating G_{n,x_0} on an arbitrary but fixed ball $B_{R_0}(x_0)$: Using Equation (3.12) we find

$$\int_{B_{R_0}(x_0)} |G_n(x,x_0)|\mathrm{d}^n x = \int_0^{R_0} \oint_{\partial B_1(x_0)} \frac{1}{(n-2)\omega_n} R^{2-n} \mathrm{d}\Omega_n R^{n-1} \mathrm{d}R = \frac{R_0^2}{2(n-2)} < +\infty \,.$$

(b) Since $G_n(x,x_0)$ is obtained from Equation (10.2), $\Delta_x G_n(x,x_0) = 0$ for all $x \neq x_0$.

(c) Notice that, since $G_{n,x_0} \in \mathcal{L}^1_{\mathrm{LOC}}(\mathbb{R}^n)$, the integral

$$\int_{\mathbb{R}^n} G_n(x,y)\Delta f(y)\mathrm{d}^n y \,,$$

is well-defined and finite, indeed

$$\left|\int_{\mathbb{R}^n} G_n(x,y)\Delta f(y)\mathrm{d}^n y\right| \leqslant \sup|\Delta f| \int_{B_{R_0}(0)} |G_n(x,y)|\mathrm{d}^n y < +\infty \,,$$

where $R_0 > 0$ is chosen so that $\mathrm{supp}(f) \subset B_{R_0}(x)$ —notice that in particular $\partial B_{R_0}(x) \cap \mathrm{supp}(f) = \varnothing$. We remark that $\sup\|\Delta f\| < +\infty$ because f (and thus Δf) is compactly supported.

To prove Equation (10.5) we apply Theorems 9.2-8.3 to get

$$\begin{aligned}
\int_{\mathbb{R}^n} G_n(x,y)\Delta f(y)\mathrm{d}^n y &= \lim_{\varepsilon\to 0^+} \int_{B_{R_0}(x)\setminus B_\varepsilon(x)} G_n(x,y)\Delta_y f(y)\mathrm{d}^n y \qquad \text{Thm. 9.2} \\
&= \lim_{\varepsilon\to 0^+} \int_{B_{R_0}(x)\setminus B_\varepsilon(x)} \cancel{\Delta_y G_n(x,y)} f(y)\mathrm{d}^n y \qquad \text{Thm. 8.3} \\
&+ \lim_{\varepsilon\to 0^+} \oint_{+\partial[B_{R_0}(x)\setminus B_\varepsilon(x)]} \left[G_n(x,y)\frac{\partial f}{\partial \nu_y} - \frac{\partial G_n}{\partial \nu_y}(x,y)f(y)\right]\mathrm{d}S(y) \\
&= \lim_{\varepsilon\to 0^+} - \oint_{+\partial B_\varepsilon(x)} \left[G_n(x,y)\frac{\partial f}{\partial \nu_y} - \frac{\partial G_n}{\partial \nu_y}(x,y)f(y)\right]\mathrm{d}S(y) \,.
\end{aligned}$$

In the last line we applied condition (b) together with the fact $\partial[B_{R_0}(x)\setminus B_\varepsilon(x)] = \partial B_{R_0}(x) \cup \partial B_\varepsilon(x)$. Notice that the orientation of $+\partial B_\varepsilon(x)$ is the opposite to $+\partial[B_{R_0}(x)\setminus B_\varepsilon(x)]$: This causes the additional minus sign. Moreover, the contribution on $\partial B_R(x)$ vanishes on account of the condition $\partial B_R(x) \cap \mathrm{supp}(f) = \varnothing$.

We now consider the two remaining contributions separately: For the former we have

$$\begin{aligned}\left|\oint_{+\partial B_\varepsilon(x)} G_n(x,y)\frac{\partial f}{\partial \nu_y}\mathrm{d}S(y)\right| &\leqslant \oint_{+\partial B_\varepsilon(x)} \left|\widetilde{G}_n(\varepsilon)\frac{\partial f}{\partial \nu_y}\right|\mathrm{d}S(y)\\ &\leqslant \sup\|\nabla f\| \oint_{+\partial B_1(x)} \frac{\varepsilon^{2-n}}{(n-2)\omega_n}\varepsilon^{n-1}\mathrm{d}\Omega_n\\ &= \sup\|\nabla f\|\frac{\varepsilon}{n-2} \underset{\varepsilon\to 0^+}{\longrightarrow} 0\,,\end{aligned}$$

where we estimated $\left|\frac{\partial f}{\partial \nu_y}\right| \leqslant \|\nabla f\|$ —notice that $\sup\|\nabla f\| < +\infty$ because f (and thus ∇f) is compactly supported. This leads to

$$\begin{aligned}\int_{\mathbb{R}^n} G_n(x,y)\Delta f(y)\mathrm{d}^n y &= \lim_{\varepsilon\to 0^+}\oint_{+\partial B_\varepsilon(x)} \frac{\partial G_n}{\partial \nu_y}(x,y)f(y)\mathrm{d}S(y)\\ &= \lim_{\varepsilon\to 0^+}\frac{1}{|\partial B_\varepsilon(x)|}\oint_{+\partial B_\varepsilon(x)} f(y)\mathrm{d}S(y) = f(x)\,,\end{aligned}$$

where we used identity (10.4) and Lemma 9.6.

(d) Well-definiteness of u and its regularity are consequences of Lemma 9.7. Moreover, Equation (9.6) leads to

$$\begin{aligned}\Delta u(x) = \int_{\mathbb{R}^n} \widetilde{G}_n(\|z\|)\Delta_x f(x-z)\mathrm{d}^n z &= \int_{\mathbb{R}^n} \widetilde{G}_n(\|z\|)\Delta_z f(x-z)\mathrm{d}^n z\\ &= \int_{\mathbb{R}^n} G_n(x,y)\Delta_y f(y)\mathrm{d}^n y = f(x)\,,\end{aligned}$$

where we applied Equation (10.5).

□

Remark 10.4:

(i) The proof of Theorem 10.3-d requires the assumption $f \in C^2_c(\mathbb{R}^n)$. However, the thesis remains true if we assume $f \in C^0_c(\mathbb{R}^n)$: Part of the argument requires additional tools from distribution theory but it is worth sketch the main ideas.

Let $u\colon \mathbb{R}^n \to \mathbb{R}$ be defined as per Equation (10.6) for $f \in C^0_c(\mathbb{R}^n)$. Notice that, on account of Theorem 11.1-a and the compactness of $\mathrm{supp}(f)$, the function u is well-defined, moreover, $u \in C^0(\mathbb{R}^n)$ by Theorem 9.2. In fact, arguments from distribution theory show that $u \in C^2(\mathbb{R}^n)$ —actually, $u \in C^\infty(\mathbb{R}^n)$, see [19]. Taking this for granted we shall now see that $\Delta u = f$. For that let $g \in C^2_c(\mathbb{R}^n)$ be arbitrary: By direct inspection we find

$$\begin{aligned}
\int_{\mathbb{R}^n} \Delta u(x) g(x) \mathrm{d}^n x &= \int_{\mathbb{R}^n} u(x) \Delta g(x) \mathrm{d}^n x && \text{Eq. (8.5) + supp}(g) \text{ compact} \\
&= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} G(x,y) f(y) \mathrm{d}^n y \right) \Delta g(x) \mathrm{d}^n x && \text{Eq. (10.6)} \\
&= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} G(x,y) \Delta g(x) \mathrm{d}^n x \right) f(y) \mathrm{d}^n y && \text{Fubini's Thm.} \\
&= \int_{\mathbb{R}^n} g(y) f(y) \mathrm{d}^n y && \text{Thm. } 10.3 - a - c\,.
\end{aligned}$$

The above chain of equalities shows that

$$\int_{\mathbb{R}^n} [\Delta u(x) - f(x)] g(x) \mathrm{d}^n x = 0 \qquad \forall g \in C^2_c(\mathbb{R}^n)\,.$$

The arbitrariness of g ensures that $\Delta u = f$ —this can be shown by a continuity argument using that $\Delta u - f \in C^0(\mathbb{R}^n)$ or, more generally, by using [16, Thm. 1.40].

(ii) The content of Equation (10.5)-(10.6) can be summarized as follows. Let $G_n \colon C^2_c(\mathbb{R}^n) \to C^2(\mathbb{R}^n)$ be the operator defined by

$$(G_n f)(x) := \int_{\mathbb{R}^n} G_n(x,y) f(y) \mathrm{d}^n y\,.$$

Then $\Delta \circ G_n = \mathrm{Id}_{C^\infty_c(\mathbb{R}^n)}$ and similarly $G_n \circ \Delta|_{C^2_c(\mathbb{R}^n)} = \mathrm{Id}_{C^\infty_c(\mathbb{R}^n)}$. Notice that, in the first identity, Δ is regarded as an operator $\Delta \colon C^2(\mathbb{R}^n) \to C^0(\mathbb{R}^n)$, whereas in the second one the domain of Δ is restricted to $C^2_c(\mathbb{R}^n) \subset C^2(\mathbb{R}^n)$.

We conclude this section with the following existence result for the outer Dirichlet problem on $\mathbb{R}^n$.

Theorem 10.5: Let $n > 2$, $u_\infty \in \mathbb{R}$, $f \in C^2_c(\mathbb{R}^n)$. Then there exists a unique solution to the outer Dirichlet problem (7.4) in $\mathbb{R}^n$ with data f, u_∞, namely, there exists a unique $u \in C^2(\mathbb{R}^n)$ such that

$$-\Delta u = f\,, \tag{10.7a}$$
$$\lim_{x\to\infty} |u(x) - u_\infty| = 0\,. \tag{10.7b}$$

In particular the solution u is given by

$$u(x) := u_\infty - \int_{\mathbb{R}^n} G_n(x,y) f(y) \mathrm{d}^n y\,. \tag{10.8}$$

Moreover, for all $x_0 \in \mathbb{R}^n$ there exists $R, C > 0$ such that $\mathrm{supp}(f) \subset B_R(x_0)$ and

$$|u(x) - u_\infty| \leqslant \frac{C}{\|x - x_0\|^{n-2}} \|f\|_{L^1(\mathbb{R}^n)} \qquad \forall x \in \mathbb{R}^n \backslash B_R(x_0)\,. \tag{10.9}$$

Proof. The uniqueness of the solution u to the problem (10.7) has been proved in Theorem 7.10. Moreover, u defined in Equation (10.8) is well-defined and solves the Laplace equation $-\Delta u = f$ on account of Theorem 10.3-d.

It remains to show that u fulfils the asymptotic Dirichlet boundary condition (7.4c) together with the estimate (10.9). Notice that the latter estimate proves the first statement, therefore, we focus on (10.9).

We observe that, for all $R > 0$, $y \in B_R(0)$ and $x \in \mathbb{R}^n \backslash B_R(0)$ it holds

$$\|x\| \leqslant \|x - y\| + \|y\| \leqslant \|x - y\| + R \quad \Rightarrow \quad \|x - y\| \geqslant \|x\| - R > 0\,.$$

Moreover, for all $\delta \in (0,1)$ there exists $R_\delta > R$ so that, if $x \in \mathbb{R}^n \backslash B_{R_\delta}(0)$, then $\|x\| - R > \delta\|x\|$ —for that it is enough to choose $R_\delta := \frac{R}{1-\delta} > R$. It follows that, for all $y \in B_R(0)$ and $x \in \mathbb{R}^n \backslash B_{R_\delta}(0)$

$$\frac{1}{\|x - y\|^{n-2}} \leqslant \frac{1}{\delta^{n-2}\|x\|^{n-2}}\,.$$

Let now $x_0 \in \mathbb{R}^n$ and $R > 0$ be such that $\mathrm{supp}(f) \subset B_R(x_0)$. For all $y \in B_R(x_0)$ and $x \in \mathbb{R}^n \backslash B_{R_\delta}(x_0)$ we have

$$|G_n(x,y)| = |G_n(x - x_0, y - x_0)| \leqslant \frac{1}{(n-2)\omega_n \delta^{n-2}} \frac{1}{\|x - x_0\|^{n-2}}\,;$$

Thus, for all $x \in \mathbb{R}^n \backslash B_{R_\delta}(x_0)$ we have

$$|u(x) - u_\infty| \leqslant \int_{B_R(x_0)} |G_n(x,y)|\,|f(y)| \mathrm{d}^n y \leqslant \frac{1}{(n-2)\omega_n \delta^{n-2}} \frac{1}{\|x - x_0\|^{n-2}} \|f\|_{L^1(\mathbb{R}^n)}\,.$$

□

Remark 10.6: Theorem 10.5 applies also for $f \in C^0_c(\mathbb{R}^n)$ —*cf.* Remark 10.4.

11 Elliptic PDEs: Mean-value formula and its consequences

In this section we shall explore the consequences of Theorem 10.3. In particular we will derive a representation formula for suitably regular functions on bounded domains and we will prove the mean-value theorem for harmonic functions —*cf.* Theorems 11.1-11.4.

The mean-value formula is essential to prove many of the properties of harmonic functions which we are going to discuss. In particular, we will obtain a refined version of the maximum principle, *cf.* Theorem 7.6. This section profits of [3, 11].

We start by proving that on a bounded domain Ω the fundamental solution leads to a representation formula for sufficiently regular u.

THEOREM 11.1 (Representation formula): Let $\Omega \subset \mathbb{R}^n$ be open, non-empty, with compact closure and such that $\partial\Omega$ is an orientable C^1-hypersurface. Then for all $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ with $\Delta u \in L^\infty(\Omega)$ it holds

$$u(x) = \oint_{+\partial\Omega} \left[\frac{\partial G_n}{\partial \nu_y}(x,y)u(y) - \frac{\partial u}{\partial \nu_y} G_n(x,y)\right] \mathrm{d}S(y) + \int_\Omega G_n(x,y)\Delta u(y)\mathrm{d}^n y\,, \qquad \forall x \in \Omega\,. \tag{11.1}$$

Proof. The proof proceeds in a way similar to the one of Theorem 10.3-c. In particular for all $x \in \Omega$ we have

$$\begin{aligned}
\int_\Omega G_n(x,y)\Delta_y u(y)\mathrm{d}^n y &= \lim_{\varepsilon\to 0^+}\int_{\Omega\setminus B_\varepsilon(x)} G_n(x,y)\Delta_y u(y)\mathrm{d}^n y \qquad \text{Thm. 9.2} \\
&= \lim_{\varepsilon\to 0^+}\int_{\Omega\setminus B_\varepsilon(x)} \cancel{\Delta_y G_n(x,y)}u(y)\mathrm{d}^n y \\
&\quad + \lim_{\varepsilon\to 0^+}\oint_{+\partial(\Omega\setminus B_\varepsilon(x))}\left[G_n(x,y)\frac{\partial u}{\partial\nu_y} - \frac{\partial G_n}{\partial \nu_y}u(y)\right]\mathrm{d}S(y) \qquad \text{Thm. 8.1 + Rmk. 8.2} \\
&= \oint_{+\partial\Omega}\left[G_n(x,y)\frac{\partial u}{\partial\nu_y} - \frac{\partial G_n}{\partial \nu_y}u(y)\right]\mathrm{d}S(y) \\
&\quad - \lim_{\varepsilon\to 0^+}\oint_{+\partial B_\varepsilon(x)}\left[G_n(x,y)\frac{\partial u}{\partial\nu_y} - \frac{\partial G_n}{\partial \nu_y}u(y)\right]\mathrm{d}S(y)\,,
\end{aligned}$$

where the additional minus sign is due to the different orientation of $+\partial B_\varepsilon(x)$ and $+\partial(\Omega\setminus B_\varepsilon(x))$. Proceeding as in the proof of Theorem 10.3-c the limit for $\varepsilon \to 0^+$ leads to $u(x)$. Rearranging the found equality leads to Equation (11.1). $\square$

REMARK 11.2: Theorem 11.1 can be used to prove that an harmonic function $u \in C^2_c(\Omega)$ with compact support $\operatorname{supp}(u) \subset \Omega$ is necessarily the zero function $u = 0$. Since we shall prove a much stronger statement —*cf.* Theorem 11.9— we will refrain from providing further details, referring to [11, Prop. 3.1].

The representation formula (11.1) has crucial consequences for the case of harmonic functions.

THEOREM 11.3 (harmonic functions are C^∞): Let $\Omega \subseteq \mathbb{R}^n$ be open and non-empty, $u \in C^2(\Omega)$ harmonic in Ω. Then $u \in C^\infty(\Omega)$.

Proof. We shall prove that $\partial^\alpha u(x_0)$ exists for all $x_0 \in \Omega$ and all $\alpha \in \mathbb{Z}^n_+$. For that, let $\varepsilon > 0$ be such that $B_\varepsilon(x_0) \subseteq \Omega$. Let $0 < \varepsilon' < \varepsilon$ and let $x \in B_{\varepsilon'}(x_0)$: The representation formula (11.1) for $\Omega = B_\varepsilon(x_0)$ leads to

$$u(x) = \oint_{+\partial B_\varepsilon(x_0)} \left[\frac{\partial G_n}{\partial \nu_y}(x,y)u(y) - G_n(x,y)\frac{\partial u}{\partial \nu_y}\right] \mathrm{d}S(y)\,.$$

Notice that, for all $x \in B_{\varepsilon'}(x_0)$ and $y \in \partial B_\varepsilon(x_0)$, $G_n(x,y)$ is smooth —*cf.* Theorem 10.3-a For all $\alpha \in \mathbb{Z}^n_+$ we find

$$\left|\partial^\alpha_x \frac{\partial G_n}{\partial \nu_y}(x,y)u(y) - \partial^\alpha_x G_n(x,y)\frac{\partial u}{\partial \nu_y}\right| \leqslant \sup_{\substack{x \in B_{\varepsilon'}(x_0)\\ y \in \partial B_\varepsilon(x_0)}} \left|\partial^\alpha_x \frac{\partial G_n}{\partial \nu_y}(x,y)u(y)\right| + \sup_{\substack{x \in B_{\varepsilon'}(x_0)\\ y \in \partial B_\varepsilon(x_0)}} \left|\partial^\alpha_x G_n(x,y)\frac{\partial u}{\partial \nu_y}\right|\,,$$

where ∂^α_x acts on the x-variable. Thus, Corollary 9.5 applies —*cf.* condition ii— and we find

$$\partial^\alpha u(x) = \oint_{+\partial B_\varepsilon(x_0)} \left[\frac{\partial \partial^\alpha_x G_n}{\partial \nu_y}(x,y)u(y) - \partial^\alpha_x G_n(x,y)\frac{\partial u}{\partial \nu_y}\right] \mathrm{d}S(y)\,,$$

which shows the claimed differentiability for all $x \in B_\varepsilon(x_0)$ and all $x_0 \in \Omega$. □

The following theorem will be the key ingredient for many of the remarkable properties of harmonic functions.

THEOREM 11.4 (Mean value formula): Let $\Omega \subseteq \mathbb{R}^n$ be open and non-empty, let $u \in C^2(\Omega)$ be harmonic in Ω. Then for all $x \in \Omega$ and $R > 0$ such that $\overline{B_R(x)} \subseteq \Omega$ it holds

$$u(x) = \frac{1}{|\partial B_R(x)|} \oint_{+\partial B_R(x)} u(y)\mathrm{d}S(y) \tag{11.2a}$$

$$= \frac{1}{|B_R(x)|} \int_{B_R(x)} u(y)\mathrm{d}^n y\,. \tag{11.2b}$$

Proof.

$\boxed{(11.2a)}$ The representation formula (11.1) for $\Omega = B_R(x)$ leads to

$$u(x) = \oint\limits_{+\partial B_R(x)} \left[\frac{\partial G_n}{\partial \nu_y}(x,y)u(y) - \frac{\partial u}{\partial \nu_y} G_n(x,y)\right] \mathrm{d}S(y)\,.$$

The second integral vanishes on account of Gauss divergence theorem 8.1: Indeed

$$\oint\limits_{+\partial B_R(x)} \frac{\partial u}{\partial \nu_y} G_n(x,y)\mathrm{d}S(y) = \widetilde{G}_n(R) \oint\limits_{+\partial B_R(x)} \frac{\partial u}{\partial \nu_y}\mathrm{d}S(y) = \widetilde{G}_n(R) \int_{B_R(x)} \cancel{\Delta u(y)}\mathrm{d}^n y\,.$$

The remaining contribution is rearranged as

$$\oint\limits_{+\partial B_R(x)} \frac{\partial G_n}{\partial \nu_y}(x,y)u(y)\mathrm{d}S(y) \overset{(10.4)}{=} \frac{1}{|\partial B_R(x)|} \oint\limits_{+\partial B_R(x)} u(y)\mathrm{d}S(y)\,.$$

$\boxed{(11.2b)}$ Integration in polar coordinates —*cf.* Equation (3.12)— leads to

$$\int_{B_R(x)} u(y)\mathrm{d}^n y = \int_0^R \oint\limits_{+\partial B_{R'}(x)} u(y)\mathrm{d}S(y)\mathrm{d}R' = \int_0^R |\partial B_{R'}(x)|u(x)\mathrm{d}R' = |B_R(x)|u(x)\,,$$

where we applied the mean-valued formula (11.2a) as well as identity (3.13).

□

Remark 11.5: Remarkably, Theorem 11.4 holds in the opposite direction: If $u \in C^0(\Omega)$ is such that either Equation (11.2a) or (11.2b) hold true, then $u \in C^2(\Omega)$ and is harmonic on Ω [18, Thm. 3.4]. The remarkable fact is that Equation (11.2) implies that $u \in C^2(\Omega)$. Once this has been proved, showing that $\Delta u = 0$ is relatively simple: Indeed, at any point $x \in \Omega$ we have

$$\begin{aligned} 0 = u(x) - \frac{1}{|\partial B_\varepsilon(x)|} \oint\limits_{+\partial B_\varepsilon(x)} u(y)\mathrm{d}S(y) &= \int_{B_\varepsilon(x)} [G_n(x,y) - \widetilde{G}_n(\varepsilon)]\Delta_y u(y)\mathrm{d}^n y \\ &= \int_0^\varepsilon \oint\limits_{+\partial B_R(x)} [\widetilde{G}_n(R) - \widetilde{G}_n(\varepsilon)]\Delta_y u(y)\mathrm{d}S(y)\mathrm{d}R\,, \end{aligned}$$

where we have chosen $\varepsilon > 0$ so that $\overline{B_\varepsilon(x)} \subset \Omega$ and applied the representation formula (11.1). If $\Delta u(x) \neq 0$, say $\Delta u(x) > 0$, there exists $\delta > 0$ such that $\Delta u(y) > 0$ for all $y \in B_\delta(x)$. Choosing $\varepsilon < \min\{\delta, 1\}$ we find that: (a) $\Delta_y u(y) > 0$ for all $y \in B_\varepsilon(x)$; (b) $\widetilde{G}_n(R) - \widetilde{G}_n(\varepsilon) \leqslant 0$ for $y \in B_\varepsilon(x)$ —*cf.* Equation (10.3). This implies $\int\limits_0^\varepsilon \oint\limits_{+\partial B_R(x)} [\widetilde{G}_n(R) - \widetilde{G}_n(\varepsilon)]\Delta_y u(y)\mathrm{d}S(y)\mathrm{d}R < 0$, a contradiction ☠.

The next Theorem provides a stronger version of the maximum principle proved in Theorem 7.6 —*cf.* [11, Lem. 3.1, Thm. 3.6].

THEOREM 11.6 (Maximum principle — strong form): Let $\Omega \subset \mathbb{R}^n$ be a non-empty, open and connected subset with $\overline{\Omega}$ compact. Let $u \in C^0(\overline{\Omega})$ be such that Equations (11.2) hold for all $x_0 \in \Omega$ and $R > 0$ such that $\overline{B_R(x_0)} \subset \Omega$. Let $x_0 \in \Omega$: If one of the following condition holds true

$$u(x_0) = \max_{\overline{\Omega}} u \tag{11.3a}$$

$$u(x_0) = \min_{\overline{\Omega}} u \tag{11.3b}$$

$$u(x_0) = \max_{\overline{\Omega}} |u| \tag{11.3c}$$

then u is constant in Ω.

Proof. Notice that $\max_{\overline{\Omega}} u$, $\min_{\overline{\Omega}} u$ and $\max_{\overline{\Omega}} |u|$ are well-defined because $u \in C^0(\overline{\Omega})$ and $\overline{\Omega}$ is compact. We shall prove the theorem within the assumption (11.3a). If (11.3b) holds true, then the proof is the same using $-u$. Similarly if (11.3c) then either (11.3a) or (11.3b) hold true on account of the equality $|u(x)| = \max\{u(x), -u(x)\}$.

We consider the set

$$U := \{x \in \Omega \,|\, u(x) = \max_{\overline{\Omega}} u\}\,.$$

Then $x_0 \in U$ so that $U \neq \varnothing$, moreover, $U = \Omega \cap u^{-1}(\{u(x_0)\})$ is Ω-closed because it is the preimage of a closed set with respect to a continuous function.

Finally, U is open (and thus Ω-open since $U \subseteq \Omega$). For that, let $x \in U$: we shall prove that there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq U$. By contradiction let assume that for all $n \in \mathbb{N}$ there exists $x_n \in B_{1/n}(x)$ such that $u(x_n) < u(x_0)$. Since $u \in C^0(\overline{\Omega})$, there exists $\delta > 0$ such that $B_\delta(x_n) \subseteq B_{1/n}(x_0)$ and $u(x) < u(x_0)$ for all $x \in B_\delta(x_n)$. The mean-value formula (11.2b) implies

$$\begin{aligned} u(x_0) = u(x) = \frac{1}{|B_{1/n}(x)|} \int_{B_{1/n}(x)} u(y) \mathrm{d}^n y \\ &= \frac{1}{|B_{1/n}(x)|} \int_{B_{1/n}(x) \setminus B_\delta(x_n)} u(y) \mathrm{d}^n y + \frac{1}{|B_{1/n}(x)|} \int_{B_\delta(x_n)} u(y) \mathrm{d}^n y < u(x_0)\,, \end{aligned}$$

a contradiction ☠ —in the last line we used the fact that $u(y) < u(x_0)$ for all $y \in B_\delta(x_n)$ together with the fact that $|B_\delta(x_n)| > 0$.

Thus, U is non-empty, Ω-open and Ω-closed: Since Ω is connected, it follows that $U = \Omega$. □

Remark 11.7:

(i) Notice that the proof of Theorem 11.6 only requires that u is continuous on $\overline{\Omega}$ together with the mean-value formulas (11.2). In fact, these conditions are equivalent to $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ and $\Delta u = 0$ on Ω [18, Thm. 3.4].

(ii) Theorem 11.6 implies Theorem 7.6. Indeed, since $\max_{\partial\Omega} u \leqslant \max_{\overline{\Omega}} u$, we have two cases. Either $\max_{\partial\Omega} u = \max_{\overline{\Omega}} u$, which is the thesis of Theorem 7.6, or $\max_{\partial\Omega} u < \max_{\overline{\Omega}} u$. However, in the latter case u is constant because of Theorem 11.6 and thus $\max_{\partial\Omega} u = \max_{\overline{\Omega}} u$, a contradiction. Thus, only the first possibility occurs.

The following result can be seen as a maximum principle for $\Omega = \mathbb{R}^n$ —*cf.* [11, Thm. 3.7].

Theorem 11.8 (Liouville): Let $u \in C^2(\mathbb{R}^n)$ be harmonic. If one of the following conditions holds true

$$\sup_{x\in\mathbb{R}^n} u(x) < +\infty \tag{11.4a}$$

$$\inf_{x\in\mathbb{R}^n} u(x) > -\infty\,, \tag{11.4b}$$

then u is constant.

Proof. We shall prove the theorem within the assumption (11.4b) —if (11.4a) holds true we can apply the same arguments to $-u$.

Let $v \in C^2(\mathbb{R}^2)$ be defined by $v(x) := u(x) - \inf_{y\in\mathbb{R}^n} u(y)$: Then v is harmonic, moreover, $v \geqslant 0$. Since Δ commutes with partial derivatives, $\frac{\partial v}{\partial x^k}$ is harmonic for all $k \in \{1, \ldots, n\}$ —notice that $v \in C^\infty(\mathbb{R}^n)$ on account of Theorem 11.3. The mean-formula (11.2b) implies

$$\begin{aligned}
\left|\frac{\partial v}{\partial x^k}\bigg|_x\right| &= \left|\frac{1}{|B_R(x)|}\int_{B_R(x)} \frac{\partial v}{\partial y^k}\mathrm{d}^n y\right| \\
&= \left|\frac{1}{|B_R(x)|}\oint_{+\partial B_R(x)} v(y)\nu^k \mathrm{d}S(y)\right| && \text{Thm. 8.1} \\
&\leqslant \frac{1}{|B_R(x)|}\oint_{+\partial B_R(x)} v(y)\mathrm{d}S(y) && v \geqslant 0\,,\ \|\nu\| = 1 \\
&= \frac{|\partial B_R(x)|}{|B_R(x)|} v(x) && \text{Eq. (11.2a)} \\
&\xrightarrow[R\to+\infty]{} 0 && \text{Eq. (3.13).}
\end{aligned}$$

Thus $\nabla v = 0$ and therefore u is constant because of Lemma 8.9. □

Theorem 11.8 shows in particular that harmonic functions in $L^\infty(\mathbb{R}^n)$ are necessarily constants. We shall prove that this generalizes to $L^p(\mathbb{R}^n)$, $p \geqslant 1$: In particular an harmonic function in $L^p(\mathbb{R}^n)$ has to vanish identically —*cf.* [11, Thm. 3.8] for the case $p = 1, 2$. To this avail we recall Hölder's inequality [16, Thm. 3.5]: If $f \in L^p(X, \mu)$ and $g \in L^{p'}(X, \mu)$ where $1/p + 1/p' = 1$ —p, p' are called **conjugate exponents**— then $fg \in L^1(X, \mu)$ and

$$\|fg\|_{L^1(X,\mu)} \leqslant \|f\|_{L^p(X,\mu)} \|g\|_{L^{p'}(X,\mu)} . \tag{11.5}$$

THEOREM 11.9: Let $p \in [1, +\infty)$, $u \in C^2(\mathbb{R}^n) \cap L^p(\mathbb{R}^n)$ be harmonic. Then $u = 0$.

Proof. By direct inspection we have

$$\begin{aligned}
0 \leqslant |u(x)| &\leqslant \frac{1}{|B_R(x)|} \int_{B_R(x)} |u(y)| \mathrm{d}^n y && \text{Eq. (11.2b)} \\
&\leqslant \frac{1}{|B_R(x)|} \|u\|_{L^p(\mathbb{R}^n)} \|1_{B_R(x)}\|_{L^{p'}(\mathbb{R}^n)} && \text{Hölder ineq. (11.5)} \\
&= \|u\|_{L^p(\mathbb{R}^n)} |B_R(x)|^{1/p'-1} \\
&= \|u\|_{L^p(\mathbb{R}^n)} |B_R(x)|^{-1/p} && \text{def. } p' \\
&\xrightarrow[R \to +\infty]{} 0
\end{aligned}$$

□

12 Elliptic PDEs: Harmonic functions in outer domains

The goal of this section is to prove a uniqueness result for the following boundary-value problem with Neumann boundary condition in outer domains.

THEOREM 12.1: Let $\Omega \subset \mathbb{R}^n$ be open with $\overline{\Omega}$ compact, with orientable C^1 boundary $\partial\Omega$ and such that $\mathbb{R}^n \backslash \Omega$ is connected. If $n \geqslant 3$ then there exists at most one solution $u \in C^2(\mathbb{R}^n \backslash \overline{\Omega}) \cap C^1(\mathbb{R}^n \backslash \Omega)$ to the outer Neumann problem

$$-\Delta u = f\,, \tag{12.1a}$$

$$\frac{\partial u}{\partial \nu} = u_1\,, \tag{12.1b}$$

$$\lim_{x\to\infty} |u(x) - u_\infty| = 0\,, \tag{12.1c}$$

where $f \in C^0(\mathbb{R}^n \backslash \overline{\Omega})$, $u_1 \in C^0(\partial\Omega)$ and $u_\infty \in \mathbb{R}$.

The proof of Theorem 12.1 relies on the decay properties of harmonic functions in unbounded domains. In particular we will see that condition (12.1c) implies stronger decay properties if $n \geqslant 3$, u is harmonic and $u_\infty = 0$. This section profits of [3, 11, 18].

We begin with a corollary of the maximum principle (in strong form) —*cf.* [18, Cor. 3.6].

COROLLARY 12.2: Let $u_0, u_0' \in C^0(\partial\Omega)$ with $u_0 \geqslant u_0'$ and $u_0 > u_0'$ in at least one point of $\partial\Omega$. Let $u, u' \in C^2(\Omega) \cap C^0(\overline{\Omega})$ be solutions to the Dirichlet inner problem (7.3) with $f = 0$ and boundary data u_0 and u_0' respectively. Then $u > u'$.

Proof. Let $v := u - u'$. Then $v \in C^2(\Omega) \cap C^0(\overline{\Omega})$, moreover, v is harmonic in Ω. We also have $v|_{\partial\Omega} \geqslant 0$ and $v|_{\partial\Omega} > 0$ in at least one point of $\partial\Omega$.

We shall now prove that $v(x) > 0$ for all $x \in \Omega$. To this avail, we observe that, for all $x \in \Omega$, either $v(x) > \min_{\overline{\Omega}} v$ or $v(x) = \min_{\overline{\Omega}} v$. In the first case the weak maximum principle implies

$$v(x) > \min_{\overline{\Omega}} v = \min_{\partial\Omega} v \geqslant 0\,.$$

In the second case the strong maximum principle implies that v is a constant. Since by assumption there exists a point in $y \in \partial\Omega$ such that $v(y) > 0$ we obtain that v is a positive constant. Thus in both cases we have $v(x) > 0$. □

We now have an analogous result for the Dirichlet problem in outer domains (7.4) —*cf.* [18, Thm. 6.1].

Proposition 12.3: Let $u \in C^2(\mathbb{R}^n \backslash \overline{\Omega}) \cap C^0(\mathbb{R}^n \backslash \Omega)$ be harmonic in $\mathbb{R}^n \backslash \Omega$ and such that $u|_{\partial\Omega} \geqslant 0$ (*resp.* $u|_{\partial\Omega} \leqslant 0$) and $\lim\limits_{x\to\infty} u(x) = 0$. Then $u \geqslant 0$ (*resp.* $u \leqslant 0$).

Proof. We shall consider the case $u|_{\partial\Omega} \geqslant 0$, the proof of the other case being analogous. By Remark 7.3 we have that $\lim\limits_{R\to+\infty} \sup\limits_{\partial B_R(0)} |u| = 0$. This implies that for all $\varepsilon > 0$ there exists $R_\varepsilon > 0$ such that for all $R > R_\varepsilon$: (a) $\overline{\Omega} \subset B_R(0)$; (b) $|u|_{\partial B_R(0)}| < \varepsilon$.

The latter inequality entails in particular $u|_{\partial B_R(0)} > -\varepsilon$. Notice that the function $u'(x) := -\varepsilon$ solves the inner Dirichlet problem (7.3) on the domain $\Omega_R := B_R(0)\backslash\overline{\Omega}$ with $f = 0$ and with boundary conditions $-\varepsilon$ on $\partial\Omega_R = \partial B_R(0) \cup \partial\Omega$. Similarly, u solves the inner Dirichlet problem on Ω_R with $f = 0$ and boundary conditions $u_0 := u|_{\partial\Omega_R} > -\varepsilon$. By Corollary 12.2 we get $u(x) > -\varepsilon$ for all $x \in \Omega_R$. Since this holds for all $R > R_\varepsilon$, it follows that $u(x) > -\varepsilon$ for all $x \in \mathbb{R}^n\backslash\Omega$. The arbitrariness of $\varepsilon > 0$ entails that $u(x) \geqslant 0$ for all $x \in \mathbb{R}^n\backslash\Omega$. □

The following result shows that, in dimension $n \geqslant 3$, an harmonic function which vanishes at infinity has also additional decay properties —*cf.* [18, Thm. 6.3].

Theorem 12.4: Let $n \geqslant 3$, $u \in C^2(\mathbb{R}^n\backslash\overline{\Omega})$ be harmonic with $\lim\limits_{x\to+\infty} u(x) = 0$. Then there exists $C, R > 0$ such that $\overline{\Omega} \subset B_R(0)$ and

$$|u(x)| \leqslant \frac{C}{\|x\|^{n-2}}\,, \qquad \left|\frac{\partial u}{\partial x^k}\right| \leqslant \frac{C}{\|x\|^{n-1}}\,, \qquad \forall x \in \mathbb{R}^n\backslash B_R(0)\,,\ \forall k \in \{1,\ldots,n\}\,. \tag{12.2}$$

Proof. Once both inequalities in (12.2) are proved for possibly different constants $C', C'' > 0$, the full inequality (12.2) is proved with $C := \max\{C', C''\}$.

$\boxed{u}$ Since $\overline{\Omega}$ is compact and $u(x) \xrightarrow[x\to\infty]{} 0$, there exists $R > 0$ such that: (a) $\overline{\Omega} \subset B_R(0)$; (b) $|u(x)| < \frac{1}{(n-2)\omega_n}$ for all $x \in \partial B_R(0)$.

We then consider $v(x) := u(x) + R^{n-2}G_n(x, 0)$: Then $v \in C^2(\mathbb{R}^n\backslash\overline{B_R(0)}) \cap C^0(\mathbb{R}^n\backslash B_R(0))$ is harmonic in $\mathbb{R}^n\backslash\overline{B_R(0)}$. Moreover, for all $x \in \partial B_R(0)$ we have

$$v(x) = u(x) + R^{n-2}\widetilde{G}_n(R) = u(x) - \frac{1}{(n-2)\omega_n} < 0\,.$$

and $v(x) \xrightarrow[x\to\infty]{} 0$ —here we used the assumption $n \geqslant 3$, *cf.* Equation (10.3). Proposition 12.3 implies that $v(x) \leqslant 0$ for all $x \in \mathbb{R}^n\backslash B_R(0)$, which entails

$$u(x) \leqslant -R^{n-2}\widetilde{G}_n(x) = \frac{R^{n-2}}{(n-2)\omega_n}\frac{1}{\|x\|^{n-2}} =: \frac{C}{\|x\|^{n-2}}\,.$$

An analogous argument considering the function $v'(x) := u(x) - R^{n-2}G_n(x, 0)$ leads to the inequality $u(x) \geqslant -C/\|x\|^{n-2}$. Thus, the first inequality in (12.2) is proved.

$\boxed{\partial u}$ Let $R > 0$ as above and let $x \in \mathbb{R}^n \backslash B_{2R}(0)$. Then there exists $m \in \mathbb{N}$, $m \geqslant 2$, such that $mR \leqslant \|x\| \leqslant (m+1)R$. It follows that $B_{(m-1)R}(x) \subset \mathbb{R}^n \backslash B_R(0)$. Indeed, by contradiction, if $y \in B_{(m-1)R}(x) \cap B_R(0)$ we would have $\|x\| \leqslant \|x - y\| + \|y\| < mR$, a contradiction ☠.

Proceeding as in the proof of Theorem 11.8 we find

$$\begin{aligned}
\left|\frac{\partial u}{\partial x^j}\bigg|_x\right| &= \left|\frac{1}{|B_{(m-1)R}(x)|}\int_{B_{(m-1)R}(x)} \frac{\partial u}{\partial x^j}\mathrm{d}^n x\right| && \text{Eq. (11.2b)} \\
&= \frac{1}{|B_{(m-1)R}(x)|}\left|\oint_{+\partial B_{(m-1)R}(x)} u\nu^j \mathrm{d}S\right| && \text{Thm. 8.1} \\
&\leqslant \frac{n}{(m-1)R}\|u\|_{L^\infty(\partial B_{(m-1)R}(x_0))} \,.
\end{aligned}$$

Since $B_{(m-1)R}(x) \subseteq \mathbb{R}^n \backslash B_R(0)$ for all $y \in \partial B_{(m-1)R}(x)$ we may apply the first inequality in (12.2),

$$|u(y)| \leqslant \frac{C}{\|y\|^{n-2}} \leqslant \frac{C}{[\|x\| - (m-1)R]^{n-2}} \leqslant \frac{C}{\|x\|^{n-2}} \,,$$

where $|\|x\| - \|y\|| \leqslant \|x - y\| = (m-1)R$. This leads to

$$\left|\frac{\partial u}{\partial x^j}\bigg|_x\right| \leqslant \frac{n}{(m-1)R}\|u\|_{L^\infty(\partial B_{(m-1)R}(x))} \leqslant \frac{nC}{(m-1)R}\frac{1}{\|x\|^{n-2}} \leqslant \frac{3nC}{\|x\|^{n-1}} \,,$$

where in the second inequality we used $3(m-1)R \geqslant (m+1)R \geqslant \|x\|$.

□

Remark 12.5: For completeness we stress that, in the hypotheses of Theorem 12.4, there is also an estimate for the second derivative of u, in particular

$$\left|\frac{\partial^2 u}{\partial x^k \partial x^j}\right| \leqslant \frac{C}{\|x\|^n} \,, \qquad \forall x \in \mathbb{R}^n \backslash B_R(0) \,.$$

We refer to [18, Thm. 6.3] for further details.

Proof of Theorem 12.1. Let $v = u - u'$ be the difference of two solutions to problem (12.1) with the same data f, u_1 and u_∞: Then v solves problem (12.1) with vanishing data. Let $N \in \mathbb{N}$ be large enough so that $\overline{\Omega} \subset B_N(0)$: Such N exists because $\overline{\Omega} \subseteq \mathbb{R}^n$ is compact and thus bounded. Let $\Omega_N := B_N(0) \backslash \overline{\Omega}$: Then Ω_N is open, $\overline{\Omega_N}$ is compact with C^1-regular oriented boundary

$\partial\Omega_N = \partial B_N(0) \cup \partial\Omega$ —notice that the orientation of $\partial\Omega$ is reversed when considered as part of the boundary of Ω_N or as the boundary of Ω. An application of Equation (8.4) leads to

$$\begin{aligned} 0 = \int_{\Omega_N} v\Delta v \,\mathrm{d}^n x &= \oint\limits_{+\partial\Omega_N} v\frac{\partial v}{\partial \nu}\,\mathrm{d}S - \int_{\Omega_N} \|\nabla v\|^2 \,\mathrm{d}^n x \\ &= \oint\limits_{+\partial B_N(0)} v\frac{\partial v}{\partial \nu}\,\mathrm{d}S - \oint\limits_{+\partial\Omega} v\frac{\cancel{\partial v}}{\cancel{\partial \nu}}\,\mathrm{d}S - \int_{\Omega_N} \|\nabla v\|^2 \,\mathrm{d}^n x\,. \end{aligned} \tag{12.3}$$

On account of Theorem 12.4 the first integral can be estimated by

$$\left|\oint\limits_{+\partial B_N(0)} v\frac{\partial v}{\partial \nu}\,\mathrm{d}S\right| \leqslant \oint\limits_{+\partial B_N(0)} |v|\,\|\nabla v\|\,\mathrm{d}S \leqslant \frac{C}{(1+N)^{2n-3}} N^{n-1}\omega_n \xrightarrow[N\to+\infty]{} 0\,.$$

We observe that the second integral can be written as

$$\int_{\Omega_N} \|\nabla v\|^2 \,\mathrm{d}^n x = \int_{\mathbb{R}^n\setminus\Omega} 1_{B_N(0)} \|\nabla v\|^2 \,\mathrm{d}^n x \xrightarrow[N\to+\infty]{} \int_{\mathbb{R}^n\setminus\Omega} \|\nabla v\|^2 \,\mathrm{d}^n x\,,$$

where we applied the monotone convergence Theorem, *cf.* [16, §1].

Summing up the limit $N \to \infty$ of (12.3) leads to

$$\int_{\mathbb{R}^n\setminus\Omega} \|\nabla v\| \,\mathrm{d}^n x = 0\,.$$

Since ∇v is continuous and $\mathbb{R}^n\setminus\Omega$ is connected we find that $\nabla v = 0$. Thus, v is a constant, moreover $v = 0$ because $v(x) \xrightarrow[x\to\infty]{} 0$. □

Remark 12.6: With minor adjustment Theorem 12.1 can be generalized to prove uniqueness of the following outer Robin problem:

$$-\Delta u = f\,, \tag{12.4a}$$

$$\left.\left(\frac{\partial u}{\partial \nu} + \alpha u\right)\right|_{\partial\Omega} = u_1\,, \tag{12.4b}$$

$$\lim_{x\to\infty} |u(x) - u_\infty| = 0\,, \tag{12.4c}$$

where $f \in C^0(\mathbb{R}^n\setminus\overline{\Omega})$, $u_1 \in C^0(\partial\Omega)$, $u_\infty \in \mathbb{R}$ and $\alpha > 0$.

13 Elliptic PDEs: Dirichlet problem on $\mathbb{Z}^d$

In this section we will discuss the discrete Dirichlet problem. Roughly speaking, the latter consists of solving the boundary-value problem introduced in Section 7 —*cf.* Definition 7.2— on a discrete lattice, namely $\mathbb{Z}^d$. As we will see, many of the results seen in the previous sections apply in the discrete case. Moreover, we will also provide an existence result for the solution to the Dirichlet problem using probabilistic techniques, see [10, §6] for further details.

In what follows we will consider the discrete lattice $\mathbb{Z}^d$, $d \geqslant 1$. Two vertexes $x, y \in \mathbb{Z}^d$ are called **nearest neighbour** if $\|x-y\|_1 = \sum_{k=1}^d |x_k - y_k| = 1$: In this case we will write $\boxed{x \sim y}$ or $\boxed{xy \in \mathscr{E}}$ where $xy = \{x, y\}$ is called an un-ordered **edge** of $\mathbb{Z}^d$. The **discrete gradient** of $f\colon \mathbb{Z}^d \to \mathbb{R}$ is the function

$$\nabla f\colon \mathbb{Z}^d \times \mathbb{Z}^d \to \mathbb{R}\,, \qquad (\nabla f)_{xy} := \begin{cases} f_y - f_x & x \sim y \\ 0 & \text{otherwise} \end{cases} \tag{13.1}$$

The **discrete Laplacian** of f is the function

$$\Delta f\colon \mathbb{Z}^d \to \mathbb{R}\,, \qquad (\Delta f)_x := \sum_{xy \in \mathscr{E}} (\nabla f)_{xy}\,. \tag{13.2}$$

A function f is called **(discrete) harmonic** if $\Delta f = 0$.

REMARK 13.1: By direct inspection we have

$$(\Delta f)_x = \sum_{xy \in \mathscr{E}} (f_y - f_x) = \sum_{xy \in \mathscr{E}} f_y - 2d f_x\,,$$

where we observed that there are $2d$ vertexes $y \in \mathbb{Z}^d$ such that $xy \in \mathscr{E}$. The latter equality has an important consequences: If f is harmonic then

$$f_x = \frac{1}{2d} \sum_{xy \in \mathscr{E}} f_y\,. \tag{13.3}$$

Equation (13.3) is called **discrete mean-value formula**.

REMARK 13.2: Equation (13.3) implies that, for $d = 2$, a discrete harmonic function is completely characterized by its value on the strips $\{z\} \times \mathbb{Z}$, where $z \in \{x, x+1\}$ for an arbitrary but fixed $x \in \mathbb{Z}$.

Indeed, for all $k \in \mathbb{Z}$, Equation (13.3) implies that

$$u_{(x+1,k)} = \frac{1}{4}\big[u_{(x+2,k)} + u_{(x,k)} + u_{(x+1,k+1)} + u_{(x+1,k-1)}\big]\,.$$

In the previous equation all values of u are known but $u_{x+2,k}$, which is therefore fixed by the discrete mean-value formula. This shows that the knowledge of $u_{z,k}$ for all $k \in \mathbb{Z}$ and $z \in \{x, x+1\}$ suffices to determine $u_{x+2,k}$ for all $k \in \mathbb{Z}$: By iterating this process u is known for all $u_{z,k}$, $k \in \mathbb{Z}$ and $z \geqslant x$. A similar argument goes for $z \leqslant x$.

The following definition is the discrete analogous of Definition 7.2. In what follows we will write $\boxed{\Lambda \Subset \mathbb{Z}^d}$ if Λ is a finite subset of $\mathbb{Z}^d$.

Definition 13.3: Let $\Lambda \Subset \mathbb{Z}^d$, $f\colon \Lambda \to \mathbb{R}$ and $v\colon \mathbb{Z}^d \to \mathbb{R}$. A function $u\colon \mathbb{Z}^d \to \mathbb{R}$ is a solution to the **discrete Dirichlet problem** if

$$\Big(-\frac{1}{2d}\Delta u\Big)_x = f_x \qquad \forall x \in \Lambda \tag{13.4a}$$

$$u_x = v_x \qquad \forall x \notin \Lambda\,. \tag{13.4b}$$

The following uniqueness result is similar to Theorem 7.8 and its proof is a discrete version of Theorem 11.6.

Proposition 13.4: There exists at most one solution to the discrete Dirichlet problem (13.4).

Proof. Let u', u'' be two solutions to (13.4). Then $u := u' - u''$ is a solution to (13.4) with $f = 0$ and $v = 0$: We will now prove that $u = 0$.

To this avail, let $x^* \in \Lambda$ be such that

$$|u_{x^*}| = \max_{x\in\Lambda} |u_x|\,;$$

Notice that such vertex exists because $\Lambda \Subset \mathbb{Z}^d$. Without loss of generality we may assume $u_{x^*} \geqslant 0$.

We now distinguish two cases: Either all nearest neighbours vertexes y of x^* are in Λ or there exists at least one $y^* \notin \Lambda$ such that $y^* \sim x^*$. In the latter case we have $u_{y^*} = 0$ because of the boundary condition (13.4b), moreover, Equation (13.3) implies

$$u_{x^*} = \frac{1}{2d}\sum_{y\sim x^*} u_y = \frac{1}{2d}\sum_{\substack{y\sim x^*\\ y\in\Lambda}} u_y \leqslant \Big(1 - \frac{1}{2d}\Big) u_{x^*}\,,$$

which implies $u_{x^*} = 0$ and thus $u = 0$. In the former case we claim that $u_y = u_{x^*}$ for all nearest neighbour vertex $y \sim x^*$. Indeed, since $y \in \Lambda$, $u_y \leqslant u_{x^*}$: Per contradiction, let $y^* \in \Lambda$ be such that $y^* \sim x^*$ and $u_{y^*} < u_{x^*}$. Since $(\Delta u)_{x^*} = 0$, Equation (13.3) entails

$$u_{x^*} = \frac{1}{2d}\sum_{y\sim x^*} u_y < u_{x^*} \qquad .$$

Thus, $u_y = u_{x^*}$ for all $y \sim x^*$. By iterating this argument we conclude that $u_x = u_{x^*}$ for all $x \in \Lambda$: This implies in particular that $u_x = u_{x^*}$ for those vertexes $x \in \Lambda$ for which there exists a nearest neighbour $y \notin \Lambda$ —these vertexes exist because $\Lambda \Subset \mathbb{Z}^d$. With the same argument as above this leads to $u_{x^*} = 0$, therefore, $u = 0$. □

REMARK 13.5: On account of Proposition 13.4 we may look for a solution u to the discrete Dirichlet problem (13.4) in the form

$$u = u^v + u^f ,$$

where u^v and u^f solve the discrete Dirichlet problems

$$\Big(-\frac{1}{2d}\Delta u^v\Big)_x = 0 \quad \forall x \in \Lambda \qquad u^v_x = v_x \quad \forall x \notin \Lambda\,, \tag{13.5}$$

$$\Big(-\frac{1}{2d}\Delta u^f\Big)_x = f_x \quad \forall x \in \Lambda \qquad u^v_x = 0 \quad \forall x \notin \Lambda\,. \tag{13.6}$$

REMARK 13.6: In the particular case of $d = 1$ we may provide an explicit solution to problem (13.5) for the choice $\Lambda = \{a, \ldots, b\}$. Indeed, it suffices to consider a solution of the form $u_x := \alpha + \beta x$, $\alpha, \beta \in \mathbb{C}$, $x \in \Lambda \cup \{a-1, b+1\}$, which is harmonic in Λ by direct inspection —we set $u_x := v_x$ for $x \notin \Lambda$. The exact values of $\alpha, \beta \in \mathbb{C}$ are determined by solving the linear system obtained by imposing the boundary conditions

$$\alpha + \beta(a-1) = v_{a-1}\,, \qquad \alpha + \beta(b+1) = v_{b-1}\,.$$

We will now provide an existence result for the discrete Dirichlet problem (13.4). The latter is based on the notion of (symmetric) random walk.

DEFINITION 13.7: Let $(\xi_n)_{n\in\mathbb{N}}$ be independent identically distributed, I.I.D. for short, random variables such that ξ_n is uniformly distributed on $\{x \in \mathbb{Z}^d \,|\, x \sim 0\}$ for all $n \in \mathbb{N}$. To wit, for all $n \in \mathbb{N}$, ξ_n is a $\mathbb{Z}^d$-valued random variable with

$$\mathbb{P}(\xi_n = x) = \begin{cases} 1/2d & x \sim 0 \\ 0 & \text{otherwise} \end{cases}$$

The **(symmetric) random walk** on $\mathbb{Z}^d$ starting at $x \in \mathbb{Z}^d$ is the sequence of $\mathbb{Z}^d$-valued random variables $(X_n)_{n\in\mathbb{Z}_+}$ defined by

$$X_0 := x\,, \qquad X_{n+1} := X_n + \xi_{n+1} = x + \sum_{k=1}^{n+1} \xi_k\,. \tag{13.7}$$

We will denote by $\mathbb{P}_x$ the law of $(X_n)_{n\in\mathbb{Z}_+}$.

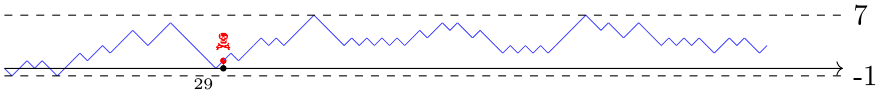


Figure 4: 100 steps of a 1D random walk. The red dot represents the end of the (killed) random walk with killing probability 2%, *cf.* Definition 14.4.

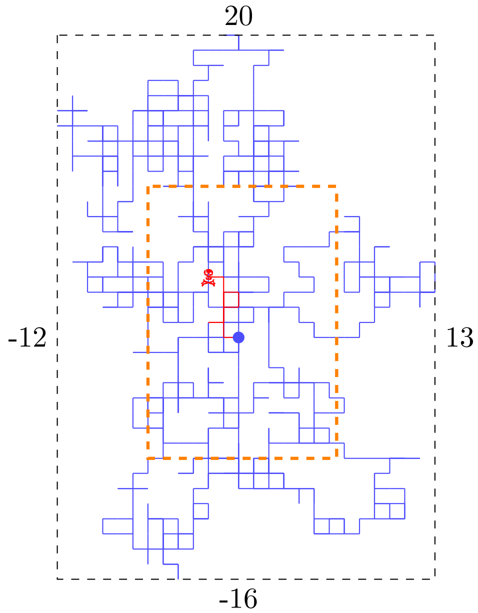


Figure 5: 1000 steps of a 2D random walk: the bullet vertex is the starting point $x = 0 \in \mathbb{Z}^2$. The green box is randomly drawn and fulfils the thesis of Lemma 13.9 The red path denotes the killed random walk with killing probability 2%, *cf.* Definition 14.4.

Informally, a random walk is a path made by nearest neighbour vertexes where at each step the new vertex is chosen uniformly at random.

Remark 13.8:

(i) Notice that, although $\xi_n \perp \xi_k$ for all $n \neq k$, the latter is not true for the sequence $(X_n)_{n\in\mathbb{Z}_+}$. In particular this latter sequence fulfils the following property:

$$\mathbb{P}_x(X_{n+1} = y \,|\, X_n = z) = \begin{cases} 1/2d & y \sim z \\ 0 & \text{otherwise} \end{cases} \qquad \forall n \in \mathbb{N}\,.$$

In particular $\mathbb{P}_x(X_{n+1} = y \,|\, X_n = z)$ does not depend neither on $n \in \mathbb{N}$ nor on $x \in \mathbb{Z}^d$. In what follows we will set $P(y, z) := \mathbb{P}_x(X_{n+1} = y \,|\, X_n = z)$: These are usually called **transition probabilities** of the random walk.

(ii) For later convenience it is worth observing that $\mathbb{P}_x(\ |X_1 = y) = \mathbb{P}_y$ for all $x, y \in \mathbb{Z}^d$ with $x \sim y$. Informally, the law of a symmetric random walk starting at $x \in \mathbb{Z}^d$ reduces to the law of a symmetric random walk starting at y once we condition on $X_1 = y$ (provided $x \sim y$) *i.e.* once we condition on the first step of the random walk.

Theorem 13.10 will provide a solution to problem (13.5) in terms of the symmetric random walk introduced in Definition 13.7. To this avail we define, for $\Lambda \Subset \mathbb{Z}^d$, the **exit time from** Λ of the random walk $(X_n)_{n\in\mathbb{Z}_+}$:

$$\tau_\Lambda := \inf\{n \in \mathbb{Z}_+ \,|\, X_n \notin \Lambda\}\,. \tag{13.8}$$

The following Lemma ensures that τ_Λ is almost surely finite, thus, a symmetric random walk leaves any finite region Λ almost surely.

LEMMA 13.9: Let $\Lambda \Subset \mathbb{Z}^d$ and let τ_Λ be the exit time of the symmetric random walk defined by Equation (13.8). Then, for all $x \in \mathbb{Z}^d$, $\tau_\Lambda < +\infty$ $\mathbb{P}_x$-almost surely, that is $\mathbb{P}_x(\tau_\Lambda < +\infty) = 1$. In particular, there exists $c_\Lambda > 0$ such that

$$\mathbb{P}_x(\tau_\Lambda > n) \leqslant e^{-c_\Lambda n}\,. \tag{13.9}$$

Proof. It is enough to prove Equation (13.9) because

$$\mathbb{P}_x(\tau_\Lambda = +\infty) = \mathbb{P}_x\Big(\bigcap_{n\in\mathbb{N}} \tau_\Lambda > n\Big) = \lim_{n\to\infty} \mathbb{P}_x(\tau_\Lambda > n) \overset{(13.9)}{=} 0\,,$$

proving that $\mathbb{P}_x(\tau_\Lambda < \infty) = 1$. Let $R > 0$ be defined by

$$R := \sup_{x\in\Lambda} \inf_{y\notin\Lambda} \|x - y\|_1\,;$$

Notice that $R < \infty$ because Λ is finite. Moreover, by definition, for all $x \in \Lambda$ there exists a nearest neighbour path γ_x starting at x which leaves Λ in at most R steps. It follows that

$$\mathbb{P}_x(\tau_\Lambda \leqslant R) \geqslant \mathbb{P}_x(\gamma_x) \geqslant (1/2d)^R\,.$$

Notice that this result does not depend on the chosen point $x \in \Lambda$. This inequality can be used to estimate $\mathbb{P}_x(\tau_\Lambda > n)$ as follows. If $\tau_\Lambda > n$ then $X_1, \ldots, X_n \in \Lambda$ and we may recollect these random variables in $\lfloor n/R \rfloor$ many R-uples

$$X_1, \ldots, X_R\,, \quad X_{R+1}, \ldots, X_{2R}\,, \quad \ldots \quad X_{(\lfloor n/R\rfloor - 1)R}, \ldots, X_{\lfloor n/R\rfloor R}\,.$$

Each R-uple is a symmetric random walk of length R which does not exist Λ, *i.e.* each R-uple fulfils $\tau_\Lambda > R$ whose probability is bounded by $1 - (1/2d)^R$. Overall we have

$$\mathbb{P}_x(\tau_\Lambda > n) \leqslant (1 - (1/2d)^R)^{\lfloor n/R\rfloor} \leqslant \exp[-c_\Lambda n]\,.$$

□

Theorem 13.10: The solution to the discrete Dirichlet problem (13.5) is

$$u\colon \mathbb{Z}^d \to \mathbb{R} \qquad u_x := \mathbb{E}_x[v_{X_{\tau_\Lambda}}]\,. \tag{13.10}$$

Proof. Equation (13.10) is well-posed because $\tau_\Lambda < \infty$ almost surely by Lemma 13.9. Moreover, if $x \notin \Lambda$ we have $\tau_\Lambda = 0$ because $X_0 = x \notin \Lambda$, therefore, $\mathbb{E}_x[v_{X\tau_\Lambda}] = v_x$. Thus, (13.10) fulfils the boundary condition $u_x = v_x$ for $x \notin \Lambda$.

Let now $x \in \Lambda$: We will prove that $(\Delta u)_x = 0$. To this avail we observe that

$$\begin{aligned} u_x = \mathbb{E}_x[v_{X_{\tau_\Lambda}}] = \sum_{xy\in\mathscr{E}} \mathbb{E}_x[v_{X_{\tau_\Lambda}} \cap X_1 = y] = \sum_{xy\in\mathscr{E}} \mathbb{E}_x[v_{X_{\tau_\Lambda}} | X_1 = y]\mathbb{P}_x(X_1 = y) \\ = \frac{1}{2d}\sum_{xy\in\mathscr{E}} \mathbb{E}_y[v_{X_{\tau_\Lambda}}] = \frac{1}{2d}\sum_{xy\in\mathscr{E}} u_y\,, \end{aligned}$$

where we used that $\mathbb{P}_x(\ |X_1 = y) = \mathbb{P}_y$ and that $X_1 = x + \xi_1$. □

Remark 13.11: An interesting application of Equation (13.10) is obtained as follows. Let consider $d = 1$, $\Lambda = \{0, \ldots, N\}$, $N \in \mathbb{N}$, and a boundary datum v such that $v_0 = 0$ and $v_N = 1$. In this case Equation (13.10) reduces to

$$u_x = \mathbb{P}_x(X_{\tau_\Lambda} = N)\,.$$

Thus, the solution u at the vertex x coincides with the probability that the random walk exits Λ from the right, *i.e.* at N. We may provide an interpretation of this results as follows: Let consider a gambling game with two players. At each turn, each player pays one fish and has a $1/2$ probability of winning one. The process describing the number of fishes of one player as the game proceeds is exactly given by the 1D random walk $(X_n)_{n\in\mathbb{Z}_+}$: The starting point X_0 identifies the initial number of fishes. Within this context we may interpret $X_n = 0$ as the bankrupt of the player, whereas $X_n = N$ coincides with the bankrupt of their opponent. Thus, within this situation, the solution u described above provides the probability that the player wins over their opponent given the initial number of fishes.

We now move to the discussion of the discrete Dirichlet problem (13.6). To this avail the following remark will be extremely relevant.

Remark 13.12: From Equation (13.2) we find

$$(\Delta u)_x = \sum_{y\in\mathbb{Z}^d} \Delta_{xy} u_y\,, \qquad \Delta_{xy} = \begin{cases} -2d & x = y \\ 1 & xy \in \mathscr{E} \\ 0 & \text{otherwise} \end{cases}\,.$$

In what follows we will set $\Delta_\Lambda = (\Delta_{xy})_{x,y\in\Lambda}$. Let now u be the solution to problem (13.6). Since $u_x = 0$ for all $x \notin \Lambda$ we find that $(\Delta u)_x = (\Delta_\Lambda u)_x$ for all $x \in \Lambda$. Thus, solving (13.6) is equivalent to prove that the matrix $-\frac{1}{2d}\Delta_\Lambda$ is invertible. By direct inspection we have

$$-\frac{1}{2d}\Delta_\Lambda = I_\Lambda - P_\Lambda\,, \qquad I_\Lambda = (\delta_{xy})_{x,y\in\Lambda}\,, \qquad P_\Lambda = (P(x,y))_{x,y\in\Lambda}\,, \tag{13.11}$$

where $P(x,y)$ denote the transition probabilities introduced in Remark 13.8.

The following Lemma will ensure the invertibility of $-\frac{1}{2d}\Delta_\Lambda$.

THEOREM 13.13: The matrix $-\frac{1}{2d}\Delta_\Lambda$ is invertible with inverse

$$G_\Lambda(x,y) = \mathbb{E}_x\Big[\sum_{n=0}^{\tau_\Lambda-1} 1_{\{X_n=y\}}\Big]\,, \tag{13.12}$$

where the empty sum is conventionally set to zero. In particular, the solution to the discrete Dirichlet problem (13.6) is given by

$$u_x = \sum_{y\in\Lambda} G_\Lambda(x,y) f_y = \mathbb{E}_x\Big[\sum_{n=0}^{\tau_\Lambda-1} f_{X_n}\Big]\,. \tag{13.13}$$

Proof. It suffices to prove that Equation (13.12) defines the inverse of $-\frac{1}{2d}\Delta_\Lambda$. If this is the case, then Equation (13.13) defines a solution u to the equation $(-\Delta u)_x = f_x$, $x \in \Lambda$ —the second equality in (13.6) is a consequence of the identity $f_{X_n} = \sum_{n=0}^{\tau_\Lambda-1} 1_{\{X_n=y\}} f_y$ for $n < \tau_\Lambda$. Moreover, Equation (13.12) implies that $G_\Lambda(x,y) = 0$ if either $x \notin \Lambda$ or $y \notin \Lambda$, therefore, $u_x = 0$ if $x \notin \Lambda$.

We observe that, for all $n \in \mathbb{N}$,

$$-\frac{1}{2d}\Delta_\Lambda(I_\Lambda + P_\Lambda + P_\Lambda^2 + \ldots + P_\Lambda^n) = (I_\Lambda - P_\Lambda)(I_\Lambda + P_\Lambda + P_\Lambda^2 + \ldots + P_\Lambda^n) = I_\Lambda - P_\Lambda^{n+1}\,.$$

We will now prove that $P_\Lambda^{n+1}(x,y) \xrightarrow[n\to\infty]{} 0$: To this avail we compute

$$\begin{aligned} P_\Lambda^{n+1}(x,y) = \sum_{x_1,\ldots,x_n\in\Lambda} P_\Lambda(x,x_1)\cdots P_\Lambda(x_n,y) &= \mathbb{P}_x(X_{n+1} = y \cap \tau_\Lambda > n+1) \\ &\leqslant \mathbb{P}_x(\tau_\Lambda > n+1) \leqslant e^{-c_\Lambda(n+1)} \xrightarrow[n\to\infty]{} 0 \quad \text{Lem. 13.9} \end{aligned}$$

Thus, $-\frac{1}{2d}\Delta_\Lambda$ is invertible with inverse

$$\begin{aligned} G_\Lambda(x,y) = \sum_{n\geqslant 0} P_\Lambda^n(x,y) &= \sum_{n\geqslant 0} \mathbb{P}_x(X_n = y \cap \tau_\Lambda > n) \\ &= \sum_{n\geqslant 0} \mathbb{E}_x(1_{\{X_n=y\}} 1_{\{\tau_\Lambda > n\}}) \qquad \mathbb{P}(E) = \mathbb{E}(1_E) \\ &= \mathbb{E}_x\Big[\sum_{n\geqslant 0} 1_{\{X_n=y\}} 1_{\{\tau_\Lambda > n+1\}}\Big] = \mathbb{E}_x\Big[\sum_{n=0}^{\tau_\Lambda-1} 1_{\{X_n=y\}}\Big]\,, \end{aligned}$$

where we exchanged the series and the expectation value on account of the monotone convergence Theorem [16, §1]. □

Remark 13.14:

(i) The matrix G_Λ is called **(discrete) fundamental solution** to $-\frac{1}{2d}\Delta_\Lambda$ —*cf.* Section 10 and Definition 10.1. For all $x, y \in \Lambda$, $G_\Lambda(x, y)$ is the expected number of visits of the vertex $y \in \Lambda$ made by a random walk starting at $x \in \Lambda$ —notice that $G_\Lambda(x, y) = G_\Lambda(y, x)$. Notice that, applying Theorem 13.13 with $f_x = \delta_{xy}$, where $y \in \Lambda$ is fixed, the resulting solution u as per Equation (13.13) is exactly given by $u_x = G_\Lambda(x, y)$ for all $x \in \Lambda$.

(ii) An interesting application of Equation (13.13) is obtained by considering $f \equiv 1$. In the latter case we have

$$u_x = \sum_{y\in\Lambda} \mathbb{E}_x\Big[\sum_{n=0}^{\tau_\Lambda-1} 1_{X_n=y}\Big] = \mathbb{E}_x\Big[\sum_{n=0}^{\tau_\Lambda-1}\sum_{y\in\Lambda} 1_{X_n=y}\Big] = \mathbb{E}_x[\tau_\Lambda]\,,$$

so that u_x coincides with the averaged exit time from Λ of the random walk starting at $x \in \Lambda$.

(iii) Similarly to Remark 7.9-ii, the Dirichlet problem (13.4) is well-posed when the data are chosen in appropriate Banach spaces. In particular, let $\boxed{\ell^\infty(\mathbb{Z}^d)}$ be the Banach space of functions $u\colon \mathbb{Z}^d \to \mathbb{R}$ such that $\|u\|_\infty := \sup_{x\in\mathbb{Z}^d}|u_x| < \infty$. Then the unique solution u to (13.4) with data $f, v \in \ell^\infty(\mathbb{Z}^d)$ (we are implicitly setting $f_x = 0$ for all $x \in \mathbb{Z}^d\backslash\Lambda$) is also in $\ell^\infty(\mathbb{Z}^d)$, moreover, we may estimate $\|u\|_\infty$ in terms of $\|f\|_\infty$ and $\|v\|_\infty$.

Indeed, Equations (13.10) and (13.13) implies that

$$|u_x| \leqslant |\mathbb{E}_x[v_{X_{\tau_\Lambda}}]| + \Big|\sum_{y\in\Lambda} G_\Lambda(x, y) f_y\Big| \leqslant \|v\|_\infty + C_\Lambda\|f\|_\infty\,,$$

where we set $C_\Lambda := \sup_{x\in\Lambda}\mathbb{E}_x[\tau_\Lambda]$ —notice that $C_\Lambda \in (0, \infty)$ because $\mathbb{E}_x[\tau_\Lambda] = 0$ for $x \notin \Lambda$. It follows that $\|u\|_\infty \leqslant \|v\|_\infty + C_\Lambda\|f\|_\infty$ which proves that $u \in \ell^\infty(\mathbb{Z}^d)$ together with the claimed estimate for $\|u\|_\infty$.

14 Elliptic PDEs: Massive outer Dirichlet problem on $\mathbb{Z}^d$

In this section we will investigate the discrete Dirichlet problem for the massive Laplacian. The results of previous section extend almost straightforwardly the only difference being a slightly different random walk representation. For this reason, rather than considering the Dirichlet problem (13.4) on finite regions $\Lambda \Subset \mathbb{Z}^d$ we will study the Dirichlet problem directly on the whole $\mathbb{Z}^d$. As we will see, the presence of a positive mass contribution $\mu^2 > 0$ provides nicer decay properties which allow to treat the case of $\mathbb{Z}^d$.

For later convenience we will denote by $\boxed{c_0(\mathbb{Z}^d)} \subset \ell^\infty(\mathbb{Z}^d)$ the space of functions $f\colon \mathbb{Z}^d \to \mathbb{R}$ which vanishes at infinity:

$$\lim_{x\to\infty} f_x = 0\,. \tag{14.1}$$

The space $c_0(\mathbb{Z}^d)$ is a discrete analogous of the space $C^0_0(\mathbb{R}^d)$ of continuous functions vanishing at infinity. In particular, $c_0(\mathbb{Z}^d)$ is a Banach space with respect to the norm $\|f\|_\infty := \sup_{x\in\mathbb{Z}^d}|f_x|$ — notice in particular that f attains its maximum. It is worth to mention that the space $\boxed{c_c(\mathbb{Z}^d)}$ of compactly supported functions is dense in $c_0(\mathbb{Z}^d)$. By definition $f \in c_c(\mathbb{Z}^d)$ if and only if $\boxed{\Lambda_f} := \{x \in \mathbb{Z}^d \,|\, f_x \neq 0\} \Subset \mathbb{Z}^d$.

Definition 14.1: Let $\mu > 0$ and let $f \in c_0(\mathbb{Z}^d)$. A function $u\colon \mathbb{Z}^d \to \mathbb{R}$ is a solution to the **discrete massive outer Dirichlet problem** if

$$\Big(\mu^2 u - \frac{1}{2d}\Delta u\Big)_x = f_x \qquad \forall x \in \mathbb{Z}^d \tag{14.2a}$$

$$\lim_{x\to\infty} u_x = 0\,. \tag{14.2b}$$

Remark 14.2:

(i) The assumption on f ensures the existence of the solution to (14.2), *cf.* Theorem 14.7. However, Proposition (14.3) does not need such assumption.

(ii) Proceeding as in Remark 13.1 we observe that any **discrete massive harmonic function** u —that is, a solution to the equation $(\mu^2 - \frac{1}{2d}\Delta)u = 0$— fulfils the following discrete massive mean-value formula:

$$u_x = \frac{1}{1+\mu^2}\frac{1}{2d}\sum_{xy\in\mathscr{E}} u_y\,. \tag{14.3}$$

The latter equation implies that, if $u \in \ell^\infty(\mathbb{Z}^d)$ is a bounded discrete massive harmonic function, then $u = 0$. Indeed, for all $x \in \mathbb{Z}^d$ we have

$$|u_x| = \frac{1}{1+\mu^2}\frac{1}{2d}\Big|\sum_{xy\in\mathscr{E}} u_y\Big| \leqslant \frac{1}{1+\mu^2}\|u\|_\infty\,,$$

from which it follows $\|u\|_\infty \leqslant (1+\mu^2)^{-1}\|u\|_\infty$, thus, $u=0$. The latter result can be regarded as a discrete version of Liouville Theorem 11.8.

(iii) If $d=1$, all massive harmonic functions on $\mathbb{Z}$ are given by $u_x = Ae^{\alpha x} + Be^{\alpha x}$ for arbitrary $A, B \in \mathbb{R}$ and $\alpha := \log(1+\mu^2+\sqrt{2\mu^2+\mu^4})$. Indeed, $u\colon \mathbb{Z} \to \mathbb{R}$ is massive harmonic if

$$0 = (1+\mu^2)u_x - \frac{1}{2}u_{x+1} - \frac{1}{2}u_{x-1} \qquad \forall x \in \mathbb{Z}\,.$$

By considering the ansaltz $u_x = Az^x$ we find

$$0 = \Big[(1+\mu^2)z - \frac{1}{2}z^2 - \frac{1}{2}\Big]z^{x-1}\,;$$

The solutions to the second order equation leads to $z_\pm = (1+\mu^2) \pm \mu\sqrt{2+\mu^2}$.

Proposition 14.3: There is a unique solution to the problem (14.2).

Proof. Let u, u' be two solutions to (14.2) and let $v := u - u'$. Then v is a solution to the discrete massive outer Dirichlet problem (14.2) with $f = 0$, that is, it a discrete massive harmonic function in $c_0(\mathbb{Z}^d) \subset \ell^\infty(\mathbb{Z}^d)$: Remark 14.2-ii ensures that $v = 0$. □

The probabilistic treatise of the massive Dirichlet problem needs the following notion of “killed” symmetric random walk.

Definition 14.4: Let $(X_n)_{n\in\mathbb{Z}_+}$ be a random walk as by Definition 13.7. Let $(Y_n)_{n\in\mathbb{N}}$ be a sequence of I.I.D. Bernoulli random variables with parameter $p_\mu := \mu^2/(1+\mu^2)$: To wit, for all $n \in \mathbb{N}$, Y_n is a $\{0,1\}$-random variable with $\mathbb{P}(Y_n = 1) = p_\mu$. The process $(Y_n)_{n\in\mathbb{N}}$ is assumed to be independent from $(X_n)_{n\in\mathbb{Z}_+}$.

Let $\mathbb{Z}^d_*$ be the lattice defined by $\mathbb{Z}^d_* := \mathbb{Z}^d \cup \{*\}$ where $*$ is called **graveyard vertex**. The **killed (symmetric) random walk** $(Z_n)_{n\in\mathbb{Z}_+}$ starting at $x \in \mathbb{Z}^d$ is the random walk on $\mathbb{Z}^d_*$ defined by:

$$Z_0 := x\,, \qquad Z_{n+1} := \begin{cases} * & \text{if } Z_n = * \\ * & \text{if } Z_n \in \mathbb{Z}^d \text{ and } Y_n = 1 \\ X_{n+1} & \text{if } Z_n \in \mathbb{Z}^d \text{ and } Y_n = 0 \end{cases}\,. \tag{14.4}$$

We will denote by $\mathbb{P}^\mu_x$ the law of $(Z_n)_{n\in\mathbb{Z}_+}$.

Remark 14.5:

(i) Let $\tau_* := \inf\{n \in \mathbb{N} \mid Z_n = *\}$ be the **killing time**. Then for all $x, y \in \mathbb{Z}^d$

$$\mathbb{P}^\mu_x(Z_n = y) = \mathbb{P}^\mu_x(\tau_* > n)\mathbb{P}_x(X_n = y) = (1+\mu^2)^{-n}\mathbb{P}_x(X_n = y)\,,$$

which shows the relation between a symmetric random walk and its killed version. Notice that

$$\mathbb{P}^\mu_x(\tau_* = \infty) = \mathbb{P}^\mu_x\Big(\bigcap_{n\in\mathbb{N}} \tau_* > n\Big) = \lim_{n\to\infty} \mathbb{P}^\mu_x(\tau_* > n) = 0\,,$$

which proves that a killed random walk ends almost surely. Notice that this is not entirely surprising because τ_* only depends on the process $(Y_n)_n$: In fact, it is distributed according to a geometric random variable with parameter p_μ no matter the choice of $x \in \mathbb{Z}^d$.

This is the core property which allows to consider $\mathbb{Z}^d$ in (14.2), instead of resorting to a finite region $\Lambda \Subset \mathbb{Z}^d$ as in (13.4).

(ii) Proceeding as in Remark 13.12 we may observe that

$$\mu^2 - \frac{1}{2d}\Delta = (1+\mu^2)I - P = (1+\mu^2)[I - P^\mu]\,, \tag{14.5}$$

where $I = (\delta_{xy})_{x,y\in\mathbb{Z}^d}$, $P^\mu = (P^\mu(x,y))_{x,y\in\mathbb{Z}^d}$, $P^\mu(x,y)$ being the transition probability associated with the killed random walk:

$$P^\mu(x,y) = \mathbb{P}^\mu_z(Z_{n+1} = y \mid Z_n = x) = \begin{cases} \frac{1}{1+\mu^2}\frac{1}{2d} & x,y \in \mathbb{Z}^d\,,\ x \sim y \\ \frac{\mu^2}{1+\mu^2} & x \in \mathbb{Z}^d\,,\ y = * \\ 1 & x = y = * \\ 0 & \text{otherwise} \end{cases}\,,$$

where $z \in \mathbb{Z}^d$ is arbitrary.

We will prove an existence result for the Dirichlet problem (14.2): The proof is similar to the one of Theorem 13.13. To begin with we start by proving that $\mu^2 - \frac{1}{2d}\Delta$ is invertible.

Proposition 14.6: The operator $\mu^2 - \frac{1}{2d}\Delta$ is invertible, with inverse

$$G^\mu(x,y) = \frac{1}{1+\mu^2}\mathbb{E}^\mu_x\Big[\sum_{n=0}^{\tau_*-1} 1_{\{Z_n = y\}}\Big]\,. \tag{14.6}$$

Moreover we have

$$|G^\mu(x,y)| \leqslant p_\mu^{\|x-y\|_1} = e^{-c_\mu\|x-y\|_1}\,, \tag{14.7}$$

where we set $c_\mu := \log p_\mu$.

Proof. Notice that, since $\tau_* < \infty$ $\mathbb{P}^\mu$-almost surely, the series in Equation (14.6) is $\mathbb{P}^\mu$-almost surely finite. We also observe that

$$\Big(\mu^2 - \frac{1}{2d}\Delta\Big)^{-1} = \frac{1}{1+\mu^2}\sum_{n\geqslant 0}(P^\mu)^n\,,$$

provided the series on the right-hand side converges. For that we observe that, if $x, y \in \mathbb{Z}^d$,

$$\begin{aligned}(P^\mu)^{n+1}(x,y) &= \sum_{x_1,\ldots,x_n\in\mathbb{Z}^d} P^\mu(x,x_1)\cdots P^\mu(x_n,y) = \mathbb{P}^\mu_x(Z_{n+1}=y)\\ &\leqslant \mathbb{P}^\mu_x(\tau_* > n+1) = (1+\mu^2)^{-(n+1)} \xrightarrow[n\to\infty]{} 0\,.\end{aligned}$$

Thus $\mu^2 - \frac{1}{2d}\Delta$ is invertible with inverse given by

$$G(x,y) = \frac{1}{1+\mu^2}\sum_{n\geqslant 0}(P^\mu)^n(x,y) = \frac{1}{1+\mu^2}\sum_{n\geqslant 0}\mathbb{P}^\mu_x(Z_n=y) = \frac{1}{1+\mu^2}\mathbb{E}_x\Big[\sum_{n=0}^{\tau_*-1} 1_{Z_n=y}\Big]\,.$$

It remains to prove the decay property (14.7): To this avail we observe that

$$\mathbb{P}^\mu_x(Z_n=y) = 0 \qquad \forall x,y\in\mathbb{Z}^d\colon \|x-y\|_1 > n\,.$$

It follows that

$$\begin{aligned}G^\mu(x,y) &= \frac{1}{1+\mu^2}\sum_{n\geqslant\|x-y\|_1}\mathbb{P}^\mu_x(Z_n=y) \leqslant \frac{1}{1+\mu^2}\sum_{n\geqslant\|x-y\|_1}\mathbb{P}^\mu_x(\tau_*>n)\\ &\leqslant \frac{1}{1+\mu^2}\sum_{n\geqslant\|x-y\|_1} p_\mu^n = p_\mu^{\|x-y\|_1}\,.\end{aligned}$$

□

THEOREM 14.7: The solution to the discrete massive outer Dirichlet problem (14.2) is given by

$$u_x = \sum_{y\in\mathbb{Z}^d} G^\mu_\Lambda(x,y) f_y = \mathbb{E}^\mu_x\Big[\sum_{n=0}^{\tau_*-1} f_{Z_n}\Big]\,. \tag{14.8}$$

Moreover, if $f \in c_c(\mathbb{Z}^d)$, then

$$|u_x| \leqslant \|f\|_\infty |\Lambda_f| p_\mu^{\mathrm{d}(\Lambda_f,x)}\,, \tag{14.9}$$

where $\mathrm{d}(\Lambda_f,x) := \inf_{y\in\Lambda_f}\|x-y\|_1$.

Proof. We observe that u is well-defined on account of the boundedness of f, in particular

$$|u_x| \leqslant \|f\|_\infty \sum_{y\in\mathbb{Z}^d} \mathbb{E}_x^\mu\Big[\sum_{n=0}^{\tau_*-1} 1_{\{Z_n=y\}}\Big] \leqslant \|f\|_\infty \mathbb{E}_x^\mu[\tau_*] = \frac{1+\mu^2}{\mu^2}\|f\|_\infty\,,$$

where we used that τ_* is a geometric random variable of parameter p_μ for all $x \in \mathbb{Z}^d$. In particular, the above inequality proves that $u \in \ell^\infty(\mathbb{Z}^d)$.

By direct inspection u solves Equation (14.2a), therefore, it remains to prove condition (14.2b). This can be proved directly or by resorting to an equivalent approximation procedure, which also proves the decay estimate (14.9). In particular, let assume $f \in c_c(\mathbb{Z}^d)$: Then

$$|u_x| \leqslant \|f\|_\infty \sum_{y\in\Lambda_f} p_\mu^{\|x-y\|_1} \leqslant \|f\|_\infty |\Lambda_f| p_\mu^{\mathrm{d}(\Lambda_f,x)}\,,$$

which proves condition (14.2b) and the decay estimate (14.9).

We now discuss the general case of $f \in c_0(\mathbb{Z}^d)$. Let $(f^{(n)})_n$ be a sequence in $c_c(\mathbb{Z}^d)$ converging to f in the $\|\,\|_\infty$-norm: The latter exists because $\overline{c_c(\mathbb{Z}^d)} = c_0(\mathbb{Z}^d)$. Let $u^{(n)} \in c_0(\mathbb{Z}^d)$ be the corresponding solution to the Dirichlet problem (14.2) with datum $f^{(n)}$. Proceeding as above, we find

$$\|u-u^{(n)}\|_\infty \leqslant \Big|\sum_{y\in\mathbb{Z}^d} G^\mu(x,y)(f_y - f_y^{(n)})\Big| \leqslant \frac{1+\mu^2}{\mu^2}\|f-f^{(n)}\|_\infty \underset{n\to\infty}{\longrightarrow} 0\,.$$

Thus, $u^{(n)} \to u$ in $\ell^\infty(\mathbb{Z})^d$: Since $c_0(\mathbb{Z}^d)$ is a Banach space, it follows that that $u \in c_0(\mathbb{Z}^d)$. □

Remark 14.8:

(i) The proof of Theorem 14.7 proves that

$$\|u\|_\infty \leqslant \frac{1+\mu^2}{\mu^2}\|f\|_\infty\,.$$

Thus, the Dirichlet problem (14.2) is well-posed for data $f \in c_0(\mathbb{Z}^d)$ and solution $u \in c_0(\mathbb{Z}^d)$.

(ii) It is worth to point out that the results presented above —*cf.* Proposition 14.3 and Theorem 14.7— immediately generalize to the case where $f \in \ell^\infty(\mathbb{Z}^d)$ and condition (14.2b) is replaced by $u \in \ell^\infty(\mathbb{Z}^d)$. In this scenario the outer Dirichlet problem is still well-posed.

It is worth asking what would happen by considering the discrete outer Dirichlet problem for the massless case $\mu = 0$. Within this setting the decay estimate (14.7) are lost because the killing random walk $(Z_n)_n$ reduces to the standard random walk $(X_n)_n$, *cf.* Definitions 13.7-14.4. In particular, the definition of G^0 may be ill-defined: Actually one would have

$$G^0(x,y) := \mathbb{E}_x\Big[\sum_{n\geqslant 0} 1_{X_n=y}\Big],$$

which is interpreted as the average number of times that a random walk starting at x visits the vertex y. Thus, the extension of Theorem 14.7 is related to the recurrence properties of the random walk on $\mathbb{Z}^d$. The latter can be studied in quite details [10]: In particular it can be shown that the event E_{xy} that a random walk starting at x visits y infinitely many time (*i.e.* that $\sum_{n\geqslant 0} 1_{X_n=y} = \infty$) is deterministic, that is, it has probability either 0 or 1, moreover, it does not depend neither on x nor on y. Thus, either the symmetric random walk returns to all vertexes infinitely many times or it returns to all vertexes finitely many times: In the first case we say that $(X_n)_n$ is **recurrent**, otherwise that it is **transient**.

Finally, $(X_n)_n$ can be shown to be recurrent for $d \in \{1,2\}$, while it is transient for $d \geqslant 3$. Thus, the results of Theorem 14.7 can be generalized for $\mu = 0$ if $d \geqslant 3$ but are problematic for lower dimensions, *cf.* Theorem 10.5

15 Hyperbolic PDEs: Uniqueness on bounded domains

In this section we begin our investigation on hyperbolic PDEs —the main references for this section are [3, 11]. We shall focus on the wave equation and the Klein-Gordon equation

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \Delta_x u = 0\,, \tag{15.1a}$$

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \Delta_x u + \mu^2 u = 0\,, \tag{15.1b}$$

where $c, \mu \in \mathbb{R}$. As seen in Section 1, the wave equation (15.1a) models the transversal vibrations of a continuum Ω, whereas the Klein-Gordon equation (15.1b) describes a quantum relativistic particle —in this latter case $\mu := mc/\hbar$, m being the mass of the particle, c the speed of light and $\hbar$ Plank's constant, *cf.* Equation (2.9). As already seen when computing particular solutions —*cf.* Example 2.10 — we expect the solutions to these equations to describe a "disturbance" which propagates at finite speed, the latter being fixed by the parameter c.

According to Definition 2.9, Equations (15.1) are of hyperbolic type. They have non-trivial characteristic surfaces, two examples being given in (6.6) and (6.7). In what follows we shall be interested in the Cauchy problem for Equations (15.1) on $\mathbb{R}^{n+1}$ with data on the hypersurface

$$\Sigma := \{(x,t) \in \mathbb{R}^{n+1} \,|\, t = 0\}\,.$$

By direct inspection the latter hypersurface is non-characteristic, therefore, Theorem 4.6 applies. Notice that $S(x,t) = t$ fulfils the properties (3.1), therefore, $\frac{\partial u}{\partial \nu} = \frac{\partial u}{\partial t}$.

We will also be interested in a slight variation of the Cauchy problem (6.1). In particular we will consider an evolution problem on a domain $\Omega_T = \Omega \times (-T,T) \subseteq \mathbb{R}^n \times \mathbb{R}$ with $T > 0$. In this latter case, initial conditions will be imposed at $\Omega \times \{0\}$ whereas boundary conditions will be imposed at $\partial\Omega \times (-T,T)$. An example of this problem is given in (1.3), where the boundary conditions are of Neumann type.

The precise definition of these problems is given in the following definition.

Definition 15.1: Let $\Omega \subset \mathbb{R}^n$ be an open non-empty subset with $\overline{\Omega}$ compact and such that $\partial\Omega$ is a C^1-regular orientable hypersurface and outward-pointing unit normal ν. Let $T > 0$, $\theta \in [0, \pi/2]$ and set $\boxed{\Omega_T} := \Omega \times (-T,T)$ and $\boxed{\overline{\Omega}_T} := \overline{\Omega} \times (-T,T)$ —notice that $\Omega_T \subset \overline{\Omega}_T \subset \overline{\Omega_T} = \overline{\Omega} \times [-T,T]$. Notice that $\Omega_T \subset \mathbb{R}^{n+1}$ is open, with compact closure $\overline{\Omega_T} = \overline{\Omega} \times [-T,T]$ and C^2-regular orientable boundary $\partial\Omega_T := [\partial\Omega \times (-T,T)] \cup [\Omega \times \{\pm T\}]$.

The **Cauchy problem with (θ-)boundary conditions** consists of finding $u \in C^2(\overline{\Omega}_T)$ such that

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \Delta_x u + \mu^2 u = f\,, \qquad \forall (x,t) \in \overline{\Omega} \times (-T,T) \tag{15.2a}$$

$$u(x,0) = u_0(x)\,, \quad \left.\frac{\partial u}{\partial t}\right|_{t=0} = u_1(x)\,, \qquad \forall x \in \overline{\Omega} \tag{15.2b}$$

$$\sin\theta \frac{\partial u}{\partial \nu}(x,t) + \cos\theta\, u(x,t) = u_{\mathrm{BC}}(x,t) \qquad \forall (x,t) \in \partial\Omega \times (-T,T) \tag{15.2c}$$

where $c, \mu \in \mathbb{R}$ while $f \in C^0(\overline{\Omega}_T)$, $u_0 \in C^2(\overline{\Omega})$, $u_1 \in C^1(\overline{\Omega})$ and $u_{\mathrm{BC}} \in C^1(\partial\Omega \times (-T,T))$ ($u_{\mathrm{BC}} \in C^2(\partial\Omega \times (-T,T))$ if $\theta = 0$) are assigned.

Remark 15.2:

(i) Notice that problem (1.3) coincides with the problem (15.2) for the choices $\Omega = (0, L)$ and $\theta = \pi/2$.

(ii) Equation (15.2c) corresponds to the choice of a boundary condition on $\partial\Omega \times (-T,T)$. For various choices of θ we fall into three different cases: (a) $\theta = 0$ corresponds to Dirichlet boundary conditions (7.3b); (b) $\theta = \pi/2$ corresponds to Neumann boundary conditions (8.6b); (c) $0 < \theta < \pi/2$ corresponds to Robin boundary conditions (8.7b).

(iii) The regularity of the data $f \in C^0(\overline{\Omega}_T)$, $u_0 \in C^2(\overline{\Omega})$, $u_1 \in C^1(\overline{\Omega})$ matches the regularity $u \in C^2(\overline{\Omega}_T)$ of the wanted the solution. The same applies to the function $u_{\mathrm{BC}} \in C^1(\partial\Omega \times (-T,T))$ with the caveat that, for $\theta = 0$, we have to require $u_{\mathrm{BC}} \in C^2(\partial\Omega \times (-T,T))$.

We stress that the data $u_0, u_1, u_{\mathrm{BC}}$ are not completely independent. Indeed evaluating Equation (15.2c) for $t = 0$ leads to

$$u_{\mathrm{BC}} = \sin\theta \frac{\partial u}{\partial \nu}\bigg|_{t=0} + \cos\theta\, u|_{t=0} = \sin\theta \frac{\partial u_0}{\partial \nu} + \cos\theta\, u_0\,, \tag{15.3a}$$

$$\frac{\partial u_{\mathrm{BC}}}{\partial t}\bigg|_{t=0} = \sin\theta \frac{\partial^2 u}{\partial \nu \partial t}\bigg|_{t=0} + \cos\theta\, \frac{\partial u}{\partial t}\bigg|_{t=0} = \sin\theta \frac{\partial u_1}{\partial \nu} + \cos\theta\, u_1\,. \tag{15.3b}$$

(iv) Definition 15.1 can be generalized by replacing the time interval $(-T,T)$ with either $(0,T)$ or $(-T,0)$ —in these cases one requires $u \in C^2(\overline{\Omega} \times (0,T)) \cap C^1(\overline{\Omega} \times [0,T))$ or $u \in C^2(\overline{\Omega} \times (-T,0)) \cap C^1(\overline{\Omega} \times (-T,0])$. In the first case we refer to the corresponding problem as the **forward** Cauchy problem with boundary conditions, while in the second case we obtain the **backward** Cauchy problem.

We stress that the possibility of addressing both forward and backward Cauchy problem is a priori not granted. Within our setting, this is ultimately ensured by the following symmetry property of Equations (15.1): If u solves either (15.1a) or (15.1b) then $\widetilde{u}(x,t) := u(x,-t)$ is again a solution. This property is called **time-reversal symmetry** and loosely speaking it ensures the equivalence between forward and backward problems. As we shall see, this is not true for the heat equation (1.7): The latter lacks of time-reversal symmetry and its backward Cauchy turns out to be way more delicate than the forwards Cauchy problem.

As usual, the proof that the problem (15.2) is well-posed —*cf.* conditions 1-2-3— is out of our league. However, similarly to Sections 7-8, we shall discuss the uniqueness of the solution.

Theorem 15.3: There exists at most one solution u to the Cauchy problem with boundary conditions (15.2).

Proof. Let u, u' be two solutions to (15.2) with the same data $f, u_0, u_1, u_{\mathrm{BC}}$ and let $v := u - u'$. Then $v \in C^2(\overline{\Omega}_T)$ solves the problem (15.2) with vanishing data. Let

$$E_v := \frac{1}{2}\left[\frac{1}{c^2}\left(\frac{\partial v}{\partial t}\right)^2 + \|\nabla_x v\|^2 + \mu^2 v^2\right] . \tag{15.4}$$

Then $E_v \in C^1(\overline{\Omega}_T)$, $E_v(x,t) \geqslant 0$ and $E_v(x,0) = 0$. Moreover, by direct inspection we have

$$\begin{aligned}
\frac{\partial E_v}{\partial t} &= \frac{1}{c^2}\frac{\partial^2 v}{\partial t^2}\frac{\partial v}{\partial t} + \nabla_x\left(\frac{\partial v}{\partial t}\right)\cdot\nabla_x v + \mu^2\frac{\partial v}{\partial t}v \\
&= \frac{\partial v}{\partial t}\Delta_x v + \nabla_x\left(\frac{\partial v}{\partial t}\right)\cdot\nabla_x v \qquad\qquad \frac{1}{c^2}\frac{\partial^2 v}{\partial t^2} = \Delta_x v - \mu^2 v \\
&= \nabla_x\cdot\left[\frac{\partial v}{\partial t}\nabla_x v\right] \qquad\qquad \nabla_x\left(\frac{\partial v}{\partial t}\right)\cdot\nabla_x v = \nabla_x\cdot\left[\frac{\partial v}{\partial t}\nabla_x v\right] - \frac{\partial v}{\partial t}\Delta_x v .
\end{aligned}$$

Let now $E_{v,\Omega}$ be the function

$$E_{v,\Omega}(t) := \int_\Omega E_v(x,t)\mathrm{d}^n x ,$$

Then $E_{v,\Omega} \in C^1((-T,T))$, $E_{v,\Omega}(0) = 0$ and

$$\frac{\mathrm{d}E_{v,\Omega}}{\mathrm{d}t} = \int_\Omega \frac{\partial E_v}{\partial t}(x,t)\mathrm{d}^n x ,$$

where we applied Corollary 9.5 —the latter applies because $\overline{\Omega}$ is compact and $\frac{\partial E_v}{\partial t} \in C^0(\overline{\Omega}_T)$. The previous computation and Gauss's divergence theorem —*cf.* Theorem 8.1— entail

$$\begin{aligned}
\frac{\mathrm{d}E_{v,\Omega}}{\mathrm{d}t} &= \int_\Omega \frac{\partial E_v}{\partial t}(x,t)\mathrm{d}^n x = \int_\Omega \nabla_x\cdot\left[\frac{\partial v}{\partial t}\nabla_x v\right]\mathrm{d}^n x \\
&= \oint_{+\partial\Omega} \nu\cdot\left[\frac{\partial v}{\partial t}\nabla_x v\right]\mathrm{d}S(x) = \oint_{+\partial\Omega} \frac{\partial v}{\partial t}\frac{\partial v}{\partial \nu}\mathrm{d}S(x) .
\end{aligned}$$

At this stage we consider two separated cases:

$\boxed{\theta \in \{0, \pi/2\}}$ In this case $\frac{\partial v}{\partial t}\frac{\partial v}{\partial \nu} = 0$ on $\partial\Omega \times (-T,T)$ because either $v|_{\partial\Omega} = 0$ (for $\theta = 0$) or $\frac{\partial v}{\partial \nu} = 0$ (for $\theta = \pi/2$). This entails that

$$\frac{\mathrm{d}E_{v,\Omega}}{\mathrm{d}t} = 0 \quad\Rightarrow\quad E_{v,\Omega}(t) = E_{v,\Omega}(0) = 0\ \forall t \in (-T,T) .$$

Since $E_v \geqslant 0$ and it is continuous, $E_{v,\Omega}(t) = 0$ implies $E_v(x,t) = 0$ for all $x \in \Omega$ and $t \in (-T,T)$. Moreover, $E_v = 0$ if and only if $\frac{\partial v}{\partial t} = 0$, $\nabla_x v = 0$ and, if $\mu \neq 0$, $v = 0$. Therefore, if $\mu \neq 0$ we find $v = 0$, *i.e.* $u = u'$. If instead $\mu = 0$ we find that v is a constant, moreover $v(0,x) = 0$, therefore $v = 0$. In either case we proved $u = u'$.

$\boxed{0 < \theta < \pi/2}$ In this case we find

$$\oint_{+\partial\Omega} \frac{\partial v}{\partial t}\frac{\partial v}{\partial \nu} \mathrm{d}S(x) = -\cot\theta \oint_{+\partial\Omega} \frac{\partial v}{\partial t} v \, \mathrm{d}S(x) = -\frac{1}{2}\cot\theta \frac{\mathrm{d}}{\mathrm{d}t} \oint_{+\partial\Omega} v^2(x,t)\mathrm{d}S(x)\,,$$

where in the latter equality we applied Corollary 9.5 —condition (ii) holds because v^2 is continuous and $\partial\Omega$ is compact.

Overall we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[E_{v,\Omega}(t) + \frac{1}{2}\cot\theta \oint_{+\partial\Omega} v^2(x,t)\mathrm{d}S(x)\right] = 0$$

$$\Rightarrow \quad E_{v,\Omega}(t) + \frac{1}{2}\cot\theta \oint_{+\partial\Omega} v^2(x,t)\mathrm{d}S(x) = 0\,.$$

Since $\theta \in (0,\pi/2)$ the latter equality entails $E_{v,\Omega}(t) = 0$ for all $t \in (-T,T)$. By proceeding as in the previous case we find $v = 0$, *i.e.* $u = u'$.

□

Remark 15.4: The function E_v defined in Equation (15.4) is called **energy density** of the function v. Considering a domain domain $V \subset \overline{\Omega}_T$ such that ∂V is an orientable C^1-hypersurface the function

$$E_{u,V}(t) := \int_V E_u(x,t)\mathrm{d}^n x\,, \tag{15.5}$$

is interpreted as the energy of the function $u \in C^1(\overline{\Omega}_T)$ within V. In the proof of Theorem 15.3 we proved that, if $u \in C^2(\overline{\Omega}_T)$ solves either (15.1a) or (15.1b), then

$$\frac{\partial E_u}{\partial t} = \nabla_x \cdot J_u\,, \qquad J_u := \frac{\partial u}{\partial t}\nabla_x u\,, \tag{15.6}$$

where $J_u \in C^1(\Omega_T, \mathbb{R}^3)$ is called the **energy flux density**. Equation (15.6) is called **continuity equation** for the energy density E_u. Integrating Equation (15.6) and applying Theorem 8.1 and Corollary 9.5 we are lead to

$$\frac{\mathrm{d}E_{u,V}}{\mathrm{d}t} = \oint_{+\partial V} \nu \cdot J_u(x,t)\mathrm{d}S(x)\,. \tag{15.7}$$

This equation is interpreted by saying that the time variation of the total energy of u in the domain V is proportional to the flux of energy across ∂V. Notice that this equation resembles the diffusion equation (1.5), which was derived by a similar argument —*cf.* Equation (1.4).

16 Hyperbolic PDEs: Finite propagation speed

In this section we shall prove that a solution to either Equation (15.1a) or to (15.1b) propagates at finite speed, whose value is c. The proof of this fact is based on the energy density introduced in Equation (15.4).

For later convenience we prove the following lemma.

LEMMA 16.1: Let $I = [t_0, t_1] \subset \mathbb{R}$ be a compact interval, $F \in C^0(I^2)$ and set

$$f(t) := \int_{t_0}^{t} F(t,s)\mathrm{d}s\,. \tag{16.1}$$

If $\frac{\partial F}{\partial t} \in C^0(I^2)$ then $f \in C^1(I)$, moreover,

$$\frac{\mathrm{d}f}{\mathrm{d}t} = F(t,t) + \int_{t_0}^{t} \frac{\partial F}{\partial t}(t,s)\mathrm{d}s\,. \tag{16.2}$$

Proof. We prove that $f \in C^0(I)$. Notice that, since I is compact F is uniformly continuous on I^2 by Weierstrass' theorem [6, Thm. 11.3]. It follows that, for all $\varepsilon > 0$, there exists $\delta_\varepsilon > 0$ such that

$$|F(t,s) - F(t',s')| < \varepsilon\,,$$

for all $t, s, t', s' \in I$ with $(t,s) \in B_{\delta_\varepsilon}(t',s')$.

Let $t' \in I$, $\varepsilon > 0$, $0 < \delta < \min\{\delta_\varepsilon, \varepsilon\}$, and $t \in (t'-\delta, t'+\delta)$: It follows that

$$\begin{aligned}
|f(t) - f(t')| &= \left|\int_{t_0}^{t} F(t,s)\mathrm{d}s - \int_{t_0}^{t'} F(t',s)\mathrm{d}s\right| \\
&\leqslant \int_{(t',t)} |F(t,s)|\mathrm{d}s + \int_{t_0}^{t} |F(t,s) - F(t',s)|\mathrm{d}s \\
&< \varepsilon\left[\sup_{t,s\in I}|F(t,s)| + |t_1 - t_0|\right]\,,
\end{aligned}$$

which shows that $\lim_{t\to t'} f(t) = f(t')$.

Similarly, for all $t \in I$ we evaluate

$$\begin{aligned}
\frac{f(t+h) - f(t)}{h} &= \frac{1}{h}\int_{t}^{t+h} F(t+h,s)\mathrm{d}s + \int_{t_0}^{t} \frac{F(t+h,s) - F(t,s)}{h}\mathrm{d}s \\
&= F(t+h, t+h') + \int_{t_0}^{t} \frac{\partial F}{\partial t}(t+h'', s)\mathrm{d}s\,,
\end{aligned}$$

where in the last equality we applied Lagrange's theorem together with the mean value formula [6], where $h', h'' \in (0,h)$. Since $F, \frac{\partial F}{\partial t} \in C^0(I^2)$, Theorem 9.2 applies and we find

$$\lim_{h\to 0^+} \frac{f(t+h)-f(t)}{h} = F(t,t) + \int_{t_0}^{t} \frac{\partial F}{\partial t}(t,s)\mathrm{d}s\,.$$

This shows that $f'(t)$ exists for all $t \in I$, moreover, Equation (16.2) holds true. The proof that $f' \in C^0(I)$ is identical to the one showing that $f \in C^0(I)$: Thus, $f \in C^1(I)$. □

The following theorem shows that the value $u(x_0,t_0)$ of a solution u to one of the equations (15.1) at a point (x_0,t_0) depends only on the **cone of dependence with vertex** (x_0,t_0) —*cf.* Figure 6. The latter is defined by

$$D_{x_0,t_0} := \{(x,t) \in \mathbb{R}^{n+1} \,|\, t \leqslant t_0\,,\ \|x-x_0\| \leqslant c(t_0-t)\}\,. \tag{16.3}$$

Although this result holds true also on a finite domain, we shall consider only the case of a

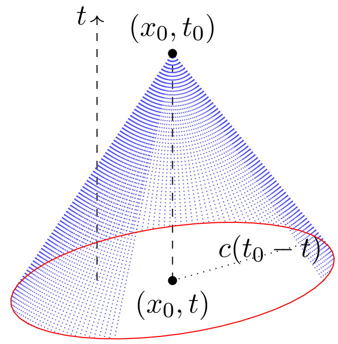


Figure 6: The cone of dependence with vertex (x_0,t_0).

solution defined on $\mathbb{R}^{n+1}$ —*cf.* [3, Thm. 6].

THEOREM 16.2: Let $u \in C^2(\mathbb{R}^{n+1})$ be a solution to either Equation (15.1a) or (15.1b). Let (x_0,t_0) be such that $u(x,0) = \frac{\partial u}{\partial t}(x,0) = 0$ for all $x \in B_{ct_0}(x_0)$. Then u vanishes on $D_{x_0,t_0} \cap (\mathbb{R}^n \times [0,t_0])$.

Proof. We consider the function $e_u\colon [0,t_0] \to \mathbb{R}$ defined by

$$e_u(t) := \int_{B_{c(t_0-t)}(x_0)} E_u(x,t)\mathrm{d}^n x = \int_0^{c(t_0-t)} \oint\limits_{+\partial B_R(x_0)} E_u(x,t)\mathrm{d}S(x)\mathrm{d}R\,,$$

where in the second equality we used identity (3.12). By Corollary 9.5 the function

$$F(t,R) := \oint\limits_{+\partial B_R(x_0)} E_u(x,t)\mathrm{d}S(x)\,,$$

is $C^0([0,t_0]^2)$ and $\frac{\partial F}{\partial t} \in C^0([0,t_0]^2)$ — Theorem 9.2 implies that $\lim_{R\to 0^+} F(t,R) = 0$. By Lemma 16.1 we find

$$\begin{aligned}
\frac{\mathrm{d}e_u}{\mathrm{d}t} &= \int_0^{c(t_0-t)} \oint_{+\partial B_R(x_0)} \frac{\partial E_u}{\partial t}(x,t)\mathrm{d}S(x)\mathrm{d}R - c \oint_{+\partial B_{c(t_0-t)}(x_0)} E_u(x,t)\mathrm{d}S(x) \\
&= \int_{B_{c(t-t_0)}(x_0)} \frac{\partial E_u}{\partial t}(x,t)\mathrm{d}^n x - c \oint_{+\partial B_{c(t_0-t)}(x_0)} E_u(x,t)\mathrm{d}S(x) && \text{Eq. (3.12)} \\
&= \int_{B_{c(t-t_0)}(x_0)} \nabla_x \cdot \left(\frac{\partial u}{\partial t}\nabla_x u\right)(x,t)\mathrm{d}^n x - c \oint_{+\partial B_{c(t_0-t)}(x_0)} E_u(x,t)\mathrm{d}S(x) && \text{Eq. (15.6)} \\
&= c \oint_{+\partial B_{c(t_0-t)}(x_0)} \left[\frac{1}{c}\frac{\partial u}{\partial t}\frac{\partial u}{\partial \nu} - E_u(x,t)\right]\mathrm{d}S(x) && \text{Thm. 8.1}\,.
\end{aligned}$$

The first contribution in the latter integral can be estimated by

$$\left|\frac{1}{c}\frac{\partial u}{\partial t}\frac{\partial u}{\partial \nu}\right| \leqslant \left|\frac{1}{c}\frac{\partial u}{\partial t}\right| \|\nabla_x u\| \leqslant \frac{1}{2}\left[\frac{1}{c^2}\left(\frac{\partial u}{\partial t}\right)^2 + \|\nabla_x u\|^2\right] \leqslant E_u(x,t)\,.$$

It follows that $\frac{\mathrm{d}e_u}{\mathrm{d}t} \leqslant 0$ and therefore $e_u(t) \leqslant e_u(0) = 0$ for all $t \geqslant 0$. Since $E_u(x,t) \geqslant 0$ and $E_u \in C^1(\mathbb{R}^{n+1})$, this implies that $E_u(x,t) = 0$ for all $x \in B_{c(t_0-t)}(x_0)$ and all $t \in [0,t_0]$. This entails $\frac{\partial u}{\partial t} = 0$ and $\nabla_x u = 0$ (as well as $u = 0$ if $\mu \neq 0$) on $D_{x_0,t_0} \cap (\mathbb{R}^n \times (0,t_0))$. The assumption $u(x,0) = \frac{\partial u}{\partial t}(x,0) = 0$ for all $x \in B_{ct_0}(x_0)$ implies $u = 0$ on $D_{x_0,t_0} \cap (\mathbb{R}^n \times [0,t_0])$. ☐

Remark 16.3: By time-reversal symmetry —*cf.* Remark 15.2— Theorem 16.2 can be strengthen as follows —*cf.* Figure 7 . If $u \in C^2(\mathbb{R}^{n+1})$ solves either Equation (15.1a) or Equation (15.1b) and moreover $u(x,0) = \frac{\partial u}{\partial t}(x,0) = 0$ for all $x \in B_{ct_0}(x_0)$ then

$$u(x,t) = 0\ \forall (x,t)\colon \|x - x_0\| \leqslant c(t_0 - |t|)\,,\ t \in (-t_0,t_0)\,.$$

Theorem 16.5 implies the following finite propagation speed property. For that we define the **future/past causal cone** of a subset $A \subset \mathbb{R}^{n+1}$ by —*cf.* Figure 8—

$$J^+(A) := \bigcup_{(x,t)\in A} J^+(\{(x,t)\})\,, \tag{16.4a}$$

$$J^+(\{(x,t)\}) := \{(y,s) \in \mathbb{R}^{n+1} \mid s \geqslant t\,, \|x-y\| \leqslant c|t-s|\}\,, \tag{16.4b}$$

$$J^-(A) := \bigcup_{(x,t)\in A} J^-(\{(x,t)\})\,, \tag{16.4c}$$

$$J^-(\{(x,t)\}) := \{(y,s) \in \mathbb{R}^{n+1} \mid s \leqslant t\,, \|x-y\| \leqslant c|t-s|\}(= D_{x,t})\,, \tag{16.4d}$$

$$J(A) := J^+(A) \cup J^-(A)\,. \tag{16.4e}$$

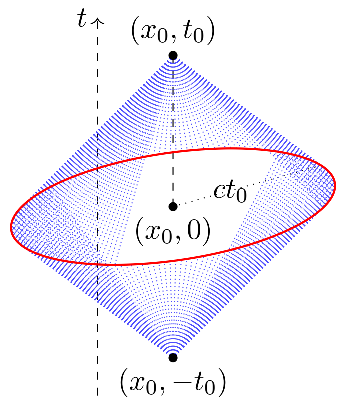


Figure 7: The "double cone" with upper vertex (x_0, t_0).

Remark 16.4:

(i) Notice that $(y, s) \in J(\{x, t\})$ if and only if the straight line from (y, s) to (x, t) has slope within the interval $(-c, c)$. The physical interpretation of this condition is that the event (y, s) can be connected to the event (x, t) by a signal which travels at most at the speed c. The event (y, s) lies in the future/past of the event (x, t) depending on whether $s > t$ or $s < t$. Conversely, points $(y, s) \notin J(\{(x, t)\})$ are called **causally disconnected** from

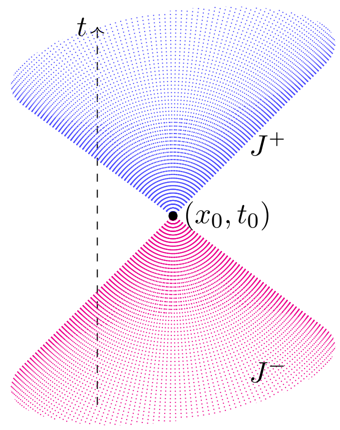


Figure 8: Future and past causal cones from (x, t).

(x, t). From the point of view of Special Relativity these points are neither in the past nor in the future of (x, t), actually, upon a Lorentz transformation the events (y, s) and (x, t) are simultaneous —*cf.* [12] for further details.

(ii) Although $J^{\pm}(\{x, t\})$ is closed, $J^{\pm}(A)$ may fail to be closed, even if A is closed. A counterexample can be constructed in $\mathbb{R}^2$ [1, App. A5] by considering the curve

$$\gamma := \{(x_0 \cosh(s), t_0 \sinh(s)) \in \mathbb{R}^2 \,|\, s \geqslant 0\} ,$$

where $x_0, t_0 > 0$ are arbitrary. The set γ is closed since it is the intersection between an hyperbola with the half-plane $t \geqslant 0$. By direct inspection $J^-(\gamma)$ is open and coincides with the set $\{(x,t) \in \mathbb{R}^2 \,|\, x < ct\}$ —*cf.* Figure 9.

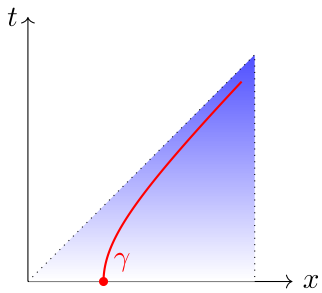


Figure 9: The red line γ is closed, however $J^-(\gamma) = \{(x,t) \in \mathbb{R}^2 \,|\, x < ct\}$ is open.

(iii) If $A \subset \mathbb{R}^{n+1}$ is compact, then $J^+(A)$, $J^-(A)$ (and thus $J(A)$) are closed. To wit, let $\{(x_n,t_n)\}_{n\in\mathbb{N}}$ be a convergent sequence of points in $J^+(A)$ and let $(x,t) := \lim_{n\to+\infty}(x_n,t_n)$: We will prove that $(x,t) \in J^+(A)$.

For all $n \in \mathbb{N}$ there exists $(y_n,s_n) \in A$ such that: (a) $s_n \leqslant t_n$; (b) $\|x_n - y_n\| \leqslant c(t_n - s_n)$. Since A is compact there exists a subsequence $\{(y_{n_k}, s_{n_k})\}_{k\in\mathbb{N}}$ which converges to a point $(y,s) \in A$. It follows that $t = \lim_{k\to+\infty} t_{n_k} \geqslant \lim_{k\to+\infty} s_{n_k} = s$ as well as

$$\|x - y\| - c(t - s) = \lim_{k\to+\infty} \|x_{n_k} - y_{n_k}\| - c(t_{n_k} - s_{n_k}) \leqslant 0 .$$

This shows that $(x,t) \in J^+(A)$, thus $J^+(A)$ is closed. A similar argument goes for $J^-(A)$ and thus $J(A) = J^+(A) \cup J^-(A)$ is closed.

(iv) Remarkably, $J^\pm(A)$ is closed whenever A is closed and there exists $t_0 \in \mathbb{R}$ such that $A \subset \{(x,t) \in \mathbb{R}^{n+1} \,|\, t = t_0\}$. Indeed let $(x_n,t_n) \in J^+(A)$ be such that $\lim_{n\to+\infty}(x_n,t_n) = (x,t) \in \mathbb{R}^{n+1}$: We will prove that $(x,t) \in J^+(A)$.

For all $n \in \mathbb{N}$ there exists $(y_n,t_0) \in A$ such that $t_n \geqslant t_0$ and $\|x_n - y_n\| \leqslant c(t_n - t_0)$. It follows that

$$\begin{aligned}
0 \leqslant \sup_n \|y_n\| &\leqslant \sup_n \{|\,\|y_n\| - \|x_n\|\,| + \|x_n\|\} \\
&\leqslant \sup_n \{\|y_n - x_n\| + \|x_n\|\} && |\,\|a\| - \|b\|\,| \leqslant \|a - b\| \\
&\leqslant \sup_n \{c(t_n - t_0) + \|x_n\|\} \\
&< +\infty && (x,t) = \lim_{n\to+\infty}(x_n,t_n) .
\end{aligned}$$

Thus, $(y_n)_n$ is a bounded sequence and therefore there exists a convergent subsequence $(y_{n_k})_k$ [6, §2]. Since A is closed we have $(y, t_0) := \lim_{k\to+\infty}(y_{n_k}, t_0) \in A$. Moreover $t = \lim_{k\to+\infty} t_{n_k} \geqslant t_0$ as well as

$$\|x - y\| - c(t - t_0) = \lim_{k\to+\infty} \|x_{n_k} - y_{n_k}\| - c(t_{n_k} - t_0) \leqslant 0\,.$$

Thus $(x, t) \in J^+(A)$, as claimed. With a similar argument $J^-(A)$ is closed and so is $J(A) = J^+(A) \cup J^-(A)$.

Theorem 16.5 (Finite propagation speed): Let $u \in C^2(\mathbb{R}^{n+1})$ be a solution to one of equations (15.1). Then

$$\operatorname{supp}(u) \subseteq J\left[(\operatorname{supp}(u_0) \cup \operatorname{supp}(u_1)) \times \{0\}\right]\,, \tag{16.5}$$

where $u_0(x) := u(x, 0)$ while $u_1(x) := \frac{\partial u}{\partial t}(x, 0)$. In particular, if u_0, u_1 are compactly supported, then the function $u(\ , t)$ is compactly supported for all $t \in \mathbb{R}$.

Proof. Let $A := (\operatorname{supp}(u_0) \cup \operatorname{supp}(u_1)) \times \{0\}$ and let $(x_0, t_0) \notin J(A)$: We will prove that $u(x_0, t_0) = 0$. Without loss of generality we may assume $t_0 > 0$, the proof being analogous for $t_0 < 0$. It follows that $(x_0, t_0) \notin J^+(A)$: Since this is equivalent to $J^-(\{x_0, t_0\}) \cap A = \varnothing$ we find $B_{ct_0}(x_0) \cap [\operatorname{supp}(u_0) \cup \operatorname{supp}(u_1)] = \varnothing$. By Theorem 16.2 we have $u = 0$ in $D_{x_0,t_0} \cap (\mathbb{R}^n \times [0, t_0])$, in particular, $u(x_0, t_0) = 0$. Thus

$$\{(x, t) \in \mathbb{R}^{n+1} \,|\, u(x, t) \neq 0\} \subset J(A) \quad \Rightarrow \quad \operatorname{supp}(u) \subseteq \overline{J(A)} = J(A)\,,$$

where in the last equality we used Remark 16.4.

If u_0, u_1 are compactly supported then, for all $t_0 \in \mathbb{R}$, $A_{t_0} := J(A) \cap \{(x, t) \,|\, t = t_0\}$ is compact. Indeed, A_{t_0} is closed and bounded because, if $R > 0$ is such that $\operatorname{supp}(u_0) \cup \operatorname{supp}(u_1) \subset B_R(0)$, then $A_{t_0} \subset B_{R+ct_0}(0)$. It follows that $u(\ , t_0)$ is compactly supported. □

Remark 16.6:

(i) The inclusion (16.5) has the following interpretation. Events (x, t) which are causally disconnected from the supports of the initial data u_0, u_1 lie outside of the support of the solution u. Stated differently, the solution u propagates the support of the initial data u_0, u_1 at most at the speed c. This property is far from obvious and, as we shall see, it is not common to all PDEs: For example, the heat equation (1.7) prescribes an (unphysical) infinite speed of propagation.

(ii) Remarkably, Theorem 16.5 applies also when a source term f is present in equations (15.1) —*e.g.* one has $\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \Delta_x u = f$. In this case the inclusion (16.5) is modified to

$$\operatorname{supp}(u) \subseteq \overline{J\left[((\operatorname{supp}(u_0) \cup \operatorname{supp}(u_1)) \times \{0\}) \cup \operatorname{supp}(f)\right]}\,.$$

17 Hyperbolic PDEs: D'Alembert's formula

In this section we shall consider the Cauchy problem for the wave equation on $\mathbb{R}^n$. The latter consists of looking for a function $u \in C^2(\mathbb{R}^{n+1})$ such that

$$\frac{1}{c^2}\frac{\partial u}{\partial t^2} - \Delta_x u = f\,, \tag{17.1a}$$

$$u(x,0) = u_0(x)\,, \qquad \frac{\partial u}{\partial t}\bigg|_{t=0} = u_1(x)\,, \tag{17.1b}$$

where $f \in C^0(\mathbb{R}^{n+1})$, $u_0 \in C^2(\mathbb{R}^n)$ while $u_1 \in C^1(\mathbb{R}^n)$.

A direct application of Theorem 16.5 leads to a uniqueness result for the problem (17.1).

Corollary 17.1: There exists at most one solution to the Cauchy problem (17.1).

Proof. Let u, u' be two solutions to (17.1) with the same data f, u_0, u_1. Then $v := u - u'$ is a solution to (17.1) with vanishing data. Theorem 16.5 —*cf.* inclusion (16.5) — leads to $v = 0$. □

Remarkably, the simple geometry of $\mathbb{R}^{n+1}$ allows to deduce exact formulas for the Cauchy problem (17.1). Although this is possible for all $n \in \mathbb{N}$ we shall consider the particular cases of $n = 1, 2, 3$ referring to [3, Thm. 2-3-4] for further details. In this Section we will focus on the case $n = 1$: The cases $n = 2, 3$ will be considered in Section 18. Finally, for simplicity we will set $f = 0$, *cf.* Remark 17.3.

In the case of $n = 1$ the wave equation (15.1a) simplifies to

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0\,,$$

where $u \in C^2(\mathbb{R}^2)$. In this situation we introduce the **light-like coordinates**

$$x_+ := x + ct\,, \quad x_- := x - ct\,, \quad x = \frac{x_+ + x_-}{2}\,, \quad t = \frac{x_+ - x_-}{2c}\,. \tag{17.2}$$

These coordinates are particularly useful because, letting $\widetilde{u} \in C^2(\mathbb{R}^2)$ be defined by $\widetilde{u}(x_+, x_-) = u(x(x_+, x_-), t(x_+, x_-))$ we have

$$\frac{\partial \widetilde{u}}{\partial x_\pm} = \frac{1}{2}\left(\frac{\partial u}{\partial x} \pm \frac{1}{c}\frac{\partial u}{\partial t}\right) \quad \Rightarrow \quad \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = -4\frac{\partial^2 \widetilde{u}}{\partial x_+ \partial x_-}\,.$$

Out of these identities we can prove the following theorem.

Theorem 17.2 (D'Alembert's formula): The solution u to the Cauchy problem (17.1) for $n = 1$ is given by

$$u(x,t) = \frac{1}{2}\left[u_0(x+ct) + u_0(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} u_1(y)\mathrm{d}y\,. \tag{17.3}$$

Proof. Considering light-like coordinates we find that

$$\frac{\partial^2 \widetilde{u}}{\partial x_+ \partial x_-} = 0\,.$$

This implies that there exists two functions $f, g \in C^2(\mathbb{R})$ such that

$$\widetilde{u}(x_+, x_-) = f(x_+) + g(x_-) \quad \Rightarrow \quad u(x,t) = f(x+ct) + g(x-ct)\,.$$

Imposing the initial conditions $u(x,0) = u_0(x)$ and $\frac{\partial u}{\partial t}(x,0) = u_1(x)$ leads to

$$\begin{aligned} &u_0(x) = f(x) + g(x) \quad (\Rightarrow\ u_0'(x) = f'(x) + g'(x)) \\ &u_1(x) = cf'(x) - cg'(x)\,. \end{aligned}$$

Rearranging we find

$$\begin{aligned} f'(x) &= \frac{1}{2c}[c\,u_0'(x) + u_1(x)] \quad \Rightarrow \quad f(x) = \frac{1}{2c}\left[c\,u_0(x) + \int_0^x u_1(y)\mathrm{d}y\right] + A\,, \\ g'(x) &= \frac{1}{2c}[c\,u_0'(x) - u_1(x)] \quad \Rightarrow \quad g(x) = \frac{1}{2c}\left[c\,u_0(x) - \int_0^x u_1(y)\mathrm{d}y\right] - A\,, \end{aligned}$$

for an arbitrary constant $A \in \mathbb{R}$ —notice that we have implicitly imposed $f + g = u_0$. Thus, we have

$$\begin{aligned} u(x,t) &= f(x+ct) + g(x-ct) \\ &= \frac{1}{2c}\left[c\,u_0(x+ct) + \int_0^{x+ct} u_1(y)\mathrm{d}y\right] + \frac{1}{2c}\left[c\,u_0(x-ct) - \int_0^{x-ct} u_1(y)\mathrm{d}y\right] \\ &= \frac{1}{2}\left[u_0(x+ct) + u_0(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} u_1(y)\mathrm{d}y\,. \end{aligned}$$

By direct inspection Equation (17.3) defines a solution to (17.1) with Cauchy data u_0, u_1. □

REMARK 17.3:

(i) Within the proof of Theorem 17.2 we proved that any solution to the wave equation (15.1a) for $n = 1$ can be written as

$$u(x,t) = f_+(x+ct) + f_-(x-ct)\,.$$

The function $u_-(x,t) := f_-(x-ct)$ represents a right-shift of the profile f_-, whereas $u_+(x,t) := f_+(x+ct)$ corresponds to a left-shift of the profile f_+. For this reason the functions u_-, u_+ are called **progressive and regressive waves** respectively.

(ii) Whenever a source term f is present, the D'Alembert's formula has to be modified to

$$u(x,t) = \frac{1}{2}\left[u_0(x+ct) + u_0(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} u_1(y)\mathrm{d}y + \frac{c}{2}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)\mathrm{d}y\mathrm{d}s\,. \quad (17.4)$$

Equation (17.4) is perfectly well-defined for all $f \in C^0(\mathbb{R}^2)$, however, it does not define a C^2 function in general — This can be seen by considering Equation (17.3) for $u_0 = u_1 = 0$ and $f(y,s) = e^{-|y|}$. A sufficient condition for Equation (17.4) to define a C^2 function is $f \in C^1(\mathbb{R}^2)$: We refer to [11, §6.1.2] for further details.

Notice that the first contribution in (17.4) is nothing but Equation (17.3), that is, the solution to the Cauchy problem (17.1) with data $f = 0$, u_0, u_1. Conversely, the second contribution coincides with the solution to (17.1) with data f, $u_0 = u_1 = 0$. The fact that the solution to (17.1) can be splitted in this way is a consequence of the linearity of the Cauchy problem (17.1).

Notice that both (17.3) and (17.4) abide by the results of Theorem 16.5, namely, they describe a propagation at finite speed of the data u_0, u_1, f.

(iii) Notice that the D'Alembert formula (17.3) still make sense if the data u_0, u_1 have lower regularity, *e.g.* $u_0, u_1 \in C^0(\mathbb{R})$. A function u defined by the D'Alembert formula with data u_0, u_1 with such lower regularity is called a **weak solution** to the wave equation.

(iv) Out of Corollary 17.1 and Theorem 17.2 we have proved existence and uniqueness for the Cauchy problem (17.1) for $n = 1$. It is worth investigating whether we may obtain well-posedness in the sense of Hadamard —*cf.* conditions 1-2-3.

For the sake of simplicity we shall investigate this aspect within the assumption $f = 0$ and $u_0 \in C^2_b(\mathbb{R})$ and $u_1 \in C^1_b(\mathbb{R})$. Here we denoted by

$$C^k_b(\mathbb{R}) := \{u \in C^k(\mathbb{R}) \mid \sup_{x\in\mathbb{R}} \left|\frac{\mathrm{d}^j u}{\mathrm{d}x^j}\right| < \infty \ \forall j \in \{0,\ldots,k\}\}\,.$$

so that $u_0 \in C^2_b(\mathbb{R})$ and $u_1 \in C^1_b(\mathbb{R})$ implies that $u_0, u_0', u_0'', u_1, u_1'$ are bounded functions. Moreover, we shall consider the solution u only for finite times, say $t \in (-T,T)$, $T > 0$.

The spaces $C^2_b(\mathbb{R})$, $C^1_b(\mathbb{R})$ and $C^2_b(\mathbb{R}_T)$, $\mathbb{R}_T := \mathbb{R} \times (-T,T)$, can be turned into complete normed spaces by considering

$$\|u_0\|_{C^2_b(\mathbb{R})} := \sup_{\substack{x\in\mathbb{R}\\ k\in\{0,1,2\}}} \left|\frac{\mathrm{d}^k u_0}{\mathrm{d}x^k}\right| \qquad \forall u_0 \in C^2_b(\mathbb{R})\,,$$

$$\|u_1\|_{C^1_b(\mathbb{R})} := \sup_{\substack{x\in\mathbb{R}\\ k\in\{0,1\}}} \left|\frac{\mathrm{d}^k u_1}{\mathrm{d}x^k}\right| \qquad \forall u_1 \in C^1_b(\mathbb{R})\,,$$

$$\|u\|_{C^2_b(\mathbb{R}_T)} := \sup_{\substack{(x,t)\in\mathbb{R}_T\\ k+j\leqslant 2\\ k,j\geqslant 0}} \left|\frac{\partial^{k+j} u}{\partial x^k \partial t^j}\right| \qquad \forall u \in C^2_b(\mathbb{R}_T)\,.$$

Let now $u, \overline{u}$ be two solutions to the Cauchy problem (17.1) for $n=1$ and with data u_0, u_1 and $\overline{u}_0, \overline{u}_1$ respectively. By direct inspection using Equation (17.3) we find

$$\begin{aligned}|u(x,t) - \overline{u}(x,t)| = \bigg|\frac{1}{2}\left[(u_0-\overline{u}_0)(x+ct) + (u_0-\overline{u}_0)(x-ct)\right] \\ + \frac{1}{2c}\int_{x-ct}^{x+ct}[u_1(y) - \overline{u}_1(y)]\mathrm{d}y\bigg| \\ \leqslant \|u_0-\overline{u}_0\|_{C^2_b(\mathbb{R})} + T\|u_1-\overline{u}_1\|_{C^1_b(\mathbb{R})}\,.\end{aligned}$$

With a similar argument we find

$$\begin{aligned}\left|\frac{\partial u}{\partial x} - \frac{\partial \overline{u}}{\partial x}\right| &= \bigg|\frac{1}{2}\left[(u_0'-\overline{u}_0')(x+ct) + (u_0'-\overline{u}_0')(x-ct)\right] \\ &\quad + \frac{1}{2c}\left[(u_1-\overline{u}_1)(x+ct) - (u_1-\overline{u}_1)(x-ct)\right]\bigg| \\ &\leqslant \|u_0-\overline{u}_0\|_{C^2_b(\mathbb{R})} + \frac{1}{c}\|u_1-\overline{u}_1\|_{C^1_b(\mathbb{R})} \\ \left|\frac{\partial u}{\partial t} - \frac{\partial \overline{u}}{\partial t}\right| &\leqslant c\|u_0-\overline{u}_0\|_{C^2_b(\mathbb{R})} + \|u_1-\overline{u}_1\|_{C^1_b(\mathbb{R})} \\ \left|\frac{\partial^{k+j}}{\partial t^k \partial x^j}(u-\overline{u})\right| &\leqslant c_{jk}\|u_0-\overline{u}_0\|_{C^2_b(\mathbb{R})} + c'_{jk}\|u_1-\overline{u}_1\|_{C^1_b(\mathbb{R})} \quad k+j=2\,,\ k,j\geqslant 0\,,\end{aligned}$$

for suitable constants $c_{jk}, c'_{jk} > 0$.

The above inequalities proves that $u \in C^2_b(\mathbb{R}_T)$, as follows by setting $\overline{u}=0$, and moreover

$$\|u-\overline{u}\|_{C^2_b(\mathbb{R}_T)} \leqslant C_0\|u_0-\overline{u}_0\|_{C^2_b(\mathbb{R})} + C_1\|u_1-\overline{u}_1\|_{C^1_b(\mathbb{R})}\,,$$

for constant $C_0, C_1 > 0$ which may depend at most linearly on T. The latter inequality entails that if the initial data u_0, u_1, $\overline{u}_0, \overline{u}_1$ are close —in the sense of the spaces $C^2_b(\mathbb{R})$, $C^1_b(\mathbb{R})$— then so are the corresponding solutions u, $\overline{u}$.

For later convenience we prove the following corollary of Theorem 17.2 which provides an analogous of the D'Alembert formula (17.3) for the Cauchy problem on the half-line. In what follows we set $\mathbb{R}_+ = (0, +\infty)$ so that $\overline{\mathbb{R}_+} := [0, +\infty)$.

COROLLARY 17.4: Let $u_0 \in C^2(\overline{\mathbb{R}_+})$ and $u_1 \in C^1(\overline{\mathbb{R}_+})$ be such that $u_0(0) = u_0''(0) = u_1(0) = 0$. Then the Cauchy problem

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0 \quad \forall (x,t) \in \overline{\mathbb{R}_+} \times \mathbb{R} \tag{17.5a}$$

$$u(x,0) = u_0(x)\,, \quad \frac{\partial u}{\partial t}(x,0) = u_1(x) \quad \forall x \in \overline{\mathbb{R}_+}\,, \tag{17.5b}$$

$$u(0,t) = 0\,, \quad \forall t \in \mathbb{R} \tag{17.5c}$$

has a unique solution $u \in C^2(\overline{\mathbb{R}_+} \times \mathbb{R})$. Moreover,

$$u(x,t) = \begin{cases} \frac{1}{2}\left[u_0(x+ct) + u_0(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} u_1(y)\mathrm{d}y & 0 \leqslant ct \leqslant x \\ \frac{1}{2}\left[u_0(x+ct) - u_0(ct-x)\right] + \frac{1}{2c}\int_{ct-x}^{x+ct} u_1(y)\mathrm{d}y & 0 \leqslant x \leqslant ct\,. \end{cases} \tag{17.6}$$

A similar formula holds for $t \leqslant 0$.

Proof. Notice that the requirements $u_0(0) = u_0''(0) = u_1(0) = 0$ are necessary compatibility conditions —*cf.* Equation (15.3)— because

$$0 = u(0,t)|_{t=0} = u_0(0)\,, \quad 0 = \frac{\partial u}{\partial t}(0,t)\Big|_{t=0} = u_1(0)\,, \quad 0 = \frac{1}{c^2}\cancel{\frac{\partial^2 u}{\partial t^2}}\Big|_{\substack{x=0\\t=0}} - \frac{\partial^2 u}{\partial x^2}\Big|_{\substack{x=0\\t=0}} = -u_0''(0)\,,$$

where we used Equation (15.1a) together with the boundary condition $u(0,t) = 0$ for all $t \in \mathbb{R}$.

Let u be a solution to (17.5) with Cauchy data u_0, u_1: We will prove that u fulfils Equation (17.6). Notice that the latter formula defines a solution to (17.5) by direct inspection.

We consider the functions $\widehat{u} \in C^2(\mathbb{R}^2)$, $\widehat{u}_0 \in C^2(\mathbb{R})$, $\widehat{u}_1 \in C^1(\mathbb{R})$ defined by odd-reflection of u, u_0, u_1 at $x = 0$, namely:

$$\widehat{u}(x,t) := \begin{cases} u(x,t) & x \geqslant 0 \\ -u(-x,t) & x < 0 \end{cases} \quad \widehat{u}_0(x) := \begin{cases} u_0(x) & x \geqslant 0 \\ -u_0(-x) & x < 0 \end{cases} \quad \widehat{u}_1(x) := \begin{cases} u_1(x) & x \geqslant 0 \\ -u_1(-x) & x < 0 \end{cases}.$$

Notice that the extension preserves the regularity also at $x = 0$ because of the conditions $u_0(0) = u_0''(0) = u_1(0) = u(0,t) = 0$. To wit, $\widehat{u}$ is C^2 for all $t \in \mathbb{R}$ and $x \in \mathbb{R}\backslash\{0\}$ because so is u. Furthermore, it is continuous at $x = 0$ because $u(0,t) = 0$, moreover,

$$\lim_{x\to 0^+} \frac{\partial \widehat{u}}{\partial x}(x,t) = \frac{\partial u}{\partial x}(0,t) = \lim_{x\to 0^-} \frac{\partial \widehat{u}}{\partial x}(x,t)\,,$$

showing that $\frac{\partial \hat{u}}{\partial x}(0,t)$ exists and is continuous at $x = 0$, thus u is C^1 at $x = 0$. With a similar argument we have

$$\lim_{x\to 0^\pm} \frac{\partial^2 \hat{u}}{\partial x^2}(x,t) = \pm\frac{\partial^2 u}{\partial x^2}(0,t) = \pm\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}(0,t) = 0\,.$$

Thus, $\hat{u}$ is C^2 at $x = 0$. A similar argument shows that $\hat{u}_0 \in C^2(\mathbb{R})$ and $\hat{u}_1 \in C^1(\mathbb{R})$.

By direct inspection $\hat{u}$ is a solution to the Cauchy problem (17.1) for $n = 1$ and initial data $\hat{u}_0, \hat{u}_1$. Thus Theorem 17.2 applies and $\hat{u}$ can be computed using Equation (17.3). Restricting to $x \geqslant 0$ leads to

$$\begin{aligned} u(x,t) &= \frac{1}{2}\left[\hat{u}_0(x+ct) + \hat{u}_0(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} \hat{u}_1(y)\mathrm{d}y \\ &= \begin{cases} \frac{1}{2}\left[u_0(x+ct) + u_0(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} u_1(y)\mathrm{d}y & 0 \leqslant ct \leqslant x \\ \frac{1}{2}\left[u_0(x+ct) - u_0(ct-x)\right] + \frac{1}{2c}\int_{ct-x}^{x+ct} u_1(y)\mathrm{d}y & 0 \leqslant x \leqslant ct \end{cases} \end{aligned}$$

where in the second line we computed

$$\int_{x-ct}^{x+ct} \hat{u}_1(y)\mathrm{d}y = \int_0^{x+ct} u_1(y)\mathrm{d}y - \int_{x-ct}^{0} u_1(-y)\mathrm{d}y = \int_{ct-x}^{ct+x} u_1(y)\mathrm{d}y\,.$$

□

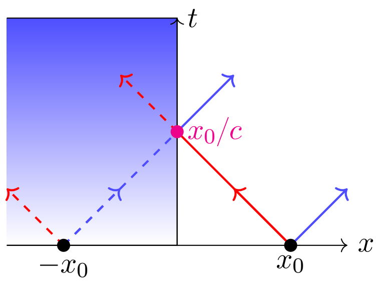


Figure 10: Reflection occuring in Equation (17.6): The progressive wave hits the boundary and is reflected to a regressive wave. Equivalently, the progressive wave from $(x,0)$ escapes the half-line $\mathbb{R}_+$ but it is simultaneously substituted by the regressive wave coming from the event $(-x,0)$.

Remark 17.5: The different expressions found in (17.6) are interpreted as a reflection across the boundary $x = 0$. Indeed, let us consider a signal u_0 "concentrated" at x_0, *i.e.* $\mathrm{supp}(u_0) \subset B_\varepsilon(x_0)$

with $\varepsilon \ll 1$. The solution to the Cauchy problem (17.5) with data u_0 and $u_1 = 0$ is given by

$$u(x,t) = \begin{cases} \frac{1}{2}\left[u_0(x+ct) + u_0(x-ct)\right] & 0 \leqslant ct \leqslant x \\ \frac{1}{2}\left[u_0(x+ct) - u_0(ct-x)\right] & 0 \leqslant x \leqslant ct \end{cases}.$$

For small times, $ct \ll x_0$, the solution behaves similarly to the standard D'Alembert formula, with a progressive and a regressive wave departing from x_0. However, at $ct = x_0$, the progressive wave hits the boundary at $x = 0$ and is reflected to a regressive wave, moving from $u_0(x - ct)$ to $u_0(ct - x)$.

18 Hyperbolic PDEs: Kirchoff's and Poisson's formulas

In this section we will discuss representation formulas for the Cauchy problem (17.1) for $n > 1$. We shall prove an analogous of formula (17.3) in dimension $n = 2, 3$ —*cf.* [3, Thm. 2-3] for the case of arbitrary $n \in \mathbb{N}$.

Let $u \in C^2(\mathbb{R}^{n+1})$ be a solution to the wave equation (15.1a) for $n \in \mathbb{N}$ arbitrary. Following [3], we introduce the **spherical mean** of u, namely the function $U\colon \mathbb{R}^{n+1} \times \overline{\mathbb{R}_+} \to \mathbb{R}$ defined by

$$U(x,t,R) := \frac{1}{|\partial B_R(x)|} \oint_{+\partial B_R(x)} u(y,t)\mathrm{d}S(y)\,. \tag{18.1}$$

where we set $U(x,t,0) := \lim_{R\to 0^+} U(x,t,0) = u(x,t)$ the last equality being a consequence of Lemma 9.6. We shall derive a PDE for U which we will solve in the particular case of $n = 2, 3$. To this avail we prove the following Lemma —*cf.* [3, Lem. 1].

Lemma 18.1 (Euler-Poisson-Darboux equation): Let $u \in C^2(\mathbb{R}^{n+1})$ be a solution to the Cauchy problem (17.1) with initial data u_0, u_1. Let U be defined as per Equation (18.1), moreover, let $U_0, U_1\colon \mathbb{R}^n \times \overline{\mathbb{R}_+} \to \mathbb{R}$ be defined by

$$U_0(x,R) := \frac{1}{|\partial B_R(x)|} \oint_{+\partial B_R(x)} u_0(y)\mathrm{d}S(y)\,, \quad U_0(x,0) := u_0(x)\,, \tag{18.2}$$

$$U_1(x,R) := \frac{1}{|\partial B_R(x)|} \oint_{+\partial B_R(x)} u_1(y)\mathrm{d}S(y)\,, \quad U_1(x,0) := u_1(x)\,. \tag{18.3}$$

Then $U \in C^2(\mathbb{R}^{n+1} \times \overline{\mathbb{R}_+})$, $U_0 \in C^2(\mathbb{R}^n \times \overline{\mathbb{R}_+})$ and $U_1 \in C^1(\mathbb{R}^n \times \overline{\mathbb{R}_+})$, moreover, U solves

$$\frac{1}{c^2}\frac{\partial^2 U}{\partial t^2} - \frac{\partial^2 U}{\partial R^2} - \frac{(n-1)}{R}\frac{\partial U}{\partial R} = 0\,, \tag{18.4a}$$

$$U|_{t=0} = U_0\,, \qquad \left.\frac{\partial U}{\partial t}\right|_{t=0} = U_1\,. \tag{18.4b}$$

Proof.

$\boxed{U \in C^2}$ We observe that, on account of Equations (3.12)

$$U(x,t,R) = \frac{1}{|\partial B_R(x)|} \oint_{+\partial B_R(x)} u(y,t)\mathrm{d}S(y) = \frac{1}{\omega_n} \oint_{+\partial B_1(0)} u(x+Rz,t)\mathrm{d}\Omega_n(z)\,.$$

A direct application of Corollary 9.5 proves the C^2-regularity of U. This is basically due to the fact that u is C^2 and the integration over z is taken over a compact set. To wit, let x_0, t_0, R_0 be arbitrary but fixed and let $(x,t,R) \in N_{x_0,t_0,R_0}$, the latter being an open neighbourhood of x_0, t_0, R_0 with compact closure $\overline{N}_{x_0,t_0,R_0}$. Then

$$\left|\frac{\partial^\alpha u}{\partial x^{\alpha_1}\partial t^{\alpha_2}\partial R^{\alpha_3}}(x+Rz,t)\right| \leqslant \sup_{\substack{(x,t,R)\in\overline{N}_{x_0,t_0,R_0}\\ z\in\partial B_1(0)}} \left|\frac{\partial^\alpha u}{\partial x^{\alpha_1}\partial t^{\alpha_2}\partial R^{\alpha_3}}(x+Rz,t)\right|,$$

for all choices of $\alpha = (\alpha_1, \alpha_2, \alpha_3) \in \mathbb{Z}_+^n \times \mathbb{Z}_+ \times \mathbb{Z}_+$ with $|\alpha| \leqslant 2$. Since $\partial B_1(0)$ is compact we have that condition (ii) is fulfilled, thus, $U \in C^2$ in a neighbourhood of x_0, t_0, R_0.

Therefore, $U \in C^2(\mathbb{R}^{n+1} \times \overline{\mathbb{R}_+})$. A similar argument shows that $U_0 \in C^2(\mathbb{R}^n \times \overline{\mathbb{R}_+})$ and $U_1 \in C^1(\mathbb{R}^n \times \overline{\mathbb{R}_+})$.

$\boxed{\text{(18.4a)}}$ Applying Corollary 9.5 we find

$$\begin{aligned}
\frac{\partial U}{\partial t} &= \frac{1}{|\partial B_R(x)|}\oint_{+\partial B_R(x)} \frac{\partial u}{\partial t}(y,t)\mathrm{d}S(y)\,,\\
\frac{\partial U}{\partial R} &= \frac{1}{\omega_n}\oint_{+\partial B_1(0)} (\nabla_x u)(x+Rz,t)\cdot z\,\mathrm{d}\Omega_n(z)\\
&= \frac{1}{\omega_n R^{n-1}}\oint_{+\partial B_R(x)} (\nabla_y u)(y,t)\cdot\frac{y-x}{R}\mathrm{d}S(y) && \text{Eq. (3.12)}\\
&= \frac{1}{\omega_n R^{n-1}}\int_{B_R(x)} \Delta_y u(y,t)\mathrm{d}^n y && \text{Thm. 8.1}\,.
\end{aligned}$$

It follows that

$$\begin{aligned}
\frac{\partial}{\partial R}\left(R^{n-1}\frac{\partial U}{\partial R}\right) &= \frac{\partial}{\partial R}\frac{1}{\omega_n}\int_{B_R(x)} \Delta_y u(y,t)\mathrm{d}^n y\\
&= \frac{1}{\omega_n}\oint_{+\partial B_R(x)} \Delta_y u(y,t)\mathrm{d}S(y) && \text{Eq. (3.12)}\\
&= \frac{1}{\omega_n}\oint_{+\partial B_R(x)} \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}(y,t)\mathrm{d}S(y) && \text{Eq. (15.1a)}\\
&= \frac{R^{n-1}}{c^2}\frac{\partial^2 U}{\partial t^2} && \text{Cor. 9.5}\,.
\end{aligned}$$

Rearranging the last equation we arrive at Equation (18.4a). The fulfilment of the initial conditions (18.4b) is an immediate consequence of Theorem 9.2 and Corollary 9.5: Indeed

$$\begin{aligned} U(x,0,R) &= \frac{1}{|\partial B_R(x)|} \oint_{+\partial B_R(x)} u(y,0)\mathrm{d}S(y) = U_0(x,R)\,, \\ \frac{\partial U}{\partial t}(x,0,R) &= \frac{1}{|\partial B_R(x)|} \oint_{+\partial B_R(x)} \frac{\partial u}{\partial t}(y,0)\mathrm{d}S(y) = U_1(x,R)\,. \end{aligned}$$

□

Remark 18.2:

(i) Equation (18.4a) is nothing but the wave equation (15.1a) for a function which depends only on $\|x\|$. Indeed let $u(x,t)$ be a solution to (15.1a) such that $u(x,t) = \widetilde{u}(\|x\|,t)$. Then we have

$$\frac{\partial u}{\partial x^k} = \frac{x^k}{R}\frac{\partial \widetilde{u}}{\partial R} \qquad \Delta_x u = \delta^{jk}\frac{\partial^2 u}{\partial x^k \partial x^j} = \frac{n-1}{R}\frac{\partial \widetilde{u}}{\partial R} + \frac{\partial^2 \widetilde{u}}{\partial R^2}\,.$$

It follows that

$$0 = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \Delta_x u = \frac{1}{c^2}\frac{\partial^2 \widetilde{u}}{\partial t^2} - \frac{\partial^2 \widetilde{u}}{\partial R^2} - \frac{(n-1)}{R}\frac{\partial \widetilde{u}}{\partial R}\,.$$

(ii) For later convenience we observe that

$$\begin{aligned} \frac{\partial U}{\partial R} &= \frac{1}{\omega_n} \oint_{+\partial B_1(0)} (\nabla_x u)(x+Rz,t)\cdot z\,\mathrm{d}\Omega_n(z) \\ &\xrightarrow[R\to 0^+]{} \frac{1}{\omega_n} \oint_{+\partial B_1(0)} (\nabla_x u)(x,t)\cdot z\,\mathrm{d}\Omega_n(z) && \text{Thm. 9.2} \\ &= \frac{1}{\omega_n}\int_{B_1(0)} \cancel{\nabla_z \cdot (\nabla_x u)(x,t)}\,\mathrm{d}^n z && \text{Thm. 8.1} \\ &= 0\,. \end{aligned}$$

Lemma 18.1 has the remarkable advantage of turning a PDE in $n+1$ variables (x,t) in a PDE in only 2 variables (R,t) which depends parametrically on $x \in \mathbb{R}^n$. Moreover, Lemma 9.6 ensures that $U(x,t,0) = u(x,t)$ so that, if we find a solution U to (18.4), chances are that we can also perform the limit $R \to 0^+$ and recover a solution to (17.1).

Theorem 18.3 (Kirchoff's formula): Let $u \in C^2(\mathbb{R}^4)$ be a solution to the Cauchy problem for $n = 3$. Then

$$u(x,t) = \frac{1}{|\partial B_{ct}(x)|} \oint\limits_{+\partial B_{ct}(x)} [u_0(y) + \nabla u_0(y) \cdot (y-x) + t u_1(y)]\, \mathrm{d}S(y)\,. \tag{18.5}$$

Proof. We consider the spherical means U, U_0, U_1 of the solution u and of the initial data u_0, u_1. By Lemma 18.1 U is a solution to

$$0 = \frac{R}{c^2}\frac{\partial^2 U}{\partial t^2} - R\frac{\partial^2 U}{\partial R^2} - 2\frac{\partial U}{\partial R} = \left[\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial R^2}\right] RU(x,t,R)\,.$$

It follows that, for all $x \in \mathbb{R}^3$, $\widetilde{U}(x,t,R) := RU(x,t,R)$ solves the Cauchy problem on the half-line (17.5) with initial data

$$\widetilde{U}_0(x,R) := RU_0(x,R)\,, \qquad \widetilde{U}_1(x,R) := RU_1(x,R)\,.$$

Notice that the hypotheses of Corollary (17.4) are met because

$$\widetilde{U}_0(x,0) = \widetilde{U}_1(x,0) = 0\,, \qquad \frac{\partial^2 \widetilde{U}_0}{\partial R^2}\bigg|_{R=0} = 2\frac{\partial U_0}{\partial R}\bigg|_{R=0} = \frac{2}{\omega_3} \oint\limits_{+\partial B_1(0)} \nabla u_0(x) \cdot z \, \mathrm{d}\Omega_3(z) = 0\,,$$

where the last equality follows from Theorem 8.1, *cf.* Remark 18.2. It follows that $\widetilde{U}$ is given by formula (17.6), in particular for $R < ct$ we have

$$U(x,t,R) = \frac{1}{2R}\left[\widetilde{U}_0(x, R+ct) - \widetilde{U}_0(x, ct-R)\right] + \frac{1}{2cR}\int_{ct-R}^{R+ct} \widetilde{U}_1(x,y)\mathrm{d}y\,. \tag{18.6}$$

We shall now consider the limit $R \to 0^+$ of Equation (18.6). The left-hand side converges to $u(x,t)$ because of Lemma 9.6. The $\widetilde{U}_0$-contribution on the right-hand side leads to

$$\begin{aligned}
&\lim_{R\to 0^+} \frac{1}{2R}\left[\widetilde{U}_0(x,R+ct) - \widetilde{U}_0(x,ct-R)\right] = \frac{\partial \widetilde{U}_0}{\partial R}(x,ct) \qquad \widetilde{U}_0 \in C^2(\mathbb{R}^3 \times \overline{\mathbb{R}_+}) \\
&= U_0(x,ct) + \frac{ct}{\omega_3} \oint\limits_{+\partial B_1(0)} \nabla_x u_0(x+ctz) \cdot z \, \mathrm{d}\Omega_3(z) \\
&= \frac{1}{|\partial B_{ct}(x)|} \oint\limits_{+\partial B_{ct}(x)} [u_0(y) + \nabla u_0(y) \cdot (y-x)]\, \mathrm{d}S(y) \qquad \text{Eq. (3.12)}\,.
\end{aligned}$$

Similarly, using the mean-value theorem [18, Thm. 9.1], the $\widetilde{U}_1$-contribution is

$$\lim_{R\to 0^+} \frac{1}{2cR}\int_{ct-R}^{R+ct} \widetilde{U}_1(x,y)\mathrm{d}y = \frac{1}{c}\widetilde{U}_1(x,ct) = \frac{t}{|\partial B_{ct}(x)|} \oint\limits_{+\partial B_{ct}(x)} u_1(y)\mathrm{d}S(y)\,.$$

Overall the limit as $R \to 0^+$ of Equation (18.6) leads to Equation (18.5). □

THEOREM 18.4 (Poisson's formula): Let $u \in C^2(\mathbb{R}^3)$ be a solution to the Cauchy problem (17.1) for $n = 2$. Then

$$u(x,t) = \frac{ct}{2|B_{ct}(x)|} \int_{B_{ct}(x)} \frac{u_0(y) + \nabla u_0(y) \cdot (y-x) + t u_1(y)}{\sqrt{(ct)^2 - \|x-y\|^2}} \mathrm{d}^2 y \,. \tag{18.7}$$

Proof. Let $\widehat{u} \in C^2(\mathbb{R}^4)$, $\widehat{u}_0 \in C^2(\mathbb{R}^3)$ and $\widehat{u}_1 \in C^1(\mathbb{R}^3)$ the functions defined by lifting u, u_0, u_1, *i.e.*

$$\widehat{u}(x_1,x_2,x_3,t) := u(x_1,x_2,t)\,, \quad \widehat{u}_0(x_1,x_2,x_3) := u_0(x_1,x_2)\,, \quad \widehat{u}_1(x_1,x_2,x_3) := u_1(x_1,x_2)\,.$$

It follows that $\widehat{u}$ solves the Cauchy problem (17.1) for $n = 3$ and with initial data $\widehat{u}_0, \widehat{u}_1$. By Theorem 18.3 we have

$$u(x,t) = \frac{1}{|\partial B^{(3)}_{ct}(x)|} \oint_{+\partial B^{(3)}_{ct}(x)} [\widehat{u}_0(y) + \nabla \widehat{u}_0(y) \cdot (y-x) + t\widehat{u}_1(y)] \,\mathrm{d}S^{(3)}(y) \qquad \forall x \in \mathbb{R}^2\,,\ \forall t \in \mathbb{R}\,.$$

where $B^{(3)}_{ct}(x)$ denotes the 3-dimensional ball of radius ct and centre x.

The latter integral can be recast to an integral over the 2-dimensional ball $B^{(2)}_{ct}(x)$ as follows. Let $\gamma\colon B^{(2)}_{ct}(x) \ni y \mapsto \sqrt{(ct)^2 - \|x-y\|^2} \in \mathbb{R}$: Then

$$\begin{aligned}\partial B^{(3)}_{ct}(x) &= \{(y,\gamma(x)) \mid y \in B^{(2)}_{ct}(x)\} \cup \{(y,-\gamma(x)) \mid y \in B^{(2)}_{ct}(x)\} \\ &= \mathrm{graph}(\gamma) \cup \mathrm{graph}(-\gamma)\,.\end{aligned}$$

i.e. $B^{(2)}_{ct}(x)$ parametrizes the north and south hemisphere of $\partial B^{(3)}_{ct}(x)$. Thus, we may use the area formula [6, §15] to compute the integral over $\partial B^{(3)}_{ct}(x)$.

To wit, let $f \in C^0(\mathbb{R}^2)$ and let $\widehat{f} \in C^0(\mathbb{R}^3)$ be the function obtained by setting $\widehat{f}(x_1,x_2,x_3) := f(x_1,x_2)$. It follows that

$$\begin{aligned}
&\frac{1}{|\partial B^{(3)}_{ct}(x)|} \oint_{+\partial B^{(3)}_{ct}(x)} \widehat{f}(y)\mathrm{d}S^{(3)}(y) \\
&= \frac{2}{4\pi(ct)^2} \int_{B^{(2)}_{ct}(x)} \widehat{f}(y)\sqrt{1 + \|\nabla\gamma(y)\|^2}\,\mathrm{d}^2 y && \partial B^{(3)}_{ct}(x) = \mathrm{graph}(\gamma) \cup \mathrm{graph}(-\gamma) \\
&= \frac{1}{2\pi ct} \int_{B^{(2)}_{ct}(x)} \frac{f(y)}{\sqrt{(ct)^2 - \|x-y\|^2}}\,\mathrm{d}^2 y && \|\nabla\gamma(y)\|^2 = \frac{\|x-y\|^2}{\gamma(y)^2} \\
&= \frac{ct}{2|B^{(2)}_{ct}(x)|} \int_{B^{(2)}_{ct}(x)} \frac{f(y)}{\sqrt{(ct)^2 - \|x-y\|^2}}\,\mathrm{d}^2 y\,.
\end{aligned}$$

Inserting this equality in Kirchoff's formula for u leads to (18.7). □

Remark 18.5:

(i) Kirchoff's and Poisson's representation formulas (18.5)-(18.7) are necessary conditions for the solution u to the Cauchy problem (17.1). However, in general they cannot be used to define the solution to such problems: In particular u may fail to be C^2. Indeed, let consider $n = 3$, $u_0 = 0$ and $u_1 \in C^1(\mathbb{R}^3)\backslash C^2(\mathbb{R}^3)$ be defined by

$$u_1(y) := \begin{cases} 1 & \|y\| \leqslant 1 \\ e^{-(1-\|y\|)^2} & \|y\| > 1 \end{cases} .$$

Then Kirchoff's formula (18.5) entails that

$$u(0,t) = \frac{t}{|\partial B_{ct}(0)|} \oint_{+\partial B_{ct}(0)} u_1(y) \mathrm{d}S(y) = t u_1(ct) ,$$

which is not C^2.

A sufficient condition which guarantees that Equations (18.5)-(18.7) define a solution to the Cauchy problem (17.1) is $u_0 \in C^3(\mathbb{R}^n)$ and $u_1 \in C^2(\mathbb{R}^n)$ *cf.* [21, §3.4].

(ii) Kirchoff's formula (18.5) and Poisson's formula (18.7) are particular cases of more general representation formulas which hold for the solution to the Cauchy problem (17.1) with $n \in \mathbb{N}$ arbitrary. Representation formulas for the non-homogeneous problems are also available: We refer to [3] for further details.

(iii) Notably Kirchoff's formula (18.5) and Poisson's formula (18.7) abide by the result of Theorem 16.5 in a slight different way. In particular, for $n = 3$ the support of u is contained in

$$\partial J[(\mathrm{supp}(u_0) \cup \mathrm{supp}(u_1)) \times \{0\}] ,$$

that is, the support is contained on the boundary of the causal cones emanated from the initial data u_0, u_1. In particular $u(x,t)$ does not vanish if and only if there exists $y \in \mathrm{supp}(u_0) \cup \mathrm{supp}(u_1)$ such that $\|x - y\| = ct$. Conversely, in $n = 2$, the support of u fulfils the standard inclusion (16.5), in particular $\mathrm{supp}(u)$ is contained in the causal cones emanated from the initial data u_0, u_1.

This slight different behaviour is known as **Huygens' principle**. Notably, it applies for all dimensions $n \in \mathbb{N}$. In particular if $n \in \mathbb{N}$ is odd then $\mathrm{supp}(u)$ is contained in the boundary of $J[(\mathrm{supp}(u_0) \cup \mathrm{supp}(u_1)) \times \{0\}]$. Conversely, for n even $\mathrm{supp}(u)$ abides by the standard inclusion (16.5).

19 Hyperbolic PDEs: The Cauchy problem on $\mathbb{S}^1$

In this section we will consider the Cauchy problem for the Klein-Gordon equation (2.9) for $n = 1$ on the circle $\mathbb{S}^1_R$ of fixed radius $R > 0$, *cf.* Definition 19.1. We will prove uniqueness for the solution u to such problem, moreover, we will also provide sufficient conditions for the existence of such solution.

A little bit of Calculus on $\mathbb{S}^1$

Before we dwell into the details of problem (19.4), it is worth introducing the definition of C^k-functions on $\mathbb{S}^1_R$ and of the Laplacian $\Delta_{\mathbb{S}^1_R}$. The following definitions are particular instances of more general framework and generalize for example to the case of a d-dimensional sphere.

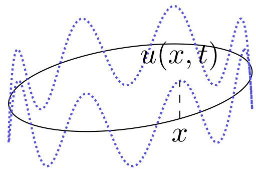


Figure 11: Transverse vibrations of a circular string.

In what follows we will denote by $\boxed{\mathbb{S}^1_R} := \{x \in \mathbb{R}^2 \,|\, \|x\| = R\}$ the 2D circle of radius $R > 0$.

By introducing polar coordinates, *cf.* Remark 3.7, we may identify $\mathbb{S}^1_R$ with the quotient $\mathbb{R}/2\pi\mathbb{Z}$, the identification being

$$\mathbb{R}/2\pi\mathbb{Z} \ni \theta \mapsto (R\cos\theta, R\sin\theta) \in \mathbb{S}^1_R\,.$$

We will also identify $\mathbb{R}^2 = \mathbb{C}$ and sloppily write $Re^{i\theta} \in \mathbb{S}^1_R$.

A function $f\colon \mathbb{S}^1_R \to \mathbb{C}$ can be extended to $\check{f}\colon \mathbb{R}^2 \to \mathbb{C}$ by the following procedure. Let $\varepsilon \in (0,1)$ and $\varrho \in C^\infty(\mathbb{R}_+)$ be such that $\operatorname{supp}(\varrho) \subset (R-\varepsilon, R+\varepsilon)$ and $\varrho|_{(R-\varepsilon/2, R+\varepsilon/2)} = 1$: Then we may define $\check{f}(x) := \varrho(\|x\|)f(Rx/\|x\|)$. By direct inspection $\check{f}$ is well-defined, moreover, $\check{f}|_{\mathbb{S}^1_R} = f$. We will say that $f \in \boxed{C^k(\mathbb{S}^1_R)}$ if $\check{f} \in C^k(\mathbb{R}^2)$ for one (hence, any) extension of f as described above. By exploiting the identification $\mathbb{S}^1_R \simeq \mathbb{R}/2\pi\mathbb{Z}$ one realizes that $C^k(\mathbb{S}^1_R)$ is isomorphic to the vector space $\boxed{C^k_\pi(\mathbb{R})}$ of C^k-functions on $\mathbb{R}$ with periodic conditions on all derivatives:

$$C^k_\pi(\mathbb{R}) := \{f \in C^k(\mathbb{R}) \,|\, f^{(\ell)}(\theta) = f^{(\ell)}(\theta + 2\pi)\ \forall \theta \in \mathbb{R}\ \forall \ell \in \{0, \ldots, k\}\}\,.$$

Explicitly the isomorphism of vector spaces $C^k(\mathbb{S}^1_R) \simeq C^k_\pi(\mathbb{R})$ is given by

$$C^k(\mathbb{S}^1_R) \ni f \mapsto f_\pi \in C^k_\pi(\mathbb{R}) \qquad f_\pi(\theta) := f(Re^{i\theta})\,. \tag{19.1}$$

In the forthcoming discussion we will frequently identify $C^k(\mathbb{S}^1_R)$ and $C^k_\pi(\mathbb{R})$ and drop the distinction between f and f_π.

The Laplacian $\boxed{\Delta_{\mathbb{S}^1_R}}$ on $\mathbb{S}^1_R$ is the differential operator $\Delta_{\mathbb{S}^1_R}\colon C^2(\mathbb{S}^1_R) \to C^0(\mathbb{S}^1_R)$ defined by

$$\Delta_{\mathbb{S}^1_R} f := (\Delta\check{f})|_{\mathbb{S}^1_R}\,, \tag{19.2}$$

here $\check{f} \in C^2(\mathbb{R}^2)$ is any extension of f to $\mathbb{R}^2$ defined as above. In fact, an explicit computation proves that

$$[\Delta_{\mathbb{S}^1_R} f]_\pi = \frac{1}{R^2}\frac{\partial^2 f_\pi}{\partial\theta^2}\,. \tag{19.3}$$

Indeed, let $\tilde{f}(r,\theta) := \check{f}(x(r,\theta), y(r,\theta))$ be the function $\check{f}$ written in polar coordinates: In particular we have $\tilde{f}(r,\theta) = \varrho(r) f(Re^{i\theta}) = \varrho(r) f_\pi(\theta)$. A lengthy but instructive computation shows that

$$\frac{\partial\check{f}}{\partial x} = \frac{\partial r}{\partial x}\frac{\partial\tilde{f}}{\partial r} + \frac{\partial\theta}{\partial x}\frac{\partial\tilde{f}}{\partial\theta} = \frac{x}{r}\frac{\partial\tilde{f}}{\partial r} - \sin\theta\frac{1}{r}\frac{\partial\tilde{f}}{\partial\theta}\,, \qquad \frac{\partial\check{f}}{\partial y} = \frac{y}{r}\frac{\partial\tilde{f}}{\partial r} + \cos\theta\frac{1}{r}\frac{\partial\tilde{f}}{\partial\theta}\,.$$

It follows that

$$[(\Delta\check{f})|_{\mathbb{S}^1_R}]_\pi = \left(\frac{\partial^2\tilde{f}}{\partial r^2} + \frac{1}{r}\frac{\partial\tilde{f}}{\partial r} + \frac{1}{r^2}\frac{\partial^2\tilde{f}}{\partial\theta^2}\right)\Bigg|_{r=R} = \frac{1}{R^2}\frac{\partial^2 f_\pi}{\partial\theta^2}\,,$$

where we used the support properties of ϱ.

We are finally in position to introduce the Cauchy problem we are interested in this section.

Definition 19.1: The Cauchy problem for Klein-Gordon equation on $\mathbb{S}^1_R$ consists of finding $u \in C^2(\mathbb{S}^1_R \times \mathbb{R})$ such that

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \Delta_{\mathbb{S}^1_R} u + \mu^2 u = 0\,, \qquad \forall (x,t) \in \mathbb{S}^1_R \times \mathbb{R} \tag{19.4a}$$

$$u(x,0) = u_0(x)\,, \quad \frac{\partial u}{\partial t}\Big|_{t=0} = u_1(x)\,, \qquad \forall x \in \mathbb{S}^1_R\,. \tag{19.4b}$$

where $c, \mu, R > 0$ while, $u_0 \in C^2(\mathbb{S}^1_R)$ and $u_1 \in C^1(\mathbb{S}^1_R)$.

Remark 19.2: The identification $C^k(\mathbb{S}^1_R) \simeq C^k_\pi(\mathbb{R})$ and Equation (19.3) allow to recast problem (19.4) in a fairly explicit form. In particular (19.4) is equivalent to find $u \in C^2_\pi(\mathbb{R}^2)$ (with a slight abuse of notation the index $_\pi$ refers to the periodicity of the first argument of u only) such that

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \frac{1}{R^2}\frac{\partial^2 u}{\partial\theta^2} + \mu^2 u = 0\,, \qquad \forall(\theta,t) \in \mathbb{R}\times\mathbb{R} \tag{19.5a}$$

$$u(\theta,0) = u_0(\theta)\,, \quad \frac{\partial u}{\partial t}\Big|_{t=0} = u_1(\theta)\,, \qquad \forall\theta \in \mathbb{R} \tag{19.5b}$$

$$\frac{\partial^k u}{\partial\theta^k}(\theta,t) = \frac{\partial^k u}{\partial\theta^k}(\theta+2\pi,t)\,, \qquad \forall(\theta,t)\in\mathbb{R}\times\mathbb{R}\ \forall k\in\{0,1,2\}\,. \tag{19.5c}$$

where $u_0 \in C^2_\pi(\mathbb{R})$ and $u_1 \in C^1_\pi(\mathbb{R})$. Notice that the third condition in (19.5c) is a consequence of Equation (15.1) together with the regularity requirement on u since

$$0 = \left(\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \frac{1}{R^2}\frac{\partial^2 u}{\partial\theta^2} + \mu^2 u\right)\Bigg|_{\theta=0}^{\theta=2\pi} = \frac{\partial^2 u}{\partial\theta^2}(0,t) - \frac{\partial^2 u}{\partial\theta^2}(2\pi,t)\,.$$

A (not so short) detour on Fourier series

To proceed further in the analysis of problem (19.4) we will need to recall some result of Fourier analysis on $\mathbb{S}^1_R$. To this avail, we recall that the identification $\mathbb{S}^1_R \simeq \mathbb{R}/2\pi\mathbb{Z}$ and the isomorphism $C^k(\mathbb{S}^1_R) \simeq C^k_\pi(\mathbb{R})$ allow to define the L^2-space $\boxed{L^2(\mathbb{S}^1_R)}$ as the L^2-space $L^2(0, 2\pi)$. In particular, the L^2-scalar product on $L^2(\mathbb{S}^1_R)$ is given by

$$(f, g)_{L^2(\mathbb{S}^1_R)} = \int_0^{2\pi} \overline{f(\theta)} g(\theta) \mathrm{d}\theta \,.$$

The space $L^2(\mathbb{S}^1_R)$ is an **Hilbert space**, *cf.* [16, §4] for further details. We recall that $C^0(\mathbb{S}^1_R, \mathbb{C})$ is a dense subspace of $L^2(\mathbb{S}^1_R)$ [16, §3], moreover, for all $f \in C^0(\mathbb{S}^1_R, \mathbb{C})$ we have

$$\|f\|_{L^2(\mathbb{S}^1_R)} \leqslant \sqrt{2\pi} \|f\|_\infty \,, \qquad \|f\|_\infty := \sup_{\mathbb{S}^1_R} |f| \,. \tag{19.6}$$

Within the framework of Hilbert spaces, the concept of **complete orthonormal basis** comes in handy. Rather than dwelling on the general case, we shall prove Theorem 19.4 —*cf.* [11, Thm. 6.5] and [16, §4]— which deals with the particular case at hand. To this avail we recall the following theorem, *cf.* [17, Thm. 5.7].

THEOREM 19.3 (Stone-Weierstrass' approximation theorem): Let X be a compact Hausdorff topological space and let $\mathcal{A} \subset C^0(X, \mathbb{C})$ be such that:

(I) $\mathcal{A}$ is a subalgebra, that is, $\alpha f + \beta g \in \mathcal{A}$ and $fg \in \mathcal{A}$ for all $f, g \in \mathcal{A}$, $\alpha, \beta \in \mathbb{C}$;

(II) $\mathcal{A}$ is self-adjoint, namely $f^* \in \mathcal{A}$ for all $f \in \mathcal{A}$ —here $f^*(x) := \overline{f(x)}$;

(III) $\mathcal{A}$ separates points in X, that is, for all $x, y \in X$, $x \neq y$, there exists $f_{xy} \in \mathcal{A}$ such that $f_{xy}(x) \neq f_{xy}(y)$;

(IV) for all $x \in X$ there exists $f_x \in \mathcal{A}$ such that $f_x(x) \neq 0$.

Then $\overline{\mathcal{A}} = C^0(X, \mathbb{C})$ where the closure is meant in the topology induced by the norm of $C^0(X, \mathbb{C})$, that is, $\|f\| := \sup_X |f|$ for all $f \in C^0(X, \mathbb{C})$.

THEOREM 19.4: For all $n \in \mathbb{Z}$ let $e_n \in L^2(\mathbb{S}^1_R)$ be the (equivalence class of the) function defined by

$$e_n(\theta) := \frac{e^{in\theta}}{\sqrt{2\pi}} \,. \tag{19.7}$$

Then the collection $\{e_n \,|\, n \in \mathbb{Z}\}$ is a complete orthonormal basis of $L^2(\mathbb{S}^1_R)$, namely it holds

$$(e_n, e_k)_{L^2(\mathbb{S}^1_R)} = \delta_{nk} \; \forall n, k \in \mathbb{Z} \,, \qquad \overline{\operatorname{span}\{e_n \,|\, n \in \mathbb{Z}\}} = L^2(\mathbb{S}^1_R) \,. \tag{19.8}$$

In particular the following properties hold true:

(i) for all $f \in L^2(\mathbb{S}^1_R)$ we have

$$\lim_{N\to+\infty}\Big\| f - \sum_{|n|\leqslant N} \widehat{f}(n)e_n \Big\|_{L^2(\mathbb{S}^1_R)} = 0\,, \qquad \widehat{f}(n) := (e_n, f)_{L^2(\mathbb{S}^1_R)} = \frac{1}{\sqrt{2\pi}}\int_0^{2\pi} f(\theta)e^{-in\theta}\mathrm{d}\theta\,. \tag{19.9}$$

$$\|f\|^2_{L^2(\mathbb{S}^1_R)} = \sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2\,. \tag{19.10}$$

(ii) The map $L^2(\mathbb{S}^1_R) \ni f \mapsto (\widehat{f}(n))_{n\in\mathbb{Z}} \in \ell^2(\mathbb{Z})$ is a unitary isomorphism of Hilbert spaces. In particular, for all $f, g \in L^2(\mathbb{S}^1_R)$ it holds

$$(f, g)_{L^2(\mathbb{S}^1_R)} = \sum_{n\in\mathbb{Z}} \overline{\widehat{f}(n)}\widehat{g}(n)\,. \tag{19.11}$$

Proof. We shall only prove (19.8): The rest of the properties follows from the general results on complete orthonormal bases, *cf.* [16, §4].

We begin by observing that for all $n, k \in \mathbb{Z}$, $n \neq k$,

$$(e_n, e_k)_{L^2(\mathbb{S}^1_R)} = \frac{1}{2\pi}\int_0^{2\pi} e^{i(k-n)\theta}\mathrm{d}\theta = \begin{cases} \dfrac{1}{2\pi}\dfrac{e^{i(k-n)\theta}}{i(k-n)}\Big|_{\theta=0}^{\theta=2\pi} & n \neq k \\ 1 & n = k \end{cases}\,.$$

This shows that the set $\{e_n \,|\, n \in \mathbb{Z}\}$ is made by orthonormal functions, therefore, they are linearly independent. We shall now prove that $\mathcal{A} := \mathrm{span}\{e_n \,|\, n \in \mathbb{Z}\}$ is dense in $L^2(\mathbb{S}^1_R)$ by applying Theorem 19.3. We observe that $\mathcal{A}$ is a subalgebra of $C^0(\mathbb{S}^1_R, \mathbb{C})$ and it is self-adjoint: These properties follows from the equalities

$$e_n e_k = e_{n+k}\,, \qquad e_n^* = e_{-n}\,.$$

Moreover, $e_0 \equiv 1 \in \mathcal{A}$ so that for all $x \in \mathbb{S}^1$ there exists an element $f_x \in \mathcal{A}$ such that $f_x(x) \neq 0$.

Finally, for all $x, x' \in \mathbb{S}^1_R$, $x \neq x'$, there exists $f_{x,x'} \in \mathcal{A}$ such that $f_{x,x'}(x) \neq f_{x,x'}(x')$. Indeed, notice that $x = Re^{i\theta}$, $x' = Re^{i\theta'}$ for $\theta, \theta' \in \mathbb{R}/2\pi\mathbb{Z}$, moreover, $\theta - \theta' \notin 2\pi\mathbb{Z}$ because otherwise $x = x'$. Let consider the function $f_{x,x'} \in \mathcal{A}$ defined by

$$f_{x,x'}(\vartheta) := \alpha\cos(\vartheta) + \beta\sin(\vartheta)\,,$$

for $\alpha, \beta \in \mathbb{C}$ to be determined —notice that $f_{x,x'} \in \mathcal{A}$ by Euler's formula (19.12). We then have

$$\begin{matrix} f_{x,x'}(x) = a \\ f_{x,x'}(x') = b \end{matrix} \quad \Leftrightarrow \quad \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ \cos(\theta') & \sin(\theta') \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}\,,$$

which is a linear system for α, β where we are free to choose $a, b \in \mathbb{R}$ with $a \neq b$. The determinant of the associated matrix is

$$\cos(\theta)\sin(\theta') - \cos(\theta')\sin(\theta) = \sin(\theta' - \theta) \neq 0\,,$$

provided $\theta - \theta' \notin \pi\mathbb{Z}$. In the latter case we may choose $a = 0$, $b = 1$, and let α, β be the unique solution to the resulting system. Whenever $\theta - \theta' = \pi(2k+1)$, $k \in \mathbb{Z}_+$ we may set $\beta = 0$ and $\alpha = 1$ and consider $f_{xx'}(\vartheta) = \cos\vartheta$: In this latter case $f_{xx'}(\theta) = f_{xx'}(\theta' + \pi) = -f_{xx'}(\theta')$ which separates between x, x' as long as $\cos(\theta') \neq 0$ — if $\cos(\theta') = 0$ we set $f_{xx'}(\vartheta) := \sin\vartheta$.

By Theorem 19.3 $\mathcal{A}$ is C^0-dense in $C^0(\mathbb{S}^1_R, \mathbb{C})$. Since $C^0(\mathbb{S}^1_R, \mathbb{C})$ is dense in $L^2(\mathbb{S}^1_R)$ it follows that $\mathcal{A}$ is dense in $L^2(\mathbb{S}^1_R)$. Indeed for all $f \in L^2(\mathbb{S}^1_R)$ and $\varepsilon > 0$ there exists $f_\varepsilon \in C^0(\mathbb{S}^1_R, \mathbb{C})$ such that $\|f - f_\varepsilon\|_{L^2(\mathbb{S}^1_R)} < \varepsilon$. Moreover, there exists $f'_\varepsilon \in \mathcal{A}$ such that $\|f_\varepsilon - f'_\varepsilon\|_{C^0(\mathbb{S}^1_R)} < \varepsilon$. It follows that

$$\begin{aligned}\|f - f'_\varepsilon\|_{L^2(\mathbb{S}^1_R)} &\leqslant \|f - f_\varepsilon\|_{L^2(\mathbb{S}^1_R)} + \|f_\varepsilon - f'_\varepsilon\|_{L^2(\mathbb{S}^1_R)} \\ &\leqslant \|f - f_\varepsilon\|_{L^2(\mathbb{S}^1_R)} + \sqrt{2\pi}\|f_\varepsilon - f'_\varepsilon\|_{C^0(\mathbb{S}^1_R)} < \varepsilon(1 + \sqrt{2\pi})\,,\end{aligned}$$

where we used (19.6). This proves the claimed density. □

Remark 19.5:

(i) The series $\sum_{n\in\mathbb{Z}} \widehat{f}(n) e_n$ is usually called **Fourier series**, while $(\widehat{f}(n))_{n\in\mathbb{Z}}$ are called **Fourier coefficients**. Notice that, on account of Euler's formula

$$e^{i\theta} = \cos\theta + i\sin\theta \qquad x \in \mathbb{R}\,, \tag{19.12}$$

we may rewrite any Fourier series as a series in sine and cosine functions:

$$\begin{aligned}\sum_{n\in\mathbb{Z}} \widehat{f}(n)\frac{e^{in\theta}}{\sqrt{2\pi}} &= \frac{\widehat{f}(0)}{\sqrt{2\pi}} + \sum_{n\geqslant 1}\Big[\frac{\widehat{f}(n) + \widehat{f}(-n)}{\sqrt{2\pi}}\Big]\cos(n\theta) + i\sum_{n\geqslant 1}\Big[\frac{\widehat{f}(n) - \widehat{f}(-n)}{\sqrt{2\pi}}\Big]\sin(n\theta) \\ &= \frac{c_0}{2} + \sum_{n\geqslant 1}\Big[c_n\cos(n\theta) + s_n\sin(n\theta)\Big]\,,\end{aligned} \tag{19.13}$$

where $(c_n)_{n\in\mathbb{Z}_+}$ and $(s_n)_{n\in\mathbb{N}}$ are given by

$$c_n = \frac{1}{\pi}\int_0^{2\pi} f(\theta)\cos(n\theta)\mathrm{d}\theta\,, \qquad s_n = \frac{1}{\pi}\int_0^{2\pi} f(\theta)\sin(n\theta)\mathrm{d}\theta\,,$$

The series in the above identities are convergent in the L^2-norm, *cf.* (19.9). This can be proved by observing that

$$\overline{\mathrm{span}\{e_n \mid n \in \mathbb{Z}\}} = \overline{\mathrm{span}\{(2\pi)^{-1/2}, \pi^{-1/2}\cos(n\theta), \pi^{-1/2}\sin(n\theta) \mid n \in \mathbb{N}\}}\,,$$

moreover the set $\{(2\pi)^{-1/2}, (2\pi)^{-1/2}\cos(n\theta), (2\pi)^{-1/2}\sin(n\theta) \,|\, n \in \mathbb{N}\}$ is made by orthonormal functions. The latter property follows from Euler's formula (19.12), *e.g.*

$$\begin{aligned}
&\frac{1}{2\pi}\int_0^{2\pi} \cos(n\theta)\cos(m\theta)\mathrm{d}\theta \\
&= \frac{1}{2}\Big[(e_n, e_m)_{L^2(\mathbb{S}^1_R)} + (e_{-n}, e_m)_{L^2(\mathbb{S}^1_R)} + (e_n, e_{-m})_{L^2(\mathbb{S}^1_R)} + (e_{-n}, e_{-m})_{L^2(\mathbb{S}^1_R)}\Big] \\
&= \begin{cases} 1 & n = m \\ 0 & n \neq m \end{cases} .
\end{aligned}$$

Finally notice that Equation (19.13) leads to the following interesting approximation formula: In the sense of convergence in the $L^2(\mathbb{S}^1_R)$-norm we have

$$\begin{aligned}
f(\theta) &= \frac{c_0}{2} + \lim_{N\to\infty}\sum_{n=1}^{N}\Big[c_n\cos(n\theta) + s_n\sin(n\theta)\Big] \\
&= \lim_{N\to\infty}\frac{1}{\pi}\int_0^{2\pi} f(\theta')\Big[\frac{1}{2} + \sum_{n=1}^{N}\Big(\cos(n\theta')\cos(n\theta) + \sin(n\theta')\sin(n\theta)\Big)\Big]\mathrm{d}\theta' \\
&= \lim_{N\to\infty}\frac{1}{\pi}\int_0^{2\pi} f(\theta')\Big[\frac{1}{2} + \sum_{n=1}^{N}\cos[n(\theta-\theta')]\Big]\mathrm{d}\theta' \\
&= \lim_{N\to\infty}\frac{1}{2\pi}\int_0^{2\pi} f(\theta')\frac{\sin[(N+\frac{1}{2})(\theta-\theta')]}{\sin[(\theta-\theta')/2]}\mathrm{d}\theta' ,
\end{aligned}$$

where in the last line we computed

$$\begin{aligned}
\frac{1}{2} + \sum_{n=1}^{N}\cos[n(\theta-\theta')] &= \frac{1}{2}\sum_{n=-N}^{N} e^{in(\theta-\theta')} \\
&= \frac{1}{2}\frac{e^{i(N+1)(\theta-\theta')} - e^{-iN(\theta-\theta')}}{e^{i(\theta-\theta')} - 1} = \frac{1}{2}\frac{\sin[(N+\frac{1}{2})(\theta-\theta')]}{\sin[(\theta-\theta')/2]} .
\end{aligned}$$

Theorem 19.4 allows to expand any element of $L^2(\mathbb{S}^1_R)$ in terms of imaginary exponential functions. However, the convergence of the series is only in terms of the $L^2(\mathbb{S}^1_R)$ norm, *cf.* Equation (19.9). It is an unfortunate fact of life that L^p convergence does not imply pointwise convergence in general, although this holds true for a subsequence, *cf.* [16, Thm 3.12]. We will now discuss sufficient criteria to conclude pointwise convergence of the Fourier series, in particular, we will also discuss the regularity of the obtained function.

The following proposition is taken from [11, Prop. 6.2] and deals with the properties of the Fourier coefficients of a regular function. In what follows we say that $f \in C^0(\mathbb{S}^1_R)$ is C^1-piecewise if the associated $f_\pi \in C^0_\pi(\mathbb{R})$ defined as per Equation (19.1) is C^1-piecewise. By definition, this implies that there exists $\ell \in \mathbb{N}$ and $\theta_1, \ldots, \theta_\ell \in \mathbb{R}$ such that $f_\pi|_{[\theta_j,\theta_{j+1}]} \in C^1([\theta_j, \theta_{j+1}])$, *cf.* Section 8.

Proposition 19.6: Let $N \in \mathbb{N} \cup \{0\}$ and let $f \in C^N(\mathbb{S}^1_R)$ be C^{N+1}-piecewise. Then:

1. For all $k \leqslant N+1$ and $n \in \mathbb{Z}$ it holds
$$\widehat{f^{(k)}}(n) = (in)^k \widehat{f}(n)\,. \tag{19.14}$$

2. For all $k \leqslant N$ it holds
$$\sum_{n\in\mathbb{Z}} |n^k \widehat{f}(n)| < +\infty\,, \tag{19.15}$$
that is, $(n^k \widehat{f}(n))_n \in \ell^1(\mathbb{Z})$.

Proof.

1 Since $C^0(\mathbb{S}^1_R, \mathbb{C}) \subset L^2(\mathbb{S}^1_R)$ the sequence $(\widehat{f^{(k)}}(n))_{n\in\mathbb{Z}}$, $k \leqslant N+1$, is well-defined. Moreover, by direct inspection we have
$$\begin{aligned}\widehat{f^{(k)}}(n) = \int_0^{2\pi} f^{(k)}(\theta) \frac{e^{-in\theta}}{\sqrt{2\pi}} \mathrm{d}\theta &= \cancel{f^{(k-1)}(\theta) \frac{e^{-in\theta}}{\sqrt{2\pi}}\Big|_{\theta=0}^{\theta=2\pi}} + \int_0^{2\pi} f^{(k-1)}(\theta)(in) \frac{e^{-in\theta}}{\sqrt{2\pi}} \mathrm{d}\theta \\ &= (in)\widehat{f^{(k-1)}}(n)\,,\end{aligned}$$
which leads to Equation (19.14). Notice that, when integrating by parts, the boundary contribution vanishes on account of the periodicity of $f^{(k)}$.

2 For all $k \leqslant N$ we have $f^{(k+1)} \in L^2(\mathbb{S}^1_R)$, therefore $(\widehat{f^{(k+1)}}(n))_n \in \ell^2(\mathbb{Z})$ by Equation (19.10). It follows that
$$\begin{aligned}|n|^k |\widehat{f}(n)| &= |n|^{k+1} |\widehat{f}(n)| \frac{1}{|n|} \\ &= |\widehat{f^{(k+1)}}(n)| \frac{1}{|n|} && \text{Eq. (19.14)}\,;\, k \leqslant N \\ &\leqslant \frac{1}{2}\Big[|\widehat{f^{(k+1)}}(n)|^2 + \frac{1}{|n|^2}\Big] && |ab| \leqslant \frac{1}{2}[a^2+b^2]\,.\end{aligned}$$
Therefore
$$\sum_{n\in\mathbb{Z}} |n^k||\widehat{f}(n)| \leqslant \frac{1}{2}\Big[\sum_{n\in\mathbb{Z}} |\widehat{f^{(k+1)}}(n)|^2 + \sum_{n\in\mathbb{Z}} \frac{1}{|n|^2}\Big] < +\infty\,.$$
□

Proposition 19.6 implies in particular that the Fourier series of a regular function is uniformly convergent.

Corollary 19.7: Let $f\colon \mathbb{S}^1_R \to \mathbb{C}$ be as in Proposition 19.6. Then the Fourier series of $f, f^{(1)}, \ldots, f^{(N)}$ converge uniformly to $f, f^{(1)}, \ldots, f^{(N)}$.

Proof. We prove that the Fourier series of f converges uniformly to f: The same argument applies for $f^{(k)}$, $k \leqslant N$. Equation (19.15) for $k = 0$ implies that $(\widehat{f}(n))_n \in \ell^1(\mathbb{Z})$. Since $|\widehat{f}(n)e_n| \leqslant |\widehat{f}(n)|/\sqrt{2\pi}$, the series

$$g(\theta) = \sum_{n\in\mathbb{Z}} \widehat{f}(n)e_n(\theta)\,,$$

converges absolutely (in fact, uniformly), moreover, it defines a continuous function $g \in C^0(\mathbb{S}^1_R)$ as a consequence of Corollary 9.4, *cf.* Lemma 9.8. In particular $g \in L^2(\mathbb{S}^1_R)$ and we may compute

$$\begin{aligned}
\widehat{g}(n) = \int_0^{2\pi} g(\theta)\overline{e_n(\theta)}\mathrm{d}\theta &= \int_0^{2\pi} \sum_{k\in\mathbb{Z}} \widehat{f}(k)e_k(\theta)\overline{e_n(\theta)}\mathrm{d}\theta \\
&= \sum_{k\in\mathbb{Z}} \widehat{f}(k)\int_0^{2\pi} e_k(\theta)\overline{e_n(\theta)}\mathrm{d}\theta && \text{Fubini's Thm.} \\
&= \widehat{f}(n) && (e_n, e_k)_{L^2(\mathbb{S}^1_R)} = \delta_{nk}\,,
\end{aligned}$$

where in order to apply Fubini's theorem we observed that the function $(k,\theta) \mapsto \widehat{f}(k)e_k(\theta)\overline{e_n(\theta)}$ is $\mathcal{L}^1(\mathbb{Z}\times\mathbb{R}/2\pi\mathbb{Z}, \sharp\times\mathrm{d}\theta)$. It follows that $[f] = [g]$ in $L^2(\mathbb{S}^1_R)$, therefore, $f = g$ almost everywhere. Finally, since f, g are continuous, the equality must hold everywhere. Indeed, if $x \in \mathbb{S}^1_R$ is such that $f(x) \neq g(x)$ then $f(y) \neq g(y)$ for all y in an open neighbourhood N_x of x: Since $|N_x| > 0$ we have a contradiction with the requirement $f = g$ almost everywhere ☠. Thus, we have

$$f(x) = g(x) = \sum_{n\in\mathbb{Z}} \widehat{f}(n)e_n(x)\,,$$

where the series is uniformly convergent. □

Remark 19.8:

(i) The assumption of Corollary 19.7 can be weaken at the price of considering only pointwise convergence for the Fourier series of f. For further details we refer to [2, §6.3].

(ii) Corollary 19.7 implies a partial converse of Proposition 19.6. Actually, if $(c_n)_{n\in\mathbb{Z}}$ is such that $(n^k c_n)_{n\in\mathbb{Z}} \in \ell^1(\mathbb{Z})$ for all $k \leqslant N$ then the function

$$f := \sum_{n\in\mathbb{Z}} c_n e_n\,,$$

is well-defined and $f \in C^N(\mathbb{S}^1_R, \mathbb{C})$.

Indeed, the series defining f is uniformly convergent on account of the estimate $|c_n e_n(\theta)| \leqslant |c_n|$, thus, $f \in C^0(\mathbb{S}^1_R, \mathbb{C})$. To prove that $f \in C^k(\mathbb{S}^1_R, \mathbb{C})$ for all $k \leqslant N$ we rely on Corollary 9.5. In particular since $(n^k c_n)_{n\in\mathbb{Z}} \in \ell^1(\mathbb{Z})$ condition ii of Corollary 9.5 is fulfilled, therefore, we have $f \in C^k(\mathbb{S}^1_R, \mathbb{C})$ and

$$\frac{\mathrm{d}^k f}{\mathrm{d}\theta^k} = \sum_{n\in\mathbb{Z}} c_n (in)^k e_n(\theta)\,.$$

Back to the circular string

We will now discuss existence and uniqueness of the solution to problem (19.5) by using the methods discussed in the previous section.

Theorem 19.9: If u is a solution to (19.4) then

$$u(\theta, t) = \sum_{n\in\mathbb{Z}} \left[\widehat{u}_0(n)\cos(\omega_n t) + \widehat{u}_1(n)\frac{\sin(\omega_n t)}{\omega_n}\right] e_n(\theta) \qquad \omega_n = c\Big(\frac{n^2}{R^2} + \mu^2\Big)^{1/2}\,. \tag{19.16}$$

In particular problem (19.4) has at most one solution. Moreover, if $u_0 \in C^2(\mathbb{S}^1_R)$, $u_1 \in C^1(\mathbb{S}^1_R)$ are such that u_0 is C^3-piecewise while u_1 is C^2-piecewise, then Equation (19.16) defines a $C^2(\mathbb{S}^1_R \times \mathbb{R})$ function which solves problem (19.5).

Proof. The proof is divided in two steps. We will first prove that any solution u to (19.5) has to fulfil (19.16): In particular this will imply uniqueness. We stress that this first part of the proof holds for all $u_0 \in C^2(\mathbb{S}^1_R)$, $u_1 \in C^1(\mathbb{S}^1_R)$ no matter the additional regularity conditions.

In the second part of the proof we will show that Equation (19.16) defines a solution to (19.5) provided the initial data u_0, u_1 fulfil the required regularity.

$\boxed{!}$ We consider the Fourier expansions of the solution u and of the initial data u_0, u_1:

$$u(\theta, t) = \sum_{n\in\mathbb{Z}} \widehat{u}(n,t) e_n(\theta)\,, \qquad u_0(\theta) = \sum_{n\in\mathbb{Z}} \widehat{u}_0(n) e_n(\theta)\,, \qquad u_1(\theta) = \sum_{n\in\mathbb{Z}} \widehat{u}_1(n) e_n(\theta)\,.$$

Notice that $u(\theta, t)$ is determined almost everywhere by the sequence $(\widehat{u}(n,t))_{n\in\mathbb{Z}}$. We now provide a closed formula for $\widehat{u}(n,t)$, $n \in \mathbb{Z}$: This will prove uniqueness. For that, we observe that Corollary 9.5 applies since $u \in C^2(\mathbb{S}^1_R \times \mathbb{R})$ and $\mathbb{S}^1_R$ is compact. In more details, let $t_0 \in \mathbb{R}$ and $\delta > 0$. Then for all $\theta \in \mathbb{R}$ and $t \in (t_0 - \delta, t_0 + \delta)$ we have

$$\left|\frac{\partial^k u}{\partial t^k}(\theta, t)\overline{e_n(\theta)}\right| \leqslant \frac{1}{\sqrt{2\pi}} \sup_{\substack{\theta\in\mathbb{R}\\ t\in[t_0-\delta, t_0+\delta]}} \left|\frac{\partial^k u}{\partial t^k}(\theta, t)\right| \qquad k \in \{0,1,2\}\,.$$

Thus assumption ii of Corollary 9.5 is fulfilled (twice) and we may conclude that $\widehat{u}(n,t) \in C^2((t_0-\delta, t_0+\delta))$ for all $t_0 \in \mathbb{R}$, thus $\widehat{u}(n, \ ) \in C^2(\mathbb{R})$ for all $n \in \mathbb{Z}$. Moreover,

$$\begin{aligned}
\widehat{u}(n,0) &= \frac{1}{\sqrt{2\pi}}\int_0^{2\pi} u_0(\theta)e^{-in\theta}\mathrm{d}\theta = \widehat{u}_0(n)\,, \\
\frac{\mathrm{d}\widehat{u}}{\mathrm{d}t}(n,0) &= \frac{1}{\sqrt{2\pi}}\int_0^{2\pi} u_1(\theta)e^{-in\theta}\mathrm{d}\theta = \widehat{u}_1(n)\,, \\
\frac{1}{c^2}\frac{\mathrm{d}^2\widehat{u}}{\mathrm{d}t^2} &= \frac{1}{\sqrt{2\pi}}\int_0^{2\pi} \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}e^{-in\theta}\mathrm{d}\theta && \text{Cor. 9.5} \\
&= \frac{1}{\sqrt{2\pi}}\int_0^{2\pi} \frac{1}{R^2}\frac{\partial^2 u}{\partial \theta^2}e^{-in\theta}\mathrm{d}\theta - \mu^2\widehat{u}(n,t) && \text{Eq. (19.4a)} \\
&= -\frac{n^2}{R^2}\widehat{u}(n,t) - \mu^2\widehat{u}(n,t) && \text{integration by parts + Eq. (19.5c)}
\end{aligned}$$

We are thus lead to countably many ordinary differential equations

$$\left(\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \omega_n^2\right)\widehat{u}(n,t) = 0\,, \qquad \widehat{u}(n,0) = \widehat{u}_0(n)\,, \qquad \frac{\mathrm{d}\widehat{u}}{\mathrm{d}t}(n,0) = \widehat{u}_1(n) \qquad \forall n \in \mathbb{Z}\,,$$

whose solutions are given by

$$\widehat{u}(n,t) = \widehat{u}_0(n)\cos(\omega_n t) + \widehat{u}_1(n)\frac{\sin(\omega_n t)}{\omega_n} \qquad \omega_n = c\Big(\frac{n^2}{R^2} + \mu^2\Big)^{1/2}\,. \tag{19.17}$$

Notice in particular that, for $n = \mu = 0$, we have

$$\widehat{u}(0,t) = \widehat{u}_0(0)\cos(c\mu t) + \widehat{u}_1(0)\frac{\sin(c\mu t)}{c\mu} = \widehat{u}_0(0) + \widehat{u}_1(0)t\,. \tag{19.18}$$

$\boxed{\exists}$ Notice that there exists $c_1, c_2 > 0$ such that $c_1|n| \leqslant \omega_n \leqslant c_2|n|$ for all $n \in \mathbb{Z}\backslash\{0\}$: For this reason in what follows we will sloppily denote $\omega_n/n = O(1) = n/\omega_n$. Proposition 19.6 implies that $(n^k\widehat{u}_0(n))_{n\in\mathbb{Z}} \in \ell^1(\mathbb{Z})$ for all $k \in \{0,1,2\}$ while $(n^k\widehat{u}_1(n))_{n\in\mathbb{Z}} \in \ell^1(\mathbb{Z})$ for all $k \in \{0,1\}$. This entails in particular, for all $n \in \mathbb{Z}$,

$$|n^k\widehat{u}(n,t)| \leqslant \left[|n^k\widehat{u}_0(n)| + \frac{n^k}{\omega_n}|\widehat{u}_1(n)|\right] = \left[|n^k\widehat{u}_0(n)| + O(1)|n^{k-1}\widehat{u}_1(n)|\right]\,.$$

which implies that $(n^k\widehat{u}(n,t))_{n\in\mathbb{Z}} \in \ell^1(\mathbb{Z})$ for all $k \in \{0,1,2\}$. It follows that the Fourier series defining u is uniformly convergent.

We now prove that $u \in C^2(\mathbb{S}^1_R \times \mathbb{R})$. To this avail we will apply twice Corollary 9.5, proving that the estimate of Corollary 9.5-ii holds true. For that we consider the sequences of time and space derivatives of $\widehat{u}(n,t)e_n(\theta)$, $n\in\mathbb{Z}$: In particular we have

$$\begin{aligned}
\left|\frac{\partial}{\partial\theta}\widehat{u}(n,t)e_n(\theta)\right| &\leqslant \frac{|n|}{\sqrt{2\pi}}[|\widehat{u}_0(n)|+\frac{1}{\omega_n}|\widehat{u}_1(n)|] \leqslant \frac{1}{\sqrt{2\pi}}[|n\widehat{u}_0(n)|+O(1)|\widehat{u}_1(n)|]\in\ell^1(\mathbb{Z})\,,\\
\left|\frac{\partial}{\partial t}\widehat{u}(n,t)e_n(\theta)\right| &\leqslant \frac{1}{\sqrt{2\pi}}[\omega_n|\widehat{u}_0(n)|+|\widehat{u}_1(n)|] \leqslant O(1)[|n\widehat{u}_0(n)|+|\widehat{u}_1(n)|]\in\ell^1(\mathbb{Z})\,,\\
\left|\frac{\partial^2}{\partial\theta\partial t}\widehat{u}(n,t)e_n(\theta)\right| &\leqslant \frac{n}{\sqrt{2\pi}}[|\omega_n\widehat{u}_0(n)|+|\widehat{u}_1(n)|] \leqslant O(1)[|n^2\widehat{u}_0(n)|+|n\widehat{u}_1(n)|]\in\ell^1(\mathbb{Z})\\
\left|\frac{\partial^2\widehat{u}}{\partial t^2}(n,t)e_n(\theta)\right| &= \frac{1}{\sqrt{2\pi}}[\omega_n^2|\widehat{u}_0(n)|+\omega_n|\widehat{u}_1(n)|] \leqslant O(1)[|n^2\widehat{u}_0(n)|+|n\widehat{u}_1(n)|]\in\ell^1(\mathbb{Z})\\
\left|\frac{\partial^2\widehat{u}}{\partial\theta^2}(n,t)e_n(\theta)\right| &\leqslant O(1)[|n^2\widehat{u}_0(n)|+|n\widehat{u}_1(n)|]\in\ell^1(\mathbb{Z})\,.
\end{aligned}$$

Thus, Corollary 9.5-ii is fulfilled so that $u\in C^2(\mathbb{S}^1_R\times\mathbb{R})$ and we may compute derivatives of u by deriving its Fourier series term by term. This proves that Equation (19.5) defines a solution to Equation (19.5a). Out of Theorem 9.2 we may also evaluate

$$u(\theta,0)=\sum_{n\in\mathbb{Z}}\widehat{u}(n,0)e_n(\theta)=u_0(\theta)\,,\qquad \frac{\partial u}{\partial t}(\theta,0)=\sum_{n\in\mathbb{Z}}\frac{\partial\widehat{u}}{\partial t}(n,0)e_n(\theta)=u_1(\theta)\,,$$

showing that u is a solution to problem (19.4).

□

Remark 19.10:

(i) Let consider $\mu=0$. By Euler formula (19.12) we may turn the coefficients $\hat{u}(n,t)$ of Equation (19.16) into complex exponentials. This leads to a modified expansion given by

$$\begin{aligned}
u(\theta,t)=\frac{1}{2\sqrt{2\pi}}\sum_{n\in\mathbb{Z}}\Big[\widehat{u}_0(n)\Big(&e^{i(n\theta+|n|ct/R)}+e^{i(n\theta-|n|ct/R)}\Big)\\
&-i\frac{\widehat{u}_1(n)}{\omega_n}\Big(e^{i(n\theta+|n|ct/R)}-e^{i(n\theta-|n|ct/R)}\Big)\Big]\,.
\end{aligned}$$

Thus, u is expanded in a series of complex exponentials $e^{in(\theta\pm ct/R)}$: Considering once again Euler formula the latter are linear combinations of real-valued progressive and regressive waves of the form

$$\sin[n(\theta\pm ct/R)]\,,\ \cos[n(\theta\pm ct/R)]\,,\qquad n\in\mathbb{Z}_+\,.$$

This is similar to the result of Section 17, *cf.* Remark 17.3.

(ii) Notice that Equation (19.5) makes sense for all $u_0, u_1 \in L^2(\mathbb{S}^1_R)$, although the resulting Fourier series is only convergent in the L^2-norm. The resulting function u is an $L^2(\mathbb{S}^1_R)$-valued function in time, *i.e*,

$$\boldsymbol{u}\colon \mathbb{R} \to L^2(\mathbb{S}^1_R) \qquad \boldsymbol{u}(t) := [x \mapsto u(x,t)]\,.$$

In this situation we refer to u as a weak solution to the Cauchy problem (19.5). This solves equation (15.1) in the following weak sense [19]. Let $f \in C^\infty_c(\mathbb{S}^1_R \times \mathbb{R}, \mathbb{C})$ be a smooth, compactly supported function. On account of Proposition 19.6 and Corollary 19.7 we have that

$$f(\theta,t) = \sum_{n\in\mathbb{Z}} \widehat{f}(n,t) e_n(\theta)\,,$$

where the series is uniformly convergent. Moreover, derivatives of f can be computed by deriving its Fourier series term by term. In this situation we can prove that

$$\int_{\mathbb{R}}\int_0^{2\pi} \left[\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{1}{R^2}\frac{\partial^2}{\partial\theta^2} + \mu^2\right)\overline{f(\theta,t)}\right] u(\theta,t)\mathrm{d}\theta\mathrm{d}t = 0\,. \tag{19.19}$$

Assuming that u defines a C^2 function the previous equality follows using integration by parts:

$$\begin{aligned}&\int_{\mathbb{R}}\int_0^{2\pi} \left[\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{1}{R^2}\frac{\partial^2}{\partial\theta^2} + \mu^2\right)\overline{f(\theta,t)}\right] u(\theta,t)\mathrm{d}\theta\mathrm{d}t \\ &\qquad = \int_{\mathbb{R}}\int_0^{2\pi} \overline{f(\theta,t)}\left[\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{1}{R^2}\frac{\partial^2}{\partial\theta^2} + \mu^2\right) u(\theta,t)\right]\mathrm{d}\theta\mathrm{d}t = 0\,,\end{aligned}$$

where the boundary contributions are either zero because f is compactly supported or they cancel out because u and f are periodic. Integration by parts is not available if u is defined by Equation (19.16) with data u_0, u_1 which do not fulfils the hypotheses of Theorem 19.9. Nevertheless Equation (19.19) still holds true, indeed

$$\begin{aligned}&\int_{\mathbb{R}}\int_0^{2\pi} \left[\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{1}{R^2}\frac{\partial^2}{\partial\theta^2} + \mu^2\right)\overline{f(\theta,t)}\right] u(\theta,t)\mathrm{d}\theta\mathrm{d}t \\ &= \int_{\mathbb{R}}\sum_{n\in\mathbb{Z}} \left[\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \frac{\omega_n^2}{c^2}\right)\overline{\widehat{f}(n,t)}\right]\widehat{u}(n,t)\mathrm{d}t && \text{Eq. (19.11)} \\ &= \int_{\mathbb{R}}\sum_{n\in\mathbb{Z}} \overline{\widehat{f}(n,t)}\left[\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \frac{\omega_n^2}{c^2}\right)\widehat{u}(n,t)\right]\mathrm{d}t && \text{Fubini Thm. + integration by parts in } t \\ &= 0\,.\end{aligned}$$

20 Hyperbolic PDEs: Stationary waves in 1D

In this section we will analyse the wave equation (15.1a) in a 1D (spatial) dimensional domain with Dirichlet boundary conditions. Specifically we will consider the Cauchy problem with boundary conditions (15.2) for a finite interval $\Omega = (0, R)$, $R > 0$, *cf.* Definition 20.1.

From a physical point of view, the 1D problem models the vibrations of a string with fixed endpoints and can be used to model the sound produced by a guitar. Similarly the 2D problem on the disk models the sound produced by a drum. In this Section we will obtain a result similar in spirit to Theorem 19.9, *cf.* Theorem 20.4. However, the resulting solution will have a slightly different behaviour: The latter will be linked to the characteristic features of musical instruments. For a more comprehensive treatment of this topic we refer to [11, §7].

We will now introduce the Cauchy problem for the wave equation on a finite string with Dirichlet boundary conditions. The problem is similar to (19.5) but with different boundary conditions: Its analysis profits of Theorem 19.9. The following definition is a particular case of Definition 15.1.

Definition 20.1: The Cauchy problem on $(0, R)$ with Dirichlet boundary conditions consists of finding $u \in C^2([0, R] \times \mathbb{R})$ such that

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0\,, \qquad \forall (x,t) \in [0,R] \times \mathbb{R}\,, \tag{20.1a}$$

$$u(x,0) = u_0(x)\,, \quad \left.\frac{\partial u}{\partial t}\right|_{t=0} = u_1(x)\,, \qquad \forall x \in [0,R]\,, \tag{20.1b}$$

$$u(0,t) = u(R,t) = 0\,, \qquad \forall t \in \mathbb{R}\,, \tag{20.1c}$$

where $u_0 \in C^2([0, R])$, $u_1 \in C^1([0, R])$ are assigned.

Remark 20.2:

(i) Problem (20.1) is similar to problem (19.5) with the identification $x = R\theta$ and with the periodic boundary conditions (19.5c) replaced by the Dirichlet boundary conditions (20.1c).

(ii) Similarly to Remark 15.2 the data u_0, u_1 have to fulfil additional compatibility conditions. In particular, proceeding as in the proof of Corollary 17.4, we find

$$u_0(0) = u_0(R) = 0\,, \qquad u_0''(0) = u_0''(R) = 0\,, \qquad u_1(0) = u_1(R) = 0\,. \tag{20.2}$$

In what follows we will prove a uniqueness and existence result for problem 20.1 under suitable assumptions, *cf.* Theorem 20.4. This results strongly relies on Theorem 19.9.

Remark 20.3: For later convenience it is worth observing that the following is an orthonormal bases of $L^2(0,R)$:

$$\Big\{S_n(x) := \sqrt{\frac{2}{R}}\sin\Big(\frac{\pi n}{R}x\Big)\Big\}_{n\in\mathbb{N}}\,, \qquad \Big\{C_n(x) := \sqrt{\frac{2}{R}}\cos\Big(\frac{\pi n}{R}x\Big)\Big\}_{n\in\mathbb{Z}_+}\,. \tag{20.3}$$

Indeed, let us focus on the first set, the proof for the second being analogous. We first observe that for all $f,g\in L^2(0,R)$,

$$(f,g)_{L^2(0,R)} = \int_0^R \overline{f(x)}g(x)\mathrm{d}x = \frac{R}{\pi}\int_0^\pi \overline{f(R\theta/\pi)}g(R\theta/\pi)\mathrm{d}\theta\,.$$

This implies that

$$L^2(0,R)\ni f\mapsto \sqrt{\frac{R}{\pi}}f\Big(\frac{R}{\pi}\,\cdot\Big)\in L^2(0,\pi)\,, \tag{20.4}$$

is a unitary isomorphism of Hilbert spaces. Thus, the basis in (20.3) is an orthonormal basis of $L^2(0,R)$ if and only if $\{s_n(\theta) := \sqrt{\frac{2}{\pi}}\sin(n\theta)\}_{n\in\mathbb{N}}$ is an orthonormal basis of $L^2(0,\pi)$.

To prove that, we observe that the orthogonality relation can be verified with the help of Euler's formula. Concerning completeness, let $f\in L^2(0,\pi)$ and let $F\in L^2(-\pi,\pi)$ be its odd extension

$$F(\theta) = \begin{cases} f(\theta) & \theta\in(0,\pi)\\ -f(-\theta) & \theta\in(-\pi,0)\end{cases}\,.$$

Notice that, since we only care about F up to $\mathrm{d}\theta$-null measure sets, the above definition makes sense. Since $L^2(-\pi,\pi)\simeq L^2(0,2\pi)$ by translation invariance of the Lebesgue measure, we may write

$$\begin{aligned}
F(\theta) &= \sum_{n\in\mathbb{Z}}\widehat{F}(n)\frac{e^{in\theta}}{\sqrt{2\pi}}\\
\widehat{F}(n) &= \frac{1}{\sqrt{2\pi}}\int_{-\pi}^{\pi}F(\theta)e^{-in\theta}\mathrm{d}\theta = \frac{1}{i}\sqrt{\frac{2}{\pi}}\int_0^\pi f(\theta)\sin(n\theta)\mathrm{d}\theta = -i(s_n,f)_{L^2(0,\pi)}\,.
\end{aligned}$$

In particular $\widehat{F}(-n) = -\widehat{F}(n)$, thus,

$$f = F|_{[0,R]} = \sum_{n\geqslant 1}(s_n,f)_{L^2(0,\pi)}s_n\,,$$

as claimed.

As an example of the previous result, the constant function 1 can be decomposed in Fourier series using the orthonormal basis $\{s_n\,|\,n\in\mathbb{N}\}$. The associated Fourier coefficients and Fourier series are given by

$$\begin{aligned}
\widehat{1}(n) &= (s_n, 1) = \sqrt{\frac{2}{\pi}} \int_0^\pi \sin(n\theta) \mathrm{d}\theta \\
&= -\sqrt{\frac{2}{\pi}} \frac{1}{n} \cos(n\theta) \Big|_{x=0}^{x=\pi} = \sqrt{\frac{2}{\pi}} \frac{1}{n} [1 - (-1)^n] = \begin{cases} 0 & n \in 2\mathbb{N} \\ \sqrt{\frac{2}{\pi}} \frac{2}{n} & n \in 2\mathbb{N}+1 \end{cases} \\
1 &= 2\sqrt{\frac{2}{\pi}} \sum_{k \geqslant 0} \frac{1}{2k+1} \sin[(2k+1)\theta] \,.
\end{aligned}$$

We stress that a priori the previous series converges only in the $L^2(0, \pi)$-norm.

Theorem 20.4: If u is a solution to (20.1) then

$$u(x,t) = \sum_{n \geqslant 1} \left[(S_n, u_0)_{L^2(0,R)} \cos(\omega_n t) + (S_n, u_1)_{L^2(0,R)} \frac{\sin(\omega_n t)}{\omega_n} \right] S_n(x) \,, \qquad \omega_n = \frac{c\pi n}{R} \,. \tag{20.5}$$

In particular problem (20.1) has at most one solution. Moreover, if $u_0 \in C^2([0,R])$ and $u_1 \in C^1([0,R])$ are such that

$$u_0(0) = u_0(R) = u_0''(0) = u_0''(R) = 0 \,, \qquad u_0 \text{ is } C^3\text{-piecewise} \tag{20.6}$$

$$u_1(0) = u_1(R) = 0 \,, \qquad u_1 \text{ is } C^2\text{-piecewise} \,, \tag{20.7}$$

then Equation (20.5) defines a $C^2([0,R] \times \mathbb{R})$ function which solves problem (20.1).

Proof. The proof rely on an argument similar to the one used in Corollary 17.4 and in Remark 19.5, together with the results of Theorem 19.9.

Let $u \in C^2([0,R] \times \mathbb{R})$ be a solution to problem (20.1): By Remark 20.2 the data u_0, u_1 must fulfil the compatibility conditions (20.2).

We apply the unitary isomorphism (20.4) and define $\tilde{u} \in C^2([0,\pi] \times \mathbb{R})$ by

$$\tilde{u}(\theta, t) := \sqrt{\frac{R}{\pi}} u\left(\frac{R}{\pi}\theta, t\right) . \tag{20.8}$$

Similarly we define $\tilde{u}_0 \in C^2([0,\pi])$, $\tilde{u}_1 \in C^1([0,\pi])$. Next, we extend $\tilde{u}$ and the initial data $\tilde{u}_0$ and $\tilde{u}_1$ by odd reflection across $\theta = 0$: In particular we define

$$\tilde{U}(\theta, t) = \begin{cases} \tilde{u}(\theta, t) & \theta \in [0, \pi] \\ -\tilde{u}(-\theta, t) & \theta \in [-\pi, 0] \end{cases} ,$$

and similarly for $\tilde{U}_0$, $\tilde{U}_1$. An argument analogous to the one of Corollary 17.4 shows that $\tilde{U} \in C^2([-\pi,\pi]\times\mathbb{R})$, $\tilde{U}_0 \in C^2([-\pi,\pi])$ and $\tilde{U}_1 \in C^1([-\pi,\pi])$. Moreover, $\tilde{U}_0 \in C^2_\pi([-\pi,\pi])$ and $\tilde{U}_1 \in C^1_\pi([-\pi,\pi])$ while $\tilde{U}$ fulfils periodic boundary conditions (19.5c): For example we have

$$\begin{aligned}
&\tilde{U}_0(\pi) = \tilde{u}_0(\pi) = 0 = -\tilde{u}_0(\pi) = \tilde{U}_0(-\pi)\,,\\
&\tilde{U}_0'(\pi) = \tilde{u}_0'(\pi) = \tilde{U}_0'(-\pi)\,,\\
&\tilde{U}_0''(\pi) = \tilde{u}_0''(\pi) = 0 = -\tilde{u}_0''(\pi) = \tilde{U}_0''(-\pi)\,.
\end{aligned}$$

Notice that the compatibility conditions (20.6) are crucial to ensure these properties on $\tilde{U}_0$, $\tilde{U}_1$.

By direct inspection the function $\tilde{U}$ solves the Cauchy problem (19.5) on $(-\pi,\pi)$ with initial data $\tilde{U}_0$, $\tilde{U}_1$ but with $R \to R/\pi$ in Equation (19.4a). In particular we have

$$\frac{\pi^2}{R^2}\frac{\partial^2\tilde{U}}{\partial\theta^2} = \frac{\pi^2}{R^2}\frac{\partial^2\tilde{u}}{\partial\theta^2} = \frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{1}{c^2}\frac{\partial^2\tilde{U}}{\partial t^2}\,.$$

Furthermore, since the hypotheses of Theorem 19.9 are fulfilled we have that,

$$\tilde{U}(\theta,t) = \sum_{n\in\mathbb{Z}}\left[\hat{\tilde{U}}_0(n)\cos(\omega_n t) + \hat{\tilde{U}}_1(n)\frac{\sin(\omega_n t)}{\omega_n}\right]\frac{e^{in\theta}}{\sqrt{2\pi}}\,,$$

where $\omega_n = c\pi|n|/R$. At this stage we proceed as in Remark 19.5 and we observe that

$$\hat{\tilde{U}}_0(n) = -i(s_n,\tilde{u}_0)_{L^2(0,\pi)}\,,\qquad \hat{\tilde{U}}_1(n) = -i(s_n,\tilde{u}_1)_{L^2(0,\pi)}\,.$$

This implies in particular that $\widehat{U}(n,t)$ is odd, thus,

$$u(x,t) = \sum_{n\in\mathbb{Z}}\left[(S_n,u_0)_{L^2(0,R)}\cos(\omega_n t) + (S_n,u_1)_{L^2(0,R)}\frac{\sin(\omega_n t)}{\omega_n}\right]S_n(x)\,.$$

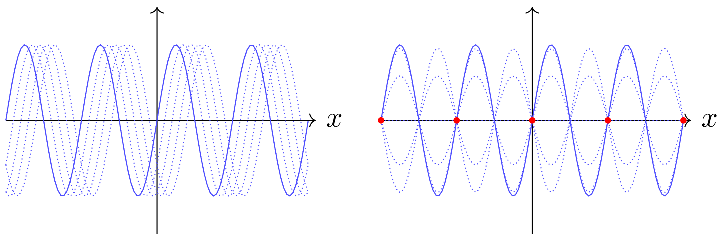
This prove Equation (20.5). For the rest of the thesis it suffices to observe that, if u_0 is C^3 piecewise and u_1 is C^2 piecewise, then $\tilde{U}_0$ is C^3-piecewise and U_1 is C^2-piecewise. The thesis then follows by Theorem 19.9. □

Remark 20.5:

(i) The expansion (20.5) is similar to Equation (19.16) but with remarkable differences. In particular, Equation (19.16) leads to an expansion of the solution u to problem (19.5) in terms of progressive and regressive waves, *cf.* Remark 19.10. This is not true for Equation (20.5).

This is due to the choice of the boundary conditions which leads to the expansion in the basis $S_n(x)$. The resulting solution can be written as a linear superposition of

$$u_n(x,t) := \left[\alpha_n\sin\left(\frac{\pi nc}{R}t\right) + \beta_n\cos\left(\frac{\pi nc}{R}t\right)\right]\sin\left(\frac{\pi n}{R}x\right) = \gamma_n\sin\left(\frac{\pi nc}{R}t+\phi_n\right)\sin\left(\frac{\pi n}{R}x\right)\,,$$



(a) (b)

Figure 12: Fig. 12a: progressive wave evolving in time. The continuous line represents the initial datum whereas dotted lines represent the wave at different $t > 0$. Fig. 12b: stationary wave evolving in time. Red circles denote the nodal points.

where we used the trigonometric identity

$$\begin{aligned}\left[\alpha_n \sin\left(\frac{\pi nc}{R}t\right) + \beta_n \cos\left(\frac{\pi nc}{R}t\right)\right] &= \gamma_n\left[\frac{\alpha_n}{\gamma_n} \sin\left(\frac{\pi nc}{R}t\right) + \frac{\beta_n}{\gamma_n} \cos\left(\frac{\pi nc}{R}t\right)\right] \qquad \gamma_n := \sqrt{\alpha_n^2 + \beta_n^2}\\ &= \gamma_n\left[\cos(\phi_n) \sin\left(\frac{\pi nc}{R}t\right) + \sin(\phi_n) \cos\left(\frac{\pi nc}{R}t\right)\right]\\ &= \gamma_n \sin\left(\frac{\pi nc}{R}t + \phi_n\right),\end{aligned}$$

where $\phi_n \in [0, 2\pi)$ is defined by $\cos\phi_n = \alpha_n/\gamma_n$ and $\sin\phi_n = \beta_n/\gamma_n$.

The obtained solutions are neither progressive nor regressive waves. Instead they are called **stationary waves** (or harmonics) because they describe an oscillating profile which does not propagate, *cf.* Figure 12b. A characteristic features of stationary waves is the presence of **nodal points**, namely points x where the solution u_n vanishes for all times t. For the case at hand this happens for

$$x \in \left\{\frac{k}{n}R \,|\, k \in \{0, \dots, n\}\right\} \qquad n \in \mathbb{N}.$$

Notice that each stationary wave has its own set of nodal points. As we shall see in Section **??**, the appearance of stationary waves is not special of the 1D situation but it is rather a characteristic feature of the chosen boundary conditions.

(ii) Both in (19.16)-(20.5), the resulting solutions are periodic in time: The frequency of the oscillation is called **resonance frequency**. By direct inspection the latter are given by

$$\text{Periodic:} \left\{\frac{nc}{R} \,|\, n \in \mathbb{N} \cup \{0\}\right\}, \quad \text{Dirichlet:} \left\{\frac{nc}{2R} \,|\, n \in \mathbb{N}\right\}.$$

Notice that, in all cases, the resonant frequencies are all integer multiples of a single frequency, called **fundamental frequency**. This latter fact has an important counterpart in our perception of the sound of musical instruments. Indeed, let us consider a guitar:

The sound produced by this instrument is mainly due to the vibrations of its strings. These vibrations are approximately transversal, thus, they fit into the framework of the 1D Cauchy problem (20.1). The vibrations of the string produce sound waves, namely perturbations of the air surrounding the string, which are those recognized by our ears as "music". (As a matter of fact the vibrations of the string induce transversal vibrations on the barrel of the guitar which are responsible for the sound waves that we hear: We shall come back to this in a later stage.) The remarkable point is that our ears are sensitive to the presence of a fundamental frequency: This is the reason why we call the sound of a guitar "harmonic". As we will see in Section 21 the presence of a fundamental frequency is characteristic 1D situation and it is lost in general in higher dimensions.

21 Hyperbolic PDEs: Resonance phenomena

To better grasp the physical importance of the resonance frequency we will now discuss resonance phenomena: These occur whenever a vibration of, say, a membrane is caused by an external periodic force. If the frequency of the external force is suitably chosen, *i.e.* if it coincides with one of the resonance frequencies, the associated vibrations turn to be wilder than expected.

Resonance phenomena are particularly relevant for building pedestrian bridges. In an extremely rough approximation, a bridge can be thought as a 1D string whose transversal vibrations can be described by the solution to problem (20.1). In this situation one would like to avoid the bridge to have the human footstep's frequency as a fundamental frequency: Indeed, in this latter case, the bridge would undergo a resonance phenomenon and collapse once filled by a crowd.

We shall now consider a concrete model for the resonance phenomenon. To this avail we consider the Cauchy problem (20.1) with zero Cauchy data $u_0 = u_1 = 0$ but with a source term f, actually

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = f\,, \qquad \forall (x,t) \in [0,R] \times \mathbb{R} \tag{21.1a}$$

$$u(x,0) = \frac{\partial u}{\partial t}\Big|_{t=0} = 0\,, \qquad \forall x \in [0,R] \tag{21.1b}$$

$$u(0,t) = u(R,t) = 0 \qquad \forall t \in \mathbb{R}\,. \tag{21.1c}$$

Here $u \in C^2([0,R] \times \mathbb{R})$ while $f \in C^0([0,R] \times \mathbb{R})$.

The following proposition is a counterpart of Theorem 20.4 to the case at hand.

Proposition 21.1: If u is a solution to (21.1) then

$$u(x,t) = c^2 \sum_{n\geqslant 1} \bigg[\int_0^t \frac{\sin[\omega_n(t-s)]}{\omega_n} \tilde{f}(n,s)\mathrm{d}s \bigg] S_n(x)\,, \qquad \omega_n = \frac{c\pi n}{R}\,, \tag{21.2}$$

where $\tilde{f}(n,t) := (S_n, f(\ ,t))_{L^2(0,R)}$. In particular problem (20.1) has at most one solution. Moreover, if $f \in C^2([0,R] \times \mathbb{R})$ and fulfils (21.1c) then Equation (20.5) defines a function $u \in C^2([0,R] \times \mathbb{R})$ which solves problem (21.1).

Proof. Let $u \in C^2([0,R]\times\mathbb{R})$ be as solution to problem (21.1) and let $\tilde{u}(n,t) := (S_n, u(\ ,t))_{L^2(0,R)}$. Proceeding as in the proof of Theorem 25.3 we apply Corollary 9.5 to conclude that $\tilde{u}_n(n,\ ) \in$

$C^2(\mathbb{R})$ for all $n \in \mathbb{N}$. Moreover,

$$\begin{aligned}
\tilde{u}(n,0) &= 0 = \frac{\mathrm{d}\tilde{u}}{\mathrm{d}t}(n,0) \\
\frac{1}{c^2}\frac{\mathrm{d}^2\tilde{u}}{\mathrm{d}t^2}(n,t) &= \int_0^R \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}(x,t)S_n(x)\mathrm{d}x && \text{Cor. 9.5} \\
&= \int_0^R \left[\frac{\partial^2 u}{\partial x^2}(x,t) + f(x,t)\right]S_n(x)\mathrm{d}x && \text{Eq. (21.1a)} \\
&= -\Big(\frac{\pi n}{R}\Big)^2\tilde{u}(n,t) + \tilde{f}(n,t) && \text{Eq. (21.1a)}\,,
\end{aligned}$$

Thus, for all $n \geqslant 1$, $\tilde{u}(n,\ )$ is a solution to the Cauchy problem

$$\frac{\mathrm{d}^2\tilde{u}}{\mathrm{d}t^2}(n,t) + \omega_n^2\tilde{u}(n,t) = c^2\tilde{f}(n,t)\,, \qquad \tilde{u}(n,0) = 0\,, \qquad \frac{\mathrm{d}\tilde{u}}{\mathrm{d}t}(n,0) = 0\,, \tag{21.3}$$

where $\omega_n = c\pi n/R$. The solution to the latter problem can be computed as

$$\tilde{u}(n,t) = c^2\int_0^t \frac{\sin[\omega_n(t-s)]}{\omega_n}\tilde{f}(n,s)\mathrm{d}s\,, \tag{21.4}$$

Indeed, by direct inspection $\tilde{u}(n,0) = 0$ and by applying Lemma 16.1 we may compute

$$\frac{\mathrm{d}\tilde{u}}{\mathrm{d}t}(n,t) = c^2\int_0^t \cos[\omega_n(t-s)]\tilde{f}(n,s)\mathrm{d}s\,,$$

which shows that the initial conditions (21.1b) are fulfilled. Another application of Lemma 16.1 shows that Equation (20.1a) is also fulfilled.

Thus, u fulfils Equation (21.2). We will now prove that Equation (21.2) defines a solution to problem (21.1) under the assumption $f \in C^2([0,R]\times\mathbb{R})$ and $f(0,t) = f(R,t) = 0$ for all $t \in \mathbb{R}$. To this avail, it is enough to prove that the hypothesis of Corollary 9.5 are fulfilled (twice) for $x \in [0,R]$ and $t \in [0,T]$, $T \in \mathbb{R}$ being arbitrary — in what follows we will consider the case $T > 0$, the other case being analogous. We preliminary observe that

$$\begin{aligned}
\Big|\frac{\partial^k}{\partial x^k}\Big[\tilde{u}(n,t)S_n(x)\Big]\Big| &\leqslant \sqrt{\frac{2}{R}}\frac{\omega_n^{k-1}}{c^{k-2}}\int_0^T |\tilde{f}(n,s)|\mathrm{d}s \qquad k \in \{0,1,2\} \\
\Big|\frac{\partial}{\partial t}\Big[\tilde{u}(n,t)S_n(x)\Big]\Big| &\leqslant \sqrt{\frac{2}{R}}c^2\int_0^T |\tilde{f}(n,s)|\mathrm{d}s \\
\Big|\frac{\partial^2}{\partial t^2}\Big[\tilde{u}(n,t)S_n(x)\Big]\Big| &\leqslant \sqrt{\frac{2}{R}}\Big[c^2|\tilde{f}(n,t)| + c^2\omega_n\int_0^T |\tilde{f}(n,s)|\mathrm{d}s\Big]|\,.
\end{aligned}$$

We also observe that

$$\begin{aligned}
\omega_n\tilde{f}(n,t) = (\omega_n S_n, f(\ ,t))_{L^2(0,R)} = -c(C_n', f(\ ,t))_{L^2(0,R)} &= c\bigg(C_n, \frac{\partial f}{\partial x}(\ ,t)\bigg)_{L^2(0,R)}\,, \\
\omega_n^2\tilde{f}(n,t) = -c^2(S_n'', f(\ ,t))_{L^2(0,R)} &= c^2\bigg(S_n, \frac{\partial^2 f}{\partial x^2}(\ ,t)\bigg)_{L^2(0,R)}\,,
\end{aligned}$$

where we used that $f \in C^2([0,R]\times\mathbb{R})$ and that it fulfils the boundary conditions (21.1c). This implies in particular that

$$|\tilde{f}(n,t)| \leqslant \frac{c^2 R}{\omega_n^2} \sup_{\substack{x\in[0,R]\\ t\in[0,T]}} \left|\frac{\partial^2 f}{\partial x^2}\right| \in \ell^1(\mathbb{N})\,.$$

Overall, we see that assumption ii of Corollary 9.5 is fulfilled provided that

$$(n^k \tilde{F}_n)_n \in \ell^1(\mathbb{N}) \quad \forall k\in\{0,1,2\} \qquad \tilde{F}_n := \frac{1}{\omega_n}\int_0^T |\tilde{f}(n,s)|\mathrm{d}s\,.$$

By direct inspection we have

$$\begin{aligned}
\sum_{n\geqslant1}|\tilde{F}_n| &= \int_0^T \sum_{n\geqslant1}\frac{1}{\omega_n}|\tilde{f}(n,s)|\mathrm{d}s && \text{Monotone convergence Thm.}\\
&\leqslant \frac{1}{2}\int_0^T \sum_{n\geqslant1}\left[\frac{1}{\omega_n^2} + |\tilde{f}(n,s)|^2\right]\mathrm{d}s\\
&\leqslant \frac{1}{2}\Big[\sum_{n\geqslant1}\frac{T}{\omega_n^2} + \int_0^T\int_0^R |f(x,s)|^2\mathrm{d}x\mathrm{d}s\Big] < +\infty && \text{Eq. (19.10)}\\
\sum_{n\geqslant1} n|\tilde{F}_n| &\leqslant C\int_0^T\sum_{n\geqslant1}|\tilde{f}(n,s)|\mathrm{d}s && n/\omega_n \leqslant C\\
&\leqslant \frac{1}{2}\int_0^T\sum_{n\geqslant1}\Big[\frac{c^2}{\omega_n^2} + \Big|\Big(C_n, \frac{\partial f}{\partial x}(\ ,s)\Big)\Big|^2\Big]\mathrm{d}s && \tilde{f}(n,t) = \frac{c}{\omega_n}\Big(C_n, \frac{\partial f}{\partial x}(\ ,t)\Big)_{L^2(0,R)}\\
&= \frac{1}{2}\Big[\sum_{n\geqslant1}\frac{c^2T}{\omega_n^2} + \int_0^T\int_0^R\Big|\frac{\partial f}{\partial x}(x,s)\Big|^2\mathrm{d}x\mathrm{d}s < +\infty && \text{Eq. (19.10)}\\
\sum_{n\geqslant1} n^2|\tilde{F}_n| &\leqslant C\int_0^T\sum_{n\geqslant1}\omega_n|\tilde{f}(n,s)|\mathrm{d}s\\
&\leqslant \frac{1}{2}\int_0^T\sum_{n\geqslant1}\Big[\frac{c^4}{\omega_n^2} + \Big|\Big(S_n, \frac{\partial^2 f}{\partial x^2}(\ ,s)\Big)\Big|^2\Big]\mathrm{d}s && \omega_n\tilde{f}(n,t) = \frac{c^2}{\omega_n}\Big(S_n, \frac{\partial^2 f}{\partial x^2}(\ ,t)\Big)_{L^2(0,R)}\\
&= \frac{1}{2}\Big[\sum_{n\geqslant1}\frac{c^4T}{\omega_n^2} + \int_0^T\int_0^R\Big|\frac{\partial^2 f}{\partial x^2}(x,s)\Big|^2\mathrm{d}x\mathrm{d}s < +\infty && \text{Eq. (19.10)}
\end{aligned}$$

This proves that $(n^k\tilde{F}_n)_n \in \ell^1(\mathbb{N})$ for $k\in\{1,2\}$. □

REMARK 21.2:

(i) If $u \in C^2([0,R]\times\mathbb{R})$ solves Problem (21.1) then

$$f(0,0) = \left[\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2}\right](0,0) = 0\,,$$

where the first factor on the right-hand side vanishes on account of the boundary conditions (21.1c) while the second one vanishes because of the initial data (21.1b).

(ii) The assumptions in the second part of Theorem 21.1 suffice to ensure the absolute convergence of the Fourier series in (21.2). In fact, they also imply that the Fourier series

$$f(x,t) = \sum_{n\geqslant 0} (S_n, f(\ ,t))_{L^2(0,R)} S_n(x)\,,$$

is uniformly convergent and that x-derivatives of f can be computed by deriving the series term by term. It is important to notice that these assumptions include $f \in C^2([0,R]\times\mathbb{R})$ *and* $f(0,t) = f(R,t) = 0$ for all $t \in \mathbb{R}$, *cf.* Remark 20.3 for and example of a smooth functions whose Fourier series in the orthonormal basis $\{S_n\}_{n\in\mathbb{N}}$ is not uniformly convergent.

(iii) Notice the solution to Equation (21.3) can be written as an integral

$$\tilde{u}(n,t) = \int_0^t \tilde{u}_n(t;s)\mathrm{d}s\,, \qquad \tilde{u}_n(t;s) = c^2\frac{\sin[\omega_n(t-s)]}{\omega_n}\tilde{f}(n,s)\,,$$

where $\tilde{u}_n$ solves the problem

$$\frac{\mathrm{d}^2\tilde{u}_n}{\mathrm{d}t^2} + \omega_n^2\tilde{u}_n = 0\,, \qquad \tilde{u}_n(s;s) = 0\,, \qquad \frac{\mathrm{d}\tilde{u}_n}{\mathrm{d}t}(s;s) = c^2\tilde{f}(n,s)\,.$$

This fact is called **Duhamel's principle** and explains why, if a particle is subjected to a force for a very short period of time, one may approximate the associated Cauchy problem by neglecting the force in favour of a non-vanishing initial velocity. In the example above, this is equivalent to approximate $\tilde{u}(n,t) \approx \tilde{u}_n(t;0)$ under the assumption that $\tilde{f}(n,s)$ is supported for $s \in (0,\varepsilon)$, $\varepsilon \ll 1$. We refer to *cf.* [3, §2.3.1] for further details.

We now apply Proposition 21.1 for the case of a source term f which is periodic in time: Specifically we will assume that

$$f(x,t) = \sin(\omega t)S_1(x)\,, \qquad \omega \in \mathbb{R}\,.$$

Such source f abides by the assumptions of Proposition 21.1 and the corresponding solution u_ω to Problem 21.1 reads

$$u_\omega(x,t) = \frac{c^2}{\omega_1}\int_0^t \sin[\omega_1(t-s)]\sin(\omega s)\mathrm{d}s\ S_1(x)\,. \tag{21.5}$$

The integral in time is computed directly by observing that

$$\begin{aligned}
&(\omega^2-\omega_1^2)\int_0^t \sin[\omega_1(t-s)]\sin(\omega s)\mathrm{d}s \\
&=\int_0^t\left[\frac{\mathrm{d}^2}{\mathrm{d}s^2}\Big(\sin[\omega_1(t-s)]\Big)\sin(\omega s)-\sin[\omega_1(t-s)]\frac{\mathrm{d}^2}{\mathrm{d}s^2}\sin(\omega s)\right]\mathrm{d}s \\
&=\left[\frac{\mathrm{d}}{\mathrm{d}s}\Big(\sin[\omega_1(t-s)]\Big)\sin(\omega s)-\sin[\omega_1(t-s)]\frac{\mathrm{d}}{\mathrm{d}s}\sin(\omega s)\right]\Bigg|_{s=0}^{s=t} \\
&=-\omega_1\sin(\omega t)+\omega\sin(\omega_1 t) \\
&=(\omega-\omega_1)\sin(\omega_1 t)+\omega_1[\sin(\omega_1 t)-\sin(\omega t)]\,.
\end{aligned}$$

Thus, we find

$$u_\omega(x,t)=\frac{c^2}{\omega_1(\omega+\omega_1)}\left[\sin(\omega_1 t)-\omega_1\frac{\sin(\omega t)-\sin(\omega_1 t)}{\omega-\omega_1}\right]S_1(x)\,.$$

The resulting solution is not stationary because there is no clear periodicity in the time variable t. This is due to the presence of the source term f, whose periodicity "disturbs" the resonance frequencies of the string. It is particularly interesting to compute the limit $\omega\to\omega_1$, *i.e.* matching the frequency of the source f with the resonant frequency ω_1: In this latter case we find

$$u_{\omega_1}(x,t)=\frac{c^2}{2\omega_1^2}\Big[\sin(\omega_1 t)-\omega_1 t\cos(\omega_1 t)\Big]S_1(x)\,.$$

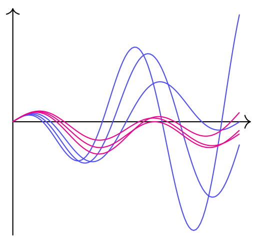


Figure 13: The resonance phenomenon: Blue waves arise from a source term with $\omega\approx\omega_1$ and are dominant for large times.

Notice that the time behaviour of the solution has changed: Indeed, u_ω was bounded, whereas for u_{ω_1} the oscillations grow linearly in time. In this situation we say that resonance is occurring.

For a generic periodic source f of period τ we may exploit the expansion, *cf.* Remark 19.5,

$$f(x,t)=\frac{c_0(x)}{2}+\sum_{\ell\geqslant 1}\left[c_\ell(x)\cos\left(\frac{2\pi\ell}{\tau}t\right)+s_\ell(x)\sin\left(\frac{2\pi\ell}{\tau}t\right)\right],$$

for suitable functions $c_0(x)$, $c_\ell(x)$ and $s_\ell(x)$. In this latter case the resulting solution u as per Equation (21.2) will be mainly determined by those values of $\ell\in\mathbb{N}$ for which $2\pi\ell/\tau$ is close to one of the resonant frequencies ω_n, *cf.* Figure 13.

Remark 21.3 (🏃):

(i) It is worth to briefly comment on the analogous version of problems (20.1)-(21.1) for 2D domains, *e.g.* the disk $B_R(0)$, $R>0$. This corresponds to investigate problem (15.2) for a

2D domain Ω and with Dirichlet boundary conditions. In this latter case the solution u can be used to model, say, the transversal vibrations of the membrane of a drum. Similarly to what happens for problem (20.1) the resulting solution u is expanded in stationary waves, in particular

$$u(x,t) = \sum_\lambda \left[(v, u_0)_{L^2(\Omega)} \cos(c\lambda t) + (v, u_0)_{L^2(\Omega)} \frac{\sin(c\lambda t)}{\lambda}\right] v_\lambda(x)\,, \tag{21.6}$$

where (λ, v_λ) is a solution to spectral problem for the Laplace operator, that is,

$$-\Delta v_\lambda = \lambda^2 v_\lambda \qquad v_{|\partial\Omega} = 0\,. \tag{21.7}$$

for $v_\lambda \in C^2(\Omega) \cap C^0(\overline{\Omega})$ —in fact, it can be shown that $v \in C^\infty(\overline{\Omega})$. It is possible to prove that there are at most countably many solutions to the latter problem. Specifically there exists a sequence $\{\lambda_n\}_n$ with

$$0 \leqslant \lambda_1 \leqslant \lambda_2 \leqslant \ldots \leqslant \lambda_n \underset{n\to\infty}{\longrightarrow} \infty\,,$$

for which there exists v_n such that (λ_n, v_n) solves problem (21.7). For any $n \neq k$ the resulting v's are orthogonal in $L^2(\Omega)$, moreover, $\{v_n\}_n$ is complete orthonormal basis for such space. We refer to [8, §8.5] for further details. The functions

$$u_n(x,t) := [\alpha_n \cos(c\lambda_n t) + \beta_n \sin(c\lambda_n t)] v_{\lambda_n}(x)\,,$$

are again called stationary waves. As for the 1D case, for fixed n, $u_n(\ , t)$ vanishes on specific connected sets called **nodal curves**. Furthermore, each u_n is periodic with resonant frequency $\omega_n := c\lambda_n$.

(ii) Despite the fact that u is a superposition of stationary waves, there is no analogous of the fundamental frequency, *cf.* Remark 20.5. There is an important caveat to this fact, which has important applications when considering musical instruments. Indeed, although the resonance frequencies ω_n fail to possess a fundamental frequency —that is, ω_n is not an integer multiple of fixed frequency ω for all n— it is nevertheless possible to consider a subset of stationary waves such that the associated frequencies do possess a fundamental frequency. This is particularly important when considering a kettledrum: In this scenario, once we restrict to those stationary waves sharing an approximated fundamental frequency, the induced vibrations produce a sound which is perceived to be harmonic. This restriction is experimentally realized by playing the kettledrum in a specific way.

(iii) Another application of the resonance phenomenon occurs when considering the sound produced by, say, a guitar. As we already mentioned in Section 20, the music we hear from a guitar is produced by the vibration of its strings, which are transmitted to the air surrounding the guitar. In fact, the transmission is slightly more complicated: Each string vibrates and induces vibrations on the barrel which then produces the air perturbation that we perceive as music. In this context a natural question arises: The sound produced by a guitar is perceived as harmonic, that is, our ears recognize the presence of the fundamental frequency. However, although the initial vibrations are produced by a string, which has a fundamental frequency, the actual sound that we hear is produced by the induced vibrations of the barrel. Since the latter is essentially a 2D membrane we should not expect a fundamental frequency. Yet, our ears perceive it.

The solution to this apparent puzzle relies on the fact that the initial vibration has a fundamental frequency together with the resonance phenomenon. In essence, the vibrations induced on the barrel are a superposition of stationary waves whose frequency is close to (a multiple of) the fundamental frequency. Thus, although the vibrations of the barrel do not possess a fundamental frequency in general, those induced by the vibrations of the string does, at least approximately.

22 Parabolic PDEs: Uniqueness on bounded domains

In this section we begin our investigation on the properties of parabolic PDEs. In particular we will focus on the heat equation

$$\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \Delta_x u = 0\,, \tag{22.1}$$

where $\varsigma > 0$.

As already discussed in Section 1, Equation (22.1) models heat propagation in a continuum. Within this setting, u represents the temperature of the continuum and $D := \varsigma^2$ is the diffusion constant —we stress that ς has the dimension of length$^2\times$time^{-1}. The solution u can also be interpreted as the concentration of a certain chemical (or population), in which case Equation (22.1) is usually called diffusion equation.

From the technical side, Equation (22.1) is a second order, linear PDE of parabolic type, *cf.* Definition 2.9. Moreover, the characteristic hypersurfaces associated to (22.1) —*cf.* Definition 6.2— coincide with

$$\Sigma_{t_0} := \{(x,t) \in \mathbb{R}^n \times \mathbb{R} \,|\, t = t_0\} \qquad t_0 \in \mathbb{R}\,.$$

This fact has an important consequence, namely, the Cauchy problem (4.2) for (22.1) on such hypersurfaces cannot be converted in normal form, *cf.* (4.4): In particular Theorem 4.6 does not apply in this setting. However, to cope with this problem it suffices to consider the Cauchy problem (4.2) dropping the requirement on $\frac{\partial u}{\partial \nu}$, which coincides with $\frac{\partial u}{\partial t}$ in this context: We shall refer to this problem as **initial-value problem**.

Concerning regularity, solutions to Equation (22.1) usually turn out to be quite regular, in fact smooth, in the domain where (22.1) is imposed. This is similar to what happens for harmonic functions —*cf.* Theorem 11.3— and it is characteristic of diffusive phenomena: The possibly highly irregular initial datum is instantaneously converted to a smooth one.

Finally, although Equation (22.1) describes a dynamical phenomenon, it shares little features with hyperbolic equations (15.1). In particular there is neither finite propagation speed nor time-reversal symmetry —*cf.* Section 16 and Remark 15.2. The first fact is widely accepted as a limitation in the model described by (22.1) and can be handle by considering suitable non-linear generalizations of the latter equation which restore the finiteness of the propagation speed, *cf.* [18, §2.10]. The lack of time-reversal symmetry implies that, if u solves (22.1), then $\widetilde{u}(x,t) := u(x,-t)$ does not solve (22.1) in general. This fact has an important consequence: The backward and forward initial-value problem for the heat equation are not related. In fact, the forward initial-value problem is usually way easier to handle than the backward Cauchy problem, the latter being of some relevance for financial mathematics [6, §2.9].

We now introduce the initial-value problem with boundary conditions, *cf.* Definition 15.1.

Definition 22.1: Let $\Omega \subset \mathbb{R}^n$ be an open non-empty subset with $\overline{\Omega}$ compact and such that $\partial\Omega$ is a C^1-regular orientable hypersurface and outward-pointing unit normal ν. Let $T > 0$, $\theta \in [0, \pi/2]$

and set $\boxed{\Omega_T^+} := \Omega \times (0,T)$ and $\boxed{\overline{\Omega}_T^+} := \overline{\Omega} \times (0,T)$ —notice that $\Omega_T^+ \subset \overline{\Omega}_T^+ \subset \overline{\Omega_T^+} = \overline{\Omega} \times [0,T]$. Notice that $\Omega_T^+ \subset \mathbb{R}^{n+1}$ is open, with compact closure $\overline{\Omega_T^+} = \overline{\Omega} \times [0,T]$ and C^1-regular orientable boundary $\partial\Omega_T^+ := [\partial\Omega \times (0,T)] \cup [\Omega \times \{0\}] \cup [\Omega \times \{T\}]$.

The **forward initial-value problem with (θ-)boundary conditions** consists of finding $u \in C^2(\Omega_T^+) \cap C^0(\overline{\Omega_T^+}) \cap C^1(\overline{\Omega}_T^+)$ —$u \in C^2(\Omega_T^+) \cap C^0(\overline{\Omega_T^+})$ if $\theta = 0$— such that

$$\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \Delta_x u = f\,, \qquad \forall (x,t) \in \Omega \times (0,T) \tag{22.2a}$$

$$u(x,0) = u_0(x)\,, \qquad \forall x \in \overline{\Omega} \tag{22.2b}$$

$$\sin\theta \frac{\partial u}{\partial \nu}(x,t) + \cos\theta\, u(x,t) = u_{\mathrm{BC}}(x,t) \qquad \forall (x,t) \in \partial\Omega \times (0,T) \tag{22.2c}$$

where $\varsigma > 0$ while $f \in C^0(\Omega_T^+)$, $u_0 \in C^0(\overline{\Omega})$, and $u_{\mathrm{BC}} \in C^0(\partial\Omega \times (0,T))$ are assigned.

Similarly the **backward initial-value problem with (θ-)boundary conditions** is defined by

$$\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \Delta_x u = f\,, \qquad \forall (x,t) \in \Omega \times (0,T) \tag{22.3a}$$

$$u(x,T) = u_0(x)\,, \qquad \forall x \in \overline{\Omega} \tag{22.3b}$$

$$\sin\theta \frac{\partial u}{\partial \nu}(x,t) + \cos\theta\, u(x,t) = u_{\mathrm{BC}}(x,t) \qquad \forall (x,t) \in \partial\Omega \times (0,T) \tag{22.3c}$$

that is, upon replacing (22.2b) with (22.3b).

Remark 22.2:

(i) By comparison with (15.2) we have dropped several regularity conditions on the data f, u_0, u_{BC}. In particular:

   (i) we are considering (22.2a) only on the open set Ω_T —as opposed to $\overline{\Omega}_T$ in (15.2)— which leads to the weaker requirement $u \in C^2(\Omega_T^+)$ and $f \in C^0(\Omega_T^+)$, rather than $u \in C^2(\overline{\Omega}_T^+)$ and $f \in C^0(\overline{\Omega}_T^+)$. A similar argument goes for the boundary conditions, for which we require $u \in C^1(\overline{\Omega}_T^+)$ and $u_{\mathrm{BC}} \in C^0(\partial\Omega \times (0,T))$. Notice that, if $\theta = 0$, we drop the requirement $u \in C^1(\overline{\Omega}_T^+)$.

   (ii) we are considering a C^0-matching with the initial data, that is, we assume $u \in C^0(\overline{\Omega_T^+})$.

Weakening these requirements does not spoil the possibility of solving (22.2): This is an insight of the regularity properties of (22.1). Notice that another consequence of the weaker regularity of the data is that there are no compatibility conditions, that is, there is no analogous of (15.3). There is only one exception, namely $\theta = 0$: In that latter case one finds the compatibility condition

$$\lim_{t\to 0^+} u_{\mathrm{BC}}(x,t) = \lim_{t\to 0^+} u(x,t) = u_0(x) \qquad \forall x \in \partial\Omega\,. \tag{22.4}$$

(ii) In [11, 18] the regularity of the solution in time and space is carefully separated. Indeed, (22.1) only requires C^1-regularity in t and C^2-regularity in x: For this reason one may introduce suitable spaces $C^{2,1}$ which take care of this different regularity. However, solutions to the heat equation are typically highly regular, therefore, we will stick with the standard C^2-regularity requirement.

We also point out that in [3, 11, 18] the set Ω_T^+ is usually defined as $\Omega \times (0, T]$ and named **parabolic cylinder**: This fits better with the proofs presented therein, however, it will not be necessary in the forthcoming discussion.

We now prove a uniqueness result for the forward initial-valued problem with boundary condition (22.2) [3, Thm. 11].

THEOREM 22.3: There exists at most one solution $u \in C^2(\Omega_T^+) \cap C^1(\overline{\Omega}_T^+) \cap C^0(\overline{\Omega_T^+})$ (this regularity applying also for $\theta = 0$) to problem (22.2).

Proof. Let $u, u' \in C^2(\Omega_T^+) \cap C^1(\overline{\Omega}_T^+) \cap C^0(\overline{\Omega_T^+})$ be two solutions to (22.2) with the same data f, u_0, u_{BC}. Let $v := u - u'$: Then $v \in C^2(\Omega_T^+) \cap C^1(\overline{\Omega}_T^+) \cap C^0(\overline{\Omega_T^+})$ solves (22.2) with vanishing data.

We now prove that $v = 0$. Let $E_{v,\Omega}(t)$ be defined by

$$E_{v,\Omega}(t) := \frac{1}{2}\int_\Omega v(x,t)^2 \mathrm{d}^n x = \frac{1}{2}\|v(\ ,t)\|^2_{L^2(\Omega)} \,. \tag{22.5}$$

Since $\overline{\Omega}$ is compact and $v \in C^1(\overline{\Omega}_T^+) \cap C^0(\overline{\Omega_T^+})$, Corollary 9.5 ensures that $E_{v,\Omega} \in C^1((0,T)) \cap C^0([0,T])$. Thus in particular

$$\begin{aligned}
E_{v,\Omega}(0) &= \frac{1}{2}\int_\Omega v(x,0)^2 \mathrm{d}^n x = 0\,, \\
\frac{1}{\varsigma^2}\frac{\mathrm{d}E_{v,\Omega}}{\mathrm{d}t} &= \int_\Omega v \frac{1}{\varsigma^2}\frac{\partial v}{\partial t}\mathrm{d}^n x && \text{Cor. 9.5} \\
&= \int_\Omega v \Delta_x v \mathrm{d}^n x && \text{Eq. (22.2a)} \\
&= \oint_{+\partial\Omega} v\frac{\partial v}{\partial \nu}\mathrm{d}S - \int_\Omega \|\nabla_x v\|^2 \mathrm{d}^n x && \text{Thm. 8.3 + Rmk. 8.2} \\
&\leqslant \oint_{+\partial\Omega} v\frac{\partial v}{\partial \nu}\mathrm{d}S\,.
\end{aligned}$$

We thus obtain

$$\frac{1}{\varsigma^2}\frac{\mathrm{d}E_{v,\Omega}}{\mathrm{d}t} \leqslant \oint_{+\partial\Omega} v\frac{\partial v}{\partial\nu}\mathrm{d}S = \begin{cases} 0 & \theta\in\{0,\pi/2\} \\ -\cot\theta \oint_{+\partial\Omega} v^2\mathrm{d}S \leqslant 0 & 0<\theta<\pi/2\ (\Rightarrow \cot\theta>0) \end{cases}$$

which implies that $0 \leqslant E_{v,\Omega}(t) \leqslant E_{v,\Omega}(0) = 0$ for all $t \in [0,T]$. Thus $v(\ ,t) = 0$ in $L^2(\Omega)$ for all $t \in [0,T]$, therefore $v = 0$ since $v \in C^0(\overline{\Omega^+_T})$. ☐

Remark 22.4: The uniqueness of the solution to (22.2) still holds true for $\theta = 0$ when dropping the higher regularity requirement $u \in C^1(\overline{\Omega}^+_T)$ [11, Thm. 8.2-8.3]. The proof is based on a maximum principle similar to Theorem 7.6: We will address this problem in Section 24.

The proof of Theorem 22.3 is similar in spirit to the one of Theorem 15.3. There is, however, an important difference which appears for the case of Dirichlet and Neumann boundary conditions $\theta \in \{0,\pi/2\}$. While in Theorem 15.3 the hyperbolic energy function $E_{u,\Omega}(t)$ defined in (15.5) is conserved, *i.e.* it is constant in time, the parabolic energy function $E_{u,\Omega}(t)$ defined (22.5) is not conserved but only non-increasing. Therefore, if the energy function is zero at $t = 0$ it vanishes for all $t \geqslant 0$. This leads to a natural question, namely if the proof of Theorem 22.3 can be extended to the case of the backward problem. Notably, this is possible as we shall see for the case of Dirichlet and Neumann boundary conditions within an additional regularity assumption on u.

Theorem 22.5: Let $\theta \in \{0,\pi/2\}$. Then there exists at most one solution $u \in C^2(\overline{\Omega}^+_T) \cap C^0(\overline{\Omega^+_T})$ to the backward initial-value problem with boundary conditions (22.3).

Proof. We proceed as in the proof of Theorem 22.3. Given two solutions u, u' to the backward problem (22.3) with the same data f, u_0, u_{BC} we define $v := u - u'$. The latter function solves (22.3) with vanishing data. We will prove that $v = 0$.

Defining $E_{v,\Omega}(t)$ as in (22.5) we have that $E_{v,\Omega} \in C^1((0,T)) \cap C^0([0,T])$, $E_{v,\Omega}(T) = 0$ and

$$\frac{1}{\varsigma^2}\frac{\mathrm{d}E_{v,\Omega}}{\mathrm{d}t} = \int_\Omega v\Delta_x v\mathrm{d}^n x = -\int_\Omega \|\nabla_x v\|^2\mathrm{d}^n x\,.$$

Since $v \in C^2(\overline{\Omega}^+_T)$ we may apply once again Corollary 9.5 to find that $E_{v,\Omega} \in C^2((0,T))$ and

$$\begin{aligned}\frac{1}{\varsigma^4}\frac{\mathrm{d}^2E_{v,\Omega}}{\mathrm{d}t^2} &= -2\int_\Omega \nabla_x v\cdot\nabla_x\frac{1}{\varsigma^2}\frac{\partial v}{\partial t}\mathrm{d}^n x \\ &= 2\int_\Omega \Delta_x v\frac{1}{\varsigma^2}\frac{\partial v}{\partial t}\mathrm{d}^n x - 2\oint_{+\partial\Omega}\cancel{\frac{\partial v}{\partial\nu}}\frac{\partial v}{\partial t}\mathrm{d}S && \text{Thm. 8.1 + Rmk. 8.2} \\ &= 2\|\Delta_x v(\ ,t)\|^2_{L^2(\Omega)} && \text{Eq. (22.3a)}\,.\end{aligned}$$

Moreover, the Cauchy-Schwartz inequality leads to

$$\begin{aligned}\left(\frac{1}{\varsigma^2}\frac{\mathrm{d}E_{v,\Omega}}{\mathrm{d}t}\right)^2 &= |(v(\ ,t), \Delta_x v(\ ,t))_{L^2(\Omega)}|^2 \\ &\leqslant \|v(\ ,t)\|^2_{L^2(\Omega)}\|\Delta_x v(\ ,t)\|^2_{L^2(\Omega)} = E_{v,\Omega}(t)\frac{1}{\varsigma^4}\frac{\mathrm{d}^2E_{v,\Omega}}{\mathrm{d}t^2}\,.\end{aligned}$$

We now prove that $v = 0$ by proving that $E_{v,\Omega}(t) = 0$ for all $t \in [0, T]$. By contradiction, let assume that there exists $t_1 \in [0, T)$ such that $E_{v,\Omega}(t_1) > 0$. Then by continuity of $E_{v,\Omega}$ there exists $\delta > 0$ such that $E_{v,\Omega}(t) > 0$ for all $t \in [t_1, t_1 + \delta)$. Let now

$$t_2 := \sup\{t \in [t_1, T) \,|\, E_{v,\Omega}(s) > 0 \;\forall s \in [t_1, s)\}\,.$$

Then $E_{v,\Omega}(t_2) = 0$. Indeed, by contradiction, if $E_{v,\Omega}(t_2) > 0$ then by continuity of $E_{v,\Omega}$ there exists $\delta > 0$ such that $E_{v,\Omega}(t) > 0$ for all $t \in [t_2, t_2 + \delta)$: But then the definition of supremum would imply $t_2 \geqslant t_2 + \delta/2$, a contradiction ☠.

Summing up we have

$$E_{v,\Omega}(t) > 0 \;\forall t \in [t_1, t_2)\,, \qquad E_{v,\Omega}(t_2) = 0\,.$$

We then consider

$$f\colon [t_1, t_2) \to \mathbb{R}\,, \qquad f(t) := \log E_{v,\Omega}(t)\,.$$

Then $f \in C^2((t_1, t_2))$, moreover,

$$\frac{\mathrm{d}^2 f}{\mathrm{d}t^2} = \frac{1}{E^2_{v,\Omega}}\left[\frac{\mathrm{d}^2E_{v,\Omega}}{\mathrm{d}t^2}E_{v,\Omega} - \left(\frac{\mathrm{d}E_{v,\Omega}}{\mathrm{d}t}\right)^2\right] \geqslant 0\,.$$

It follows that f is convex, thus

$$f[\lambda t_1 + (1-\lambda)t] \leqslant \lambda f(t_1) + (1-\lambda)f(t)\,,$$

for all $\lambda \in [0, 1]$ and $t \in [t_1, t)$. Exponentiation of the latter inequality leads to

$$E_{v,\Omega}[\lambda t_1 + (1-\lambda)t] \leqslant E_{v,\Omega}(t_1)^\lambda E_{v,\Omega}(t)^{1-\lambda} \underset{t\to t_2^-}{\longrightarrow} 0\,.$$

Thus, for all $\lambda \in (0, 1)$ we find $E_{v,\Omega}[\lambda t_1 + (1-\lambda)t_2] = 0$ which is a contradiction with the fact that $E_{v,\Omega}(t) > 0$ for all $t \in [t_1, t_2)$ ☠. $\square$

23 Parabolic PDEs: Initial-value problem in 1D

In this section we will consider the initial-value problem for the heat equation (22.1) on $\mathbb{R}^n \times \mathbb{R}_+$, *cf.* Problem 1.8b. For the sake of simplicity we will also consider $n = 1$, although all results extends to $n > 1$ almost straightforwardly.

Definition 23.1: The initial-value problem for the heat equation (1.7) on $\mathbb{R}$ consists of determining $u \in C^2(\mathbb{R} \times \mathbb{R}_+) \cap C^0(\mathbb{R} \times \overline{\mathbb{R}_+})$ such that

$$\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0 \qquad \forall (x,t) \in \mathbb{R} \times \mathbb{R}_+ \,, \tag{23.1a}$$

$$u(x,0) = u_0 \qquad \forall x \in \mathbb{R} \,, \tag{23.1b}$$

$$\exists C > 0 \colon \|u(\ ,t)\|_{L^1(\mathbb{R})} \leqslant C \qquad \forall t \in \mathbb{R}_+ \,, \tag{23.1c}$$

where $u_0 \in C^0(\mathbb{R}) \cap L^1(\mathbb{R})$ has been assigned.

The initial-value problem (23.1) requires a mild boundary condition at infinity, *cf.* (23.1c). This is in line with the results of Sections 7-8-15. However, we have also seen in Sections 17-18 that for the particularly simple unbounded domain given by $\mathbb{R}^n$, the initial data suffice to determine uniquely the solution to (17.1) —*cf.* Corollary 17.1 and Theorems 17.2-18.3-18.4. This nicer behaviour is not shared by the heat equation (22.1) as shown by the following counterexample due to Tychonoff [5, §7.1].

Example 23.2 (Tychonoff's counterexample): Consider $u_0 = 0$: Then $u = 0$ is a solution to (23.1). We will now present another solution: To this avail we consider the following ansatz

$$u(x,t) = \sum_{k\geqslant 0} \frac{1}{k!} v_k(t) x^k \,,$$

where the functions v_k will be determined by imposing (23.1a). In particular, a formal computation leads to

$$\begin{aligned}
0 = \left[\frac{1}{\varsigma^2}\frac{\partial}{\partial t} - \frac{\partial^2}{\partial x^2}\right] u &= \sum_{k\geqslant 0} \frac{1}{k!}\frac{1}{\varsigma^2}\dot{v}_k(t) x^k - \sum_{k\geqslant 2} \frac{1}{(k-2)!} v_k(t) x^{k-2} \\
&= \sum_{k\geqslant 0} \frac{1}{k!}\left[\frac{1}{\varsigma^2}\dot{v}_k(t) - v_{k+2}(t)\right] x^k \\
&\Rightarrow v_{k+2} = \frac{\dot{v}_k}{\varsigma^2} \qquad k \geqslant 0 \,.
\end{aligned}$$

In the previous relations we are free to choose v_0, v_1 and this leads to

$$u(x,t) = \sum_{k\geqslant 0} \frac{1}{(2k)!} v_0^{(k)}(t) \left(\frac{x}{\varsigma}\right)^{2k} + x \sum_{k\geqslant 0} \frac{1}{(2k+1)!} v_1^{(k)}(t) \left(\frac{x}{\varsigma}\right)^{2k} \,. \tag{23.2}$$

Tychonoff's solution is determined by the choices

$$v_0(t) = \begin{cases} \exp(-1/t^\alpha) & t > 0 \\ 0 & t = 0 \end{cases}, \qquad v_1 = 0\,, \tag{23.3}$$

where $\alpha > 1$. Notice that $v_0 \in C^\infty(\overline{\mathbb{R}_+})$ but it is not real-analytic since $v_0^{(k)}(0) = 0$ for all $k \in \mathbb{N} \cup \{0\}$. In fact, for $t > 0$, v_0 extends to a holomorphic function on $\mathbb{C}\backslash\{0\}$. This allows to use Cauchy's Theorem to compute

$$v_0^{(k)}(t) = \frac{k!}{2\pi i} \oint\limits_{+\partial B_{t\theta}(t)} \frac{v_0(z)}{(t-z)^{k+1}} \mathrm{d}z = \frac{k!}{2\pi (t\theta)^k} \int_0^{2\pi} v_0(t + t\theta e^{i\varphi}) \mathrm{d}\varphi\,,$$

where $\theta \in (0,1)$ so that $\partial B_{t\theta}(t) \subset \{z \in \mathbb{C} \,|\, \Re z > 0\}$. Since $v_0(t + t\theta^{i\varphi}) = \exp[-t^{-\alpha}\Re(1 + e^{i\varphi})] \leqslant \exp[-2t^{-\alpha}]$ we conclude that

$$|v_0^{(k)}(t)| \leqslant \frac{k!}{(\theta t)^k} \exp\Big[-\frac{1}{2t^\alpha}\Big]. \tag{23.4}$$

This implies in particular that

$$\begin{aligned}\Big|\sum_{k\geqslant 0} \frac{1}{(2k)!} v_0^{(k)}(t) \left(\frac{x}{\varsigma}\right)^{2k}\Big| &\leqslant \sum_{k\geqslant 0} \frac{1}{k!} \frac{|x|^{2k}}{(\theta\varsigma^2 t)^k} \exp\Big[-\frac{1}{2t^\alpha}\Big] \qquad \text{Eq. (23.4)}\,,\ k!^2 \leqslant (2k)! \\ &= \exp\left[\frac{|x|^2}{\theta\varsigma^2 t} - \frac{1}{2t^\alpha}\right] = \exp\left[\frac{1}{t}\left(\frac{|x|^2}{\theta\varsigma^2} - \frac{1}{2t^{\alpha-1}}\right)\right],\end{aligned}$$

where the latter function is well-defined and smooth on $\mathbb{R} \times \overline{\mathbb{R}_+}$ on account of the hypothesis $\alpha > 1$.

Thus, the function u defined by (23.2) with (23.3) is well-defined for all $x \in \mathbb{R}$ and $t \geqslant 0$, actually $\lim\limits_{t\to 0^+} u(x,t) = 0$. In fact, iterated application of Corollary 9.5 and (23.4) shows that $u \in C^\infty(\mathbb{R} \times \overline{\mathbb{R}_+})$. To wit, let us prove that $u \in C^1(\mathbb{R} \times \overline{\mathbb{R}_+})$. We begin by proving that $u \in C^1(\mathbb{R} \times \mathbb{R}_+)$: For that let $(x_0, t_0) \in \mathbb{R} \times \mathbb{R}_+$ and let $R_{t_0}, R_{x_0} > 0$ such that $t_0 - R_{t_0} > 0$. Then for all $x \in (x_0 - R_{x_0}, x_0 + R_{x_0})$ ad $t \in (t_0 - R_{t_0}, t_0 + R_{t_0})$ —notice that $0 \notin (t_0 - R_{t_0}, t_0 + R_{t_0})$—

we have

$$
\begin{aligned}
\left|\frac{\partial}{\partial x}\frac{1}{(2k)!}v_0^{(k)}(t)\left(\frac{x}{\varsigma}\right)^{2k}\right| &= \frac{1}{\varsigma}\left|\frac{1}{(2k-1)!}v_0^{(k)}(t)\left(\frac{x}{\varsigma}\right)^{2k-1}\right| \\
&\leqslant \frac{1}{\varsigma\theta t}\frac{1}{(k-1)!}\left(\frac{|x|^2}{\varsigma^2\theta t}\right)^{k-1} \qquad \text{Eq. (23.4)}\,,\ (2k-1)! \geqslant k!(k-1)! \\
&\leqslant \frac{1}{\varsigma\theta(t_0+R_{t_0})}\frac{1}{(k-1)!}\left(\frac{(|x_0|+R_{x_0})^2}{\varsigma^2\theta(t_0-R_{t_0})}\right)^{k-1} \in \ell^1(\mathbb{N}) \\
\left|\frac{\partial}{\partial t}\frac{1}{(2k)!}v_0^{(k)}(t)\left(\frac{x}{\varsigma}\right)^{2k}\right| &= \left|\frac{1}{(2k)!}v_0^{(k+1)}(t)\left(\frac{x}{\varsigma}\right)^{2k}\right| \\
&\leqslant \frac{1}{k!}\frac{1}{(\theta t)^{k+1}}\left(\frac{|x|}{\varsigma}\right)^{2k} \qquad \text{Eq. (23.4)}\ ,\ (2k)! \geqslant (k+1)!k! \\
&\leqslant \frac{1}{\theta(t_0+R_{t_0})}\frac{1}{k!}\left(\frac{(x_0+R_{x_0})^2}{\varsigma^2\theta(t_0-R_{t_0})}\right)^{k} \in \ell^1(\mathbb{N})\,.
\end{aligned}
$$

Thus ii is fulfilled and therefore $u \in C^1(\mathbb{R}\times\mathbb{R}_+)$, moreover,

$$
\begin{aligned}
\left|\frac{\partial u}{\partial x}\right| &= \frac{1}{\varsigma}\left|\sum_{k\geqslant 1}\frac{1}{(2k-1)!}v_0^{(k)}(t)\left(\frac{x}{\varsigma}\right)^{2k-1}\right| \leqslant \frac{1}{\varsigma\theta t}\sum_{k\geqslant 1}\frac{1}{(k-1)!}\left(\frac{|x|^2}{\varsigma^2\theta t}\right)^{k-1}\exp\left[-\frac{1}{2t^\alpha}\right] \\
&= \frac{1}{\varsigma\theta t}\exp\left[\frac{1}{t}\left(\frac{|x|^2}{\theta\varsigma^2}-\frac{1}{2t^{\alpha-1}}\right)\right] \underset{t\to 0^+}{\longrightarrow} 0\,, \\
\left|\frac{\partial u}{\partial t}\right| &= \left|\sum_{k\geqslant 0}\frac{1}{(2k)!}v_0^{(k+1)}(t)\left(\frac{x}{\varsigma}\right)^{2k}\right| \leqslant \sum_{k\geqslant 0}\frac{1}{k!}\frac{1}{(\theta t)^{k+1}}\left(\frac{|x|}{\varsigma}\right)^{2k}\exp\left[-\frac{1}{2t^\alpha}\right] \\
&= \frac{1}{\theta t}\exp\left[\frac{1}{t}\left(\frac{|x|^2}{\theta\varsigma^2}-\frac{1}{2t^{\alpha-1}}\right)\right] \underset{t\to 0^+}{\longrightarrow} 0\,,
\end{aligned}
$$

which shows that $u \in C^1(\mathbb{R}\times\overline{\mathbb{R}_+})$. Finally u is a non-vanishing solution to the initial-valued problem (23.1).

Remark 23.3:

(i) The striking difference between the Cauchy problem for the wave equation in 1D (17.1) and the initial-value problem for the heat equation (1.8b) can be traced back to the non-physical propagation described by the heat equation as opposed to the finiteness of the propagation predicted by the wave equation, *cf.* Section 16. Despite this hurdle, it is enough to impose a mild condition on the initial data u_0 and on u to restore well-posedness of the problem, *cf.* (23.1c).

(ii) The integrability assumption on u_0 can be interpreted by recalling that Equation (23.1a) describes a diffusive phenomenon. In particular u_0 is interpreted as the initial density of a given population of interest: Thus, the quantity $\|u_0\|_{L^1(\mathbb{R})}$ represents the total population and the requirement $u_0 \in L^1(\mathbb{R})$ ensures that the total population is finite. From this point of view, condition (23.1c) implies that the total population $\|u(\ ,t)\|_{L^1(\mathbb{R})}$ at each time $t \geqslant 0$ is finite and bounded. On physical grounds one may also argue that we should obtain conservation of the total population, *i.e.* $\|u(\ ,t)\|_{L^1(\mathbb{R})} = \|u_0\|_{L^1(\mathbb{R})}$: We will come back to this in Remark 23.15.

From a mathematical point of view assumption (23.1c) allows to discuss (23.1) by means of techniques of Fourier analysis, which are interesting by their own.

A (not so) short detour on Fourier analysis

We shall now recollect a few relevant results on Fourier analysis which will be useful in the analysis of the initial-value problem (23.1). The treatise is focussed to the results relevant for the application to (23.1): For a broader overview of the topic we refer to [16, §. 9].

The main tools we are interested in this section are the notion of Fourier transform and of convolution product. The latter is a suitable product on $L^1(\mathbb{R})$, the former is a map from $L^1(\mathbb{R})$ to $C^0_0(\mathbb{R})$ which decomposes a given function f in its "frequency components". Here $\boxed{C^0_0(\mathbb{R})}$ denotes the space of continuous functions $f \in C^0(\mathbb{R})$ vanishing at infinity, that is, such that for all $\varepsilon > 0$ there exists a compact $K_\varepsilon \subset \mathbb{R}$ with $|f(x)| < \varepsilon$ for $x \notin K_\varepsilon$.

For later convenience we recall the following technical result, which descends from the dominated convergence Theorem 9.2.

Lemma 23.4: Let $T_x \colon L^1(\mathbb{R}^n) \to L^1(\mathbb{R}^n)$, $x \in \mathbb{R}^n$, be the linear operator defined by $(T_x f)(y) := f(y-x)$. Then T_x is strongly continuous, that is,

$$\lim_{x\to 0} \|T_x f - f\|_{L^1(\mathbb{R}^n)} = 0 \qquad \forall f \in L^1(\mathbb{R}^n)\,. \tag{23.5}$$

Proof. The proof of this result requires the density of $C^0_c(\mathbb{R}^n)$ in $L^1(\mathbb{R}^n)$, which we will take for granted, *cf.* [16, Thm. 3.14].

To begin with we observe that $T_x f \in L^1(\mathbb{R}^n)$, moreover, $\|T_x f\|_{L^1(\mathbb{R}^n)} = \|f\|_{L^1(\mathbb{R}^n)}$. We now prove Equation (23.5) for $f \in C^0_c(\mathbb{R}^n)$. To this avail we observe that, if $R > 0$ is such that $\operatorname{supp}(f) \subset B_R(0)$, then $\operatorname{supp}(T_x f) \subset B_R(x)$. Moreover, we may find $R_1 > 0$ such that $\operatorname{supp}(T_x f - f) \subset B_{R_1}(0)$ for all $x \in B_1(0)$. We then have

$$\|T_x f - f\|_{L^1(\mathbb{R}^n)} = \int_{\mathbb{R}^n} |f(y-x) - f(x)| \mathrm{d}y = \int_{B_{R_1}(0)} |f(y-x) - f(x)| \mathrm{d}y\,,$$

where, since $f \in C^0_c(\mathbb{R}^n)$ we have $|f(y-x) - f(x)| \xrightarrow[x\to 0]{} 0$ for all $y \in \mathbb{R}^n$. Moreover, $|f(y-x) - f(x)| \leqslant 2\|f\|_\infty 1_{B_{R_1}(0)}(y)$, thus, Theorem 9.2 implies that $\|T_x f - f\|_{L^1(\mathbb{R}^n)} \xrightarrow[x\to 0]{} 0$.

Let now $f \in L^1(\mathbb{R}^n)$ and $\varepsilon > 0$. Then there exists $f_\varepsilon \in C^0_c(\mathbb{R}^n)$ with $\|f - f_\varepsilon\|_{L^1(\mathbb{R}^n)} < \varepsilon$. Moreover, there exists $\delta_\varepsilon > 0$ such that $\|T_x f_\varepsilon - f_\varepsilon\|_{L^1(\mathbb{R}^n)} < \varepsilon$ for $|x| < \delta_\varepsilon$. Overall we have that, for all $|x| < \delta_\varepsilon$,

$$\|T_x f - f\|_{L^1(\mathbb{R}^n)} \leqslant \|T_x(f - f_\varepsilon)\|_{L^1(\mathbb{R}^n)} + \|T_x f_\varepsilon - f_\varepsilon\|_{L^1(\mathbb{R}^n)} + \|f_\varepsilon - f\|_{L^1(\mathbb{R}^n)} \leqslant 3\varepsilon\,,$$

which proves Equation (23.5). □

Definition 23.5: Let $f \in L^1(\mathbb{R})$. The **Fourier transform** of f is the function $\hat{f}$ defined by

$$\hat{f}\colon \mathbb{R} \to \mathbb{C} \qquad \hat{f}(\xi) := \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(x) e^{-i\xi x} \mathrm{d}x\,. \tag{23.6}$$

The following Lemma proves that Definition 23.5 is well-posed and recollect useful properties.

Lemma 23.6: For all $f \in L^1(\mathbb{R})$, the Fourier transform is well-defined. Moreover, $\hat{f} \in C^0_0(\mathbb{R})$ and

$$\|\hat{f}\|_\infty \leqslant \frac{1}{\sqrt{2\pi}} \|f\|_{L^1(\mathbb{R})} \tag{23.7}$$

$$\int_{\mathbb{R}} \hat{f}(x) g(x) \mathrm{d}x = \int_{\mathbb{R}} f(x) \hat{g}(x) \mathrm{d}x \qquad \forall g \in L^1(\mathbb{R})\,. \tag{23.8}$$

Proof. By direct inspection we have

$$|\hat{f}(\xi)| \leqslant \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} |f(x)| \mathrm{d}x = \frac{1}{\sqrt{2\pi}} \|f\|_{L^1(\mathbb{R})}\,.$$

Moreover, a direct application of Corollary 9.4 ensures that $\hat{f} \in C^0(\mathbb{R})$. It remains to prove that $\hat{f}(\xi) \xrightarrow[\xi\to\infty]{} 0$. To this avail we observe that

$$\hat{f}(\xi) = -\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(x) e^{-i\xi(x+\pi/\xi)} \mathrm{d}x = -\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(x - \pi/\xi) e^{-i\xi x} \mathrm{d}x\,,$$

which implies

$$2|\hat{f}(\xi)| \leqslant \frac{1}{\sqrt{2\pi}} \|T_{\pi/\xi} f - f\|_{L^1(\mathbb{R})} \xrightarrow[\xi\to\infty]{} 0\,, \tag{23.9}$$

where we applied Corollary 23.4.

Concerning Equation (23.8), let $f, g \in L^1(\mathbb{R})$: Then $f\hat{g}, \hat{f}g \in L^1(\mathbb{R})$ because $\hat{f}, \hat{g}$ are bounded: Thus, applying Fubini's Theorem once again, we have

$$\int_{\mathbb{R}} \hat{f}(x) g(x) \mathrm{d}x = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}^2} f(y) e^{-ixy} g(x) \mathrm{d}y \mathrm{d}x = \int_{\mathbb{R}} f(y) \hat{g}(y) \mathrm{d}y\,.$$

□

Example 23.7:

(i) Let $f(x) := e^{-|x|}$. Then $f \in L^1(\mathbb{R})$, moreover,

$$\hat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_0^{+\infty} e^{-(1+i\xi)x} \mathrm{d}x + \frac{1}{\sqrt{2\pi}} \int_0^{+\infty} e^{-(1-i\xi)x} \mathrm{d}x = \sqrt{\frac{2}{\pi}} \frac{1}{1+\xi^2} \,.$$

(ii) Let $g(x) := e^{-x^2}$. By applying Corollary 9.5 we find that $\hat{g} \in C^\infty(\mathbb{R})$, moreover,

$$\begin{aligned} \frac{\mathrm{d}\hat{g}}{\mathrm{d}\xi} = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-x^2}(-ix) e^{-ix\xi} \mathrm{d}x &= \frac{i}{2} \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \frac{\mathrm{d}}{\mathrm{d}x}\Big[e^{-x^2}\Big] e^{-ix\xi} \mathrm{d}x \\ &= \frac{i}{2} \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-x^2}(i\xi) e^{-ix\xi} \mathrm{d}x = -\frac{\xi}{2} \hat{g}(\xi) \,. \end{aligned}$$

The latter equation implies

$$\hat{g}(\xi) = \hat{g}(0) e^{-\xi^2/4} = e^{-\xi^2/4}/\sqrt{2} \,,$$

where we made use of Equation (3.14).

Definition 23.8: The **convolution** of $f, g \in L^1(\mathbb{R})$ is the function $f * g$ defined by

$$(f * g)(x) := \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(x-y) g(y) \mathrm{d}y \,, \tag{23.10}$$

for almost all $x \in \mathbb{R}$.

Lemma 23.9: For all $f, g \in L^1(\mathbb{R})$ the convolution $f * g$ is well-defined and $f * g \in L^1(\mathbb{R})$, moreover,

$$\|f * g\|_{L^1(\mathbb{R})} \leqslant \frac{1}{\sqrt{2\pi}} \|f\|_{L^1(\mathbb{R})} \|g\|_{L^1(\mathbb{R})} \tag{23.11}$$

$$\widehat{f * g} = \hat{f} \hat{g} \,. \tag{23.12}$$

Proof. By direct inspection we have

$$\int_{\mathbb{R}} |(f * g)(x)| \mathrm{d}x \leqslant \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}^2} |f(x-y) g(y)| \mathrm{d}y \mathrm{d}x = \frac{1}{\sqrt{2\pi}} \|f\|_{L^1(\mathbb{R})} \|g\|_{L^1(\mathbb{R})} \,,$$

where we used Fubini's Theorem as well as the invariance under translation of the Lebesgue measure. It follows that $f * g$ is well-defined and in $L^1(\mathbb{R})$. Moreover,

$$\begin{aligned}\widehat{f * g}(\xi) &= \frac{1}{2\pi}\int_{\mathbb{R}^2} f(x-y)g(y)\mathrm{d}y e^{i\xi x}\mathrm{d}x \\ &= \frac{1}{2\pi}\int_{\mathbb{R}^2} f(x-y)g(y)\mathrm{d}y e^{i\xi(x-y)}e^{i\xi y}\mathrm{d}x = \hat{f}(\xi)\hat{g}(\xi)\,.\end{aligned}$$

□

The following Theorem proves that, if $\hat{f} = 0$ then $f = 0$. Moreover, it also provides a sufficient condition to recover f from its Fourier transform.

Theorem 23.10: For all $\lambda > 0$ let $h(x) := \sqrt{\frac{2}{\pi}}\frac{1}{1+x^2}$ and set $h_\lambda(x) := h(x/\lambda)/\lambda$. Then, for all $f \in L^1(\mathbb{R})$ it holds

$$\|f * h_\lambda - f\|_{L^1(\mathbb{R})} \xrightarrow[\lambda\to 0]{} 0\,. \tag{23.13}$$

Moreover, for almost all $x \in \mathbb{R}$ it holds

$$(f * h_\lambda)(x) = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} \hat{f}(\xi)e^{-\lambda|\xi|}e^{i\xi x}\mathrm{d}\xi\,. \tag{23.14}$$

In particular, $f = 0$ if and only if $\hat{f} = 0$. Moreover, if $\hat{f} \in L^1(\mathbb{R})$ then

$$f(x) = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} \hat{f}(z)e^{i\xi x}\mathrm{d}\xi = \check{\hat{f}}(x)\,, \tag{23.15}$$

for almost all $x \in \mathbb{R}$ where we set $\boxed{\check{f}}(\xi) := \hat{f}(-\xi)$.

Proof. We observe that

$$\int_{\mathbb{R}} h_\lambda(x)\mathrm{d}x = \int_{\mathbb{R}} h(x)\mathrm{d}x = 1\,.$$

By direct inspection we have

$$\begin{aligned}\|f * h_\lambda - f\|_{L^1(\mathbb{R})} &= \int_{\mathbb{R}} |(f * h_\lambda)(x) - f(x)|\mathrm{d}x \\ &= \int_{\mathbb{R}} \Big|\int_{\mathbb{R}} f(x-y)h_\lambda(y) - \int_{\mathbb{R}} f(x)h_\lambda(y)\mathrm{d}y\Big|\mathrm{d}x \\ &\leqslant \int_{\mathbb{R}}\int_{\mathbb{R}} |f(x-y) - f(x)|h_\lambda(y)\mathrm{d}y\mathrm{d}x \\ &\leqslant \int_{\mathbb{R}} \|T_{\lambda y}f - f\|_{L^1(\mathbb{R})}h(y)\mathrm{d}y \xrightarrow[\lambda\to 0]{} 0\,,\end{aligned}$$

where in the last line we used Corollary 23.4 together with Corollary 9.4 —notice that assumption ii is fulfilled by observing that $\|T_y f - f\|_{L^1(\mathbb{R})} h_\lambda(y) \leqslant 2\|f\|_{L^1(\mathbb{R})}|h_\lambda(y)|$.

Recalling Example 23.7 we have

$$h_\lambda(\xi) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-\lambda|x|} e^{i\xi x} \mathrm{d}x \,.$$

It follows that

$$\begin{aligned}(f * h_\lambda)(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(x-y) h_\lambda(y) \mathrm{d}y &= \frac{1}{2\pi} \int_{\mathbb{R}} f(x-y) \int_{\mathbb{R}} e^{-\lambda|\xi|} e^{i\xi y} \mathrm{d}\xi \mathrm{d}y \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-\lambda|\xi|} f(z) e^{i\xi(x-z)} \mathrm{d}\xi \mathrm{d}z \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-\lambda|\xi|} \hat{f}(\xi) e^{i\xi x} \mathrm{d}\xi \,.\end{aligned}$$

Equation (23.15) follows from Corollary 9.4 together with the bound $|e^{-\lambda|\xi|}\hat{f}(\xi)e^{ix\xi}| \leqslant |\hat{f}(\xi)|$. □

Remark 23.11: Theorem 23.10 has particular useful consequences when applied to a suitable class of functions. To this avail we introduce the space of **Schwartz functions** $\boxed{S(\mathbb{R})}$ by

$$S(\mathbb{R}) := \left\{ f \in C^\infty(\mathbb{R}) \,|\, \forall k, \ell \in \mathbb{Z}_+ \; \exists C_{k,\ell} > 0\colon \sup_{x \in \mathbb{R}} |x^k f^{(\ell)}(x)| \leqslant C_{k,\ell} \right\} . \tag{23.16}$$

In other words $f \in S(\mathbb{R})$ if it is smooth and all its derivatives decay faster than any polynomials as $x \to \infty$. For later convenience it will be useful to denote by $\boxed{\|f\|_{k,\ell}} := \sup_{x\in\mathbb{R}} |x^k f^{(\ell)}(x)|$.

We now recollect a few relevant property of the space $S(\mathbb{R})$:

(i) Clearly $C^\infty_c(\mathbb{R}) \subset S(\mathbb{R})$, in fact, for all $f \in S(\mathbb{R})$ there exists $(f_n)_n \subset C^\infty_c(\mathbb{R})$ such that $\|f - f_n\|_{k,\ell} \underset{n\to\infty}{\longrightarrow} 0$ for all $k, \ell \in \mathbb{Z}_+$. For that, it suffices to consider $g \in C^\infty_c(\mathbb{R})$ such that $g|_{(-1,1)} = 1$ and set $f_n(x) := f(x)g(x/n)$. Then, considering for simplicity $\ell = 0$, for all $\varepsilon > 0$ and $\alpha \in \mathbb{Z}_+$ there exists $\delta_\varepsilon > 0$ such that $|x^k f_n(x)| < \varepsilon$ whenever $|x| > \delta_\varepsilon$, moreover,

$$\|f - f_n\|_{k,0} \leqslant 2\varepsilon + \sup_{|x|<\delta_\varepsilon} |x^k f(x)(1 - g(x/n)| = 2\varepsilon \,,$$

where we choose $n \geqslant \delta_\varepsilon$.

(ii) If $f \in S(\mathbb{R})$ then $f^{(\ell)} \in L^1(\mathbb{R})$ for all $\ell \in \mathbb{Z}_+$. Moreover, Corollary 9.5 implies that $\hat{f} \in C^\infty(\mathbb{R})$ —this follows because $x^k f(x)$ is integrable for all $k, \ell \in \mathbb{Z}_+$. Furthermore, for all $k \in \mathbb{Z}_+$ and $\xi \in \mathbb{R}$ we have

$$\begin{aligned}(i\xi)^k \frac{\mathrm{d}^\ell \hat{f}}{\mathrm{d}\xi^\ell}(\xi) &= \frac{(-1)^k}{\sqrt{2\pi}} \int_{\mathbb{R}} f(x)(-ix)^\ell \frac{\mathrm{d}^k}{\mathrm{d}x^k} e^{-i\xi x} \mathrm{d}x \\ &= \frac{(-i)^\ell}{\sqrt{2\pi}} \int_{\mathbb{R}} \frac{\mathrm{d}^k}{\mathrm{d}x^k}[x^\ell f(x)] e^{-i\xi x} \mathrm{d}x = (\widehat{M_\ell f)^{(k)}}(\xi) \,,\end{aligned}$$

where we set $\boxed{M_\ell f}(x) := x^\ell f(x)$. The latter equation implies that $\hat{f} \in S(\mathbb{R})$, therefore, $\hat{}: S(\mathbb{R}) \to S(\mathbb{R})$.

In particular $\hat{f} \in L^1(\mathbb{R})$ and Equation (23.15) applies. Moreover, the decay property of $\hat{f}$ and Corollary 9.5 also prove that

$$\frac{\mathrm{d}^\ell f}{\mathrm{d}x^\ell}(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \hat{f}(\xi)(i\xi)^\ell e^{ix\xi} \mathrm{d}\xi = (-i)^\ell (\widehat{M_\ell \check{f}})(x) .$$

We conclude that the Fourier transform is a 1-1 linear map from $S(\mathbb{R})$ to itself.

(iii) The following observation will be relevant for the proof of Theorem 25.3. Let $u \in L^1_{\mathrm{LOC}}(\mathbb{R})$ be polynomially bounded —that is, there exist $n \in \mathbb{Z}_+$ and $C > 0$ such that $|f(x)| \leqslant C|x|^n$ for all $x \in \mathbb{R}\backslash\{0\}$— and such that

$$\int_{\mathbb{R}} u(x)\hat{f}(x)\mathrm{d}x = 0 \qquad \forall f \in C^\infty_c(\mathbb{R}) .$$

Then $u = 0$. Indeed, let $f \in S(\mathbb{R})$: Then there exists $f_n \in C^\infty_c(\mathbb{R})$ such that $\|f_n - f\|_{k,\ell} \underset{n\to\infty}{\longrightarrow} 0$ for all $k, \ell \in \mathbb{Z}_+$. The dominated convergence Theorem 9.2 implies that $\hat{f}_n - \hat{f} \underset{n\to\infty}{\longrightarrow} 0$ uniformly, moreover,

$$\int_{\mathbb{R}} u(x)\hat{f}(x)\mathrm{d}x = \lim_{n\to\infty} \int_{\mathbb{R}} u(x)\hat{f}_n(x)\mathrm{d}x = 0 .$$

Since $\hat{S}(\mathbb{R}) = S(\mathbb{R})$ it follows that

$$\int_{\mathbb{R}} u(x)f(x)\mathrm{d}x = 0 \qquad \forall f \in S(\mathbb{R}) ,$$

which implies $u = 0$ by approximating characteristic functions over open subsets with Schwartz functions.

Back to the heat topic

We shall now provide an existence and uniqueness result for Problem (23.1).

Theorem 23.12: There exists a unique solution u to Problem (23.1). In particular, for all $t \in \overline{\mathbb{R}_+}$ the Fourier transform of $u(\ ,t)$ is given by

$$\hat{u}(\xi, t) = \hat{u}_0(\xi) e^{-\varsigma^2 \xi^2 t} . \tag{23.17}$$

Moreover, u is given by

$$u(x,t) := \begin{cases} u_0(x) & t = 0 \\ (G_t * u_0)(x) & t > 0 \end{cases} , \tag{23.18}$$

defines the solution to Problem (23.1) where for all $t > 0$ we set

$$G_t(x) := \frac{1}{\sqrt{2\varsigma^2 t}} e^{-x^2/4\varsigma^2 t} . \tag{23.19}$$

Before going into the details of the proof of Theorem 23.12 we will state and prove the following technical Lemma of independent interest.

LEMMA 23.13: Let $u \in L^1_{\mathrm{LOC}}(\mathbb{R}_+)$ and $a > 0$ be such that

$$\int_{\mathbb{R}_+} u(t)[\dot{\tau}(t) - a\tau(t)]\mathrm{d}t = 0 \,, \tag{23.20}$$

for all $\tau \in C^\infty_c(\mathbb{R}_+)$. Then there exists $u_0 \in \mathbb{R}$ such that $u(t) = u_0 e^{-at}$ for almost all $t \in \mathbb{R}_+$.

Proof. Let $u_a(t) := e^{at}u(t) \in L^1_{\mathrm{LOC}}(\mathbb{R}_+)$. For all $\tau \in C^\infty_c(\mathbb{R}_+)$ let $\tau_a(t) := e^{at}\tau(t) \in C^\infty_c(\mathbb{R}_+)$. Then Equation (23.20) simplifies to

$$0 = \int_{\mathbb{R}_+} u(t)[\dot{\tau}_a(t) - a\tau_a(t)]\mathrm{d}t = \int_{\mathbb{R}_+} u_a(t)\dot{\tau}(t)\mathrm{d}t \,. \tag{23.21}$$

We will now prove that (23.21) implies that u_a is constant almost everywhere.

To this avail let $f \in C^\infty_c(\mathbb{R}_+)$: Then $f = \dot{\tau}$ for

$$\tau(t) := \int_0^t f(s)\mathrm{d}s \,.$$

Notice that $\tau \in C^\infty(\mathbb{R}_+)$ is compactly supported if and only if

$$\int_{\mathbb{R}_+} f(t)\mathrm{d}t = 0 \,. \tag{23.22}$$

Therefore, we may rewrite Equation (23.21) as

$$\int_{\mathbb{R}_+} u_a(t)f(t)\mathrm{d}t = 0 \qquad \forall f \in C^\infty_c(\mathbb{R}_+) \colon I(f) = 0 \,, \tag{23.23}$$

where we set $\boxed{I(f)} := \int_{\mathbb{R}_+} f(t)\mathrm{d}t$.

Let now $h \in C^\infty_c(\mathbb{R}_+)$ be such that $I(h) = 1$. Then, for all $f \in C^\infty_c(\mathbb{R}_+)$ we may write

$$f = [f - I(f)h] + I(f)h \,.$$

We observe that $f - I(f)h \in C_c^\infty(\mathbb{R}_+)$ is such that $I[f - I(f)h] = 0$, therefore,

$$\int_{\mathbb{R}_+} u_a(t) f(t) \mathrm{d}t = \int_{\mathbb{R}_+} u_a(t)[f - I(f)h](t)\mathrm{d}t + I(f)\int_{\mathbb{R}_+} u_a(t)h(t)\mathrm{d}t = u_0 \int_{\mathbb{R}_+} f(t)\mathrm{d}t\,,$$

where we applied Equation (23.23) and set $u_0 := \int_{\mathbb{R}_+} u_a(t)h(t)\mathrm{d}t$. It follows that

$$\int_{\mathbb{R}_+} [u_a(t) - u_0] f(t)\mathrm{d}t = 0\,,$$

for all $f \in C_c^\infty(\mathbb{R}_+)$. The arbitrariness of f leads to $u_a(t) = u_0$ for almost all $t \in \mathbb{R}_+$, hence the thesis. □

Remark 23.14:

(i) Assuming $u \in L^1_{\mathrm{LOC}}(\mathbb{R}_+) \cap C^1(\mathbb{R}_+)$ leads to a straightforward proof of Lemma 23.13. Indeed, for all $\tau \in C_c^\infty(\mathbb{R}_+)$ integration by parts would lead to

$$0 = \int_{\mathbb{R}_+} u(t)[\dot{\tau}(t) - a\tau(t)]\mathrm{d}t = -\int_{\mathbb{R}_+} [\dot{u}(t) + au(t)]\tau(t)\mathrm{d}t\,.$$

The arbitrariness of τ would imply $\dot{u} + au = 0$, hence the thesis. The crucial point of (23.20) is that it allows to deduce the same result under a mild local integrability assumption on u. Equation (23.20) is called the **weak form** of the equation $\dot{u} + au = 0$, *cf.* Remark 19.10.

(ii) It is worth pointing out that the assumption of Lemma 23.13 can be greatly weakened. In particular, let $U \colon C_c^\infty(\mathbb{R}_+) \to \mathbb{C}$ be any linear operator such that $U(\dot{\tau} - a\tau) = 0$ for all $\tau \in C_c^\infty(\mathbb{R}_+)$. By following the same steps of the proof of Lemma 23.13 we find that there exists $u_0 \in \mathbb{R}$ such that

$$U(\tau) = \int_{\mathbb{R}_+} u_0 e^{-at}\tau(t)\mathrm{d}t\,.$$

Proof of Thm. 23.12. To begin with, we will prove uniqueness to (23.1) by computing explicitly the Fourier transform of any solution u. This will imply uniqueness on account of Theorem 23.10.

$\boxed{!}$ Let $u \in C^2(\mathbb{R} \times \mathbb{R}_+) \cap C^0(\mathbb{R} \times \overline{\mathbb{R}_+})$ be a solution to (23.1). Then, for all $\tau \in C_c^\infty(\mathbb{R}_+)$ and

$\chi \in C_c^\infty(\mathbb{R})$ we find

$$\begin{aligned}
0 &= \int_{\mathbb{R}_+}\int_{\mathbb{R}}\left[\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2}\right](x,t)\tau(t)\chi(x)\mathrm{d}x\mathrm{d}t && \text{Eq. (23.1a)}\\
&= -\int_{\mathbb{R}_+}\int_{\mathbb{R}} u(x,t)\left[\frac{1}{\varsigma^2}\dot\tau(t)\chi(x) + \tau(t)\chi''(x)\right]\mathrm{d}x\mathrm{d}t \\
&= -\int_{\mathbb{R}_+}\int_{\mathbb{R}} u(x,t)\left[\frac{1}{\varsigma^2}\dot\tau(t)\hat{\check{\chi}}(x) - \tau(t)\widehat{M_2\check{\chi}}(x)\right]\mathrm{d}x\mathrm{d}t && \text{Eq. (23.15)}\\
&= -\int_{\mathbb{R}_+}\int_{\mathbb{R}} \hat u(\xi,t)\left[\frac{1}{\varsigma^2}\dot\tau(t)\hat\chi(-\xi) - \xi^2\tau(t)\hat\chi(-\xi)\right]\mathrm{d}\xi\mathrm{d}t && \text{Eq. (23.8)}\\
&= -\int_{\mathbb{R}}\left[\int_{\mathbb{R}_+} \hat u(\xi,t)\left[\frac{1}{\varsigma^2}\dot\tau(t) - \xi^2\tau(t)\right]\mathrm{d}t\right]\hat\chi(-\xi)\mathrm{d}\xi\,.
\end{aligned}$$

In the last equality we applied Fubini-Tonelli's Theorem which applies on account of the estimate

$$\begin{aligned}
&\int_{\mathbb{R}}\int_{\mathbb{R}_+} |\hat u(\xi,t)[\dot\tau(t) - \varsigma^2\xi^2\tau(t)]\hat\chi(-\xi)|\mathrm{d}t\mathrm{d}\xi \\
&\qquad\qquad \leqslant \frac{C}{\sqrt{2\pi}}\int_{\mathbb{R}}\int_{\mathbb{R}_+}[|\dot\tau(t)| + \varsigma^2|\xi|^2|\tau(t)|]|\hat\chi(-\xi)|\mathrm{d}t\mathrm{d}\xi < \infty\,,
\end{aligned}$$

where $C > 0$ is such that $\|u(\ ,t)\|_{L^1(\mathbb{R})} \leqslant C$ while we made use of inequality (23.7). Lemma 23.6 and Theorem 9.2 implies that

$$\xi \mapsto \int_{\mathbb{R}_+} \hat u(\xi,t)[\dot\tau(t) - \varsigma^2\xi^2\tau(t)]\mathrm{d}t\,,$$

is continuous and polynomially bounded, thus, the arbitrariness of χ and Remark 23.11 imply that

$$U_\xi(\dot\tau - \varsigma^2\xi^2\tau) = 0 \quad \text{where} \quad U_\xi(\tau) := \int_{\mathbb{R}_+}\hat u(\xi,t)\tau(t)\mathrm{d}t\,,$$

for almost all $\xi \in \mathbb{R}$ and all $\tau \in C_c^\infty(\mathbb{R}_+)$. Lemma 23.13 and Remark 23.14 imply Equation (23.17). Uniqueness of u follows from Theorem 23.10.

$\boxed{\exists}$ By direct inspection we find $G_t \in L^1(\mathbb{R})$ with $\|G_t\|_{L^1(\mathbb{R})} = \sqrt{2\pi}$. Moreover, the results of Example 23.7 entail $\hat G_t(\xi) = e^{-\varsigma^2\xi^2 t}$.

It follows that the function u defined by Equation (23.18) is well-defined, moreover, $u(\ ,t) \in L^1(\mathbb{R})$ for all $t > 0$ with $\|u(\ ,t)\|_{L^1(\mathbb{R})} \leqslant \|u_0\|_{L^1(\mathbb{R})}$ on account of inequality (23.11). Thus, condition (23.1c) is fulfilled. Moreover, Equation (23.12) implies that $\hat u(\xi,t) = \hat u_0(\xi)e^{-\varsigma^2\xi^2 t}$

which is compatible with Equation (23.17) and shows that $\hat{u}(\ ,t) \in L^1(\mathbb{R})$: Thus, Equation (23.15) implies

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \hat{u}_0(\xi) e^{-\varsigma^2 \xi^2 t} e^{ix\xi} \mathrm{d}\xi \qquad \forall (x,t) \in \mathbb{R} \times \mathbb{R}_+ . \tag{23.24}$$

We will now prove that u is a solution to (23.1). To this avail let $a > 0$: Then, for all $x \in \mathbb{R}$ and $t \geqslant a$ we have, for all $k, \ell \in \mathbb{Z}_+$,

$$\left| \frac{\partial^{k+\ell}}{\partial x^k \partial t^\ell} \hat{u}_0(\xi) e^{-\varsigma^2 \xi^2 t} e^{i\xi x} \right| \leqslant \varsigma^{2\ell} |\hat{u}_0(\xi)| |\xi|^{k+2\ell} e^{-\varsigma^2 \xi^2 a} \in L^1(\mathbb{R}) .$$

The latter inequality implies that condition (ii) of Corollary 9.5 is fulfilled (multiple times), therefore, $u \in C^\infty(\mathbb{R} \times [a, +\infty))$. Since this holds for all $a > 0$ we have $u \in C^\infty(\mathbb{R} \times \mathbb{R}_+)$. Equation (23.1a) follows from Corollary (9.5) and Equation (23.17).

It remains to prove condition (23.1b): To this avail we compute

$$(G_t * u_0 - u_0)(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} G_t(x-y)[u_0(y) - u_0(x)] \mathrm{d}y .$$

Let $R > 0$ and $\chi \in C^\infty_c(\mathbb{R})$ be such that $\mathrm{supp}(\chi) \subset B_{2R}(0)$, $\chi|_{B_R(0)} = 1$ and $0 \leqslant \chi \leqslant 1$. Then

$$\begin{aligned} (G_t * u_0 - u_0)(x) &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} G_t(x-y)[\chi(x-y) u_0(y) - u_0(x)] \mathrm{d}y \\ &+ \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} G_t(x-y)(1 - \chi(x-y)) u_0(y) \mathrm{d}y \\ &= I_1 + I_2 . \end{aligned}$$

Concerning I_1 we have

$$\begin{aligned} I_1 &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} G_t(x-y)[\chi(x-y) u_0(y) - u_0(x)] \mathrm{d}y \\ &= \int_{\mathbb{R}} e^{-z^2} [\chi(\sqrt{4\varsigma^2 t} z) u_0(x + \sqrt{4\varsigma^2 t} z) - u_0(x)] \mathrm{d}z \xrightarrow[t \to 0]{} 0 , \end{aligned}$$

where we applied Corollary 9.4 because

$$\begin{aligned} &[\chi(\sqrt{4\varsigma^2 t} z) u_0(x + \sqrt{4\varsigma^2 t} z) - u_0(x)] \xrightarrow[t \to 0]{} 0 \\ &|[\chi(\sqrt{4\varsigma^2 t} z) u_0(x + \sqrt{4\varsigma^2 t} z) - u_0(x)]| \leqslant 2 \sup_{B_{2R}(x)} |u_0| . \end{aligned}$$

Concerning I_2 we observe that

$$\left| \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} G_t(x-y)(1 - \chi(x-y)) u_0(y) \mathrm{d}y \right| \leqslant \frac{e^{-R^2/4\varsigma^2 t}}{\sqrt{\pi \varsigma^2 t}} \|u_0\|_{L^1(\mathbb{R})} \xrightarrow[t \to 0]{} 0 ,$$

where we used that the integral is over $y \notin B_R(x)$, therefore, $|x - y| > R$.

□

Remark 23.15 (🏃):

(i) If $u_0 \in L^1(\mathbb{R})$ Equation (23.18) still makes sense and defines a function $u \in C^\infty(\mathbb{R} \times \mathbb{R}_+)$ fulfilling Equation (23.1a) and condition (23.1c). Condition (23.1b) holds in a weaker form, in particular

$$\|u(\ ,t) - u_0\|_{L^1(\mathbb{R})} \underset{t \to 0^+}{\to} 0\,.$$

(ii) Equation (23.18) can also be rewritten as

$$u(x,t) = \int_{\mathbb{R}} g_t(x-y)u_0(y)\mathrm{d}y \qquad g_t(x) := \frac{1}{\sqrt{4\pi\varsigma^2 t}}e^{-x^2/4\varsigma^2 t}\,.$$

We recognize that g_t is the probability distribution of a Gaussian random variable of zero mean and covariance $\varsigma^2 t$. The function g_t is called **heat kernel**.

(iii) The proof of Theorem 23.12 shows that the unique solution u fulfils $\|u(\ ,t)\|_{L^1(\mathbb{R})} \leqslant \|u_0\|_{L^1(\mathbb{R})}$. Recalling the interpretation of u as the density of a given population, the norm $\|u(\ ,t)\|_{L^1(\mathbb{R})}$ is interpreted as the total population at time t. Therefore, the previous inequality states that the total population is lower or at most equal to the initial one. However, since the heat equation (23.1a) ought to describe a diffusion phenomenon, one would expect to get conservation of the total population, that is, $\|u(\ ,t)\|_{L^1(\mathbb{R})} = \|u_0\|_{L^1(\mathbb{R})}$.

This apparent mismatch is solved by simply considering a positive initial datum u_0: Only in this case u_0 can be truly interpreted as a population density. From Equation (23.18) it also follows that u is positive, moreover, inequality (23.11) is an equality for positive functions, *cf.* the proof of Lemma 23.9. Thus, for a positive initial datum u_0 we find $\|u(\ ,t)\|_{L^1(\mathbb{R})} = \|u_0\|_{L^1(\mathbb{R})}$, *i.e.* conservation of the total population.

(iv) The solution u defined by (23.18) is a smooth function for all $t > 0$. This smoothing behaviour is typical of the heat equation.

Moreover, the convolution appearing in Equation (23.18) leads to a diffusion effect. Indeed the initial datum u_0 is averaged with the heat kernel g_t. This process spreads the support of u_0: In particular, $u(\ ,t)$ is not compactly supported for $t > 0$ even if u_0 is. This is a glimpse of the infinite propagation predicted by the heat equation.

The heat equation predicts an infinite propagation speed, thus, it fails to correctly describe a diffusion process which, on physical ground, is bound to have a finite propagation speed. Non-linear contributions to the heat equation may restore the finiteness of the propagation speed, *cf.* [18, §2.10]. However, heuristic consideration may allow to identify an approximated finite propagation speed for the heat equation. For that, let consider the solution u to problem (23.1) with a compactly supported datum $u_0 \in C^0_c(\mathbb{R}) \cap L^1(\mathbb{R})$, such that:

(a) $u_0 \geqslant 0$ and (b) $\mathrm{supp}(u_0) \subset (-a, a)$, $0 < a \ll 1$. Equation (23.18) implies that $u \geqslant 0$, moreover,

$$u(x,t) \leqslant U_0 \int_{x-a}^{x+a} g_t(z)\mathrm{d}z \leqslant aU_0 g_t(x+a)\,,$$

where $U_0 := \|u_0\|_\infty$. Let assume we place a probe at distance $R \gg a$ from the origin: Such device records the value $u(R,t)$ of u at R. However, due to its physical limitation, the probe cannot record anything below a threshold value U. Thus, the probe will not record the value of u for all $t \geqslant 0$ such that

$$aU_0 g_t(R+a) \leqslant U \quad \Leftrightarrow \quad t \geqslant Rv_{a,U}\,,$$

where $v_{a,U}$ is estimated numerically. Thus, any probe will detect a finite propagation speed with velocity $v_{a,U}$ depending on the extension of the support of u_0 and on the threshold value U.

24 Parabolic PDEs: Maximum principle

In this section we will prove a maximum principle for the solutions to the heat equation (22.1). This is similar to Theorem 7.6 and will lead to a uniqueness result for problem (22.2) with Dirichlet boundary conditions.

The following definition is inspired by Definition 7.4.

DEFINITION 24.1: Let $\Omega \subset \mathbb{R}^n$ be open and $T > 0$ and let $\Omega_T^+ := \Omega \times (0,T)$. A function $u \in C^2(\Omega_T^+)$ is called:

(I) **subcaloric** (*resp.* **supercaloric**) in Ω_T^+ if $\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \Delta_x u \leqslant 0$ (*resp.* $\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \Delta_x u \geqslant 0$);

(II) **caloric** in Ω_T^+ if it is both subcaloric and supercaloric in Ω_T^+, thus it solves the heat equation (22.1).

THEOREM 24.2 (Parabolic maximum principle —weak form): Let $\Omega \subset \mathbb{R}^n$ be a non-empty open subset with $\overline{\Omega}$ compact. For $T > 0$ we define the **parabolic boundary**

$$\partial_{\mathrm{P}}\Omega_T^+ := [\overline{\Omega} \times \{0\}] \cup (\partial\Omega \times (0,T]) . \tag{24.1}$$

Notice that $\partial_{\mathrm{P}}\Omega_T^+ \subset \partial\overline{\Omega_T^+}$. Let $u \in C^2(\Omega_T^+) \cap C^0(\overline{\Omega_T^+})$.

1. if u is subcaloric in Ω_T^+ then

$$\max_{\overline{\Omega_T^+}} u = \max_{\partial_{\mathrm{P}}\Omega_T^+} u ; \tag{24.2}$$

2. if u is caloric in Ω_T^+ then

$$\max_{\overline{\Omega_T^+}} u = \max_{\partial_{\mathrm{P}}\Omega_T^+} u , \qquad \min_{\overline{\Omega_T^+}} u = \min_{\partial_{\mathrm{P}}\Omega_T^+} u , \qquad \max_{\overline{\Omega_T^+}} |u| = \max_{\partial_{\mathrm{P}}\Omega_T^+} |u| . \tag{24.3}$$

Proof.

$\boxed{1}$ Let $u \in C^2(\Omega_T^+) \cap C^0(\overline{\Omega}_T^+)$ be subcaloric. Notice that, since $u \in C^0(\overline{\Omega_T^+})$ and since $\overline{\Omega_T^+}$ and $\partial_{\mathrm{P}}\Omega_T^+$ are compact, then $\max_{\overline{\Omega_T^+}} u$, $\max_{\partial_{\mathrm{P}}\Omega_T^+} u$ exist and are finite. Moreover $\partial_{\mathrm{P}}\Omega_T^+ \subset \overline{\Omega_T^+}$, therefore,

$$\max_{\partial_{\mathrm{P}}\Omega_T^+} u \leqslant \max_{\overline{\Omega_T^+}} u .$$

It remains to show the opposite inclusion.

At first, let us assume a stronger condition, namely

$$\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \Delta_x u < 0\,. \tag{24.4}$$

Let $\varepsilon > 0$ so that $T - \varepsilon > 0$ and consider on $\Omega^+_{T-\varepsilon}$. We now prove that

$$\max_{\overline{\Omega^+_{T-\varepsilon}}} u = \max_{\partial_{\mathrm{P}}\Omega^+_{T-\varepsilon}} u\,. \tag{24.5}$$

Indeed, let $(x_0, t_0) \in \overline{\Omega^+_{T-\varepsilon}}$ be such that $u(x_0, t_0) := \max_{\overline{\Omega^+_{T-\varepsilon}}} u$. If $(x_0, t_0) \in \partial_{\mathrm{P}}\Omega^+_{T-\varepsilon}$ we are done, otherwise one of the following cases occur:

(a) $(x_0, t_0) \in \Omega^+_{T-\varepsilon}$: In this case we have

$$\frac{1}{\varsigma^2}\frac{\partial u}{\partial t}(x_0, t_0) = 0\,, \qquad \nabla_x u(x_0, t_0) = 0\,.$$

Moreover, $u(x_0, t_0)$ is a maximum for the function $\Omega \ni x \mapsto u(x, t_0) =: u_{t_0}(x)$, therefore, the Hessian matrix of u_{t_0} at x_0 is negative definite. Notice that this implies in particular $\Delta_x u_{t_0}(x_0) = \Delta_x u(x_0, t_0) \leqslant 0$. Overall we have

$$0 > \frac{1}{\varsigma^2}\cancel{\frac{\partial u}{\partial t}}(x_0, t_0) - \Delta_x u(x_0, t_0) \geqslant 0\,,$$

a contradiction. ☠

(b) $t_0 = T - \varepsilon$: In this case $x_0 \notin \partial\Omega$ —otherwise $(x_0, t_0) \in \partial_{\mathrm{P}}\Omega^+_{T-\varepsilon}$— and therefore $\nabla_x u(x_0) = 0$ together with $\Delta_x u(x_0, t_0) \leqslant 0$. Moreover we also have

$$\frac{\partial u}{\partial t}(x_0, t_0) \geqslant 0\,.$$

Indeed, by contradiction, if $\frac{\partial u}{\partial t}(x_0, t_0) < 0$ then by continuity there exists $\delta > 0$ such that $\frac{\partial u}{\partial t}(x_0, t) < 0$ for all $t \in (t_0 - \delta, t_0]$. However, this would imply that $u(x_0, t_0) < u(x_0, t_0 - \delta)$ a contradiction ☠.

Overall we have

$$0 > \frac{1}{\varsigma^2}\frac{\partial u}{\partial t}(x_0, t_0) - \Delta_x u(x_0, t_0) \geqslant 0\,,$$

a contradiction. ☠

Thus we have proved $\max_{\overline{\Omega^+_{T-\varepsilon}}} u = \max_{\partial_{\mathrm{P}}\Omega^+_{T-\varepsilon}} u$. Since for all $(x,t) \in \Omega^+_T$ there exists $\varepsilon > 0$ such that $(x,t) \in \Omega^+_{T-\varepsilon}$ we find

$$u(x,t) \leqslant \max_{\overline{\Omega^+_{T-\varepsilon}}} u = \max_{\partial_{\mathrm{P}}\Omega^+_{T-\varepsilon}} u \leqslant \max_{\partial_{\mathrm{P}}\Omega^+_T} u\,.$$

Since $u \in C^0(\overline{\Omega^+_T})$ we also have

$$u(x,T) = \lim_{t\to T^-} u(x,t) \leqslant \max_{\partial_{\mathrm{P}}\Omega^+_T} u\,,$$

which thus proves (24.2) within the assumption (24.4).

In the general case we consider $\widetilde{u}(x,t) = u(x,t) - \varepsilon t$. Then $\widetilde{u} \in C^2(\Omega^+_T) \cap C^0(\overline{\Omega^+_T})$, moreover,

$$\frac{1}{\varsigma^2}\frac{\partial \widetilde{u}}{\partial t} - \Delta_x \widetilde{u} \leqslant -\frac{\varepsilon}{\varsigma^2} < 0\,.$$

Thus, Equation (24.2) applies to $\widetilde{u}$, moreover,

$$\begin{aligned}
\max_{\overline{\Omega^+_T}} u &\leqslant \max_{\overline{\Omega^+_T}} \widetilde{u} + \varepsilon T && u = \widetilde{u} + \varepsilon t \\
&= \max_{\partial_{\mathrm{P}}\Omega^+_T} \widetilde{u} + \varepsilon T && \text{Eq. (24.2)} \\
&\leqslant \max_{\partial_{\mathrm{P}}\Omega^+_T} u + \varepsilon T && \widetilde{u} = u - \varepsilon t \\
&\xrightarrow[\varepsilon\to 0^+]{} \max_{\partial_{\mathrm{P}}\Omega^+_T} u\,.
\end{aligned}$$

proving (24.2).

$\boxed{2}$ Let $u \in C^2(\Omega^+_T) \cap C^0(\overline{\Omega}^+_T)$ be caloric. Then the first two equalities in (24.3) are nothing but (24.2) for u and $-u$ respectively. The last identity is a consequence of $|u| = \max\{u, -u\}$ together with the chain of identities

$$\begin{aligned}
\max_{\overline{\Omega^+_T}} |u| = \max_{\overline{\Omega^+_T}} \max\{u,-u\} &= \max\left\{ \max_{\overline{\Omega^+_T}} u, \max_{\overline{\Omega^+_T}} -u \right\} \\
&= \max\left\{ \max_{\partial_{\mathrm{P}}\Omega^+_T} u, \max_{\partial_{\mathrm{P}}\Omega^+_T} -u \right\} = \max_{\partial_{\mathrm{P}}\Omega^+_T} \max\{u,-u\} = \max_{\partial_{\mathrm{P}}\Omega^+_T} |u|\,.
\end{aligned}$$

□

Out of Theorem 24.2 we have the following uniqueness result for problem (22.2) with Dirichlet boundary conditions.

Corollary 24.3: There exists at most one solution $u \in C^2(\Omega_T^+) \cap C^0(\overline{\Omega_T^+})$ to the forward initial-value problem (22.2) with Dirichlet boundary conditions —*i.e.* $\theta = 0$ in (22.2c).

Proof. Let $u, u' \in C^2(\Omega_T^+) \cap C^0(\overline{\Omega_T^+})$ be two solutions to (22.2) with the same data u_0, f, u_{BC}. Then $v := u - u'$ is a solution to the same problem with vanishing data. In particular Theorem 24.2 leads to

$$\max_{\overline{\Omega_T^+}} |v| = \max_{\partial_{\mathrm{P}}\Omega_T^+} |v| = 0\,,$$

because $v = 0$ vanishes on $\partial\Omega \times [0, T]$ as well as on $\Omega \times \{0\}$. Therefore $u = u'$. □

Remark 24.4: There exists a result analogous of Theorem 24.2 for unbounded domains: We refer to [11, Thm. 8.4] for further details.

25 Parabolic PDEs: Initial-value problem on $\mathbb{S}^1$; Fourier's wine cellar problem

Initial-value problem on $\mathbb{S}^1$

In this section we will investigate the forward initial-value problem for the heat equation (22.1) on $\mathbb{S}^1$. This will lead to an existence and uniqueness result which is very similar to Theorem 19.9, *cf.* Section 19.

DEFINITION 25.1: The forward initial-value problem on $\mathbb{S}^1_R$, $R > 0$, for the heat equation (1.7) consists of determining $u \in C^2(\mathbb{S}^1_R \times (0,T)) \cap C^0(\mathbb{S}^1_R \times [0,T])$ such that

$$\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \Delta_{\mathbb{S}^1_R} u = f\,, \qquad \forall (x,t) \in \mathbb{S}^1_R \times (0,T) \tag{25.1a}$$

$$u(x,0) = u_0(x) \qquad \forall x \in \mathbb{S}^1_R\,, \tag{25.1b}$$

where $f \in C^0(\mathbb{S}^1_R \times (0,T))$ and $u_0 \in C^0(\mathbb{S}^1_R)$.

REMARK 25.2: Similarly to what we discussed in Section 19, the isomorphism $C^k(\mathbb{S}^1_R) \simeq C^k_\pi(\mathbb{R})$ allows to formulate problem (25.1) as follows: We look for $u \in C^2_\pi(\mathbb{R} \times (0,T)) \cap C^0_\pi(\mathbb{R} \times [0,T])$ such that

$$\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \frac{1}{R^2}\frac{\partial^2 u}{\partial \theta^2} = f\,, \qquad \forall (\theta,t) \in \mathbb{R} \times (0,T) \tag{25.2a}$$

$$u(\theta,0) = u_0(\theta) \qquad \forall \theta \in \mathbb{R}\,, \tag{25.2b}$$

$$u(\theta,t) = u(\theta+2\pi,t)\,, \qquad \frac{\partial u}{\partial \theta}(\theta,t) = \frac{\partial u}{\partial \theta}(\theta+2\pi,t)\,, \qquad \forall (\theta,t) \in \mathbb{R} \times (0,T)\,, \tag{25.2c}$$

where $f \in C^0_\pi(\mathbb{R} \times (0,T))$ and $u_0 \in C^0_\pi(\mathbb{R})$.

The following result will prove existence and uniqueness of (25.1) under some regularity assumptions on the data u_0 and f. In fact, for the sake of simplicity we will treat the case $f = 0$ although non-vanishing source terms can be included.

THEOREM 25.3: If u is a solution to the problem (25.1) (with $f = 0$) then

$$u(\theta,t) = \sum_{n\in\mathbb{Z}} \widehat{u}_0(n) e^{-\varsigma_n^2 t} \frac{e^{in\theta}}{\sqrt{2\pi}}\,, \qquad \varsigma_n := \frac{|n|\varsigma}{R}\,. \tag{25.3}$$

In particular, the problem (25.1) has at most one solution. Moreover, if $u_0 \in C^0(\mathbb{S}^1_R)$ is C^1-piecewise, then Equation (19.16) defines a solution to problem (25.1).

Proof.

$\boxed{!}$ We prove that the solution to (25.2) is unique. As a matter of fact, for this part of the proof we will only use the assumption $u_0 \in C^0(\mathbb{S}^1_R)$. Let u be a solution to (25.2). Since $u \in C^0(\mathbb{S}^1_R \times [0,T])$, then $u(\ ,t) \in L^2(\mathbb{S}^1_R)$ for all $t \in [0,T]$. Theorem 19.4 implies that

$$u(\theta,t) = \sum_{n\in\mathbb{Z}} \widehat{u}(n,t)\frac{e^{in\theta}}{\sqrt{2\pi}} \qquad \widehat{u}(n,t) = \frac{1}{\sqrt{2\pi}}\int_0^{2\pi} u(x,t)e^{-in\theta}\mathrm{d}x\,,$$

where the series converges in the $L^2(\mathbb{S}^1_R)$-norm for all $t \in (0,T)$. Notice that the coefficients $\{\widehat{u}(n,t)\}_{n\in\mathbb{Z}}$ determine u uniquely on account of Theorem 19.4. Indeed, if $u, u' \in C^0(\mathbb{S}^1_R \times [0,T])$ are such that $\widehat{u}(n,t) = \widehat{u'}(n,t)$ for all $t \in [0,T]$ and $n \in \mathbb{Z}$, then $u(\ ,t) = u'(\ ,t)$ for all $t \in [0,T]$ and almost all $x \in \mathbb{S}^1_R$. Since $u, u' \in C^0(\mathbb{S}^1_R \times [0,T])$ this implies that $u = u'$ everywhere on $\mathbb{S}^1_R \times [0,T]$.

We will prove that $\{\widehat{u}(n,t)\}_{n\in\mathbb{Z}}$ are uniquely determined by (25.1a) and (25.1b). To begin with we observe that, since $u \in C^2(\mathbb{S}^1_R \times (0,T))$ then for all $n \in \mathbb{Z}$ we have $\widehat{u}(n,\cdot) \in C^1((0,T)) \cap C^0([0,T])$ as a consequence of Corollary 9.5. To wit, for all $t_0 \in (0,T)$ and $t \in [t_0-\delta, t_0+\delta] \subset (0,T)$, $\delta > 0$, we have

$$\left|\frac{\partial^k u}{\partial t^k}(\theta,t)e^{-in\theta}\right| \leqslant \sup_{\substack{t\in[t_0-\delta,t_0+\delta]\\ \theta\in\mathbb{R}}} \left|\frac{\partial^k u}{\partial t^k}(\theta,t)\right|, \qquad k \in \{0,1\}\,,$$

therefore the assumption ii is fulfilled and we can apply Corollary 9.5 to conclude that $\widehat{u}(n,\ ) \in C^1(0,T)$. Moreover, the estimate for $k=0$ holds also for $t=0$ and $t=T$, thus, $\widehat{u}(n,\cdot) \in C^0([0,T])$. In particular we have

$$\begin{aligned}
\widehat{u}(n,0) &= \frac{1}{\sqrt{2\pi}}\int_0^{2\pi} u_0(\theta)e^{-in\theta}\mathrm{d}\theta = \widehat{u}_0(n)\,, && \text{Thm. 9.2 + Eq. (25.1b)}\\
\frac{1}{\varsigma^2}\frac{\mathrm{d}\widehat{u}}{\mathrm{d}t}(n,t) &= \frac{1}{\sqrt{2\pi}}\int_0^{2\pi} \frac{1}{\varsigma^2}\frac{\partial u}{\partial t}e^{-in\theta}\mathrm{d}\theta && \text{Cor. 9.5}\\
&= \frac{1}{\sqrt{2\pi}}\int_0^{2\pi} \frac{1}{R^2}\frac{\partial^2 u}{\partial\theta^2}e^{-in\theta}\mathrm{d}\theta && \text{Eq. (25.1a)}\\
&= -\frac{n^2}{R^2}\widehat{u}(n,t) && \text{Prop. 19.6}\,.
\end{aligned}$$

where we used the fact that, the Fourier coefficients $(\widehat{u}_0(n))_{n\in\mathbb{Z}}$ are well-defined because $u_0 \in C^0(\mathbb{S}^1_R)$. Thus, we have

$$\left[\frac{\mathrm{d}}{\mathrm{d}t} + \varsigma_n^2\right]\widehat{u}(n,t) = 0\,, \qquad \widehat{u}(n,0) = \widehat{u}_0(n)\,,$$

which is explicitly solved by

$$\widehat{u}(n,t) = \widehat{u}_0(n)e^{-\varsigma_n^2 t}\,.$$

Thus $\{\widehat{u}(n,t)\}_{n\in\mathbb{Z}}$ are uniquely determined and so is u.

$\boxed{\exists}$ We now prove that, under the stated assumptions on u_0, Equation (25.3) defines a solution to (25.1). To this avail we observe that

$$|\widehat{u}(n,t)e^{in\theta}| \leqslant |\widehat{u}_0(n)| \in \ell^1(\mathbb{Z})\,,$$

where we used that $u_0 \in C^0(\mathbb{S}^1_R)$ is C^1-piecewise, therefore, $(\widehat{u}_0(n))_{n\in\mathbb{Z}} \in \ell^1(\mathbb{Z})$ because of Proposition 19.6. The above estimate implies that the Fourier series defining u is uniformly convergent on $\mathbb{S}^1_R \times [0,T]$. Moreover, $u \in C^0(\mathbb{S}^1_R \times [0,T])$ and $u(x,0) = u_0(x)$.

We now prove that $u \in C^\infty(\mathbb{S}^1_R \times (0,T))$. For that, let $a > 0$. Then for all $t \geqslant a$, $\theta \in \mathbb{R}$ and $k \in \mathbb{N} \cup \{0\}$ we have

$$|n^k \widehat{u}(n,t)e^{in\theta}| \leqslant |n|^k |\widehat{u}_0(n)| e^{-\varsigma_n^2 a} \in \ell^1(\mathbb{Z})\,.$$

Iterated application of Corollary 9.5 implies that $u \in C^\infty(\mathbb{S}^1_R \times (a,T))$.

To wit, let us prove that $u \in C^1(\mathbb{S}^1_R \times (a,T))$. To this avail we observe that, for all $t \in (a,T)$ and $\theta \in \mathbb{R}$

$$\left|\frac{\partial}{\partial\theta}\widehat{u}(n,t)e^{in\theta}\right| \leqslant |n||\widehat{u}_0(n)|e^{-\varsigma_n^2 a} \in \ell^1(\mathbb{Z})\,, \qquad \left|\frac{\partial}{\partial t}\widehat{u}(n,t)e^{in\theta}\right| \leqslant \varsigma_n^2|\widehat{u}_0(n)|e^{-\varsigma_n^2 a} \in \ell^1(\mathbb{Z})\,.$$

Thus assumption ii is fulfilled and we may apply Corollary 9.5 to conclude $u \in C^1(\mathbb{S}^1_R \times (0,T))$. Since $u \in C^\infty(\mathbb{S}^1_R \times (a,T))$ for all $a > 0$ we have $u \in C^\infty(\mathbb{S}^1_R \times (0,T))$. Moreover, by direct inspection u solves (25.1), concluding the proof.

□

Remark 25.4 (): Equation (25.3) can be written in a slightly different form. For that, let

$$g_t(\theta) := g(\theta,t) := \sum_{n\in\mathbb{Z}} e^{-\varsigma_n^2 t}\frac{e^{in\theta}}{\sqrt{2\pi}} \qquad \forall(\theta,t) \in \mathbb{R}\times\mathbb{R}_+\,. \tag{25.4}$$

Multiple applications of Corollary 9.5 prove that $g \in C^\infty(\mathbb{S}^1_R \times (0,T))$, moreover, g solves the heat equation (25.1a) without source. The polarized Parseval's identity (19.11) implies that Equation (25.3) can be written as

$$u(\theta,t) = \sum_{n\in\mathbb{Z}} \widehat{u}_0(n)e^{-\varsigma_n^2 t}\frac{e^{in\theta}}{\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi}}\int_0^{2\pi} g_t(\theta-\theta')u_0(\theta')\mathrm{d}\theta'\,,$$

where we observed that the Fourier coefficients of the function $\theta' \mapsto g_t(\theta-\theta')$ are $\widehat{g}_t(n)e^{in\theta}$. The function g_t is called **heat kernel with periodic boundary conditions**, *cf.* Remark 23.15.

Parabolic PDEs: Fourier's wine cellar problem

In this Section we will discuss a model firstly introduced by Fourier to address the wine cellar problem. The latter is the problem to determine the optimal depth for a wine cellar so that it stays cold during summer. This and other problems concerning heat propagation have been the main reason for Fourier to develop his heat's theory [4] which ultimately led to the heat equation (22.1). Moreover, in an attempt to solve the latter equation, Fourier introduced the series expansion we have encountered in Section 19.

We will now present the model of the wine cellar problem. The optimal depth where to place a wine cellar can be determined once we know the ground temperature as a function of time t and depth x. With a slight idealization we consider a 1D setting where the spatial variable $x \in \overline{\mathbb{R}_+}$ denotes the depth of the ground, so that $x = 0$ at the Earth's surface. Our goal is to determine the temperature $u(x,t)$ of the ground once we known the surface temperature $u(0,t) = u_0(t)$. A characteristic feature of this problem is that u_0 is assumed to be time-periodic, with period T. This is consistent with the fact that the surface temperature u_0 is season-dependent and periodic over 1 year. We shall require the same property for the ground temperature u, although we expect a delay to occur once we increase the depth x. Finally we also assume that u stays bounded in time and space as expected from the ground temperature —*cf.* Figure 14.

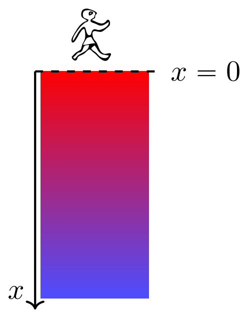


Figure 14: 2D picture of the wine cellar problem. Heat diffuses from the surface $x = 0$: The larger the depth, the lower the temperature.

Overall the problem is to determine $u \in C^2(\mathbb{R}_+ \times \mathbb{R}) \cap C^0_b(\overline{\mathbb{R}_+} \times \mathbb{R})$, such that

$$\frac{1}{\varsigma^2}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0\,, \qquad \forall (x,t) \in \mathbb{R}_+ \times \mathbb{R}\,, \tag{25.5a}$$

$$u(0,t) = u_0(t) \qquad \forall t \in \mathbb{R}\,, \tag{25.5b}$$

$$u(x,t) = u(x,t+T)\,, \qquad \forall (x,t) \in \overline{\mathbb{R}_+} \times \mathbb{R}\,. \tag{25.5c}$$

Remark 25.5:

(i) Notice that Equation (25.5c) implies that $u_0 \in C^0(\mathbb{R})$ is T-periodic, in particular $u_0(T) = u(0,T) = u(0,0) = u_0(0)$.

(ii) It is worth observing that any solution u to problem (25.5) has T-periodic time derivative on $\mathbb{R}_+$: Indeed, for all $x > 0$,

$$\frac{\partial u}{\partial t}(x,T) = \varsigma^2\frac{\partial^2 u}{\partial x^2}(x,T) = \varsigma^2\frac{\partial^2 u}{\partial x^2}(x,0) = \frac{\partial u}{\partial t}(x,0)\,,$$

where we used Equation (25.5c).

We have the following result.

THEOREM 25.6: Let $u_0 \in C^0(\mathbb{R})$ be T-periodic and C^1-piecewise. Then there exists a unique solution u to the wine problem (25.5). In particular the latter is given by

$$u(x,t) = \frac{1}{\sqrt{T}} \sum_{n\in\mathbb{Z}} \widehat{u}_0(n) e^{-\sqrt{\frac{\pi|n|}{\varsigma^2 T}}[1+i\,\mathrm{sgn}(n)]x} e^{i2\pi nt/T} . \tag{25.6}$$

Moreover, $u \in C^\infty(\overline{\mathbb{R}_+} \times \mathbb{R})$.

Proof. Let u be a solution to the wine cellar problem (25.5). By (25.5c) u is T-periodic in time, moreover, $u(x,\cdot) \in L^2(0,T)$, $x \geqslant 0$, because $u \in C^0(\overline{\mathbb{R}_+} \times \mathbb{R})$. It follows from Theorem 19.4 that

$$u(x,t) = \frac{1}{\sqrt{T}} \sum_{n\in\mathbb{Z}} \widehat{u}(x,n) e^{i2\pi nt/T} , \qquad \widehat{u}(x,n) := \frac{1}{\sqrt{T}} \int_0^T u(x,t) e^{-i2\pi nt/T} \mathrm{d}t ,$$

where, for all $x \geqslant 0$, the series converges in the L^2-norm of $(0,T)$. The Fourier coefficients $\{\widehat{u}(x,n)\}_{n\in\mathbb{Z}}$ determine uniquely u. Indeed, if $u, u' \in C^0(\overline{\mathbb{R}_+} \times \mathbb{R})$ are such that $\widehat{u}(x,n) = \widehat{u'}(x,n)$ for all $x \geqslant 0$ and $n \in \mathbb{Z}$, then $u(x,\cdot) = u'(x,\cdot)$ in the sense of $L^2(0,T)$, namely, $u(x,t) = u'(x,t)$ for all $x \geqslant 0$ and almost all $t \in (0,T)$. However, $u, u' \in C^0(\overline{\mathbb{R}_+} \times \mathbb{R})$ so that $u = u'$ everywhere on $\overline{\mathbb{R}_+} \times \mathbb{R}$.

Corollary 9.5 implies that $\widehat{u}(\cdot,n) \in C^2(\mathbb{R}_+) \cap C^0_b(\overline{\mathbb{R}_+})$ together with

$$\begin{aligned}
\widehat{u}(0,n) &= \frac{1}{\sqrt{T}} \int_0^T u_0(t) e^{-i2\pi nt/T} \mathrm{d}t = \widehat{u}_0(n) , && \text{Thm. 9.2 + Eq. (25.5b)} \\
\frac{\mathrm{d}^2 \widehat{u}}{\mathrm{d}x^2}(x,n) &= \frac{1}{\sqrt{T}} \int_0^T \frac{\partial^2 u}{\partial x^2} e^{-i2\pi nt/T} \mathrm{d}t && \text{Cor. 9.5} \\
&= \frac{1}{\sqrt{T}} \int_0^T \frac{1}{\varsigma^2} \frac{\partial u}{\partial t} e^{-i2\pi nt/T} \mathrm{d}t && \text{Eq. (25.5a)} \\
&= i \frac{2\pi n}{\varsigma^2 T} \frac{1}{\sqrt{T}} \int_0^T u(x,t) e^{-i2\pi nt/T} \mathrm{d}t && \text{Prop. 19.6} \\
&= i \frac{2\pi n}{\varsigma^2 T} \widehat{u}(x,n) .
\end{aligned}$$

It follows that $\{\widehat{u}(x,n)\}_{n\in\mathbb{Z}}$ fulfils

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} \widehat{u}(x,n) = i \frac{2\pi n}{\varsigma^2 T} \widehat{u}(x,n) , \qquad \widehat{u}(0,n) = \widehat{u}_0(n) . \tag{25.7}$$

Moreover, the boundedness requirement $u \in C^0_b(\overline{\mathbb{R}_+} \times \mathbb{R})$ implies that $\widehat{u}(x,n)$ is bounded for all $n \in \mathbb{Z}$. For $n = 0$ the unique bounded solution to (25.7) is $\widehat{u}(x,0) = \widehat{u}_0(0)$. For $n \neq 0$ we have

$$\widehat{u}(x,n) = A_n e^{z_n x} + B_n e^{-z_n x}\,,$$

where $z_n \in \mathbb{C}$ is defined by

$$\begin{aligned} z_n^2 &= i\frac{2\pi n}{\varsigma^2 T} = \frac{2\pi|n|}{\varsigma^2 T} e^{i\pi\,\mathrm{sgn}(n)/2} \qquad \Re z_n \geqslant 0\,,\\ \Rightarrow z_n &= \sqrt{\frac{2\pi|n|}{\varsigma^2 T}} e^{i\pi\,\mathrm{sgn}(n)/4} = \sqrt{\frac{\pi|n|}{\varsigma^2 T}}[1 + i\,\mathrm{sgn}(n)]\,. \end{aligned}$$

where $\mathrm{sgn}(n)$ denotes the sign of $n \in \mathbb{Z}\backslash\{0\}$. The boundedness condition $\widehat{u}(\cdot,n) \in C^0_b(\overline{\mathbb{R}_+})$ together with $\widehat{u}(0,n) = \widehat{u}_0(n)$ leads to

$$\widehat{u}(x,n) = \widehat{u}_0(n)\exp\left[-\sqrt{\frac{\pi|n|}{\varsigma^2 T}}[1 + i\,\mathrm{sgn}(n)]x\right].$$

The latter formula determines uniquely the Fourier coefficients of u: This implies that there is at most one solution to (25.5). Moreover such solution, if it exists, is given by (25.6)

Finally we have to prove that Equation (25.6) defines a solution to (25.5). By direct inspection we find

$$|\widehat{u}(x,n)| \leqslant |\widehat{u}_0(n)| \in \ell^1(\mathbb{Z})\,,$$

where we used Proposition 19.6 and the fact that u_0 is C^1-piecewise by assumption. This implies that Equation (25.6) defines a continuous and bounded function $u \in C^0(\overline{\mathbb{R}_+} \times \mathbb{R})$. Moreover, Theorem 9.2 entails that $u(0,t) = u_0(t)$.

Finally $u \in C^2(\mathbb{R}_+ \times \mathbb{R})$, in fact, $u \in C^\infty(\mathbb{R}_+ \times \mathbb{R})$. Indeed let $a > 0$: we may prove that $u \in C^\infty((a,+\infty) \times \mathbb{R})$ by observing that, for all $k \in \mathbb{N}$ and $x > a$,

$$|n^k \widehat{u}(x,n) e^{i2\pi nt/T}| \leqslant |n^k \widehat{u}_0(n)| e^{-\sqrt{\frac{\pi|n|}{\varsigma^2 T}}a} \in \ell^1(\mathbb{Z})\,.$$

An iterated application of Corollary 9.5 implies that $u \in C^\infty((a,+\infty) \times \mathbb{R})$. To wit, let us prove that $u \in C^1((a,+\infty) \times \mathbb{R})$. To this avail we observe that

$$\begin{aligned} \left|\frac{\partial}{\partial t}\widehat{u}(x,n)e^{i2\pi nt/T}\right| &= \frac{2\pi|n|}{T}|\widehat{u}(n,x)| \leqslant \frac{2\pi|n|}{T}|\widehat{u}_0(n)|e^{-\sqrt{\frac{\pi|n|}{\varsigma^2 T}}a} \in \ell^1(\mathbb{Z})\\ \left|\frac{\partial}{\partial x}\widehat{u}(x,n)e^{i2\pi nt/T}\right| &= \sqrt{\frac{\pi|n|}{\varsigma^2 T}}|\widehat{u}(x,n)| \leqslant \sqrt{\frac{\pi|n|}{\varsigma^2 T}}|\widehat{u}_0(n)|e^{-\sqrt{\frac{\pi|n|}{\varsigma^2 T}}a} \in \ell^1(\mathbb{Z})\,. \end{aligned}$$

Since the latter estimates are uniform in $t \in \mathbb{R}$ and $x \in (a,+\infty)$, Corollary 9.5 applies and we conclude that $u \in C^1((a,+\infty) \times \mathbb{R})$, moreover

$$\frac{\partial u}{\partial t} = \sum_{n\in\mathbb{Z}} \frac{i2\pi n}{T}\widehat{u}(x,n)e^{i2\pi nt/T}\,, \qquad \frac{\partial u}{\partial x} = -\sum_{n\in\mathbb{Z}} \sqrt{\frac{\pi|n|}{\varsigma^2 T}}(1 + i\,\mathrm{sgn}(n))\widehat{u}(x,n)e^{i2\pi nt/T}\,.$$

A similar argument applies for higher order derivatives, showing $u \in C^\infty((a,+\infty)\times\mathbb{R})$. Since this holds true for all $a>0$ we have $u \in C^\infty(\mathbb{R}_+\times\mathbb{R})$. Finally, Equation (25.5a) is a direct consequence of Corollary 9.5 together with (25.7). □

Remark 25.7: Notice that the solution u to (25.5) is actually way more regular than expected within the domain $\mathbb{R}_+\times\mathbb{R}$. This is an example of the regularizing behaviour of the heat equation we mentioned so far and shares similarities with the Poisson equation (7.1), *cf.* Theorem 11.3. Notice that this regularity is not extended to the boundary $x=0$ unless the data have higher regularity, *cf.* [11, Thm. 8.1].

Since u_0 is real-valued, the solution u can be written as a series in sine and cosine with real coefficients, *cf.* Remark 19.5. Indeed we have

$$\begin{aligned}
u(x,t) &= \frac{1}{\sqrt{T}}\sum_{n\in\mathbb{Z}}\widehat{u}_0(n)e^{-\sqrt{\frac{\pi|n|}{\varsigma^2T}}[1+i\,\mathrm{sgn}(n)]x}e^{-i2\pi nt/T}\\
&= \frac{1}{\sqrt{T}}\widehat{u}_0(0)+\frac{2}{\sqrt{T}}\sum_{n\geqslant1}e^{-\sqrt{\frac{\pi|n|}{\varsigma^2T}}x}\Re\left[\widehat{u}_0(n)e^{i\left(\frac{2\pi n}{T}t-\sqrt{\frac{\pi n}{\varsigma^2T}}x\right)}\right] \qquad \widehat{u}_0(-n)=\overline{\widehat{u}_0(n)}\\
&= \frac{1}{\sqrt{T}}\widehat{u}_0(0)+\frac{2}{\sqrt{T}}\sum_{n\geqslant1}e^{-\sqrt{\frac{\pi|n|}{\varsigma^2T}}x}\Re[\widehat{u}_0(n)]\cos\left(\frac{2\pi n}{T}t-\sqrt{\frac{\pi n}{\varsigma^2T}}x\right)\\
&\quad -\frac{2}{\sqrt{T}}\sum_{n\geqslant1}e^{-\sqrt{\frac{\pi|n|}{\varsigma^2T}}x}\Im[\widehat{u}_0(n)]\sin\left(\frac{2\pi n}{T}t-\sqrt{\frac{\pi n}{\varsigma^2T}}x\right)\,.
\end{aligned}$$

The found solution have two main properties. To begin with, the temperature is damped by an exponential factor $\exp\Big[-\sqrt{\frac{\pi|n|}{\varsigma^2T}}x\Big]$. Considering $n=1$ for simplicity and inserting realistic values for ς and T one finds $\sqrt{\frac{\pi}{\varsigma^2T}}\approx\frac{2\pi}{9}\mathrm{m}^{-1}$: At a depth of $x=4.5$m this implies a damping of $\exp\Big[-\sqrt{\frac{\pi}{\varsigma^2T}}x\Big]\approx e^{-\pi}\approx 1/16$. This means that when the surface temperature is, say, 40°C then at 4.5m meter depth we have a temperature of approximately 2.5°C.

The second feature of u is that it is periodic in t (as we required in (25.5c)) but "delayed" by a factor $\sqrt{\frac{\pi n}{\varsigma^2T}}x$ which depends on the depth x. Considering $n=1$ for simplicity and inserting the previous values for ς, T and x leads to $\sqrt{\frac{\pi}{\varsigma^2T}}x\approx\pi$, that is, a delay of $T/2$, namely six months.

26 A first glimpse into non-linear PDEs: Burgers' equation

In this section we will investigate the effect of non linear terms in a partial differential equation. The presence of a non-linear contribution usually leads to a loss in the regularity of the resulting solution. In particular it may happen that the solution generated by a regular initial datum develops a discontinuity in finite time. Making sense of this phenomenon leads to the notion of weak solution. Neede less to say, the topic is vast and we will content ourself to scratch the surface by focusing on Burgers' equation (1.11) and by considering explicit examples. The latter will provide a heuristic understanding of the more relevant ideas: We refer to [3, §3.4] for a detailed discussion and further result.

A function $u \in C^1(\mathbb{R} \times \mathbb{R}_+)$ is called a solution to the **Burgers' equation** if

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0 \qquad \forall (x,t) \in \mathbb{R} \times \mathbb{R}_+ \,. \tag{26.1}$$

We refer to Section 1 for a physical derivation of this equation. For later convenience we observe that u has the dimension of a velocity, *i.e.* length$\times$time^{-1}. (In what follows we will work with dimensionless quantities, *i.e.* x, t, u will not have a physical dimension, *cf.* Equation (26.8). A dimension-full solution can be obtained by rescaling, *i.e.* consiering as $\bar{u}(\bar{x}, \bar{t}) := (L/T)u(\bar{x}/L, \bar{t}/T)$, where L, T are a arbitrary but fixed length and time scales.) From a physical point of view u may model the velocity of a fluid confined in a thin 1D pipe, so that $u(x,t)$ is the velocity of the particles of the fluid passing through $x \in \mathbb{R}$ at time $t \in \mathbb{R}_+$ —in particular if $u > 0$ then the particles are moving in the direction of increasing x.

The initial-value problem for the Burgers equation consists of looking for a solution u to Equation (26.1) together with an initial condition

$$u(x,0) = u_0(x) \qquad \forall x \in \mathbb{R}\,, \tag{26.2}$$

where $u_0 \colon \mathbb{R} \to \mathbb{R}$ is assigned. Here we are have been purposely vague on the regularity of the solution u as well as of the initial datum u_0. Classically, one may expect to look for $u \in C^1(\mathbb{R} \times \mathbb{R}_+) \cap C^0(\mathbb{R} \times \overline{\mathbb{R}_+})$ with $u_0 \in C^0(\mathbb{R})$. However, Equation (1.11) is non-linear and this implies that a regular solution u may develop discontinuities in finite times. For this reason we will introduce a weak formulation for the initial-value problem of interest.

Definition 26.1: A function $u \in L^1_{\mathrm{LOC}}(\mathbb{R} \times \mathbb{R}_+)$ is called a **weak solution** to the initial-value problem for the Burgers' equation (26.1) with initial datum $u_0 \in L^1_{\mathrm{LOC}}(\mathbb{R})$ if

$$\int_{\mathbb{R}\times\mathbb{R}_+} \left[u\frac{\partial \tau}{\partial t} + \frac{u^2}{2}\frac{\partial \tau}{\partial x}\right]\mathrm{d}x\mathrm{d}t + \int_{\mathbb{R}} \tau_0 u_0 \mathrm{d}x = 0 \qquad \forall \tau \in C^\infty_c(\mathbb{R} \times \overline{\mathbb{R}_+})\,, \tag{26.3}$$

where we denoted by $\boxed{\tau_0} \in C^\infty_c(\mathbb{R})$ the restriction $\tau_0(x) := \tau(x,0)$.

REMARK 26.2: Let assume that $u_0 \in C^0(\mathbb{R})$ and let $u \in C^1(\mathbb{R}\times\mathbb{R}_+) \cap C^0(\mathbb{R}\times\overline{\mathbb{R}_+})$ be a weak solution to (26.3). Then integration by parts leads to

$$0 = -\int_{\mathbb{R}\times\mathbb{R}_+} \tau\left[\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x}\right]\mathrm{d}x\mathrm{d}t + \int_{\mathbb{R}} \tau_0(u_0 - u)\mathrm{d}x\,.$$

The arbitrariness of τ implies that u solves the Burgers equation (1.11), moreover, it satisfies the initial condition (26.2). Therefore, regular weak-solutions as per Definition 26.1 are solutions to the “classical” initial-value problem.

In what follows we will not solve problem (26.3) in full generality. Instead we will focus on particular cases for the initial datum u_0, finding explicit solutions and providing relevant examples of the physical phenomena described by Burgers' equation (26.1). In particular, we will see that u may suffer a loss of regularity in a finite time, a phenomenon proper of non-linear PDEs.

We now investigate problem (26.3) employing the method of characteristics. The latter is a general tool which is at disposal for any quasi-linear first order PDE and which simplifies greatly for the case Equation (26.1).

To begin with we consider a C^1 curve $\gamma\colon I \to \mathbb{R}\times\overline{\mathbb{R}_+}$, $\gamma(t) := (x(t), t)$, where $I \subseteq \mathbb{R}_+$ contains 0. Given a regular weak solution $u \in C^1(\mathbb{R}\times\mathbb{R}_+)$ to (26.3) and setting $\boxed{u_\gamma}(t) := u(\gamma(t)) = u(x(t), t)$ a direct inspection shows that

$$\dot{u}_\gamma = \frac{\partial u}{\partial t}\Big|_\gamma + \dot{x}\frac{\partial u}{\partial x}\Big|_\gamma\,.$$

The latter equation is particularly relevant if

$$\dot{x}(t) = u_\gamma(t)\,, \tag{26.4}$$

in which case we find that, on account of Equation (26.1), u_γ is constant, that is, u is constant along γ. This result is rather remarkable and for this reason curves γ abiding by Equation (26.4) are called **characteristic curves** (associated with the Burgers equation (26.1)). In fact, these characteristic curves are rather simple: Indeed, since u is constant along γ we find

$$\dot{x}(t) = u_\gamma(t) = u_\gamma(0) = u_0(x_0)\,,$$

where $x_0 = x(0)$. This implies in particular that characteristic curves are straight lines, moreover,

$$u(x_0 + u_0(x_0)t, t) = u_0(x_0)\,. \tag{26.5}$$

Equation (26.5) determines the solution u to (26.3) in terms of the initial datum u_0. In fact, u is determined only for those points (x, t) such that

$$x = x_0 + u_0(x_0)t\,, \tag{26.6}$$

for a unique $x_0 \in \mathbb{R}$, that is, only if x lies on a characteristic curve emanated from one (and only one!) point $x_0 \in \mathbb{R}$. The possibility of solving Equation (26.6) for all x, t is not granted and ultimately depends on the properties of u_0.

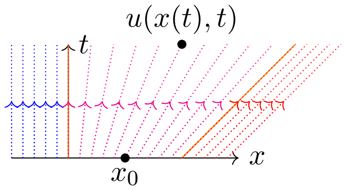


Figure 15: The stack of characteristic curves for the initial datum (26.7). Three colors are used to describe the three branches appearing in (26.8). Orange lines denote the curves where the weak solution (26.8) is not C^1.

Example 26.3: Let consider the initial datum

$$u_0(x) := \begin{cases} 0 & x < 0 \\ x & 0 \leqslant x \leqslant 1 \\ 1 & x > 1 \end{cases} \tag{26.7}$$

By direct inspection Equation (26.6) is uniquely solved for all x, t, *cf.* Figure 15. Equation (26.5) leads to

$$u(x,t) = \begin{cases} 0 & x < 0 \\ \frac{x}{1+t} & 0 \leqslant x \leqslant 1+t \\ 1 & x \geqslant 1+t \end{cases} , \tag{26.8}$$

which can be shown to define a weak solution of (26.3) by direct inspection. Notice that $u \in C^0(\mathbb{R} \times \overline{\mathbb{R}_+})$ is only C^1-piecewise, inheriting the mild irregularity of the initial datum (26.7).

We stress that the characteristic curve emanated from x_0 is a straight line with slope $u_0(x_0)$ which depends on x_0: This may lead to focusing of the characteristic curves, a phenomenon which spoils the possibility of defining a weak solution u.

Example 26.4: Let consider the initial datum

$$u_0(x) := \begin{cases} 1 & x < 0 \\ 1-x & 0 \leqslant x \leqslant 1 \\ 0 & x > 1 \end{cases} . \tag{26.9}$$

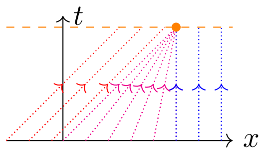


Figure 16: The stack of characteristic curves for the initial datum (26.9). Since the characteristic focuses starting from $t = 1$, the solution u is only determined for $t \in [0, 1)$.

The method of characteristic determines the weak solution u only for $t \in [0, 1)$, *cf.* Figure 16,in particular

$$u(x,t) = \begin{cases} 1 & x < t \\ \frac{1-x}{1-t} & t \leqslant x \leqslant 1 \quad 0 \leqslant t \leqslant 1 \\ 0 & x > 1 \end{cases} \quad \Rightarrow \quad u(x,1) = \begin{cases} 1 & x < 1 \\ 0 & x > 1 \end{cases} \tag{26.10}$$

We may provide a physical interpretation of the discontinuity at $(x,t) = (1,1)$. Indeed, the initial datum (26.9) describes a fluid which moves with different velocities. In particular for $x < 0$ the fluid is moving faster than for $x \in (0,1)$ or for $x > 0$, where particles are stationary. Thus, eventually the faster part of the fluid reaches the slower one, concentrating the fluid and creating a discontinuity. Notice that the same argument explains why the datum (26.7) of Example 26.3 does not lead to the formation of a discontinuity: In that case the faster portion of the fluid simply moves away from the slower one, stretching the resulting solution, *cf.* (26.8).

For the forthcoming discussion we will focus on Example 26.4. Therein the solution developed a discontinuity in finite time, leading to a solution which, at $t = 1$, is C^1 up to a point. From a physical point of view we may expect that, at later times, the developed discontinuity is preserved without any further loss in regularity. In other words, we expect the solution u to (26.3) to be C^1 up to a point for any fixed $t \geqslant 1$.

This motivates the quest for weak solutions to (26.3) with a particular form. Specifically let $\sigma\colon I \to \mathbb{R} \times \overline{\mathbb{R}_+}$, $\sigma(t) := (x_\sigma(t), t)$, be a C^1 curve. We look for a weak solution u to (26.3) with the properties that

$$u \in C^1(\overline{\Omega_\sigma^-}) \cap C^1(\overline{\Omega_\sigma^+}) \qquad \Omega_\sigma^\pm := \{(x,t) \in \mathbb{R} \times \mathbb{R}_+ \,|\, x > \pm x_\sigma(t)\}\,. \tag{26.11}$$

Thus, for all $t \in I$, $u(\ ,t) \in C^1([x_\sigma(t), +\infty)) \cap C^1((-\infty, x_\sigma(t)])$. The curve σ models the motion of the discontinuity associated to u.

By imposing (26.3), condition (26.11) implies that u solves Equation (26.1) within $\Omega_\sigma^+ \cup \Omega_\sigma^-$. For that, it suffices to consider (26.3) for $\tau \in C_c^\infty(\mathbb{R} \times \overline{\mathbb{R}_+})$ with $\operatorname{supp}(\tau) \cap \sigma(I) = \varnothing$ —*i.e.* for τ supported away from the discontinuity curve— and to apply Remark 26.2.

Let now $\tau \in C_c^\infty(\mathbb{R}\times\mathbb{R}_+)$ be such that $\mathrm{supp}(\tau)\cap\sigma(I)\neq\varnothing$. By direct inspection we have

$$0=\int_{\mathbb{R}\times\mathbb{R}_+}\left[u\frac{\partial\tau}{\partial t}+\frac{u^2}{2}\frac{\partial\tau}{\partial x}\right]\mathrm{d}x\mathrm{d}t=\int_{\Omega_\sigma^+}\left[u\frac{\partial\tau}{\partial t}+\frac{u^2}{2}\frac{\partial\tau}{\partial x}\right]\mathrm{d}x\mathrm{d}t+\int_{\Omega_\sigma^-}\left[u\frac{\partial\tau}{\partial t}+\frac{u^2}{2}\frac{\partial\tau}{\partial x}\right]\mathrm{d}x\mathrm{d}t\,.$$

At this stage we observe that $\Omega_\sigma^\pm$ are both orientable with C^1-boundary given by $\partial\Omega_\sigma^\pm=\sigma(I)$, though taken with opposite orientation. This implies that

$$\begin{aligned}\int_{\Omega_\sigma^-}\left[u\frac{\partial\tau}{\partial t}+\frac{u^2}{2}\frac{\partial\tau}{\partial x}\right]\mathrm{d}x\mathrm{d}t&=\int_{\Omega_\sigma^-}\left[\frac{\partial}{\partial t}(u\tau)+\frac{\partial}{\partial x}\Big(\frac{u^2}{2}\tau\Big)\right]\mathrm{d}x\mathrm{d}t && u|_{\Omega_\sigma^-}\text{ solves Eq. (26.1)}\\&=\int_\sigma\left[u_\sigma^-\nu^t+\frac{1}{2}(u^2)_\sigma^-\nu^x\right]\tau_\sigma\mathrm{d}\sigma && \text{Thm. (8.3)}\\&=\int_\sigma\frac{1}{\sqrt{1+\dot{x}_\sigma^2}}\left[-\dot{x}_\sigma u_\sigma^-+\frac{1}{2}(u^2)_\sigma^-\right]\tau_\sigma\mathrm{d}\sigma && \nu=\frac{1}{\sqrt{1+\dot{x}_\sigma^2}}(e_x-\dot{x}_\sigma e_t)\,.\end{aligned}$$

In the second equality we applied Gauss Theorem 8.1 to $X:=u\tau e_t+\frac{1}{2}u^2\tau e_t$ while $\boxed{u_\sigma^-}(t):=\lim_{x\to x_\sigma(t)^-}u(x,t)$ is the left value of u on σ. Similarly we have

$$\int_{\Omega_\sigma^+}\left[u\frac{\partial\tau}{\partial t}+\frac{u^2}{2}\frac{\partial\tau}{\partial x}\right]\mathrm{d}x\mathrm{d}t=-\int_\sigma\frac{1}{\sqrt{1+\dot{x}_\sigma^2}}\left[-\dot{x}_\sigma u_\sigma^++\frac{1}{2}(u^2)_\sigma^+\right]\tau_\sigma\mathrm{d}\sigma\,,$$

the minus sign being a consequence of the different (opposite) orientation of $\Omega_\sigma^\pm$. Overall we are left with

$$0=\int_{\mathbb{R}\times\mathbb{R}_+}\left[u\frac{\partial\tau}{\partial t}+\frac{u^2}{2}\frac{\partial\tau}{\partial x}\right]\mathrm{d}x\mathrm{d}t=\int_\sigma\frac{1}{\sqrt{1+\dot{x}_\sigma^2}}\left[[u]_\sigma\dot{x}_\sigma-\frac{1}{2}[u^2]_\sigma\right]\tau_\sigma\mathrm{d}\sigma\,,$$

where $\boxed{[u]_\sigma}:=u_\sigma^+-u_\sigma^-$ is the jump of u across σ and similarly $[u^2]_\sigma:=(u^2)_\sigma^+-(u^2)_\sigma^-$. The arbitrariness of τ leads to

$$[u]_\sigma\dot{x}_\sigma=\frac{1}{2}[u^2]_\sigma\,,\tag{26.12}$$

which is called **Rankine-Hugoniot condition**. The latter is a necessary condition for a function u abiding by condition (26.11) to solve problem (26.3). It is an extremely relevant condition as it relates the speed $\dot{x}_\sigma$ of the discontinuity with the jump of the solution u across the discontinuity itself. Solving (26.12) is hard because it is an ODE for x_σ which depends on u, the weak solution (26.3) we ought to determine by first computing x_σ. However, condition (26.12) may be used to construct ad hoc weak solutions of (26.3).

EXAMPLE 26.5: Let us consider once again Example 26.3. We wish to extend the solution (26.10) after $t\geqslant 1$ by means of the Rankine-Hugoniot condition (26.12). If we look for a solution of the form (26.11) then the method of characteristics implies $u|_{\Omega_\sigma^-}=1$ and $u|_{\Omega_\sigma^+}=0$. Thus,

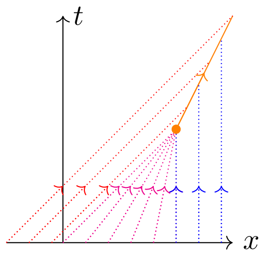


Figure 17: The weak solution (26.13) with initial datum (26.9). The orange corresponds to the discontinuity curve σ.

$[u]_\sigma = -1 = [u^2]_\sigma$, therefore, the Rankine-Hugoniot condition (26.12) boils down to the Cauchy problem

$$\dot{x}_\sigma(t) = 1/2 \qquad x_\sigma(1) = 1\,.$$

which identifies the curve $x_\sigma(t) = (1+t)/2$. Considering the characteristic curves associated to the initial datum (26.9) we end up with

$$u(x,t) = \begin{cases} 1 & x < t \\ \frac{1-x}{1-t} & t \leqslant x \leqslant 1 \quad 0 \leqslant t \leqslant 1 \\ 0 & x > 1 \end{cases} \qquad u(x,t) = \begin{cases} 1 & x < (1+t)/2 \\ 0 & x > (1+t)/2 \end{cases} t \geqslant 1 \tag{26.13}$$

By direct inspection Equation (26.13) defines a weak solution to (26.3) for all $t \geqslant 0$: For this it is enough to check that u solves (26.1) in the "regular" regions while (26.12) is fulfilled along σ.

From the physical point of view, the weak solution (26.13) describes the formation and the subsequent evolution of a discontinuity due to the interaction of a "fast" portion of fluid with the stationary one, *cf.* Equation (26.9) and Figure 17.

The Rankine-Hugoniot condition (26.12) is a necessary condition which may help with the identification of a weak solution of (26.3) in the presence of discontinuities. However, there are situations where the latter does not suffice, for example because (26.6) does not have any solution.

Example 26.6: Let consider (26.3) with initial datum

$$u_0(x) := \begin{cases} 0 & x < 0 \\ 1 & x > 1 \end{cases} . \tag{26.14}$$

In this case the method of characteristic fails in determining the solution u within the region $0 \leqslant x \leqslant t$ because Equation (26.6) cannot be solved for such (x,t), *cf.* Figure 18. This is due to the discontinuity of the initial datum.

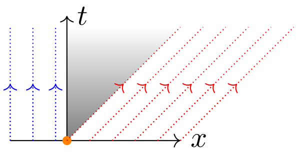


Figure 18: The stack of characteristic for the initial datum (26.14): The method of characteristics fails within the region $0 < x < t$.

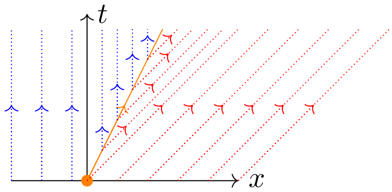


Figure 19: A weak solution to (26.3) with initial datum (26.14) with discontinuity curve $\sigma(t) := (t/2, t)$.

Notably, one may find two weak solutions to (26.3) with initial datum (26.14). The first one enjoys the form (26.11) with a discontinuity curve given by $\sigma(t) := (t, t/2)$, *cf.* Figure 19. Explicitly we have

$$u(x,t) = \begin{cases} 0 & x < t/2 \\ 1 & x > t/2 \end{cases} . \tag{26.15}$$

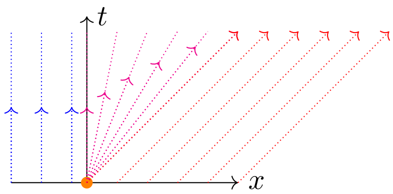


Figure 20: A continuous weak solution to (26.3) with initial datum (26.14).

The second solution is a continuous weak solution to (26.3), *cf.* Figure 20. In particular we have

$$u(x,t) := \begin{cases} 0 & x < 0 \\ x/t & 0 < x < t \\ 1 & x > t \end{cases} \tag{26.16}$$

Thus, the initial-value problem (26.3) has no unique solution for generic initial datum. To select a unique solution we need to add a physically motivated requirement to (26.3).

To introduce the latter we begin with a heuristic argument which is based on the physical interpretation of the Burger's equation (26.1). The initial datum (26.14) represents the velocity of a fluid which is divided in two pieces: a stationary part for $x < 0$; a "fast" part for $x > 0$. Within this situation is hard to imagine that the discontinuity in the initial datum could propagate any further in time (in fact, one may argue that the discontinuity appearing in (26.14) is already an idealization, a better model being a scaled version of the initial datum (26.7)). One should expect the "fast" component of the fluid to propagate away from the stationary part which would be left untouched. This is in fact what happens for the solution (26.16): The discontinuity disappears for $t > 0$ and it is substituted by a linear interpolation between the stationary and "fast" components of the fluid. Solutions of the form (26.16) are called **rarefaction waves**.

Conversely, solution (26.15) represents the propagation of the discontinuity, with the stationary part of the fluid following the "fast" one: This is not a physically reasonable behaviour.

From a mathematical point of view the weak solutions (26.15)-(26.16) can be distinguished with the help of the **entropy condition**, which for the particular case of Equation (26.1) reads

$$u_\sigma^- > \dot{x}_\sigma > u_\sigma^+ \,. \tag{26.17}$$

where $u_\sigma^\pm$ are the left/right value of the weak solution across any regular curve $\sigma(t) = (x_\sigma(t), t)$ such that $u \in C^1(\overline{\Omega_\sigma^-}) \cap C^1(\overline{\Omega_\sigma^+})$. In particular, solution (26.16) fulfils (26.17) while (26.15) does not. Weak solutions to (26.3) of the form (26.11) and fulfilling condition (26.17) are called **shock waves**.

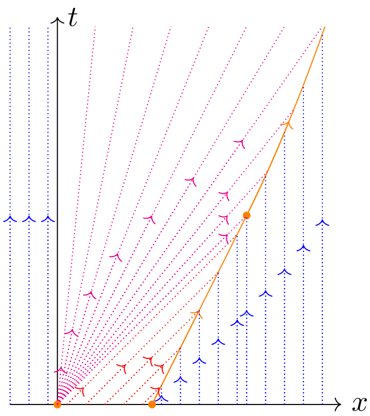


Figure 21: The unique weak solution to (26.3) with initial datum (26.18). A rarefaction wave originates from the discontinuity at $x = 0$, while a shock wave arises out of the discontinuity at $x = 1$. The rarefaction wave propagates faster than the shock wave and hits it at $(x, t) = (2, 2)$. After this collision a new discontinuity forms, with discontinuity curve $\sigma(t) = ((2t)^{1/2}, t)$.

EXAMPLE 26.7: We close this section by investigating (26.3) for an initial datum which recollects those already encountered in Examples 26.4-26.5-26.6. Specifically let

$$u_0(x) := \begin{cases} 0 & x < 0 \\ 1 & 0 < x < 1 \\ 0 & x > 1 \end{cases} . \tag{26.18}$$

We are interested in describing a weak solution u abiding by the entropy condition (26.17). We observe that the initial datum has two discontinuities identical to those of the previous examples. Specifically the discontinuity at $x = 0$ can be treated as in Example 26.6, leading to a rarefaction wave solution. Conversely, the discontinuity at $x = 1$ leads to a shock wave solution as in Example 26.5. This leads to the weak solution

$$u(x,t) = \begin{cases} 0 & x < 0 \\ x/t & 0 \leqslant x \leqslant t \\ 1 & t \leqslant x < 1 + t/2 \\ 0 & x > 1 + t/2 \end{cases} \qquad 0 \leqslant t \leqslant 2 \, . \tag{26.19}$$

It is important to observe that (26.19) makes sense only for $t \leqslant 2$. From a physical point of view this is natural, because the shock wave is "pushing" the stationary part of the fluid, therefore, it is slower than the rarefaction wave. At $t = 2$ the rarefaction wave reaches the shock wave at $x = 2$, leading to a new discontinuity, whose evolution can be determined by means of the Rankine-Hugoniot condition (26.12). Indeed, we may argue as follows: At $t = 2$ the third branch in (26.19) disappears. Thus, if a weak solution to (26.3) in the form (26.11) has to be found, the method of characteristic would lead to a discontinuity curve $\sigma(t) = (x_\sigma(t), t)$, $t \geqslant 2$, with $x_\sigma(2) = 2$ and such that u is a rarefaction wave on Ω_σ^- while it vanishes on Ω_σ^+. This would imply that

$$[u]_\sigma = -x_\sigma(t)/t \, , \qquad [u^2]_\sigma = -x_\sigma(t)^2/t^2 \, .$$

The Rankine-Hugoniot condition leads to the Cauchy problem

$$\dot{x}_\sigma(t) = \frac{x_\sigma(t)}{2t} \qquad x_\sigma(2) = 2 \, ,$$

whose unique solution is $x_\sigma(t) = (2t)^{1/2}$. We thus find

$$u(x,t) = \begin{cases} 0 & x < 0 \\ x/t & 0 \leqslant x < (2t)^{1/2} \\ 0 & x > (2t)^{1/2} \end{cases} \qquad t \geqslant 2 \, , \tag{26.20}$$

which also fulfils the entropy condition (26.17). From a physical point of view, after the rarefaction wave reaches the shock wave at $t = 2$, a new discontinuity is formed. The latter represents the interaction between the stationary part of the fluid and the rarefaction wave. As for the case of the shock wave, the rarefaction wave is pushing away the stationary part of the fluid but at a speed slower than the one of the shock wave. This can be inferred by observing that the speed of the discontinuity is slower than the one of the discontinuity associated to the shock wave: The reason behind this difference is related to the time decay behaviour of the rarefaction wave, *cf.* (26.16), which leads to a “weaker push” with respect to the more discontinuous (thus, “stronger”) shock wave.

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